UPSC CSE 2027 · GS Foundation

GEOGRAPHY

Geomorphology · Climatology · Oceanography · Indian geography · Human geography · Economic geography

Handwritten revision notes

Built from the Sarrthi IAS GS Foundation Geography handouts (pictures and the mind-map pages are in the textbook edition).
Bracket trees · diagrams · mnemonics · PYQ tags · current affairs
Right column and left binding margin are kept blank for your own use.

Personal study edition · for printing

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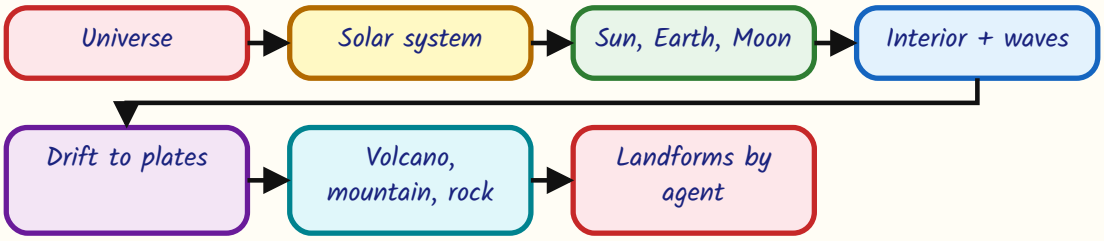
GEOMORPHOLOGY

PYQ Prelims 2025, 2024 ×3, 2023, 2022, 2019, 2018, 2015, 2014, 2013 ×2, 2012 ×2, 2011 + 4 with no year · Mains 2022 (primary rocks, 10m), 2017 (Juno mission, 10m)

✎ *From the Big Bang to the Sun, Earth, its interior, plates, rocks and every landform: 19 Prelims PYQs make this a HIGH chapter.*

At a glance: the Geomorphology route



Hot clusters

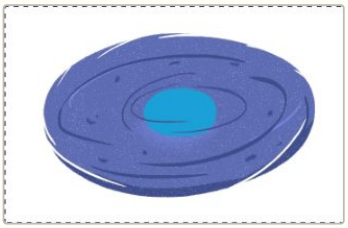
→ Earth–Sun geometry (pole star, day length, 21 June, time zones)
→ solar storm effects
→ P vs S waves
→ weathering by rain
→ drift evidences
→ volcanic products
→ chalk vs clay. No teacher ink: priority comes from handout PYQ counts only.

1 · UNIVERSE: ORIGIN THEORIES

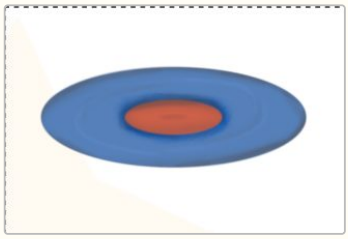
Understand it first: Every origin theory answers one question: how did a cloud of gas and dust turn into a star with planets round it? Read them as a chain where each new theory repairs the weak point of the one before it, ending with the **Big Bang** for the universe as a whole. For each, hold three things: who gave it, the idea in one line, and its drawback or proof.

Theory	By	Core idea	Drawback
Gaseous	Kant	Scattered primordial matter collided → heat → solid→liquid→gas = nebula; rotation → bulge; rings separated; residual centre = Sun, rings = planets; same process → moons	Source of primordial matter unexplained; collisions can't give rotation; how moons formed?
Nebular	Laplace, 1796 (modified Kant)	Faster spin → rings detach, each ring condenses into gas mass, cools → planets; remaining part = Sun	Source of nebula unexplained; Sun should spin fastest but doesn't

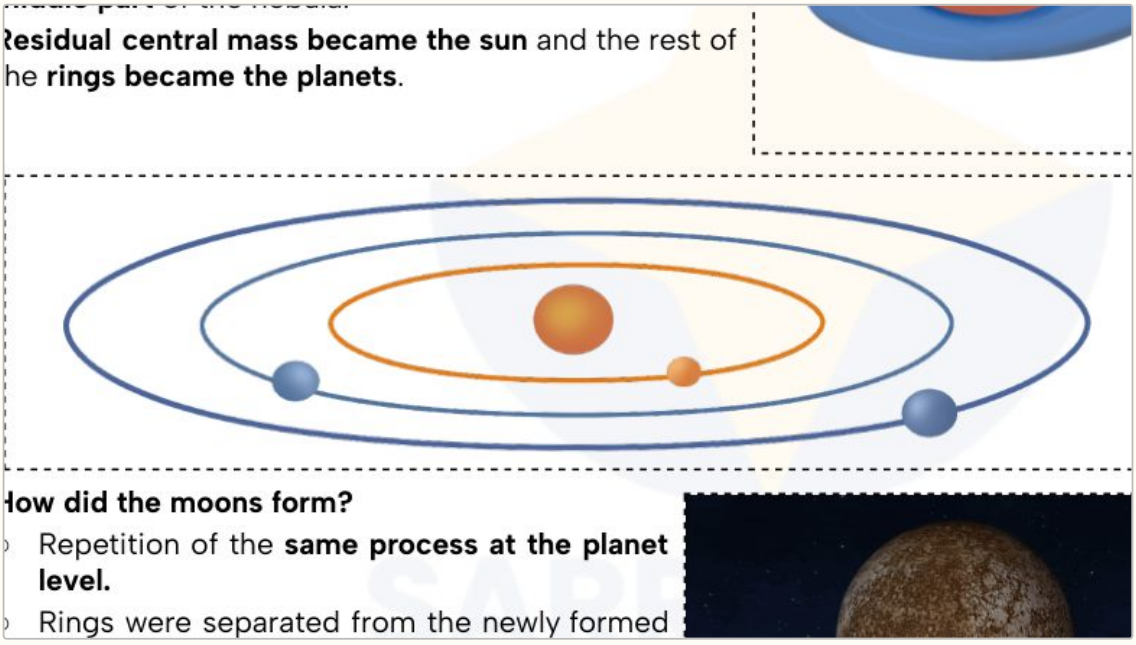
Planetesimal	Chamberlin–Moulton	Star passed close to young Sun → tidal bulges, solar prominences → ejected material; spiral arms; cooled → planetesimals + larger protoplanets → planets + moons; comets and asteroids = leftover debris	–
Accretion	Otto Schmidt 1944, Viktor Safronov 1969	Nebula → dust grains → boulders → planetesimals → planets (constant accretion)	–
Big Bang	Georges Lemaitre, 1920s	Universe began from a single primordial atom; violent explosion; dust scattered; denser regions pulled together over millions of years → gas clouds, stars, galaxies	Validated 1972



Gaseous Hypothesis · Given by Kant



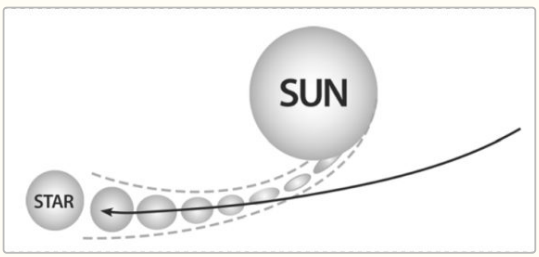
Gaseous Hypothesis · Collision resulted in the generation of heat.



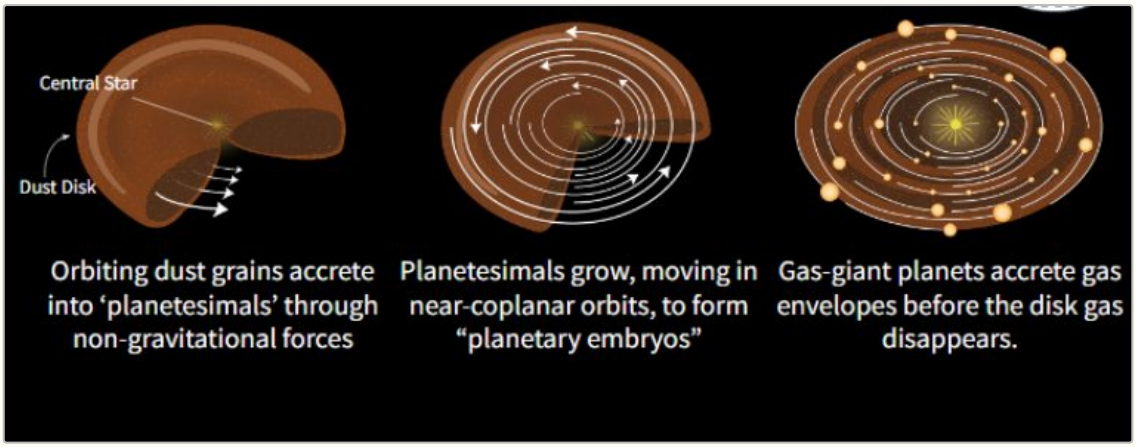
Gaseous Hypothesis · Collision resulted in the generation of heat.

Gaseous Hypothesis · How did the moons form?

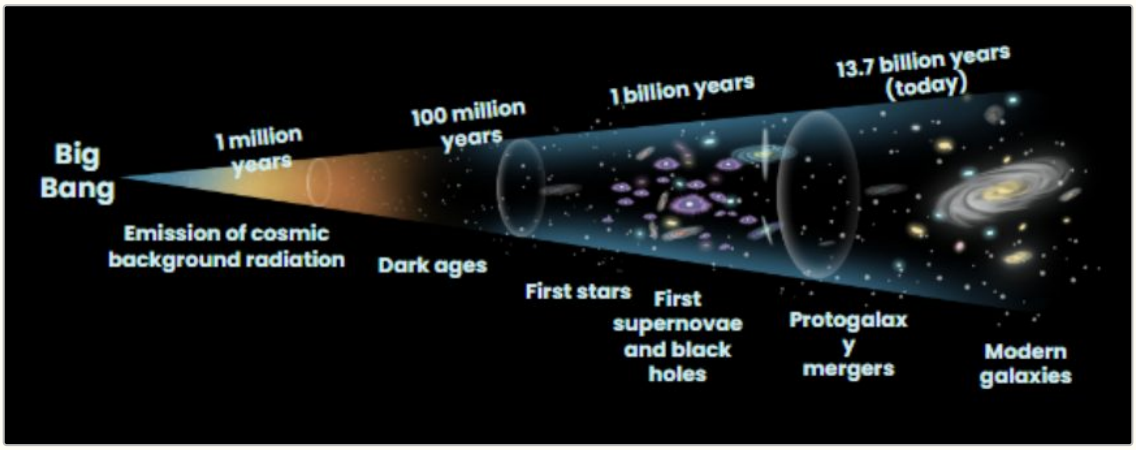
Nebular Hypothesis



Chamberlin–Moulton planetesimal hypothesis



Accretion Model

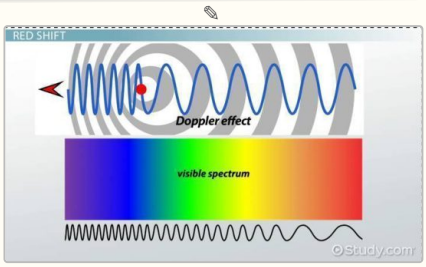


Big Bang Hypothesis · Georges Lemaitre proposed in 1920s.

KLS – Kant = Kollision · Laplace = Loses heat (cools, shrinks, spins faster) · Schmidt–Safronov = Small Steps (dust → planet)

Proofs of Big Bang

→ Hubble's law (Hubble–Lemaitre): farther galaxies recede faster = expanding universe (basis: redshift); Edwin Hubble

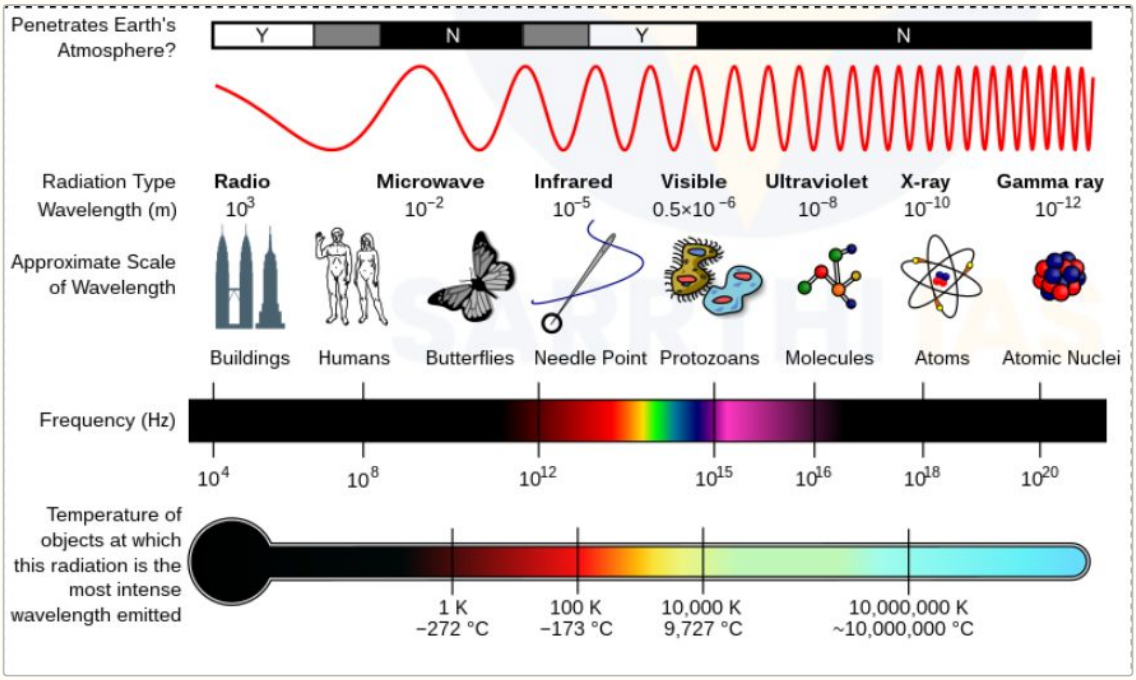


Proof

→ CMB: 1960s discovery, Penzias & Wilson, 1965; oldest light, ~380,000 yrs after Big Bang; at ~3000 K electrons joined protons (recombination) so photons flew free = "baby photo of the universe"

→ Hotter body = shorter wavelength (more energy → faster motion → higher frequency)

→ Universe age 13.8 bn yrs



Proof

Gaseous Hypothesis

→ They started colliding against each other .

→ Collision resulted in the generation of heat.

→ Nebula started rotating around its axis, creating a bulge in the center of the gaseous mass.

→ When this bulge increased in size, the rings started forming one by one and were separated from the middle part of the nebula.

How did the moons form?

→ Repetition of the same process at the planet level.

→ Material of each ring condensed to form satellites of the concerned planets.

⚠ Drawback:

→ According to the science of the law of motion, the collision of particles doesn't generate rotatory motion .

Nebular Hypothesis

Postulates:

→ A huge and hot gaseous nebula is present in the space. Thus, he solved the problem of heat of the nebula through this assumption.

→ From the very beginning, a huge and hot nebula was rotating (spinning) on its axis.

→ A huge and hot gaseous matter called a nebula existed in space which was continuously rotating on its axis.

→ Nebula was losing heat from its outer surface due to the process of radiation and was thus reducing in size and volume due to contraction.

→ As its size reduced, its velocity of rotatory motion began to increase .

→ With increased velocity, the rings started detaching .

Chamberlin–Moulton planetesimal hypothesis

→ Gravitational effects from the passing star created spiral arms of ejected material, with some of it remaining in orbit .

Accretion Model

→ Constant accretion of particles convert them from a nebula to planets eventually.

Big Bang Hypothesis

→ The idea received major boosts from Edwin Hubble's observations that galaxies are speeding away from us in all directions, as well as from the 1960s discovery of cosmic microwave radiation—interpreted as echoes of the big bang—by Arno Penzias and Robert Wilson.

→ Over several million years, slightly denser regions become gravitationally attracted to each other.

→ They grew even denser, forming gas clouds, stars, galaxies , and other astronomical structures.

Proof

→ Hubble's Law (Hubble–Lemaitre law): galaxies farther away from us are moving away at faster speeds, indicating the expansion of the universe.

→ – Basis of Hubble's law is redshift , which is the elongation of light waves to longer wavelengths, often toward the red end of the spectrum, caused by an object moving away from the observer , commonly observed in distant galaxies due to the expansion of the universe.

Cosmic Microwave Background Emissions:

→ Discovered by American physicists Arno Penzias and Robert Wilson, who detected it in 1965 .

→ It is the faint glow of electromagnetic radiation that fills the universe. It is the oldest light in the universe, dating back to about 380,000 years after the Big Bang.

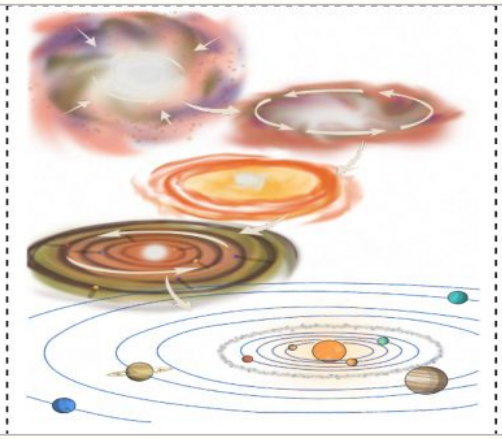
→ In the early universe, it was so hot that matter and radiation were constantly interacting — a plasma of electrons and nuclei. There were no atoms due t the heat.

→ When the universe cooled enough (~3000 K), electrons combined with protons to form neutral atoms — a process called recombination.

→ At that point, photons could travel freely — this is the moment the CMB was released.

→ CMB is like a baby photo of the universe , showing us how it looked just after birth. Its existence, temperature, uniformity, and small irregularities are exactly what the Big Bang Theory predicts , and no other theory explains it so well.

crease.
city, the **rings started**
at a point in the form of
started **rotating around**
cooled and formed the
the nebula thus became
source of **origin of the**



Nebular Hypothesis · Postulates

2 · DARK ENERGY & DARK MATTER

Understand it first: What we can see (stars, gas, planets) is only a small part of the universe. Dark matter is the unseen mass that holds galaxies together by gravity; dark energy is the unseen push that makes the universe expand faster. One pulls in, the other pushes out.

Together ≈ 95% of universe

→ Dark matter ≈ 27%: no light emitted, absorbed or reflected; inferred from gravity (galaxy rotation, gravitational lensing = light bending near massive object); invisible scaffold holding galaxies

→ Dark energy ≈ 68%: accelerating expansion; repulsive force pushing space apart (not matter or gravity)

Dark Energy & Dark Matter

→ Dark energy and dark matter are two mysterious components that together make up about 95% of the universe, yet they remain largely unexplained.

→ Though both are invisible and not directly detectable, their effects are crucial in explaining the structure, behavior, and fate of the cosmos.

3 · GALAXIES

Understand it first: A galaxy is a huge family of stars, gas and dust held together by gravity. They are sorted by shape, so learn the shapes with one example each and where our own Milky Way fits.

Galaxy types

→ Spiral (Milky Way)

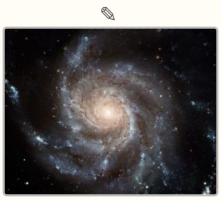
→ Elliptical (old, cool stars, little gas; minimal star formation)

→ Lenticular (bulge + disk, no arms)

→ Irregular (chaotic, gas-rich)

Galaxies · Types of Galaxies

Galaxies · Formation



Galaxies · Formation

Galaxies · Types of Galaxies

Active galaxies ~10%

→ Centre >100× brighter than all stars combined; supermassive black hole + accretion disk; radiation infrared → X-rays; jets near speed of light (AGN)

→ Seyfert (most common, lowest energy)

→ Quasar (brightest)

→ Blazar (jet points at Earth)

→ Milky Way is NOT active

Galaxies

→ After the Big Bang (13.8 billion years ago): Initial fluctuations in the density of matter led to the congregation of material at those places. The role of Dark matter's gravity is considered here.

→ First Galaxies (about 1 billion years post-Big Bang) : Gas clouds collapse under gravity, forming the first stars and galaxies.

→ **Galaxy Evolution (1 billion years to present):** Galaxies grow and evolve through mergers and interactions.

Types of Galaxies

→ **Elliptical Galaxies:** Ranging from nearly spherical to elongated shapes, elliptical galaxies **have a smooth, featureless light profile** . They contain **older, cooler stars and possess minimal gas and dust** , resulting **in low rates of new star formation.**

→ **Irregular Galaxies:** Lacking a distinct shape, irregular galaxies often **appear chaotic** , possibly due to gravitational interactions or collisions with other galaxies. They typically **contain abundant gas and dust, with active regions** of star formation.

→ **Active Galactic Nuclei: highly energetic regions** at the **centers of some galaxies** , powered by **supermassive black holes** accreting matter. These regions emit enormous amounts of radiation across the electromagnetic spectrum and are categorized based on their observational characteristics.

Types of galaxies on basis of activity:

→ **Seyfert Galaxies Most common and lowest energy** active galaxies. **Appear normal in visible light but emit strongly in infrared and sometimes X-rays.** Usually radio-quiet, though **some produce jets** .

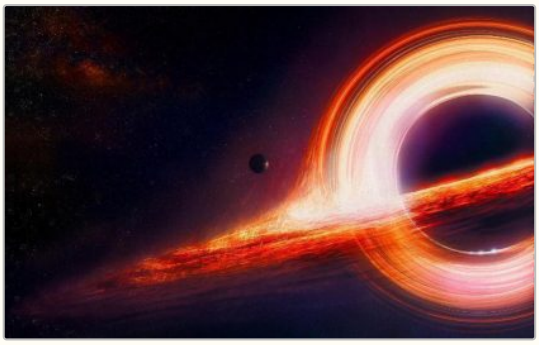
→ **Quasars Brightest and most energetic** type of active galaxy. **Can outshine entire galaxies** and are visible across great cosmic distances. **Powered by massive accretion activity** ; may be episodic, lasting ~10 million years. Useful for studying the early universe and black hole growth.

→ **Blazars** A type of active galaxy whose **jets point almost directly at Earth** . Appear **extremely bright** and are **sources of high-energy particles like neutrinos.** Give insight into the extreme environments near black holes.

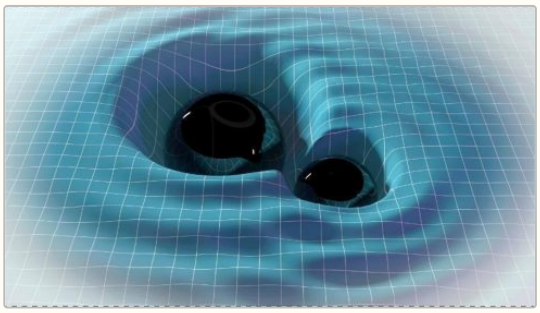
4 · STARS

Understand it first: A star lives by burning hydrogen. Its **mass decides its death:** a small star ends quietly, a heavy one explodes and leaves a very dense remnant. Follow the life line once from nebula to the end states and the terms fall into place.

Lifecycle of stars



Black Holes



Black Holes

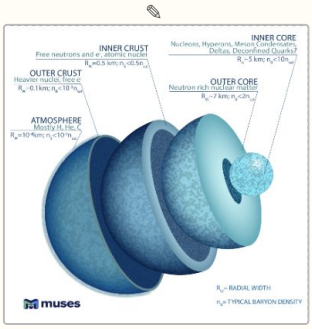
Black Holes

"Small stars retire quietly, big stars quit with a bang" ⭐

Mass	Path	End
🌐 < 8 × Sun	Main sequence → Red giant → Planetary nebula	White dwarf
🌐 > 8 × Sun	Red supergiant → Supernova (iron core)	Neutron star (8–20/25×) or Black hole (>25×)

Chandrasekhar limit

→ core < 1.44 × Sun = white dwarf
→ 1.44–3 = neutron star
→ >3 = black hole. Source quotes both 8–20 and 8–25× for neutron stars: use 25 as the cut-off to black hole.



Population 3, 2 and 1 stars

Populations (by metallicity)

→ Pop III: hypothetical, very first stars, hundreds of millions of yrs after Big Bang, no heavy metals
→ Pop II: oldest observed, 5–10 bya, halo/globular clusters
→ Pop I: younger, metal-rich, 4–5 bya, disk and arms, our Sun

Neutron star

→ Collapsed core of massive star (8–25× Sun) after supernova; almost all neutrons; strong gravity + magnetic fields
→ Can accrete mass → collapse into black hole
→ Pulsar = rapidly rotating neutron star, beams from magnetic poles, regular pulses ("lighthouse")

🌐 Black hole (>25× Sun)

→ Nothing, not even light, escapes (Earth's escape velocity printed 11.8 m/s: unit looks a typo, don't memorise)
→ Event horizon: boundary of no return
→ Accretion disk: gas/dust spiralling in, hot from friction (disk → horizon → void)
→ Gravitational waves: ripples in spacetime; LIGO (Laser Interferometer Gravitational-Wave Observatory, Nobel 2017) and Virgo
→ Spaghettification: stretched by tidal forces

→ Hawking radiation: theoretical particle emission → slow evaporation

Stars

Their formation after the Big Bang within the galaxies.

→ Approximately 380,000 years post-Big Bang, the universe cools enough for electrons to combine with protons and form neutral atoms, releasing the CMB.

→ Formation of Gas Clouds (100 million to 200 million years post-Big Bang) : Gravity causes hydrogen and helium gas to clump together, forming large clouds.

→ Protostar Formation (200 million to 400 million years post-Big Bang): These gas clouds collapse under gravity, increasing temperature and pressure, forming protostars.

→ Nuclear Fusion Ignition (200 million to 400 million years post-Big Bang): When core temperatures are high enough, nuclear fusion ignites, and stars are born.

Lifecycle of stars:

→ Stellar Nebula : Stars form from collapsing clouds of gas and dust.

→ Protostar: As the nebula collapses, a protostar forms and heats up.

→ Groups of stars that have newly formed from molecular clouds are commonly referred to as stellar clusters .

→ Molecular clouds that contain many of these clusters are known as stellar nurseries .

→ Main Sequence: Star enters a stable phase, fusing hydrogen into helium.

→ White Dwarf: Core remains, cools over time. It may convert into a black dwarf over time, but we do not have any solid proof of this as of now.

→ · High-Mass Stars (mass > 8 times that of the sun): Red Supergiant: Similar process but on a larger scale.

Population 3, 2 and 1 stars

→ Stars are classified into populations based on their metallicity, which refers to the abundance of elements heavier than hydrogen and helium.

→ Population III Stars (Hypothetical): These are theorized to be the very first stars born in the universe, hundreds of millions of years after the Big Bang.

→ – Extremely Metal-Poor: They are believed to have been almost entirely composed of hydrogen and helium, with virtually no heavier elements because those elements hadn't been forged in abundance yet.

→ – Massive and Short-Lived: Due to their large size, they are thought to have been incredibly massive and short-lived, burning out rapidly in a few million years.

→ – Direct evidence for their existence has not yet been found.

→ – Do not have a planetary system.

→ Population II Stars (Oldest Observed; 5-10 bya): These are the oldest stars still observable in the Milky Way galaxy.

→ – Metal-Poor: They contain very little of the heavier elements compared to younger stars.

→ – Location: Population II stars are typically found in the galactic halo (spherical region surrounding the disk) and globular clusters (dense collections of millions of old stars).

→ – Slower Evolution: Due to their lower metallicity, they evolve more slowly and have longer lifespans than stars with higher metal content.

→ – Metal-Rich: They contain a higher proportion of heavier elements compared to Population II stars. This enrichment is due to previous generations of stars that lived, died, and scattered these elements into the interstellar medium.

→ – **Location:** Population I stars are found in the galactic disk (flattened disc where most stars and gas reside) and spiral arms of the Milky Way.

→ – **Faster Evolution:** Their higher metallicity can lead to faster stellar evolution and shorter lifespans compared to Population II stars.

Neutron Stars

→ Sometimes they can accrete mass from their surroundings to collapse into black holes.

Black Holes

→ **Accretion Disks:** Disks of **gas and dust spiraling into black holes**, emitting intense radiation due to friction and heating.

→ **Spaghettification:** Spaghettification is the process where **objects are stretched and torn apart** by **intense tidal forces** as they approach a black hole.

5 · SOLAR SYSTEM PYQ 2015 PYQ 2011

Understand it first: The solar system is the Sun plus everything its gravity holds: planets, their moons and the small bodies. Exam questions here are about telling the **small bodies apart** (asteroid, comet, meteor), so learn what each is made of and where it lives.

Components

→ **Sun**: central star, H + He, nuclear fusion

Planets

→ **Terrestrial**: Mercury, Venus, Earth, Mars

→ **Gas giants**: Jupiter, Saturn

→ **Ice giants**: Uranus, Neptune

→ **Planet Nine/Planet X**: hypothesised, explains Kuiper Belt orbits; gas or ice giant, **5–7× Earth mass**, ~200× farther than Earth

Dwarf planets (5)

→ **Pluto** (biggest), **Ceres** (biggest in asteroid belt), **Eris**, **Haumea**, **Makemake** (newest)

→ Criteria: orbits Sun · round (spherical) · not a moon · has not cleared its orbit

Body	Moons
Mercury, Venus	**0** (Venus has quasi-satellite **Zoozve**)
Earth / Mars	**1 / 2**
Jupiter / Saturn	**101 / 274** (Jupiter has biggest moon **Ganymede**)
Uranus / Neptune	**28 / 16**
Pluto	**5**

Handout conflict: Saturn's moons printed **274** in the moons list but **74** in the inner/outer planets table: use **274**.

Region/body	Key facts
Quasi-moon	Asteroid orbiting Sun but staying close to a planet; orbit oblong, less stable, may drift away
Kuiper Belt	Beyond Neptune, **30–50 AU** (1 AU = **149.6 million km**); icy bodies, dwarf planets; rock, water ice, ammonia, methane; source of comets
Oort Cloud	Spherical shell of icy objects; starts **2,000–5,000 AU**, outer edge **10,000–100,000 AU**; long-period comets
Asteroids	Rocky; belt between **Mars and Jupiter**; **Vesta** largest ~**530 km** (329 miles); smallest under **10 m** (33 feet); total mass < Earth's Moon; **Trojans** at L4/L5 Lagrange points
Comets	Icy; gas + dust → glowing **coma and tail** near Sun; from Kuiper Belt or Oort Cloud

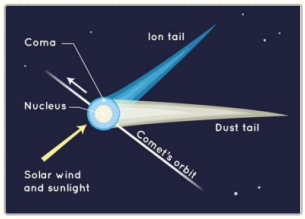
Meteoroid → meteor → meteorite	Space particle (rocky/metallic) → burns in air ("shooting star") → survives, hits ground. Of 50,000+ meteorites 99.8% from asteroids; 0.2% split between Mars and Moon

Edge of the Sun's reach

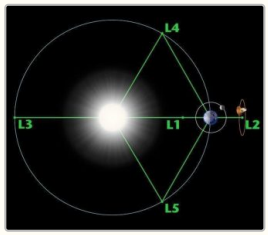
Heliosphere (solar-wind bubble) → *Termination shock: wind supersonic → subsonic* → *Heliosheath: turbulent, field irregular* → *Heliopause: solar wind = interstellar pressure* → *Bow shock: existence contested by Voyager 1 and 2*

Goldilocks zone (2015, ans c): distance from a star where water stays liquid; search for Earth-like planets (Earth; Kepler-186f). Asteroid vs comet (2011, ans B = 1 and 3): asteroids rock/metal, between Mars and Jupiter (Jupiter–Mars), orbit like planets, no tail; comets ice + dust, tail. "Comets between Venus and Mercury" is wrong. Both formed ~4.5 bn yrs ago. ✅

Components of the Solar System



Components of the Solar System



Components of the Solar System

→ The Sun: The central star of the solar system, a massive ball of hydrogen and helium undergoing nuclear fusion.

→ Dwarf Planets: Celestial bodies orbiting the Sun that are spherical but have not cleared their orbital path. The solar system has 5 known dwarf planets: Pluto (the biggest), Ceres (the biggest object in the asteroid belt), Eris, Haumea, and Makemake (the newest). What are the criteria for calling a body a dwarf planet?

→ – In orbit around the Sun : The object must be gravitationally bound to the Sun and travel in an orbit around it.

→ – Massive enough for hydrostatic equilibrium (nearly round shape) : The object's own gravity must be strong enough to pull it into a roughly round shape, overcoming the forces of its rigid body. This is in contrast to asteroids, which are irregularly shaped due to their weaker gravity.

→ – Has not cleared the neighborhood around its orbit: This is the key distinction between a dwarf planet and a full-fledged planet. The object's gravity must not be strong enough to have cleared the path around its orbit of other objects of comparable size. (Orbit of Pluto and Neptune intersects.)

→ Kuiper Belt: A region beyond Neptune filled with icy bodies and dwarf planets. In addition to rock and water ice, objects in the Kuiper Belt also contain a variety of other frozen compounds like ammonia and methane. Distance from the sun is about 30-50 AU (1 Astronomical Unit = 149.6 million km).

→ – The Kuiper Belt is gradually wearing down as objects within it occasionally collide, breaking into smaller fragments, which can include comets and dust that gets carried out of the solar system by solar wind.

→ Meteor: When a meteoroid enters Earth's atmosphere , it heats up due to friction with air molecules. This burning object, streaking across the sky, is what we call a meteor, commonly known as a "shooting star." Most meteors are completely vaporized by the heat before reaching the ground.

→ Meteorite: If a meteoroid is large and sturdy enough to survive its fiery passage through the atmosphere and impact Earth's surface, it becomes a meteorite. Meteorites can be of various sizes, from a pebble to a massive rock. Of more than 50,000 meteorites found on

Earth, 99.8% come from asteroids. The remaining small fraction (0.2%) of meteorites is split roughly equally between meteorites from Mars and the Moon.

→ **Bow Shock** : The shockwave formed as the heliosphere moves through the interstellar medium . It is analogous to the shock wave that forms in front of a moving boat or a supersonic aircraft. As the heliosphere plows through the interstellar medium, it creates this shockwave, where the interstellar medium is abruptly heated and slowed down due to the presence of the heliosphere. However, the existence of such a region has been contested by the data received from the Voyager 1 and 2 missions by NASA.

6 · PLANETS

Understand it first: Two families: the inner rocky planets close to the Sun and the outer giant planets beyond the asteroid belt. Compare them feature by feature, then keep one or two special facts for each planet.

Planet facts

→ Venus and Uranus rotate East → West

→ Uranus axis parallel to its orbit

→ Jupiter: Great Red Spot 16,000 km, Little Red Spot 3,500 km (storms)

- **Inner (rocky)** → dense, small, few/no moons, weak field, short orbits: Mercury 88 d, Earth 365, Mars 687
- **Outer (gaseous)** → big, many moons, rings, strong fields: Jupiter 12 y, Saturn 29, Uranus 84, Neptune 165
- **Atmospheres**
  - → Venus CO_2, super-hot (greenhouse)
  - → Mars thin CO_2
  - → Jupiter/Saturn H + He
  - → Uranus/Neptune methane + ammonia

Feature	Inner Planets	Outer Planets
Number	4 (Mercury, Venus, Earth, Mars)	4 (Jupiter, Saturn, Uranus, Neptune)
Composition	Primarily rocky (iron, silicon)	Primarily gaseous (hydrogen, helium) with some icy components
Size	Smaller diameter and mass	Significantly larger diameter and mass
Density	Higher density	Lower density
Surface	Solid surfaces with craters, volcanoes, and canyons (except Venus which has a thick atmosphere)	No solid surface; thick atmosphere composed mostly of hydrogen and helium
Magnetic Field	Weaker magnetic fields (except Earth)	Strong magnetic fields
Rings	No rings	Most have rings composed of dust and ice particles (Jupiter: 1, Saturn: 2, Uranus: 2, Neptune: 6)
Moons	Fewer moons (Earth: 1, Mars: 2)	Many moons (Jupiter: 101, Saturn: 274, Uranus: 28, Neptune: 16)
Temperature	Wider range of temperatures, with hot days and cold nights (except Venus with a constant hot surface)	Colder temperatures due to distance from the Sun

Planets

Formation	Closer to the Sun, formed from denser material in the early solar system	Farther from the Sun, formed from leftover icy material and gases
Orbit	Shorter orbital periods (Mercury: 88 days, Earth: 365 days, Mars: 687 days)	Longer orbital periods (Jupiter: 12 years, Saturn: 29 years, Uranus: 84 years, Neptune: 165 years)

Planets

Handout alert: Planet Nine only hypothesised (5–7 × Earth mass).

Planets

→ The Giant and the Little red spots are a defining feature of Jupiter. They are huge storms. The former is 16000 km in diameter while the latter is 3500 km in diameter.

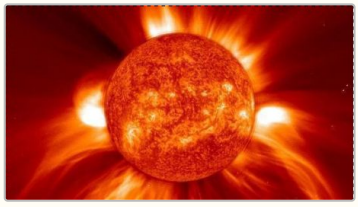
Atmosphere of different Planets

→ Venus : thick atmosphere, mainly made of CO2, making the planet super-hot due to the greenhouse effect.
→ Mars: Very thin, rich in CO2
→ Jupiter and Saturn: Hydrogen and Helium with the addition of some sulphurous and carbon gases.

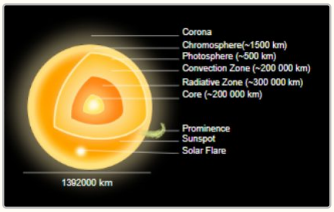
7 · THE SUN PYQ 2022

Understand it first: Think of the Sun as layers, from the core where energy is made to the outer atmosphere we see only in an eclipse. Its outbursts (flares, solar wind, storms) matter because of what they do on Earth: auroras and trouble for satellites, radio and power grids.

Sun

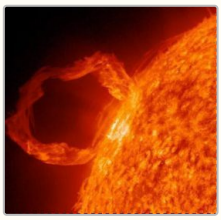


The Sun · Photosphere



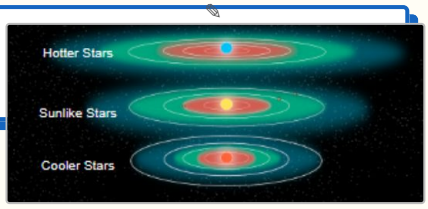
The Sun · Photosphere

→ Composition: 71% H, 27.1% He, 0.97% O, 0.4% C, traces
→ No "surface": gas gets denser toward the core
→



The Sun · Photosphere

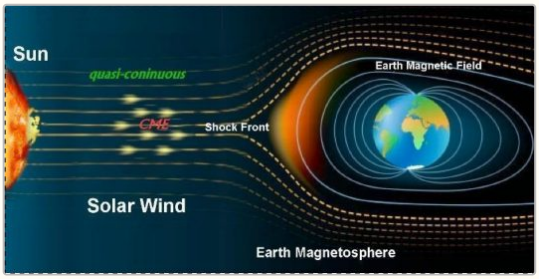
Sun = a samosa (inside → out)



The Sun · Goldilocks Zone

Core: fusion, >15 million °C → Radiative zone: energy by radiation → Convective zone: convection cells → Photosphere: turns opaque (golden crust) → Chromosphere: outer ~10,000 °C → Corona: hot plasma, millions of km

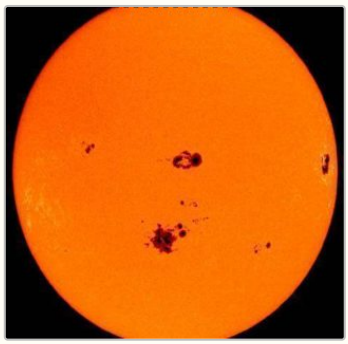
Space-weather term	What it is	Remember
Solar wind	Charged particles streaming from corona	plasma too hot for gravity to hold
Solar flare	Explosion from twisted magnetic fields; X-rays + UV	classes A, B, C, M, X (X biggest); may or may not come with CME
CME (coronal mass ejection)	Plasma + magnetic field blobs from corona	15–18 h to reach Earth
Geomagnetic storm	Earth in path of a CME	Starlink: 40 satellites lost (2022) · Quebec blackout 1989 (GICs damage transformers) · GPS errors (aviation, defence) · polar flights rerouted (radiation) · ISS shielding · radar/missile-warning disruption
Aurora	Solar-wind particles excite O and N → photons	Borealis (N), Australis (S); at poles (magnetic poles near)
Van Allen belts	Charged particles trapped by Earth's field, doughnut-shaped; shield Earth, risk to spacecraft	inner: protons, 1,000–6,000 km · outer: electrons, 13,000–60,000 km
Sunspots	Darker, cooler areas	11-year (and 400-year) cycle; now in Maxima (Solar Maxima/Minima)



The Sun · Solar Winds

The Sun · Aviation and Space

The Sun · Auroras



The Sun · Why are auroras only on the poles?

The Sun

→ Composition: 71% Hydrogen; 27.1% Helium; 0.97% Oxygen; 0.4% Carbon, trace amount of other elements.

Interior:

→ Core: Nuclear fusion takes place here. The temperature here is over 15 million degrees C.

→ Convective Zone: The convection cells move in this zone for energy transportation. (Convection: transfer of heat through the movement of a fluid (such as air or water) caused by differences in temperature and density.)

→ The Sun is made up of hot gases, so it lacks any "surface" like earth.

Atmosphere:

→ Photosphere:

→ – Sun's outer atmosphere is transparent, but when we try to look through the atmosphere deeper into the Sun, our view is blocked.

→ – The photosphere is the layer that begins where the Sun starts to look opaque

Coronal Mass Ejection:

→ Expand as they move away from the sun.

Solar Winds:

→ Solar wind is created by the outward expansion of plasma.

→ Plasma is continually heated to the point that the Sun's gravity can't hold it down.

Solar flares:

→ Tremendous explosions due to the release of energy in twisted magnetic fields (Like when rubber bands come back to their original size.)

→ It causes a sudden flash of increased brightness on the Sun.

→ Can be, but not always, associated with CMEs .

→ Flares are categorized according to their strength (energy output). The smallest ones are A-class, followed by B, C, M and X, the largest.

Geomagnetic Storms:

→ Impacts:

→ – Technological Systems:

→ Navigation errors: GPS signal degradation affects civil aviation and defence navigation.

→ · Power Infrastructure:

→ Grid failure risks: The 1989 Quebec blackout exemplifies how CMEs induce geomagnetically induced currents (GICs), damaging transformers.

→ Polar flights face increased radiation exposure, forcing rerouting.

→ Astronauts aboard the ISS require radiation shielding during intense solar events.

→ · Military and Strategic Systems:

→ Ballistic missile early warning systems and radar operations can be temporarily disabled.

Auroras:

→ Aurora (polar light) is a natural phenomenon which is characterised by a display of lights in the Polar region skies.

→ When Oxygen and Nitrogen return from excited states to the ground state, they release Photons, which are displayed as the Aurora lights.

→ Northern Hemisphere: Aurora Borealis

→ Southern Hemisphere: Aurora Australis

→ Why are auroras only on the poles?

→ – Earth is a magnet. Currently, the North and the South Poles of this magnet are near our geographical North and South Poles.

→ – This causes auroras at the geographical north and south poles.

→ Van Allen Radiation Belt: The Van Allen radiation belts consist of charged particles trapped by Earth's magnetic field , forming doughnut-shaped regions around the planet. They protect Earth from solar radiation but can pose risks to spacecraft passing through them.

→ – The inner and outer radiation belts are two distinct regions within the Van Allen radiation belts surrounding Earth. The inner belt is composed mainly of protons and extends from about 1,000 to 6,000 kilometers above the Earth's surface, while the outer belt consists mostly of electrons and extends from about 13,000 to 60,000 kilometers above the surface.

Solar Maxima & Solar Minima:

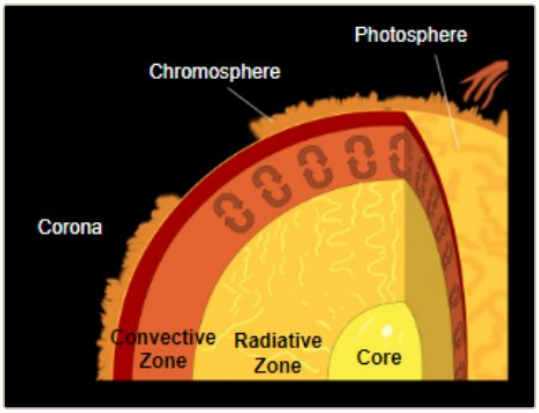
→ Solar maximum: The highest solar activity is observed during this period. The Sun has many sunspots and eruptions of solar flares. We also receive the highest insolation during Solar Maxima, because the Sun emits more energy due to increased sunspot and solar flare activity, boosting total solar irradiance reaching Earth. Right now sun is in the Maxima Phase.

Goldilocks Zone:

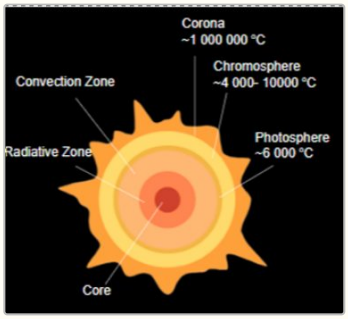
→ Distance from the star where a planet's atmospheric conditions are such that water remains liquid.

→ Distance from the star where the Goldilocks zone exists depends on the size and the strength of the star.

→ E.g. Earth; Kepler-186f.



The Sun · Interior



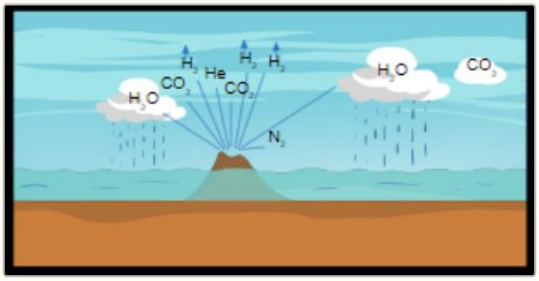
The Sun · Interior

8 · EVOLUTION OF EARTH PYQ 2018 PYQ 2012

Understand it first: The Earth began hot and molten. As it cooled, heavy material sank and light material rose, so it became layered; gases from inside made the first atmosphere, and rain filled the oceans; life then changed the air by adding oxygen. One sequence: layers, air, water, life.

Order of formation

Crust → Atmosphere → Ocean → Life



Phases · Ocean Formation

Atmosphere: 3 stages

→ (1) H and He, blown away by solar wind

→ (2) Degassing: volcanoes give CO_2, N and S oxides, water vapour → clouds → rain → oceans

→ (3) Life raises oxygen (photosynthesis)

4.5 bya
Earth forms

~4.2 bya
LUCA (as printed)

3.8 bya
simple life near deep-sea vents (blue-green algae)

1953
Miller–Urey: organic from inorganic

Source oddity: LUCA ~4.2 bya is printed but is earlier than the 3.8 bya first life (verify).

Phases

Crust formation:

→ Light substances floated up from deep inside , and then they changed into hard rocks after cooling.

→ As the earth's interior continued to cool, it contracted, and the outer crust wrinkled forming ridges and basins.

Atmosphere Formation:

→ Degassing: lighter substances swarmed up above the earth's surface forming an atmosphere of gases.

→ Happened in 3 phases:

→ – Early Atmosphere was rich in Hydrogen and Helium . These gases were eventually stripped away by solar wind.

→ – Stage 3: As life evolved on Earth, the configuration of the atmosphere changed, and the concentration of Oxygen started to increase.

Ocean Formation:

→ Atmospheric Gases formed the Clouds, which cause heavy rains for thousands of year

→ This water was collected in basins, forming oceans.

Life on Earth:

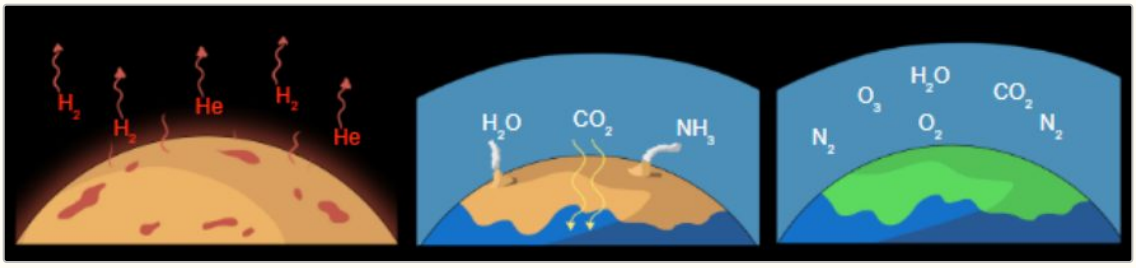
→ 4.5 Billion Years Ago: Earth forms, hot and bombarded by asteroids.

→ Over the next billion years: As Earth cools, conditions become favorable for complex molecules to form in the oceans.

→ Around 3.8 Billion Years Ago: Simple life emerges , possibly near deep-sea vents, utilizing chemicals for energy (not sunlight). These were likely single-celled organisms (blue-green algae) with basic building blocks of life.

→ Proof: Chemical Biogenesis or the Miller-Urey experiment, conducted in 1953 by Stanley Miller and Harold Urey. It proved that in plausible early Earth conditions, organic molecules important for life could arise naturally. Inorganic molecules could convert to organic ones. This can lead to the creation of necessary amino acids, essential for the creation of single-celled organisms that could divide.

→ LUCA: Last Universal Common Ancestor. Might have existed about 4.2 billion years ago.



Phases · Happened in 3 phases

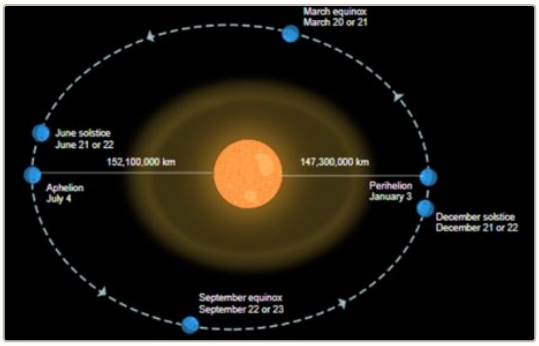
9 · EARTH FACTS, MAGNETISM, DAY-NIGHT

PYQ 2012 PYQ 2013 PYQ 2019 PYQ 2024

Understand it first: Three ideas: the Earth's size and shape figures, the Earth as a magnet (and why that protects us), and the two motions, rotation (day and night) and revolution with a tilted axis (seasons, solstices, equinoxes).

Earth fact	Value
Axial tilt	23.5° from vertical (towards Pole/North Star)
Rotation (sidereal day)	23 h 56 m 4.1 s, West → East (counter-clockwise from above North Pole); slowing (wind/wave friction, Moon receding, core slowdown)

Revolution (sidereal year)	365 d 5 h 48 m, elliptical orbit; aphelion = farthest from Sun, perihelion = nearest
Pole drift	Groundwater pumping (NW India, W North America) shifted pole ~80 cm east (0.00000719°), 1993–2010; ice melting + water moving to low latitudes also shifts axis
Magnetic field	Flips every 2–3 lakh years (chrons: normal/reversed); 183 reversals in 83 million yrs (random); magnetic north drifting Canada → Arctic → Siberia since 1990s



Earth Facts

Why magnetic field matters

→ Blocks solar wind, stops atmospheric loss
→ Creates magnetosphere (holds Van Allen belts)
→ Causes auroras; helps compasses; guides migrating animals
→ Protects satellites and grids

Pole star 2012 (ans C): village 5 km east → keep pole star on left. Day length 2013 (ans B): revolution on a tilted axis + latitude; no tilt = no seasons, equal days. 21 June 2019 (ans A): Sun does not set at Arctic Circle; equator overhead only at equinoxes. 2024 (ans d): >12 h daylight at Tropic of Cancer and Arctic Circle (2 and 4). Northern days longest at June solstice, shortest at December. ✅

No-year PYQ, axis shift (ans b): melting polar ice → water moves toward equator (mass redistribution, "true polar wander") explains shift; solar flares/CMEs are true but don't affect tilt or rotation. ✅

Magnetic field shifts: the field keeps moving and reverses about every 2,00,000 to 3,00,000 years; caused by material inside the Earth, so reversal shows that this material can move (an indirect source on the interior). 💡

Earth Facts

→ According to a study, between 1993 and 2010, excessive groundwater extraction, particularly in regions like northwestern India and western North America, caused Earth's rotational pole to drift approximately 80 centimeters eastward (0.00000719 degrees).

→ Melting of ice caps in higher latitudes, and migration of water to lower latitudes, is also causing changes in Earth's axis.

Speed of rotation is reducing due to

→ Movement of moon away from us.
→ Slowdown of the Core, and its movement in opposite direction.

→ Looking down at the North Pole from space, Earth is rotating in a counterclockwise direction.

Geomagnetism

→ Paleomagnetic studies of rocks and ocean sediment have demonstrated that the orientation of the earth's magnetic field has frequently alternated over geologic time .

→ Periods of "normal" polarity (i.e., when the north-seeking end of the compass needle points toward the present north magnetic pole, as it does today) have alternated with periods of "reversed" polarity (when the north-seeking end of the compass needle points southward).

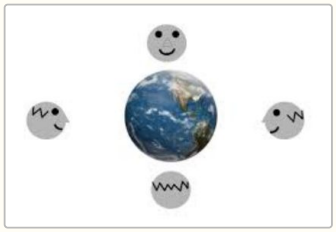
Importance of Earth's magnetic Field

→ Protects Earth from harmful solar wind and charged particles.
→ Prevents excessive atmospheric loss into space.
→ Creates the magnetosphere around Earth, which contains the Van Allen Belts. (Magnetic field = force; magnetosphere = zone of influence of that force)
→ Causes auroras (Northern and Southern Lights) near polar regions.
→ Important for migratory animals and birds for orientation.
→ Protects satellites, communication systems, and power grids from solar storms.
→ Maintains conditions suitable for life on Earth.

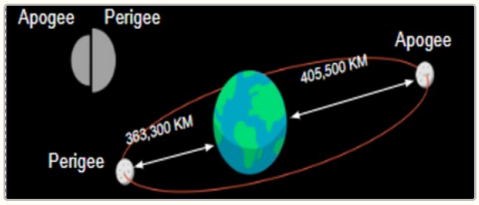
10 · MOON

Understand it first: For the Moon, learn how it is thought to have formed, why we always see the **same face**, and how its phases follow from where it stands relative to the Sun and Earth.

Moon

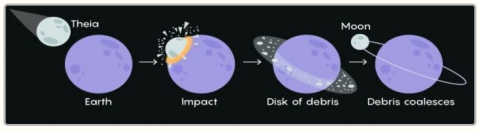


Moon · Tidally locked to the Earth



Moon

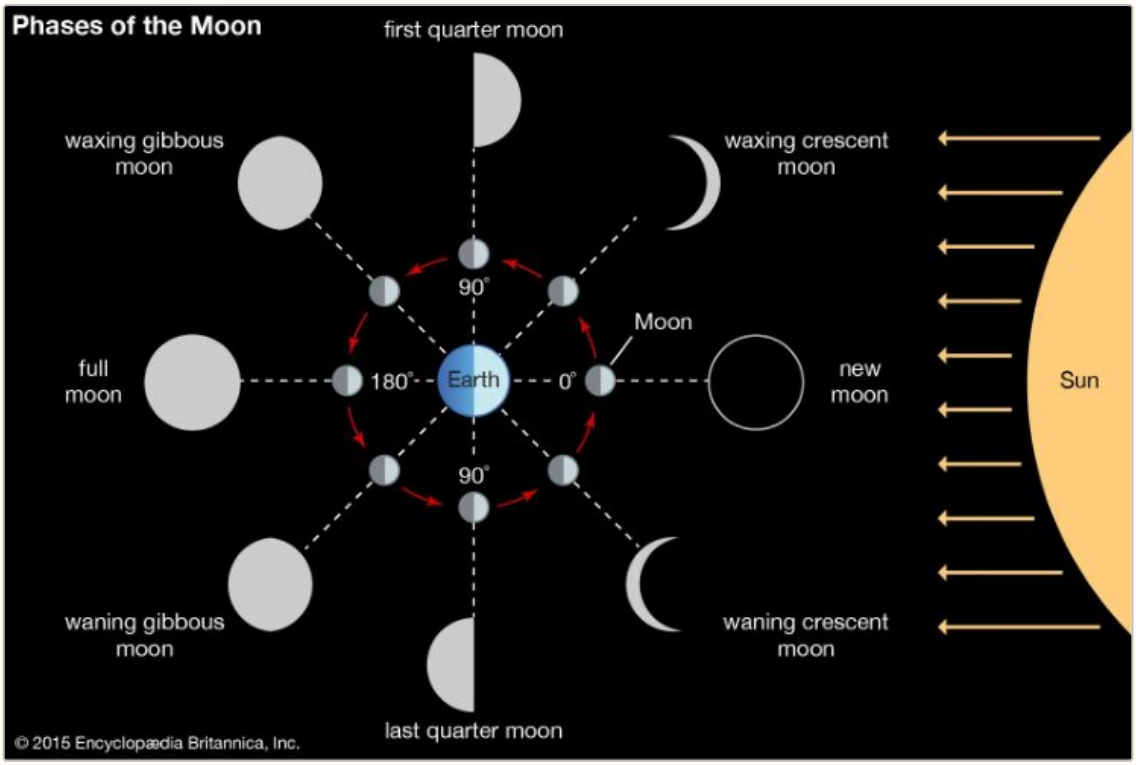
→ Earth's only natural satellite; 384,400 km; 27% (a bit over one-fourth) of Earth's size; gravity drives tides
→ **Perigee** = closest, **Apogee** = farthest
→ **Tidally locked** (same face always)



Moon · Creation

Origins

→ **Fission** (George Darwin): rejected
→ **Giant Impact** (Theia, Mars-sized, 4.5 bya): accepted
→ **Binary-exchange capture**: proposed



Moon · Phases of the Moon

Pear – Perigee = Pear-close · Apogee = Away

Moon

→ It orbits the Earth at an average distance of 384,400 km.

→ Apogee: The point on the Moon's orbit farthest away from the Earth .

Tidally locked to the Earth:

→ Earth's gravitational interaction slowed down the moon's rotation until it became tidally locked.

Creation:

→ Fission Theory, George Darwin (1838): The Moon once formed part of the rapidly spinning Earth , which flung off a large chunk due to the centrifugal force, creating the Moon. This explains the Moon's receding path from Earth and Earth's slightly bulging equatorial region. Not accepted anymore.

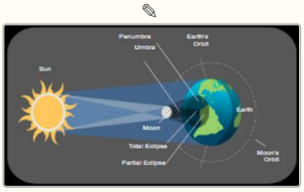
→ Giant Impact Hypothesis (Big Splash): collision between the proto-Earth and a Mars-sized planet (called Theia) , around 4.5 bn years ago created the Moon.

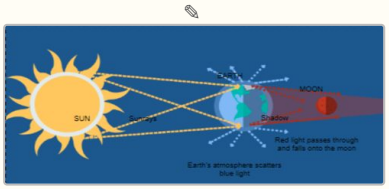
→ Binary Exchange Capture Theory (Proposed): The moon was not born out of Earth, but was infact part of a binary system of celestial objects. When they came close to the Earth, earth captured one and expelled away the other part of the binary system.

11 · ECLIPSES, LATITUDE, TIME PYQ 2024

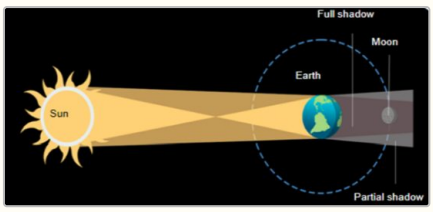
Understand it first: An eclipse is just a shadow: Sun, Earth and Moon fall in one line. Latitude tells how far north or south (and so climate), longitude tells how far east or west (and so time). Time questions are simple arithmetic once you know which way the Earth turns.

Eclipses



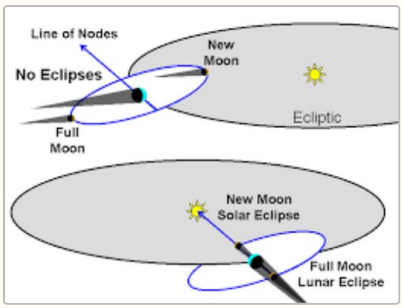


Eclipses · Solar Eclipse: Moon Between Earth and the Sun.

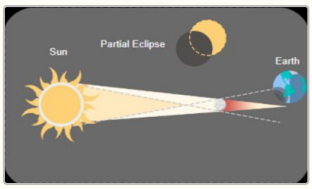


Eclipses

→ **Lunar**: Earth between Sun and Moon; **Blood Moon** = whole Moon in umbra; **Blue Moon** = 2nd full moon in a month (no eclipse link)



Eclipses · Partial Solar eclipse can be due to 2 reasons



Eclipses · Solar Eclipse: Moon Between Earth and the Sun.

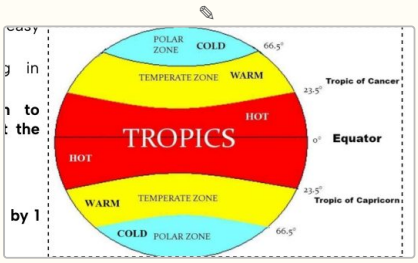
Solar: Moon between Sun and Earth

→ **Annular**: Moon farther → ring
→ **Total**: Moon closer, diamond-ring effect
→ **Partial**: penumbra only

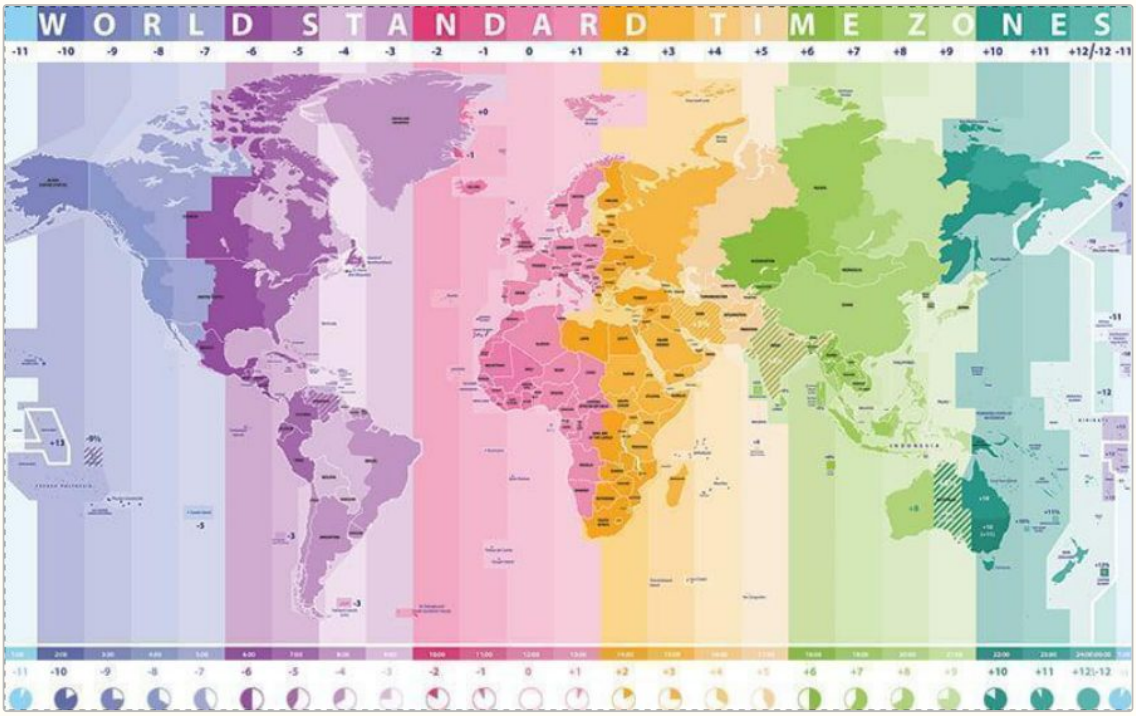
→ Not every full/new moon gives eclipse: Moon's orbit tilted 5°

Latitude and longitude

→ Imaginary lines to locate places
→ **Latitudes**: parallel, east-west
→ **Longitudes**: north-south, meet at poles; also climatic zones and time zones
→ **Time**: 4 min per 1° longitude; Earth rotates W → E so east is ahead; **International Date Line** keeps calendar day consistent



Latitudes and Longitudes



Latitudes and Longitudes · Time Zones

A-A-U — Most time zones: France 12 · Russia & USA 11 ("two elevens, a cricket team each") · A-A-U = Antarctica, Australia, UK 9 · Canada 6 · Denmark 5

No-year PYQ	Answer
Anadyr (Siberia, UTC+12) vs Nome (Alaska, UTC-9), ~800 km apart across Bering Strait	Only Statement I right (ans a): different days at breakfast; but Monday in Anadyr is Sunday in Nome, not Tuesday
UK, Denmark, NZ, Australia, Brazil: more than 4 zones?	Four (ans b): Denmark 5 (mainland UTC+1 CET + Greenland), NZ 5 (mainland NZST/NZDT UTC+12/+13 + territories), Australia 6+, UK up to 9 with territories; Brazil only 4 (UTC-2 to UTC-5; now 3 official)

PYQ: more than four time zones?	Time zones (handout answer: only four of the five)
United Kingdom	mainland GMT/BST (1 zone); with overseas territories (Bermuda, Falklands, Gibraltar) up to 9
Denmark	mainland UTC+1; Greenland: Danmarkshavn UTC+0, Ittoqqortoormiit UTC-1, Nuuk UTC-3, Thule UTC-4: 5
🆕 New Zealand	mainland UTC+12/+13; Tokelau UTC+13, Cook Islands UTC-10, Niue UTC-11: 5
Australia	mainland AWST UTC+8, ACST UTC+9:30, AEST UTC+10; Lord Howe UTC+10:30, Norfolk UTC+11, Cocos UTC+6:30, Christmas UTC+7: 6+
Brazil	historically 4 zones (UTC-2 to UTC-5), now 3 official: not more than 4

Eclipses

→ Umbra: darkest region of a shadow

→ Penumbra: region in which only a portion of the light source is blocked by the occluding body.

🌐 Solar Eclipse: Moon Between Earth and the Sun.

→ The moon blocks the light of the sun from reaching the Earth's surface and casts a shadow (Umbra) on the Earth.

→ Annular Solar Eclipse: When the Moon passes directly between the Earth and the Sun but does not completely cover the Sun's disk , as the Moon is farther away from Earth, making its shadow on Earth smaller.

→ Total Solar Eclipse: When the moon completely covers the sun , as it is closer to the Earth. It also leads to the Diamond ring effect (As the Moon moves across the face of the Sun, a point comes when only a single bright spot of sunlight remains visible, looking much like a shining diamond.)

Latitudes and Longitudes

Time Zones

→ The International Date Line exists to maintain a standard calendar day globally by marking the point where the date changes by one day when crossing time zones.

Rank	Country	Time Zones
1	France	12
2	Russia	11
2	USA	11
3	Antarctica*	9
3	Australia	9
3	UK	9
4	Canada	6
5	Denmark	5

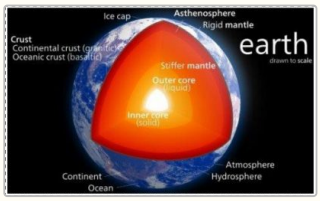
List of Countries with the most timezones

🎯 12 · INTERIOR OF THE EARTH PYQ 2017

Understand it first: Nobody has seen the inside of the Earth; we infer it, mainly from how earthquake waves bend and stop. The result is three layers by composition (crust, mantle, core) with sharp boundaries (discontinuities) between them.

Why study interior

→ Changes over and below surface; volcanism, earthquakes
→ Internal structure of solar-system objects
→ Evolution of atmosphere; deep-sea mineral exploration

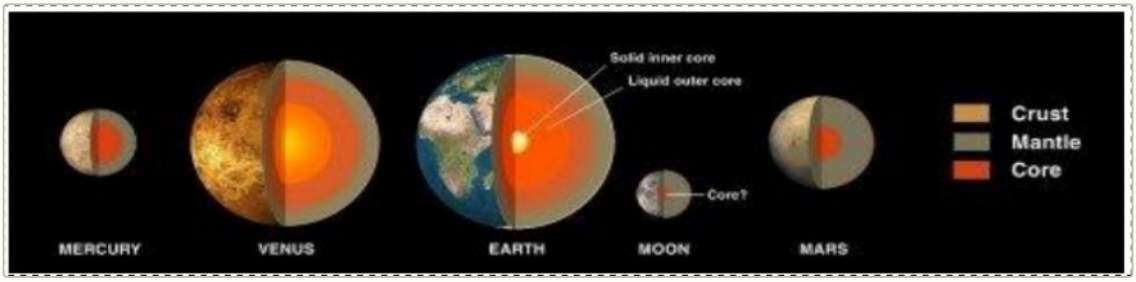


Layers Of Earth

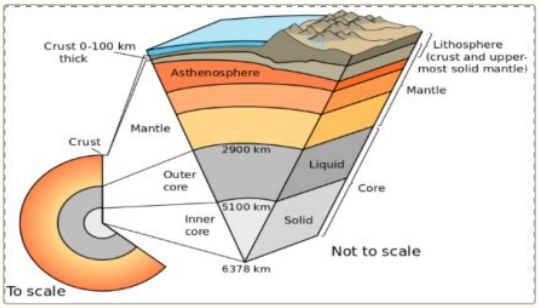
How we know

→ **Direct**: deepest mines (Mponeng, TauTona 3.6 km), deepest drill (Kola, 12 km), volcanic eruptions
→ **Indirect**: meteorites, gravity anomalies (heavy material = higher gravity), magnetic field shifts, best of all earthquake waves

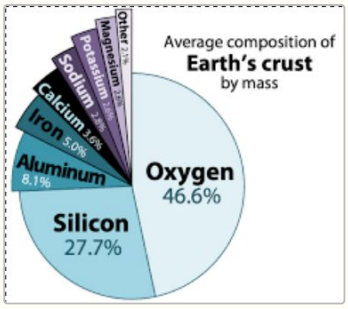
Layer	Facts
👫 Crust 30–50 km; 0.5–1.0% of volume	Oceanic (SiMa): denser, thinner, 5–30 km, basaltic · Continental (SiAl): lighter, thicker, 50–70 km (70–100 km under Himalaya), granitic · separated by Conrad discontinuity · outer = sedimentary cover, then crystalline igneous + metamorphic (acidic); lower = basaltic and ultra-basic
Mantle Moho → 2900 km; 83%	Density upper 2.9–3.3, lower 3.3–5.7, avg 4.5 g/cm³ · Lithosphere = crust + uppermost mantle, 10–200 km, brittle · Asthenosphere to 400 km, semi-solid, source of magma · lower mantle solid · upper/lower mantle by Repetti
Core 2900–6400 km; 16%	Heaviest, densest; nickel + iron [NiFe] · outer liquid, inner solid, separated by Lehmann's discontinuity



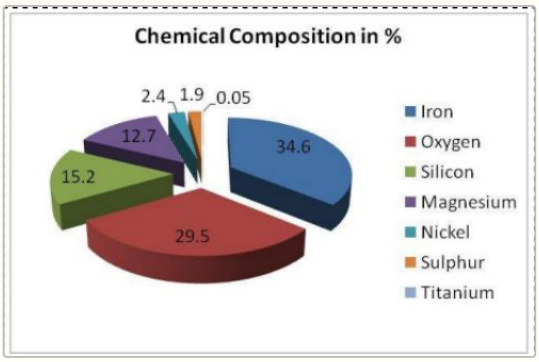
Sources



Crust: 30-50 km



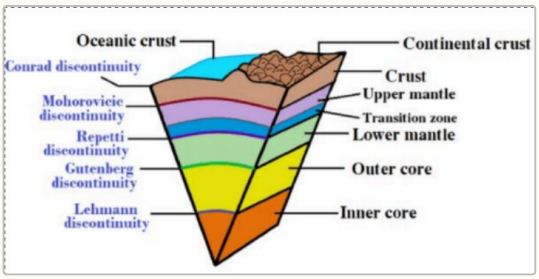
Core: 2900-6400 km · They are separated by Lehman's Discontinuity.



Core: 2900-6400 km · They are separated by Lehman's Discontinuity.

Sl. No.	Elements	By Weight(%)
1.	Oxygen	46.60
2.	Silicon	27.72
3.	Aluminium	8.13
4.	Iron	5.00
5.	Calcium	3.63
6.	Sodium	2.83
7.	Potassium	2.59
8.	Magnesium	2.09
9.	Others	1.41

Core: 2900-6400 km · They are separated by Lehman's Discontinuity.



Core: 2900-6400 km · They are separated by Lehman's Discontinuity.

CoolMenReallyGetLove – Conrad (continental/oceanic crust) · Moho / Mohorovičić (crust/mantle) · Repetti (upper/lower mantle) · Gutenberg (mantle/core, 2,900 km) · Lehmann (outer/inner core)

Discontinuity: place where seismic waves change velocity and/or direction abruptly.
Earth = boiled egg: shell = crust, white = mantle, yolk = core.

Interior Of The Earth

→ Why do we need to know about this?

→ Understanding the earth's interior is essential to understanding the nature of changes that take place over and below the earth's surface.

→ To understand geophysical phenomena like volcanism, earthquakes, etc.

Sources

→ Direct Sources: Facilitate direct observation of what's under the surface.

→ – Deep Earth mining and Drilling (deepest mine: Mponeng gold mine and TauTona gold mine in South Africa at 3.6km; deepest drilling 12km in Kola peninsula, Russia).

→ Indirect Sources : Help us statistically extrapolate what's inside the Earth.

→ – Meteors and other celestial bodies (other planets, moons, etc): Similar formation Process as Earth. So, the internal structure of these bodies will be similar to that of the Earth. Helps in expanding our imagination as to how Earth would look from the inside.

→ – Gravitation : Gravity value is not the same everywhere, and this is known as a gravitational anomaly, helps us identify material distribution inside the earth. Wherever heavier materials have accumulated, gravitational force will be higher as it is directly proportional to the mass of the bodies in question.

→ – Shifts in our magnetic fields: Our magnetic field constantly keeps moving. It reverses in approximately 200000 to 300000 years. It is the material inside the earth that is responsible for the magnetic field. This reversal gives us the idea that the material inside the earth can move.

Other minor sources and observations:	→ Volcanic eruptions and existence of hot springs, geysers etc. point to an interior which is very hot. → The high temperatures are attributed to automatic disintegration of the radioactive substances. → Gravitation and the diameter of the earth helps in estimating pressures deep inside

Layers Of Earth

→ Broadly based on chemical properties, three layers of earth can be identified– crust, mantle, and core.

Crust: 30-50 km

→ It has 2 parts: the denser oceanic (basaltic) and the lighter continental (granitic) crust.

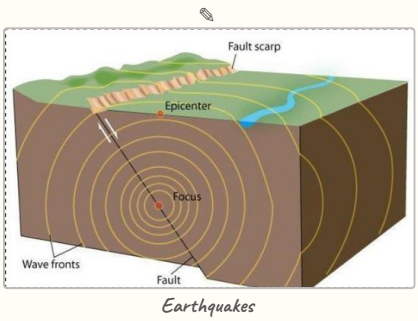
Mantle

→ The lower mantle extends beyond the asthenosphere. It is in solid state.

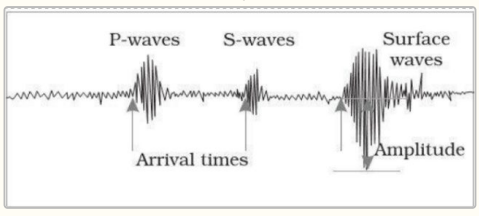
13 · EARTHQUAKES AND SEISMIC WAVES PYQ 2023

Understand it first: An earthquake is stored strain released suddenly at the focus; the point above it is the epicentre. The energy travels as body waves (P and S) and surface waves, and the way each wave behaves tells us about the interior and creates shadow zones.

Earthquake	→ Abrupt energy release along a fault (sharp break in crustal rock); friction holds, pressure builds, rock slips → shockwaves → Focus (hypocentre) = where energy released → Epicentre = surface point nearest focus



Earthquakes



Earthquakes

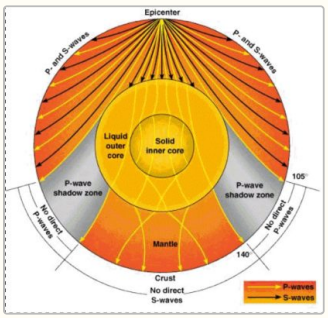
Distribution	→ Mostly plate boundaries; two belts: Circum-Pacific, Mid-continental → Deep-focus at convergent, shallow-focus at divergent boundaries

Scale	Measures
Richter	Magnitude from surface-wave amplitude, log scale 0–10 (5 → 6 = ×10 amplitude)
Mercalli	Intensity (damage) 1–12
Moment magnitude	Energy (fault area, slip, rock rigidity), log, most widely used now

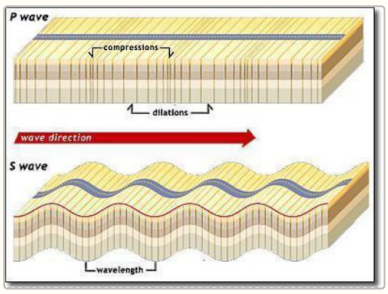
Wave	Type	Motion	Medium	Order
P (primary)	Body, longitudinal	to-and-fro along travel	solid, liquid, gas (faster in solids)	1st, fastest
S (secondary)	Body, transverse	up-down, right angles	solids only	2nd
L (Love)	Surface, shear	horizontal shaking; fastest surface wave	–	3rd; collapses buildings
R (Rayleigh)	Surface, rolling ellipse	vertical + horizontal	–	slowest; tsunamis

Push-pull – P = Push-pull, Prompt. S = Sideways, Solids only. Love shakes Left-right; Rayleigh Rolls like a surf.

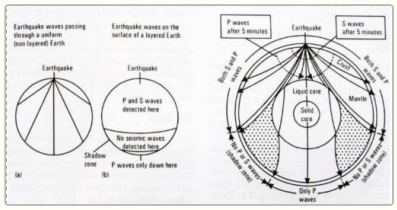
Wave behaviour



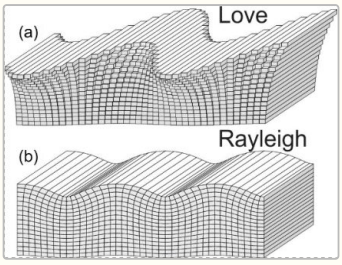
Seismic waves · Fastest, first to arrive ; Primary Waves



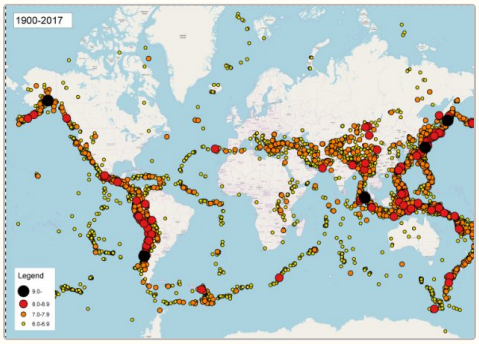
Seismic waves



Seismic waves · Rayleigh Waves.



Seismic waves · Cannot pass through gases or liquids.

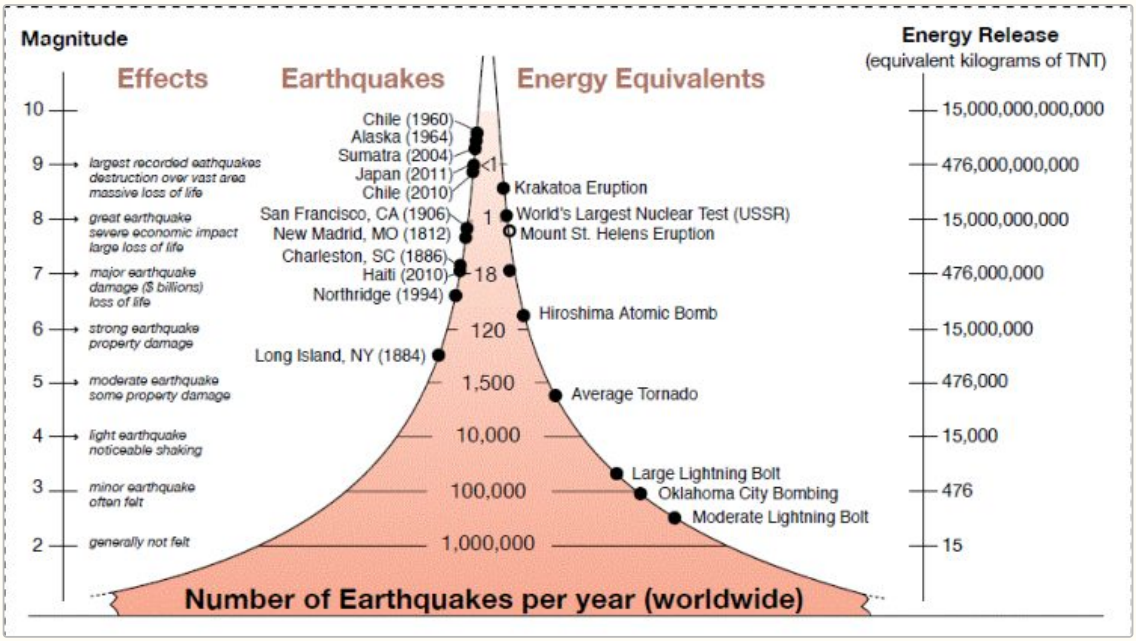


Seismic waves · Slowest surface waves

→ Recorded by seismographs; body waves (P, S) from focus; they generate surface waves at surface rocks

→ Velocity rises with elasticity; direction changes (reflect/refract) at different densities; cooler rigid areas transmit faster than hotter

Shadow zones	→ P: 103°–143° band round Earth → S: everything beyond 103°, much larger, a little over 40% of surface → S vanish → liquid outer core; P-shadow clue → solid inner core



Seismic waves · Also, a logarithmic scale .

Earthquakes

→ Rocks along a fault tend to move in opposite directions. But the friction exerted by the overlying rock strata prevents the movement of the rock layer. With time pressure builds up.

→ Under intense pressure, the rock layer, at a certain point, overcomes the friction offered by the overlying layer and undergoes an abrupt movement generating shockwaves.

Seismic waves

→ Recorded by Seismographs installed across the world in Seismic stations.

P-waves	→ Particles of the medium vibrate along the direction of propagation of the wave . → Their velocity depends on the shear strength or elasticity of the material. → Shadow zone: 103–143 degrees on both sides .
S-waves:	→ Analogous to water ripples or light waves. → Travel at varying velocities (proportional to shear strength) through the solid part of the Earth's crust, mantle. → Cannot pass through gases or liquids. → Particles move perpendicular to the direction of propagation of the wave. → Shadow zone: between 103 degrees and 180 degrees .
L-Waves:	→ Have the highest velocity of all surface waves and can cause significant horizontal shaking of the ground. → Cause displacement of rocks , and hence, the collapse of structures occurs. → Responsible for Tsunamis due to the vertical shift.

→ 8. Shadow Zones: (i) The Body waves get recorded in seismographs located at far-off locations. However, there exist some specific areas where the waves are not reported. Such a zone is called the 'shadow zone'. (ii) The study of different events reveals that an altogether different shadow zone exists for each earthquake. (iii) It was observed that seismographs located at any distance within 103° from the epicentre, recorded the arrival of both P and S-waves. (iv) However, the seismographs located beyond 143 ° from the epicentre, record the arrival of P-waves, but not that of S-waves.

⏳ How does this help us determine Earth's interior ?

→ The variations in the direction of waves are inferred with the help of their record on seismographs.

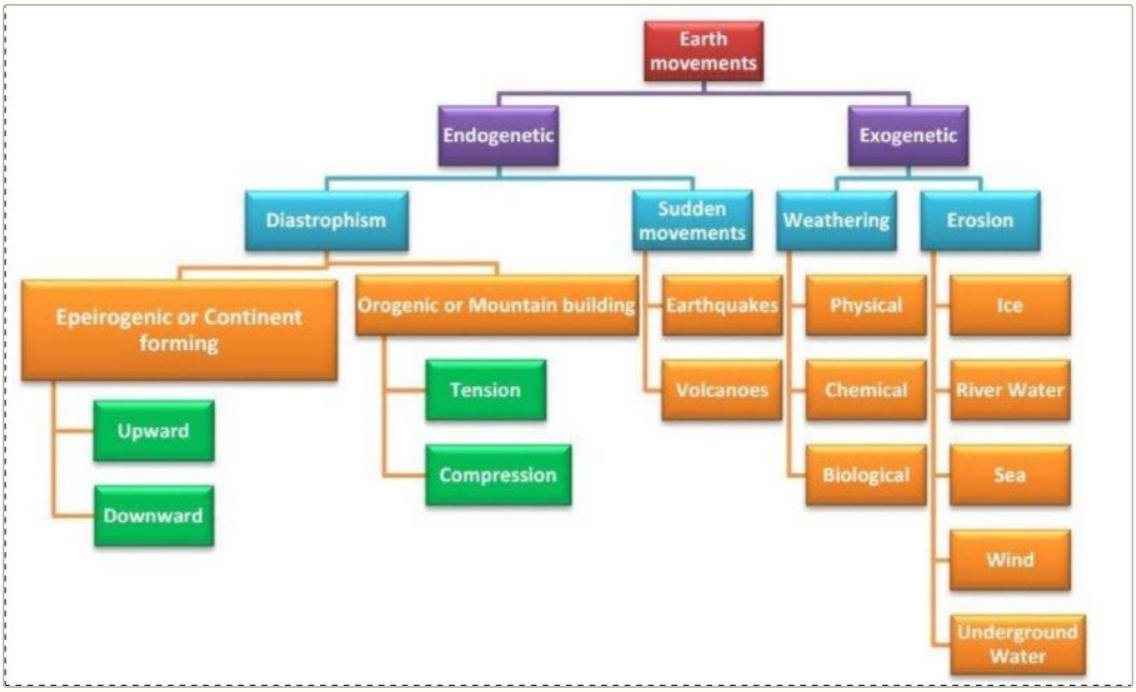
🔧 How are Earthquakes Measured ?

→ Richter's Scale : Measures earthquake magnitude by calculating the amplitude of the surface waves generated during the event.
→ – The number indicating magnitude ranges between 0 to 10 .
→ Mercalli's Scale : Measures the intensity of the earthquakes, based on the actual level of destruction caused by the event.
→ – The number indicating intensity ranges between 1 to 12
→ Moment Magnitude Scale:
→ – Also, a logarithmic scale .

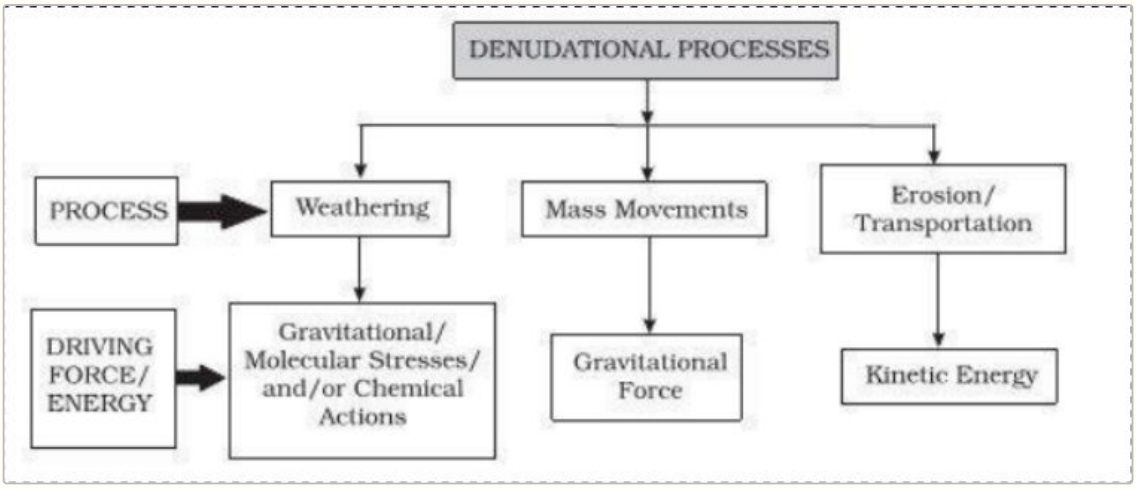
🎯 14 · EARTH MOVEMENTS AND WEATHERING PYQ 2024

Understand it first: Two sets of forces shape the land. Endogenic forces from inside build relief up (slow uplift, or sudden quakes and volcanoes). Exogenic forces from outside wear it down: weathering breaks rock where it stands, erosion carries it away.

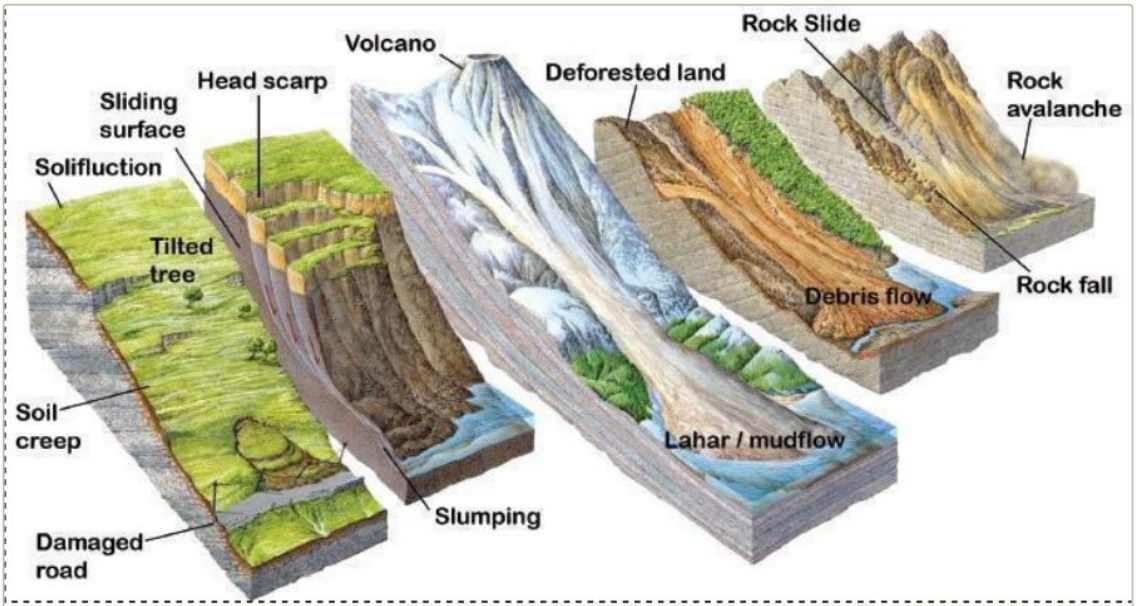
	Endogenic (radioactivity, tidal/rotational friction, primordial heat)	Exogenic (Sun-driven: temperature + rain)
Slow	Diastrophism (bending, folding, warping, fracturing): Orogenic = mountain-building, horizontal force → vertical effect, long narrow belts; Epeirogenic = continent-building, vertical force → horizontal effect, radial up/down	Weathering · Erosion · Deposition · Mass movement = denudation
Sudden	Earthquakes, volcanism	–



Earth Movements



Denudation



Failure Type	Type of Material	Type of Motion	Rate of Motion

Mass Movements

Failure Type	Type of Material	Type of Motion	Rate of Motion
Rock fall	Rock fragments	Vertical or near-vertical fall (plus bouncing in many cases)	Very fast (>10s m/s)
Rock slide	A large rock body	Motion as a unit along a planar surface (translational sliding)	Typically very slow (mm/y to cm/y), but some can be faster
Rock avalanche	A large rock body that slides and then breaks into small fragments	Flow (at high speeds, the mass of rock fragments is suspended on a cushion of air)	Very fast (>10s m/s)
Creep or solifluction	Soil or other overburden; in some cases, mixed with ice	Flow (although sliding motion may also occur)	Very slow (mm/y to cm/y)
Slump	Thick deposits (m to 10s of m) of unconsolidated sediment	Motion as a unit along a curved surface (rotational sliding)	Slow (cm/y to m/y)
Mudflow	Loose sediment with a significant component of silt and clay	Flow (a mixture of sediment and water moves down a channel)	Moderate to fast (cm/s to m/s)
Debris flow	Sand, gravel, and larger fragments	Flow (similar to a mudflow, but typically faster)	Fast (m/s)

Mass Movements

Mass movement: only under gravity. **Denudation gives:** soil formation, ore concentration (iron, manganese, aluminium, copper), landforms. Orogeny, epeirogeny, earthquakes, plate tectonics → faulting and fracturing.

Epeirogenic examples

→ Uplift: Colorado Plateau, Brazilian Highlands, Australian Shield, Siberian Platform

→ Subsidence: Siberian Basin, **Konkan**

→ Sudden: **2004 quake submerged Indira Point**; NZ 1885 uplift up to 3 m; Japan 1891 subsidence 6 m

Weathering

Types of Weathering · Onion Peeling

Types of Weathering · Chemical: Change in chemical composition

Types of Weathering · Onion Peeling

Types of Weathering · Onion Peeling

Chemical (composition changes)

→ Solution/hydration

→ Carbonation (CO_2 + water → carbonic acid)

→ Oxidation (iron, manganese, sulphur rocks)

→ Reduction (red iron oxide → green/blue-grey)

Physical

→ unloading

→ granular disintegration

→ **exfoliation** (diurnal temperature, "onion peeling", smooth rounded, dry/high places)

→ block separation (jointed rocks)

→ **shattering** (angular pieces)

→ **frost wedging** (water freezes, expands)

→ **salt weathering** (crystallisation, deserts)

Biological

→ burrowing, roots

→ decaying matter gives **humic, carbonic** and other acids

→ minerals removed by algae and plants

S-C-O-R – "Some Chefs Often Roast" = Solution/hydration · Carbonation · Oxidation · Reduction

Endogenic Forces

→ Caused **due to the internal energy of the Earth** .

→ This energy is mostly **generated by radioactivity, rotational and tidal friction and primordial heat** from the origin of the earth. This energy due to geothermal gradients and heat flow from within induces diastrophism and volcanism in the lithosphere.

→ Can be of 2 types: Diastrophic and Sudden movements.

→ Diastrophism: bending, folding, warping (bending and twisting of a large area) and fracturing .

→ – Epeirogenic processes involving uplift/subsidence or warping of large parts of the earth's crust. Vertical forces and horizontal effects.

→ · Epeirogenic or continent-forming movements act along the radius of the earth ; therefore, they are also called radial movements . Their direction may be towards (subsidence) or away (uplift) from the centre. The results of such movements may be clearly defined in the relief.

Sudden Movements:	→ Include earthquakes and volcanism involving local relatively minor movements. → Due to their sudden-ness, cause a large amount of destruction , and are included under natural disasters.

Exogenic Forces

→ A direct result of stress induced in earth materials due to various forces that come into existence due to the sun's heat .

→ Temperature and precipitation are the two important climatic elements that control various processes by inducing stress in earth materials.

Denudation

→ All the exogenic geomorphic processes are covered under a general term, denudation.

Types of Weathering

→ Solution/Hydration: Rocks formed of water-soluble minerals. Soluble rock-forming minerals like nitrates, sulphates, potassium, etc. are affected.

→ Carbonation : CO2 and water form Carbonic acid , which reacts with rocks.

→ Oxidation: combination of a mineral with oxygen to form oxides . Iron, Manganese, Sulphur rich rocks experience this.

→ Reduction : When oxidized minerals are placed in an environment where oxygen is absent . E.g. Red iron oxides turn greenish or bluish grey.

Physical:	→ Granular Disintegration (sedimentary rocks): Rocks composed of coarse mineral grains commonly fall apart grain by grain or undergo granular disintegration. → Exfoliation (diurnal temperature changes): → – With rise in temperature, every mineral expands and pushes against its neighbor and as temperature falls, a corresponding contraction takes place. → – This process is most effective in dry climates and high elevations where diurnal temperature changes are drastic. → – The surface layers of the rocks tend to expand more than the rock at depth and this leads to the formation of stress within the rock resulting in heaving and fracturing parallel to the surface. → Block separation: → – This type of disintegration takes place in rocks with numerous joints acquired by mountain-making pressures or by shrinkage due to cooling. → – This type of disintegration in rocks can be achieved by comparatively weaker forces. → Shattering (angular pieces):

→ – A huge rock may undergo disintegration along weak zones to produce highly angular pieces with sharp corners and edges through the process of shattering.

→ – This occurs when there is a temperature change ; when the rock is exposed to the effects of wind, rain and waves also contributes to disintegration.

→ Frost wedging:

→ – During the warm season, the water penetrates the pore spaces or fractures in rocks.

→ – During the cold season, the water freezes into ice and its volume expands as a result.

→ – This exerts tremendous pressure on rock walls to tear apart even where the rocks are massive.

→ – Frost weathering occurs due to the growth of ice within pores and cracks of rocks during repeated cycles of freezing and thawing (melting) .

→ Salt weathering (crystallization):

→ – Salts in rocks expand due to thermal action, hydration, and crystallisation .

→ – Many salts like calcium, sodium, magnesium, potassium, and barium tend to expand.

→ – Salt crystals in near-surface pores cause the splitting of individual grains within rocks, which eventually fall off.

→ – This process of falling off of individual grains may result in granular disintegration or granular foliation.

→ Decaying plant and animal matter help in the production of humic, carbonic, and other acids which enhance decay and solubility of some elements.

Mass Movements

→ – Movement of material ONLY under the impact of gravity.

→ Significance of Denudation:

→ Helps in the enrichment and concentrations of certain valuable ores of iron, manganese, aluminium, copper etc.

→ Landform Creation

15 · DRIFT AND SEA-FLOOR SPREADING PYQ 2014

Understand it first: Wegener said the continents moved (fit of coasts, fossils, rocks) but could not say what moved them. Later evidence from the sea floor (young rock at ridges, magnetic stripes) showed new crust forming and spreading. Drift gave the idea, sea-floor spreading gave the mechanism.

1858 Pellegrini

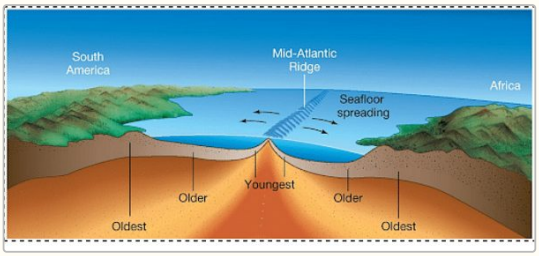
1912 Wegener drift

1930s Holmes convection

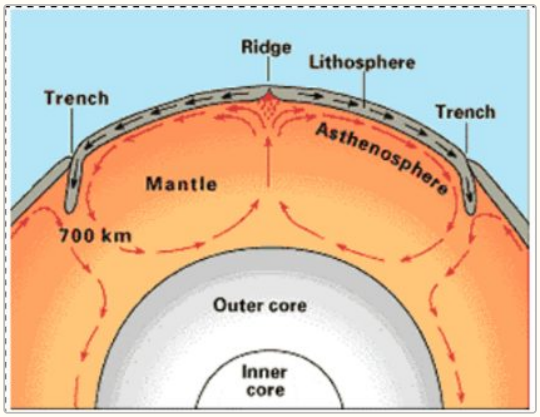
1960 Hess spreading

1967 McKenzie & Parker

1968 Morgan



Seafloor Spread Theory · By Harry Hess in 1960 .



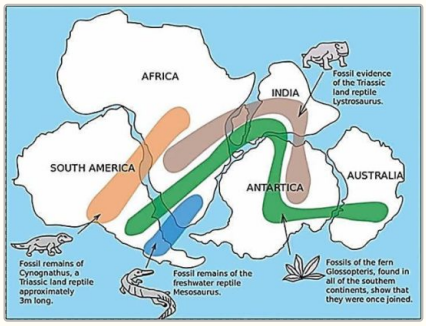
Seafloor Spread Theory · By Harry Hess in 1960 .

Earlier ideas

→ Antonio Pellegrini 1858: identical fossil plants in N American and European coal → once joined

→ Edward Suess (late 19th c.): big ancient continents split into present ones

→ F. B. Taylor: arcuate (bow-shaped) mountain belts of Asia, Europe from creep of continents toward Equator



Continental Drift Theory · Jigsaw fit of shape

Wegener 1912

→ One landmass Pangaea + one ocean Panthalassa

→ Tethys Sea split it: Laurentia (Laurasia) north, Gondwanaland south

→ Forces: tidal (westward) + gravity and polar-fleeing (equator-ward)

→ Break-up in Triassic (251–199.6 mya); ~200 mya all continents one supercontinent

Evidence	Detail
Jigsaw fit	Brazil bulge ↔ Gulf of Guinea; Greenland ↔ Ellesmere/Baffin; India–Madagascar–Africa
Rocks/ fossils	Ghana gold placers from Brazil plateau; Mesosaurus fossils only in South Africa and Iraver formations of Brazil (4,800 km apart); 2-bn-yr-old rocks on both coasts
Climate	Tillite; India's Gondwana sediments have counterparts in six southern landmasses
Mountains	Caledonian/Hercynian (Europe) continue as Appalachians

⚠ Drawbacks

→ No free float of SiAl on SiMa: lithosphere moves over asthenosphere

→ Ocean floor isn't passive; tides/gravity too weak (convection currents the engine)

→ Only equator/west movement (but Antarctica moved away, Australia east, India NE)

→ No reason for Triassic break-up

Hess 1960: sea-floor spreading

→ War-time research: ocean floor full of relief (ranges, trenches)

→ Mid-oceanic ridges most volcanically active; ocean floors not static

→ Rocks equidistant either side of ridge crest similar in constituents and age

→ Shallow-focus quakes at ridges (deep-focus at Himalayan belt)

→ Oceanic rocks much younger than continental (subduct)

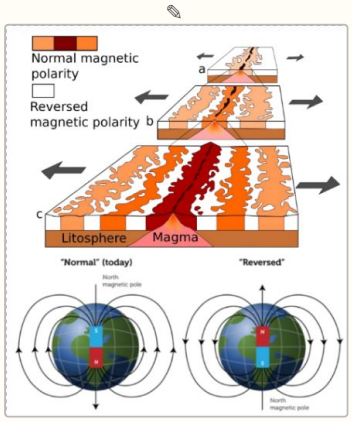
Holmes 1930s: convection

→ Radioactive heat in mantle (100–2900 km) seeks escape → convection currents

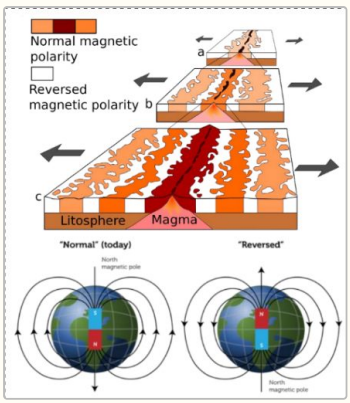
→ Rising limbs → divergence; falling limbs → convergence

Paleomagnetism

→ Rock minerals lock in



Paleomagnetism



Paleomagnetism

direction + intensity of magnetic field (age and location of formation); basalt on sea floor

→ Ocean floor each side of ridges has magnetic stripes: normal, then reversed

→ New rock at ridge pushes older (reversed-era) rock away; polarity reversals recorded in magma

Earlier Theories

→ Permanency Theory

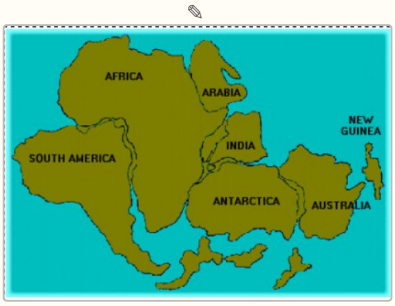
Continental Drift Theory

Assumptions:

→ Forces responsible: Tidal force (caused the Westward Movement) & Gravity + Polar Fleeing Force (caused the Equator-ward Movement).

→ – Tidal forces played a bigger role .

→ Disintegration of Pangea in the Triassic Period (251-199.6 mya; Mesozoic Era).



Continental Drift Theory · Tidal forces played a bigger role .

Arguments supporting the theory:

→ Jigsaw fit of shape:

→ – Greenland seems to fit in well with Ellesmere and Baffin islands .

→ – Mesosaurus was a small reptile adapted to shallow brackish water .

⚠ Drawbacks:

→ Tidal and Gravitational Forces are not enough to cause such massive land masses to move. Convective Currents cause this movement.

→ Did not mention why the Pangea suddenly broke only precisely in the Triassic period. Why did nothing happen before it?

Seafloor Spread Theory

→ The mid-oceanic ridges were found to be most active in terms of volcanic eruptions.

Convectional Current Theory

→ According to this theory, the intense heat generated by radioactive substances in the mantle (100-2900 km below the earth's surface) seeks a path to escape and gives rise to the formation of convection currents in the mantle.

Paleomagnetism

→ Basaltic rocks formed in the sea floor spread regions contain magnetic material that locks in the information.

→ When Earth's geomagnetic field undergoes a reversal, the change in polarity is recorded in the magma, which contributes to the alternating pattern of magnetic striping in the rocks on the seafloor.

16 · PLATE TECTONICS

Understand it first: The outer shell is broken into rigid plates that ride on the softer layer below. Everything interesting happens at their edges: they move apart, come together or slide past, and each kind of edge makes its own landforms, quakes and volcanoes.

Theory

Plate Tectonics Theory · Major Plates

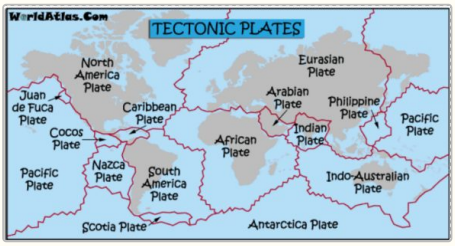


Plate Tectonics Theory · First outlined by Morgan in 1968 .

→ Suggested McKenzie and Parker 1967; outlined Morgan 1968

→ Lithosphere broken into plates (7 major, many minor) floating on asthenosphere under convection currents (Holmes 1930s)

→ Oceanic: Simatic, thinner (Pacific); continental: Sialic, thicker (Arabian); both (Indo-Australian)

→ Minor plates from stress where majors converge (Mediterranean: Eurasian + African)

→ Explains earthquakes, volcanoes, fold and block mountains

7 major plates

→ Antarctica + ocean
→ North American
→ South American
→ Pacific
→ India-Australia-NZ
→ Africa + E. Atlantic floor
→ Eurasia + ocean

Ring of Fire (Pacific boundary, NZ → Chile)

→ 70% earthquakes, 90% volcanoes (75% of active)
→ 20% earthquakes in Mid-continental belt
→ Slowest ridge Arctic (<2.5 cm/yr); fastest East Pacific Rise (>15 cm/yr, ~3,400 km west of Chile)

DCS – Divorce · Collide · Slide (divergent · convergent · transform)

Boundary	Plates...	Creates	Example
Divergent (constructive)	move apart	new crust, mid-ocean ridges, rift valleys, volcanic islands, shallow quakes	Mid-oceanic ridges, Great African Rift (Lebanon → Mozambique), Chagos
Convergent C–C	continent meets continent	fold mountains, no volcanoes (no subduction)	India–Eurasia → Himalayas
Convergent O–C	dense ocean plate subducts	trench, volcanoes, fold mountains (Cordilleran)	Pacific–S American → Andes
Transform (conservative)	slide past	only deformation; usually ⊥ ridges	San Andreas Fault

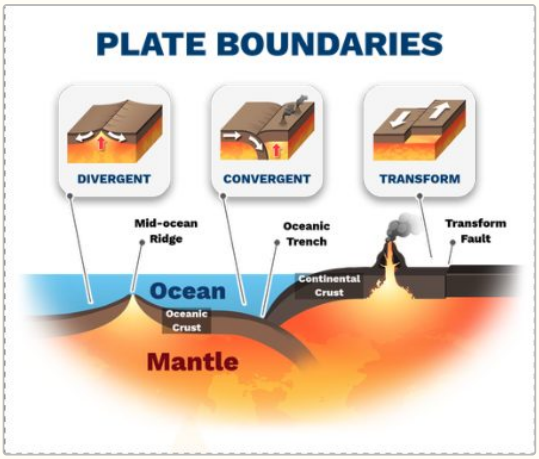


Plate Boundaries · Diverging/Constructive Boundaries

Himalayas have no volcanoes; Andes and Alps do.

~225 mya
Tethys separates India from Asia

~200 mya
moves north (Pangaea breaks)

~140 mya
at 50°S

~65 mya
Deccan Traps

40–50 mya
collision (some say 60)

Indian plate

→ island off Australia first; Himalayas still rising. Margins: E Arakan Yoma → Java Trench
→ W Kirthar
→ N Himalaya. **Limits of theory:** doesn't explain **hotspots**; can't place ancient plates or predict future boundaries.

Plate Tectonics Theory

→ Both 'convectional current theory' and 'seafloor spreading' paved the way for the Theory of Plate Tectonics.

→ **Forces for plate movement: the convectional currents** . Arthur Holmes gave the idea about these in 1930s.

Circum-Pacific Ring of Fire:

→ From **New Zealand to Chile.**

Plate Boundaries

Diverging/ Constructive Boundaries:

→ The sites where the **plates move away** from each other are called **spreading sites** .
→ Such edges are sites of earth **crust formation (hence constructive)** and **volcanic landforms** are common along such edges.
→ **Earthquakes (shallow focus)** are common along divergent edges.

Converging/ Destructive Boundaries:

→ The zone of collision may undergo crumpling and folding, and **folded mountains may emerge.**
→ The subducted material gets heated up and is thrown out forming volcanic islands and dynamic equilibrium is achieved.

Transforms/ Conservative:

→ **Generally perpendicular to oceanic ridges** .
→ The San Andreas Fault along the western coast of the USA is the best example.

Indian Plate:

→ Movement of Indian Plate:

→ – India was a large island situated off the Australian coast, in a vast ocean.

→ – The Tethys Sea separated it from the Asian continent till about 225 million years ago.

→ – India is supposed to have started her northward journey about 200 million years ago at the time when Pangaea broke.

→ – India collided with Asia about 40–50 million years ago causing rapid uplift of the Himalayas.

→ – About 140 million years before the present, the subcontinent was located as south as 50° S. latitude. The two major plates were separated by the Tethys Sea and the Tibetan block was closer to the Asiatic landmass.

→ – During the movement of the Indian plate towards the Asiatic plate, a major event that occurred was the outpouring of lava and formation of the Deccan Traps. This started somewhere around 65 million years ago and continued for a long period.

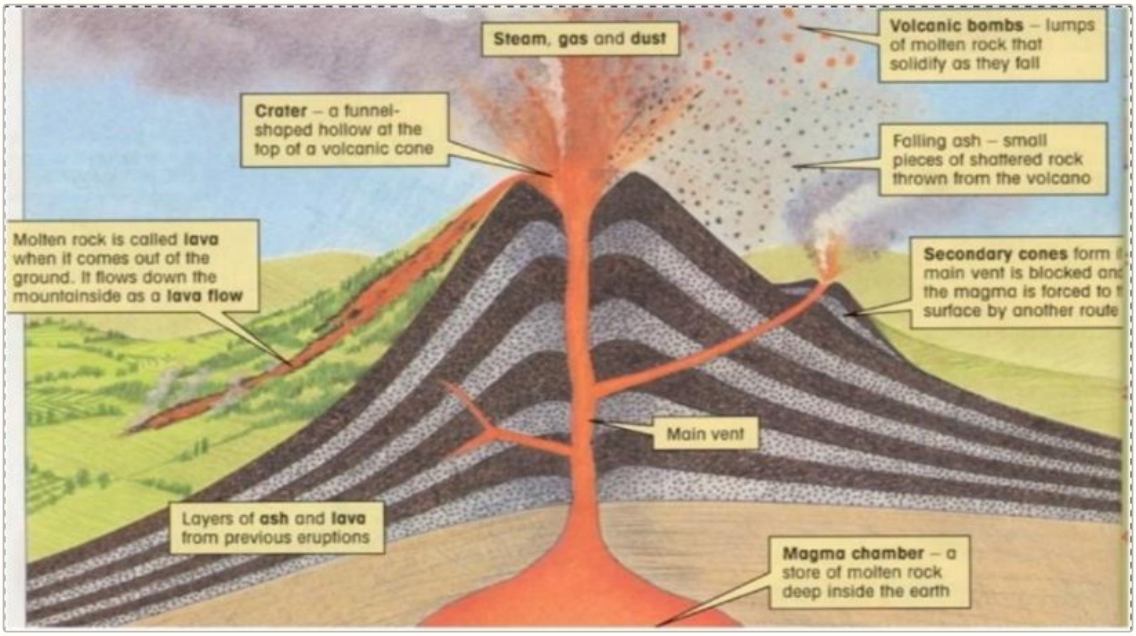
→ – Note that the subcontinent was still close to the equator. From 40–50 million years ago (disputed by some experts who say it started around 60 million years ago) and thereafter, the event of formation of the Himalayas took place.

→ – Scientists believe that the process is still continuing and the height of the Himalayas is rising even to this date.

🎯 17 · VOLCANISM PYQ 2024

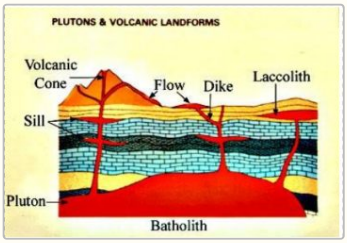
Understand it first: A volcano is an opening where **magma** reaches the surface. How runny the magma is decides everything: runny lava spreads quietly into broad shields and plateaus, sticky lava traps gas and explodes into steep cones. Then learn the intrusive forms that cool inside the crust.

Volcanism: molten rock from **asthenosphere** enters lithosphere, may or may not reach surface. **Volcano:** vent in crust erupting magma, gases, ash. Under Earth **magma**; after eruption **lava**. 💡



Volcanism

Volcanic Landforms · Extrusive: above earth's surface



Volcanic Landforms · Intrusive: Inside Earth's surface

Acidic lava

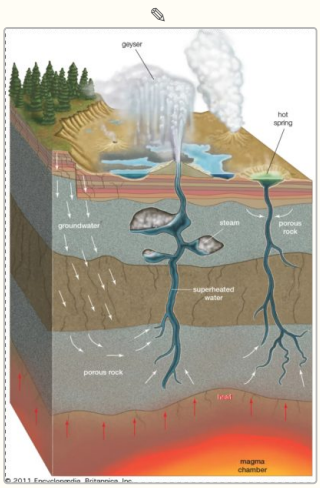
→ Light-coloured, low density, high silica
→ Highly viscous, flows slowly, little spread
→ Solidifies in vent → explosive, pyroclastic material

Basic lava

→ Dark, high density, low silica
→ Highly fluid, spreads widely
→ Lava plateaus, gentle volcanic mountains

Vents and gases

→ Fumarole: steam + gas (SO_2, CO_2); sulphur-rich = solfatara, CO_2-rich = mofette
→ Gases: carbon, sulphur, nitrogen oxides, water vapour, H_2S, trace chlorine
→ Geysers only in Iceland, New Zealand, Yellowstone (USA)



Geysers and Hot Springs

Geyser	Hot water spring
Steam or water at high pressure, along its path, gets accumulated in small reservoirs, fissures and fractures. Once the pressure exceeds the threshold limit, the steam bursts out to the surface disrupting the water at the mouth. Hence the name geyser. Usually a carter like structure is created at the mouth. Silicate deposits at mouth gives them their distinct colours Found in very few regions. **Iceland** is famous for its geysers.	Steam or water at high pressure smoothly flows to the top through the vent and condense at the surface giving rise to a spring. Usually a carter like structure is created at the mouth of the spring. Some springs are very colorful because of the presence of **cyanobacteria** of different colors. Found all across the world

Geysers and Hot Springs

Distribution

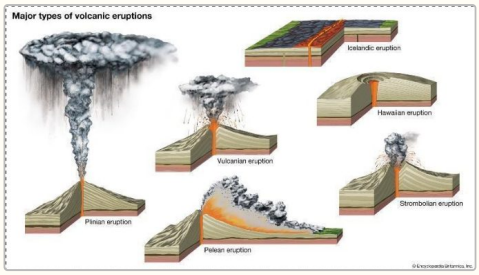
→ Only 10–20% of volcanic activity is above sea
→ Ring of Fire 75% active

→ Mid-Atlantic ridge: Iceland

→ Great African Rift: Kilimanjaro, Kenya

→ Cameroon (only active in W Africa); Pelée (West Indies); Mediterranean: Vesuvius, Stromboli, Ararat, Elburz

→ None in Australia; none in Himalaya



Types of Volcanoes · Icelandic : fissure type; basaltic lava

South Asia / India

→ Barren Island: only active lava volcano (last erupted 2022)

→ Narcondam extinct

→ Baratang: India's only mud volcano

→ Chandragup (Pakistan): mud, no lava/ pyroclast, "not a true volcano"

Extrusive (surface)	Intrusive (inside; seen after erosion)
Conical and fissure vents, basalt plateaus · mid-ocean ridges · pyroclastic debris · caldera and crater lakes (Crater Lake, Oregon)	Batholith/pluton (huge; Mt Wicklow) · Sill (flat; Great Whin Sill) · Laccolith (dome; Henry Mts) · Lopolith (saucer; Transvaal) · Phacolith (wavy; Cordon Hill) · Dyke (vertical; Alaska)

Sill=sill – Sill = window-sill, flat · Dyke = wall, upright · Laccolith = Loaf (dome) · Lopolith = Lota (bowl/saucer) · Phacolith = wavy

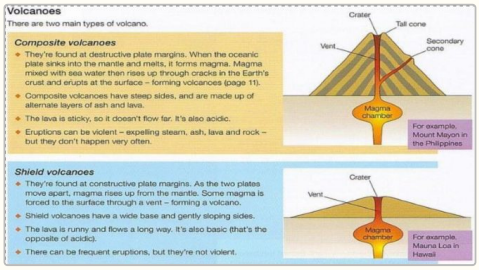
I HaveSomeVada-Pav,Please – Eruption styles calm → crazy: Icelandic (fissure, basalt) → Hawaiian (central vent, most lava) → Strombolian (short bursts, pasty lava) → Vulcanian (dark cauliflower clouds) → Pelean (most dangerous, glowing clouds) → Plinian (ash up to 50,000 ft, deadly pyroclastic flows)

Types of volcano

→ By frequency: Active (St Helens, Merapi) · Dormant (Vesuvius before AD 79) · Extinct (Narcondam)

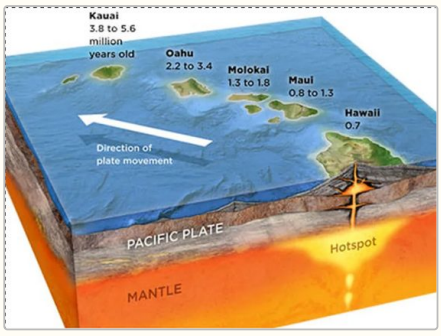
→ By shape: cinder cone · composite (strato) · shield

→ Vulcanian/Pelean/Plinian ash = hygroscopic nuclei → rain + ash → lahars (mudflows)



Types of Volcanoes · On the basis of the frequency of eruption

Mantle plume and hotspot



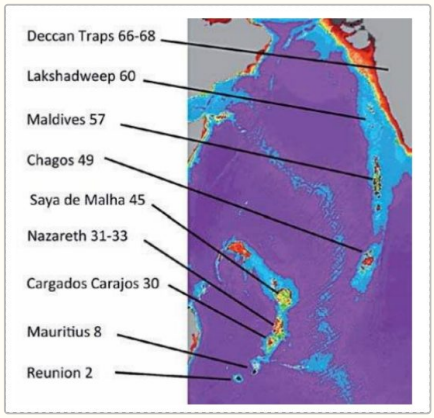
Mantle Plumes & Hotspots · Cause Hotspot Volcanism

→ Rising heat arm in mantle; intra-plate (not at boundaries)

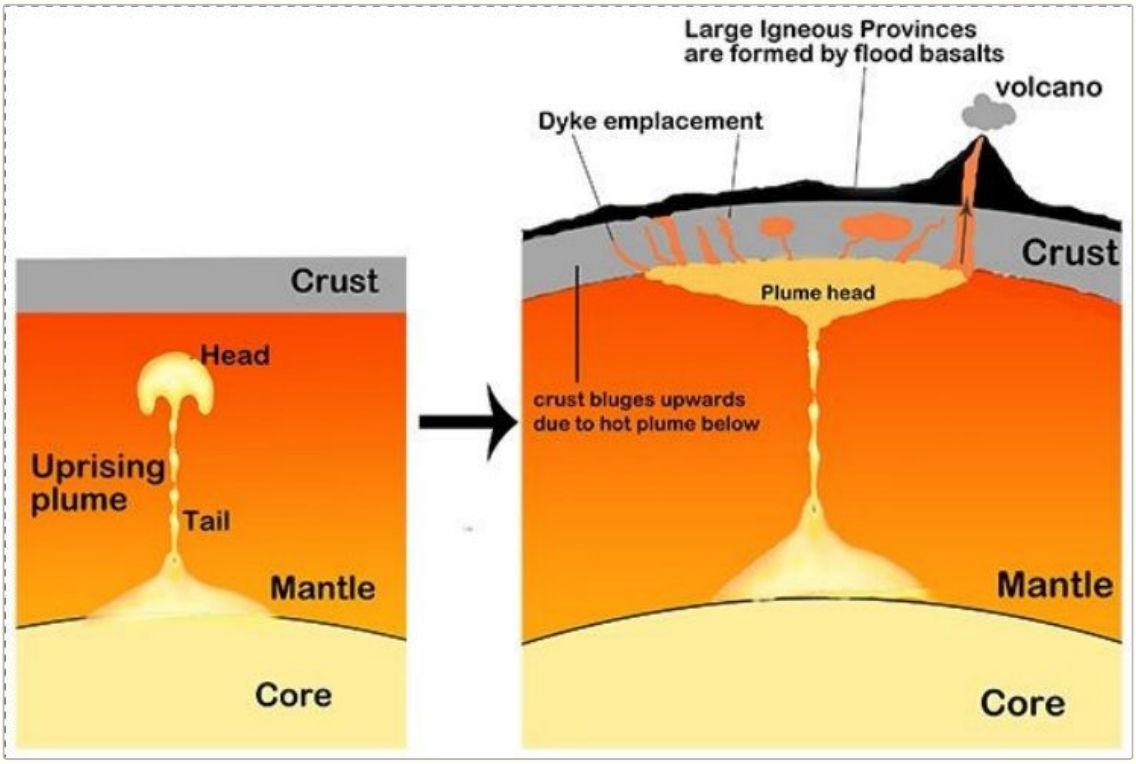
→ Plate moves over it → island chains (Hawaii); cut off from heat → extinct

→ Azores: hotspot + Mid-Atlantic ridge

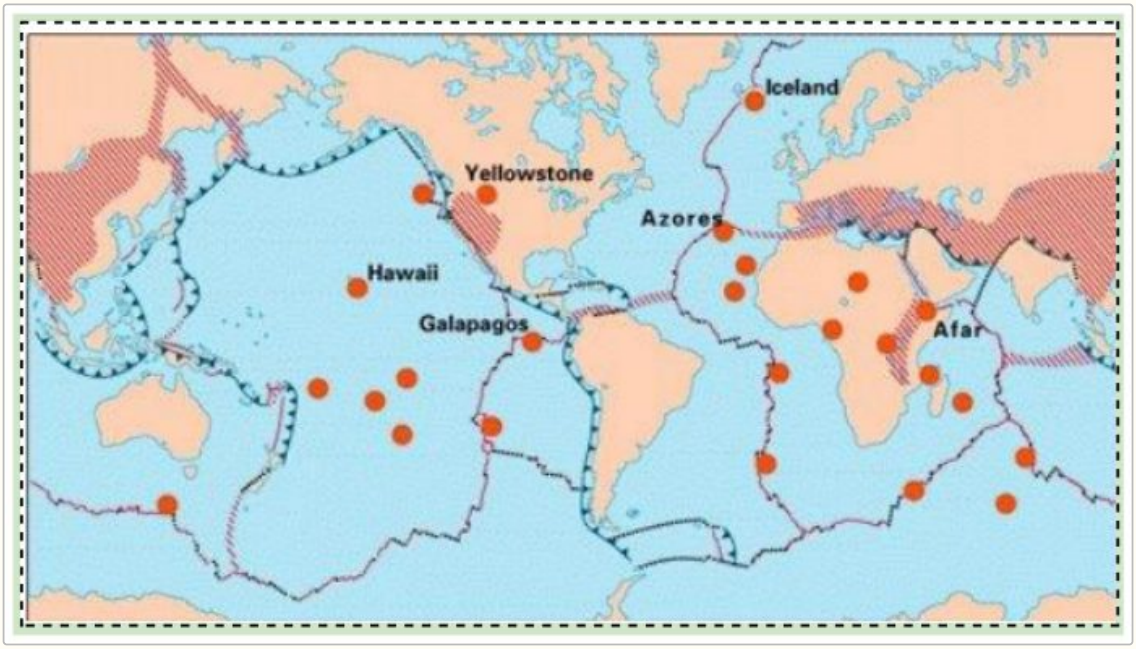
→ Réunion hotspot: active >65 million yrs; Indian plate passing over it → Deccan Traps



Mantle Plumes & Hotspots · Reunion Hotspot



Mantle Plumes & Hotspots



Mantle Plumes & Hotspots · Cause Hotspot Volcanism

Types of Magmas

→ 1. Light-coloured, of low density , and have a high percentage of silica .
→ 2. Highly Viscous ; flow slowly; do not spread much.
→ 3. Get solidifies in the vent, causing explosive eruptions and release of Pyroclastic material.
→ 1. Dark, high density, less percentage of silica
→ 2. Highly fluid ; flow readily; spread to large areas
→ 3. Lava plateaus ; gently sloping volcanic mountains
→ 1. A fumarole rich in sulphur gases is called a solfatara ; a fumarole rich in carbon dioxide is called a mofette .

Geysers and Hot Springs

→ Almost all the world's geysers are confined to three major areas: Iceland, New Zealand and Yellowstone Park of U.S.A.

Distribution of Volcanoes in the World

Volcanoes in India:	→ No volcanoes in the Himalayas, as it is a continent-continent collision boundary.

Volcanic Landforms

→ Appear on Earth's surface after erosion has removed the upper layers.
→ Massive: Batholiths or Plutons . Large rock masses formed due to the cooling down and the solidification of hot magma inside the earth. E.g. Mt. Wicklow of Ireland

Types of Volcanoes

👥 On the basis of structure:	→ Cinder cones : Small, steep-sided, conical volcanic landforms built primarily from pyroclastic fragments like cinders. These volcanoes form through explosive eruptions of gas-rich basaltic magma, creating a distinctive and relatively simple structure. Cinder cones often have relatively short lifespans compared to other types of volcanoes, as their eruptions tend to be singular events rather than prolonged periods of volcanic activity.
On the basis of Eruption	→ Hawaiian : similar to Icelandic. Difference: Hawaiian eruptions have the greatest quantity of lava pouring out of the main vent at the volcano's summit, not alongside fissures. → Vulcanian, Pelean and Plinian volcanoes release huge amount of Pyroclastic material that acts as Hygroscopic Nucleii, helping in creation of raindrops. The force of the volcano, pushes the air up, causing updraft. A combination of these processes, causes often thunderstorms and rainfall after the volcanoes take place. The rainwater gets mixed with the Pyroclastic mud, and create mudflows called LAHARS.

Mantle Plumes & Hotspots

→ Region within the Earth's mantle from which heat rises through the process of convection.
→ This does not occur at plate boundaries. It is an intra-plate event .

Cause Hotspot Volcanism	→ A volcano above a hotspot does not erupt forever . Attached to the tectonic plate below, the volcano moves and is eventually cut off from the hot spot. → Without any source of heat, the volcano becomes extinct and cools.

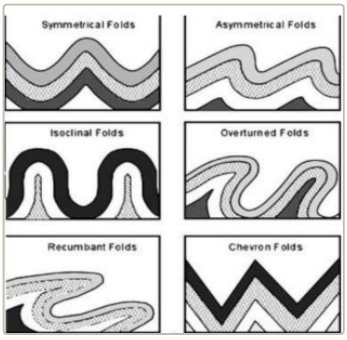
→ According to the latest studies, which are yet to be accepted by geologists, hotspots may be the first stage in the formation of diverging plate boundaries.

Reunion Hotspot: → Reunion Island is over reunion hotspot.

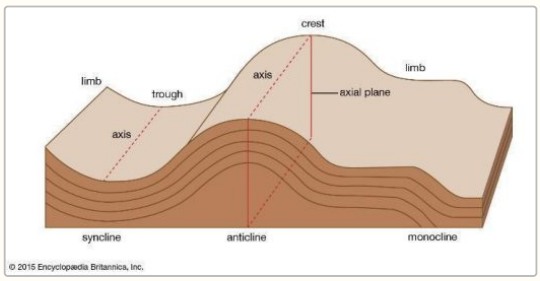
18 · OROGENY: MOUNTAINS, ISOSTASY

Understand it first: Mountains are sorted by **how they were made**: squeezed into folds, lifted or dropped as blocks along faults, piled up by volcanoes, or left standing after erosion. **Isostasy** is the balance idea: higher mountains have deeper roots, like a bigger iceberg sitting deeper in water (memory aid).

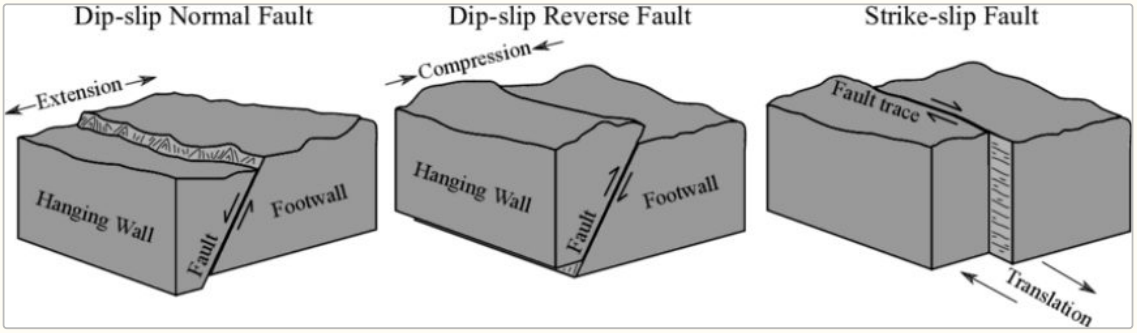
Orogeny = mountain building. By location: continental, coastal, oceanic. By formation: fold, fault (block), volcanic, residual.



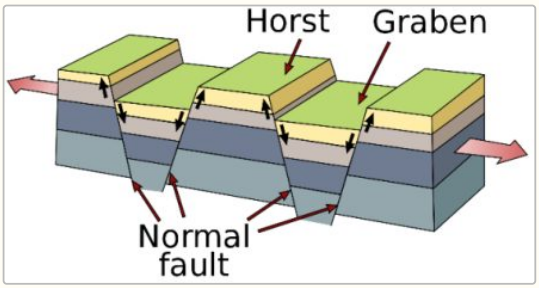
Folds And Faults



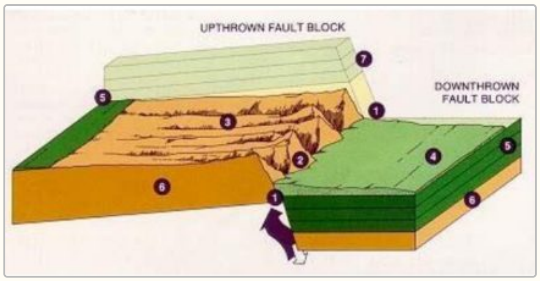
Folds And Faults



Folds And Faults



Folds And Faults



Folds And Faults

Mountain	Formed by	Notes / examples
Fold	Compression of sedimentary strata; loftiest; young near converging boundaries	Himalaya (India–Eurasia), Alps and Atlas (Eurasia–Africa), Andes (S America–Pacific): quakes + volcanoes · old: Appalachians (USA), Aravallis
Block (fault)	Dip-slip faults (vertical displacement), tension/ compression; horst = raised, graben = lowered	Great African Rift (floor graben; Mitumba and Muchinga horsts), Rhine valley–Vosges
Volcanic	Lava, bombs, cinders, ash, dust, mud in layers around vent	"Mountains of accumulation"; Circum-Pacific: Fuji, Mayon, Merapi (Sumatra), Etna

Residual	Left after denudation of fold/fault/volcanic mountains	Monadnocks (USA)

Isostasy

→ Equilibrium between lithosphere and asthenosphere (buoyancy vs gravity)
→ Crust subsides/uplifts when load added or removed (erosion, mountain building, snow/ice melt)
→ Causes: tectonic (subsidence, uplift) and climatic
→ Can cause earthquakes; ice melt → rebound → cracks → may reactivate Antarctic volcanoes

16-7-48 — Longest ranges (km): Mid-Atlantic Ridge 16,000 · Andes 7,000 · W. Cordillera 4,830 · Great Dividing Range & Trans-Antarctic 3,500 · Urals & Atlas 2,500 · Appalachians 2,414 · Himalayas 2,400 · Altai 2,000. Say "16, 7, 48, 35-35, 25-25, 24-24, 20".

Mountain Building

→ Process known as Orogeny .

Fold Mountains

→ There are old fold mountains too which were formed millions of years ago at the active plate boundaries of that time. E.g. Appalachians (USA), Aravallis (India)

Block Mountains

There are two basic types .

→ Tilted block mountains have one steep side contrasted by a gentle slope on the other side.
→ Lifted block mountains have a flat top and extremely steep slopes

Volcanic Mountains

→ Materials including molten lava, volcanic bombs, cinders, ashes, dust and liquid mud fall around the vent in successive layers, building up a characteristic volcanic cone.
→ They are common in the Circum-Pacific belt and include such volcanic peaks as Mt. Fuji (Japan) Mt. Mayon (Philippines), Mt. Merapi (Sumatra), Mt. Etna (Italy) etc.

Residual Mountains

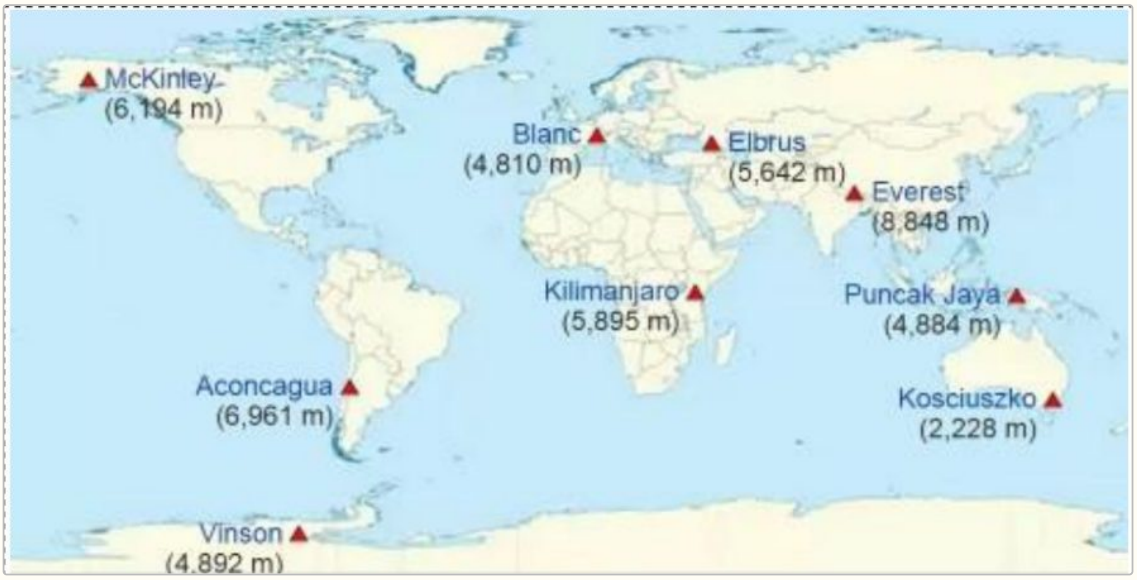
→ The mountains left after erosional activities have eroded the majority portion of the erstwhile fold/fault/volcanic mountains.
→ Basically, they evolve from denudation.

Isostasy

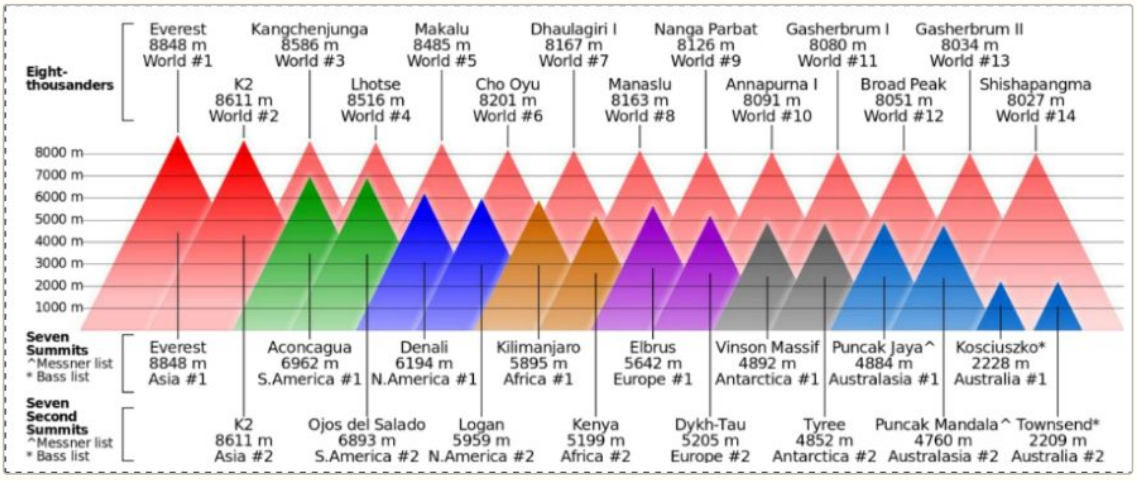
→ The Isostatic re-adjustments can be due to both Tectonic factors (subsidence & Upliftment) , as well as Climatic Factors (addition or melting of snow) .
→ Can cause earthquakes . Scientists have predicted that they can also be responsible for the reactivation of volcanoes in Antarctica , as the ice melts, the lithosphere rebounds, cracks get created, and magma might pour out.

S. No	Mountain	Location	Length
1.	Mid-Atlantic Ridge	Atlantic Ocean Basin	16000 km
2.	Andes	South America	7000 km
3.	Western Cordillera	North America	4830 km
4.	Great Dividing Range	Australia	3500 km
5.	Trans-Antarctic Mountains	Antarctica	3500 km
6.	Ural Mountains	Eurasia	2500 km
7.	Atlas Mountains	Africa	2500 km
8.	Appalachian Mountains	North America	2414 km
9.	Himalayas	Asia	2400 km
10.	Altai Mountains	Asia (Mongolia)	2000 km

Important Mountain Ranges



Important Mountain Ranges



Important Mountain Ranges

19 · ROCKS AND ROCK CYCLE PYQ 2022

Understand it first: Three rock families by origin: **igneous** (cooled from melt), **sedimentary** (laid down in layers) and **metamorphic** (changed by heat and pressure). The rock cycle only says that any one can become any other.

	Igneous (primary)	Sedimentary	Metamorphic
Origin	Magma/lava cooling	Denudation → deposits → lithification	PVT (pressure, volume, temperature) change of existing rock; recrystallise
Features	No fossils; plutonic/intrusive (slow, deep, big grains) vs extrusive (fast, surface, small smooth grains); acid (granite) light/less dense vs basic	Layered/strata, fossils, porous; 75% of crust area, 5% of volume; tillite (ice), loess (wind)	Foliation, lineation, banding; may/may not have strata and fossils; heat = thermal, pressure = dynamic types
Examples	Granite, gabbro, pegmatite, basalt	Sandstone, shale, limestone, coal, chalk, chert, halite	Gneiss, slate, schist, marble, quartzite
Value	Metal ores (iron, nickel, copper, lead, zinc, chromite, manganese, gold, diamond, platinum); building stone; black soil	Petroleum, gas, coal, phosphates, hematite iron ore, limestone; placers (bauxite, manganese, tin); rich soils	–

Sedimentary Rocks

Some Examples of Metamorphism

- Granite —Pressure→ Gneiss
- Clay, Shale —Pressure→ Schist
- Sandstone —Heat→ Quartzite
- Clay, Shale —Heat→ Slate —Heat→ Phyllite
- Coal —Heat→ Anthracite, Graphite
- Limestone —Heat→ Marble

Metamorphic Rocks · May or may not have strata and fossils.

Sedimentary by mode

- → Mechanical: sandstone, conglomerate, shale, loess
- → Organic: chalk, limestone, coal, geyserite
- → Chemical: chert, limestone, halite, potash

Amygdales: almond bubbles in basalt filled with minerals (copper). Metamorphism: rocks forced down by tectonics or contacted by rising magma.

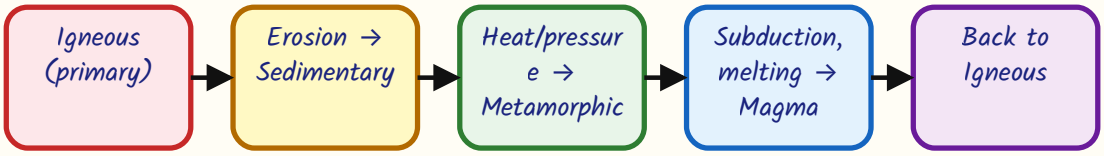
Plutonic rocks	Volcanic rocks
Intrusive rocks	Extrusive rocks
Granite	Basalt
Slow cooling allows big-sized crystals (large grains)	Rapid cooling prevents crystallization, as a result such rocks are fine-grained
Less dense and are lighter in colour than basic rocks	Denser and Darker in colour

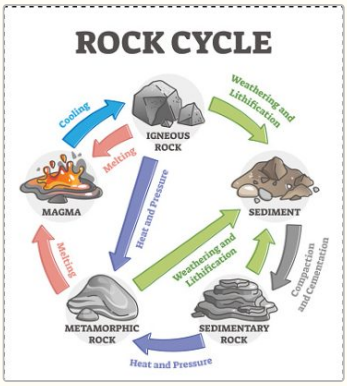
Rocks

Mineral	Chemical Composition	Hardness/ Softness	Colour	Places on Earth's Surface Where Found	Uses
Feldspar	KAlSi3O8 or NaAlSi3O8	6-7	White, pink, grey, or green	Abundant in igneous, metamorphic, and sedimentary rocks	Glass, ceramics, electronics
Quartz	SiO2	7	Colourless, white, grey, purple, pink, yellow, or black	Abundant in igneous, metamorphic, and sedimentary rocks	Jewellery, gemstones, sandblasting, electronics
Bauxite	Al2O3•2H2O	1-3	Red, orange, or yellow	Tropical and subtropical regions	Aluminium production
Cinnabar	HgS	2-3	Reddish-brown or black	Mercury deposits worldwide	Mercury production
Dolomite	CaMg(CO3)2	3.5-4	White, gray, or pink	Abundant in sedimentary rocks	Construction materials, agriculture, pharmaceuticals
Gypsum	CaSO4•2H2O	2	White, gray, or yellow	Deposition basins in arid and semi-arid regions	Construction materials, plaster, drywall
Hematite	Fe2O3	5.5-6	Reddish-brown or black	Abundant in sedimentary and metamorphic rocks	Iron production, pigments, abrasives
Magnetite	Fe3O4	5.5-6	Black	Igneous and metamorphic rocks	Iron production, magnets, electronics

Some Important Minerals

Rock cycle (continuous; any rock can become any other)





Rock Cycle

Mineral	Mohs hardness (1 soft – 10 hard) and use
Feldspar	6–7 (glass, ceramics)
Quartz	7 (jewellery, electronics)
Bauxite	1–3 (aluminium)
Cinnabar	2–3 (mercury)
Dolomite	3.5–4
Hematite and Magnetite	5.5–6 (iron)

Rocks

→ Based on place and time taken in cooling of the molten matter, igneous rocks can be divided into:

→ If molten material is cooled slowly at great depths (plutonic/intrusive), mineral grains may be very large.

Economic Significance:

→ Many igneous rocks like granite are used as building material as they come in beautiful shades.

Sedimentary Rocks

→ These rocks have fossils of plants and animals.

→ These rocks are generally porous and allow water to percolate through them.

→ Formed as a result of denudation (weathering and erosion).

→ These deposits through compaction turn into rocks . This process is called lithification.

→ Cover 75 percent of the earth's crust but volumetrically occupy only 5 per cent.

→ Depending upon the mode of formation ,Sedimentary rocks are classified into

→ Economic Significance:

→ – But some important minerals such as hematite iron ore, phosphates, building stones, coals, petroleum and material used in cement industry like limestone are found. The decay of tiny marine organisms yields petroleum.

→ – Petroleum occurs in suitable structures only.

→ – Important minerals like bauxite, manganese, tin are derived from other rocks but are found in gravels and sands carried by water. (PLACER DEPOSITS)

Metamorphic Rocks

→ The word metamorphic means 'change of form' . Form under the action of pressure, volume and temperature (PVT) changes.

→ Metamorphism is a process by which already consolidated rocks undergo recrystallization and reorganization of materials takes place.

→ Sometimes minerals or materials of different groups are arranged into alternating thin to thick layers . Such a structure is called banding.

→ On the basis of the agency of metamorphism, metamorphic rocks can be of two types:

→ – Dynamic Metamorphism : This refers to the formation of metamorphic rocks under the stress of pressure . Sometimes high pressure is accompanied by high temperatures.

Rock Cycle

→ Rock cycle is a continuous process through which old rocks are transformed into new ones .

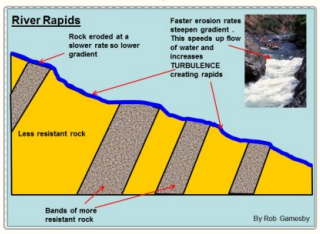
Some Important Minerals

→ The numbers in the "Hardness/Softness" column of the table refer to the Mohs Hardness Scale, a qualitative scale for measuring the scratch resistance of minerals. The scale ranges from 1 to 10, with 1 being the softest and 10 being the hardest.

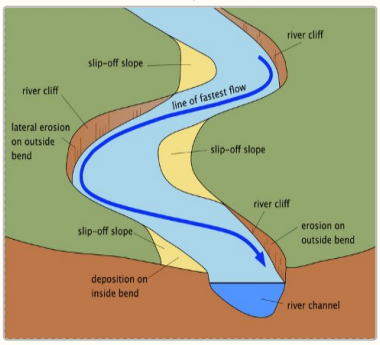
20 · EROSION, FLUVIAL CYCLE

Understand it first: Running water is the main sculptor of land. A river cuts in its upper course, carries in the middle and drops its load in the lower course, and each stage leaves its own landforms. Davis read this as youth, maturity and old age.

Erosion

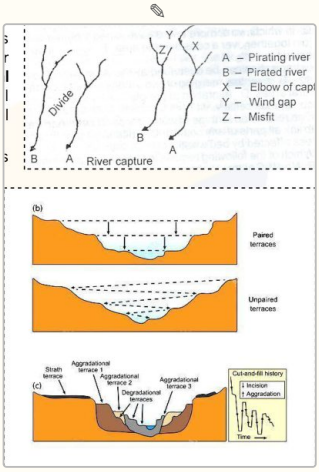


Fluvial Erosion · Major Erosional Fluvial Landforms

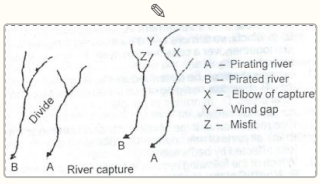


Fluvial Erosion · Terms associated with Fluvial Erosion

Fluvial Erosion · Major Erosional Fluvial Landforms



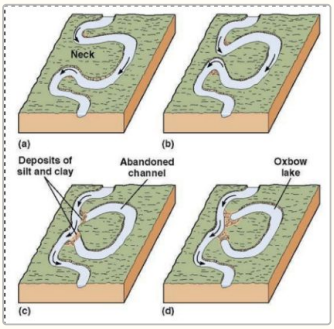
Fluvial Erosion · Major Erosional Fluvial Landforms



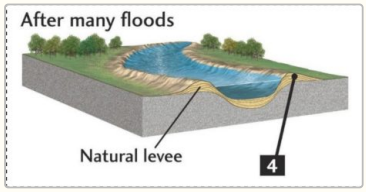
Fluvial Erosion · Major Erosional Fluvial Landforms

→ = removal +

Fluvial Erosion · Peneplains: End Product of the erosion cycle.



Fluvial Erosion · Major Erosional Fluvial Landforms



Fluvial Erosion · Major Depositional Fluvial Landforms

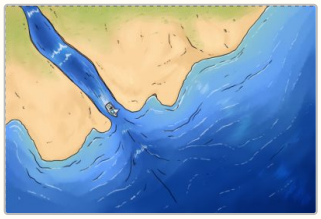
transportation + deposition (transport separates it from weathering). Agents: flowing water, underground water, wind, glaciers, coastal waves. Five cycles: fluvial

→ glacial

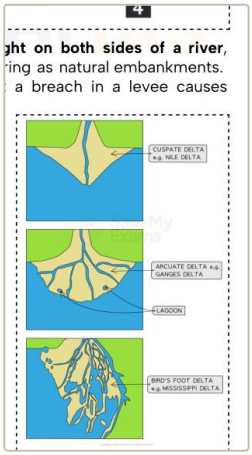
→ marine

→ arid

→ karst.



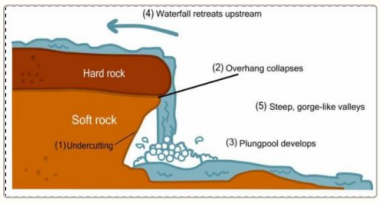
Fluvial Erosion · Major Depositional Fluvial Landforms



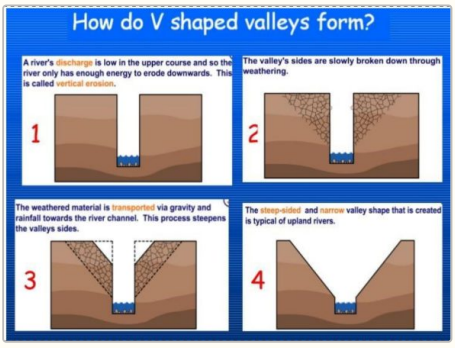
Fluvial Erosion · Major Depositional Fluvial Landforms

YMO – River life: Young – V, Mature – meander, Old – delta. Youth: fast, V-valley, waterfalls, rapids, headward erosion · Mature: lateral erosion, meanders, U-profile · Old: deposition, braiding, delta

Fluvial erosion



Fluvial Erosion · Valley Formation



Fluvial Erosion · Valley Formation

→ Types: hydraulic action, corrosion (chemical), attrition (load vs load), corrasion/abrasion (load vs rock)

→ Downcutting deepens, lateral widens, headward lengthens → river piracy (Saraswati; Shyok captured Nubra)

→ Rejuvenation: stage change from tectonic (submergence/emergence) or climatic (sea level) activity → terraces

→ Valley: young = deep, narrow, V, vertical downcutting; mature = U, broad base, concave slope; tributary valley hangs above main valley (falls/rapids)

Erosional forms

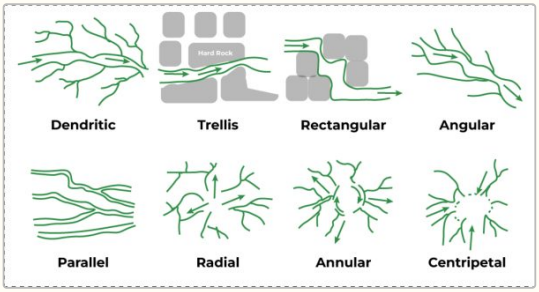
→ Gorge (narrow, steep; Indus at Nanga Parbat, Brahmaputra at Namche Barwa: antecedent rivers) vs canyon (broader; Grand Canyon, Colorado, Arizona: 483 km long, 2.88 km deep)

→ Waterfall (youth; Jog/Gersoppa on Sharavati 260 m), rapids/ cataracts
→ Potholes (swirling boulders drill)
→ Meander: outer = cliff-slope, erosion, concave; inner = slip-off, deposition, point bars
→ Ox-bow lake (cut-off meander; Ganga shifted south leaving ox-bows north)
→ Peneplain (end product) with monadnocks

Depositional forms

→ Alluvial fans/cones
→ Natural levees (breach = flood)
→ Delta: arcuate fan; bird's-foot weak tides; cuspate strong waves
→ Braided channel
→ Estuary: drowned valley; ports, fisheries

Drainage pattern	Description	Example
Dendritic/ Pinnate	irregular tree	Indus, Godavari, Mahanadi, Cauvery, Krishna
Trellis	short tributaries at right angles; soft-rock erosion	Seine, Paris basin
Rectangular	main stream and tributaries bend at right angles	Colorado (USA); Mahanadi
Angular	tributaries at acute angles	Himalayan foothills
Parallel	parallel streams, uniform slope	rivers of lesser Himalayas
Radial	from a summit in all directions	Central French Plateau, Kilimanjaro, Amarkantak, Saurashtra
Annular	ring around summit (soft outer stratum)	Black Hills, S. Dakota
Centripetal	all converge into low basin	Ladakh, Tibet, Bagmati (Nepal)



Drainage Patterns

Handout conflict: Mahanadi listed under both dendritic and rectangular. Take dendritic unless the question says rectangular.

Fluvial Erosion

⏳ Terms associated with Fluvial Erosion

→ How does water erode?
→ – Hydraulic Action : the force of running water wearing down rocks.
→ – Attrition : river load particles striking, colliding against each other and breaking down in the process.
→ – Corrasion or abrasion : solid river load striking against rocks and wearing them down.
→ Classification of erosion based on the location:

→ – Headward erosion : erosion at the origin of a stream channel, which causes the origin to move back away from the direction of the stream flow, and so causes the stream channel to lengthen. It is responsible for river Piracy .

Evolution of Fluvial Erosion:

→ Splash : A drop of water falls on the surface and disturbs the sediments, creating a tiny crater due to its kinetic energy.

→ Sheet: A sheet of water flows over the surface, and carries away the loose sediments of the topsoil.

→ Rills, Gullies & Valleys: After the sheet erosion, the areas with weaker rocks, or with rock joints, get more eroded than the rest. This creates small, narrow channels called Rills. Rills grow to form Gullies, which grow to form Valleys.

Stages of Erosion:

→ Youth stage: Young rivers close to their source tend to be fast-flowing, high-energy environments with rapid headward erosion , despite the hardness of the rock over which they may flow. Steep-sided "V-shaped' valleys, waterfalls, and rapids are characteristic features.

→ Mature stage: Mature rivers are lower-energy systems . Erosion takes place on the outside of bends, creating looping meanders in the soft alluvium of the river plain. Deposition occurs on the inside of the bends and the riverbed.

→ Old Stage: At a river's mouth (C), sediment is deposited as the velocity of the river slows. As the river becomes shallower more deposition occurs, forming islands and braiding the main channel into multiple, narrower channels. As the sediment is laid down, the actual mouth of the river moves away from the source into the sea or lake, forming a delta.

→ Valley Formation:

→ The extended depression on the ground through which a stream flows throughout its course is called a river valley. At different stages of the erosional cycle, the valley acquires different profiles.

→ The profile of the valley here is typically 'V' shaped.

→ It is characterized by cliffs or steep walls, and it is usually formed by more intense erosional forces, such as the rapid erosion of a river through resistant rock.

→ A gorge is typically considered a narrower and steeper landform , whereas a canyon is broader with more open valley walls. However, there is no universally strict scientific boundary between the two, and regional usage often overlaps.

→ As the cycle attains maturity, the lateral erosion becomes prominent, and the valley floor flattens out . The valley profile now becomes typically 'U' shaped with a broad base and a concave slope.

Major Erosional Fluvial Landforms:

→ Waterfall: A waterfall is simply the fall of an enormous volume of water from a great height. They are mostly seen in the youth stage of the river. Relative resistance of rocks, relative difference in topographic reliefs, fall in the sea level and related rejuvenation, earth movements etc. are responsible for the formation of waterfalls. For example, Jog or Gersoppa Falls on Sharavati has a fall of 260 meters.

→ Rapids: Formed due to a sporadic presence of soft and hard rocks along the river valley. The hard rocks don't get eroded as much as soft rocks, causing rivers to jump around, creating rapids. Called Cataracts in larger dimensions.

→ Potholes: The kettle-like small depressions in the rocky beds of the river valleys are called potholes which are usually cylindrical in shape.

→ – Potholing or pothole-drilling is the mechanism through which the grinding tools (fragments of rocks, e.g. boulders and angular rock fragments) when caught in the water eddies or swirling water start dancing circularly and grind and drill the rock beds of the valleys like a drilling machine.

→ – They thus form small holes which are gradually enlarged by the repetition of the said mechanism. The potholes go on increasing in both diameter and depth.

→ Terraces: Stepped benches along the river course in a floodplain are called terraces. Terraces represent the level of former valley floors and remnants of former (older) flood plains. Can be paired or unpaired. Created due to rejuvenation.

→ Meanders : A meander is defined as a pronounced curve or loop in the course of a river channel.

→ Ox-Bow Lake:

→ – Sometimes, because of intensive erosion action, the outer curve of a meander gets accentuated to such an extent that the inner ends of the loop come close enough to get disconnected from the main channel and exist as independent water bodies.

→ – These water bodies are temporary and get converted into swamps in due course of time.

→ – In the Indo-Gangetic plains, southwards shifting of Ganga has left many ox-bow lakes to the north of the present course of the Ganga

→ Peneplains: End Product of the erosion cycle.

→ – This refers to an undulating featureless plain punctuated with low-lying residual hills of resistant rocks.

→ – Gently undulating (wave-like), almost featureless plain that, in principle, would be produced by fluvial erosion that would, in the course of geologic time, reduce the land almost to base level (sea level), leaving so little gradient that essentially no more erosion could occur.

→ Monadnocks: An isolated hill of bedrock standing conspicuously above the general level of the surrounding area. Monadnocks are left as erosional remnants because of their more resistant rock composition; commonly they consist of quartzite or less jointed massive volcanic rocks.

Major Depositional Fluvial Landforms:

→ Alluvial fans and cones: When a stream leaves the mountains and comes down to the plains, its velocity decreases due to a lower gradient. As a result, it sheds a lot of material, which it had been carrying from the mountains, at the foothills. This unconsolidated deposited material acquires a conical shape and appears as a series of continuous fans. These are called alluvial fans or cones.

→ Natural Levees: These are narrow ridges of low height on both sides of a river , formed due to deposition action of the stream, appearing as natural embankments. These act as a natural protection against floods but a breach in a levee causes sudden floods in adjoining areas.

→ Delta : A delta is a tract of alluvium at the mouth of a river where it deposits more material than can be carried away. The river gets divided into distributaries which may further divide and rejoin to form a network of channels. A delta is formed by a combination of two processes:

→ – The load-bearing capacity of a river is reduced as a result of the check to its speed (due to reduced gradient) as it enters a sea or lake

→ – Clay particles carried in suspension in the river coagulate in the presence of salt water and are deposited. The finest particles are carried farthest to accumulate as bottom-set beds.

→ – Depending on the conditions under which they are formed, deltas can be of many types.

→ – Arcuate (fan-shaped); Bird's Foot (elongated , as in regions with weak tides and waves); Cuspate (Triangle ; strong wave action; almost nil distributaries)

→ Braided Channel: A river channel that splits into multiple smaller, intertwining channels separated by sediment bars or islands, typically formed due to excessive sediment load and variable flow rates

→ Estuaries: Sometimes the mouth of the river appears to be submerged. This may be due to a drowned valley because of a rise in sea level. Here fresh water and the saline water get mixed.

→ – When the river starts 'filling its mouth' with sediments, mud bars, marshes and plains seem to be developing in it.

→ – These are ideal sites for fisheries, ports and industries because estuaries provide access to deep water, especially if protected from currents and tides. Example: Hudson estuary.

Drainage Patterns

→ The typical shape of a river course as it completes its erosional cycle is referred to as the drainage pattern of a stream. A drainage pattern reflects the structure of basal rocks, resistance and strength, cracks or joints and tectonic irregularity, if any.

21 · KARST CYCLE PYQ 2025

Understand it first: Karst is what happens when slightly acidic water dissolves limestone: holes and caves below, sinks and dry valleys above, and dripstone features inside the caves.

Groundwater words

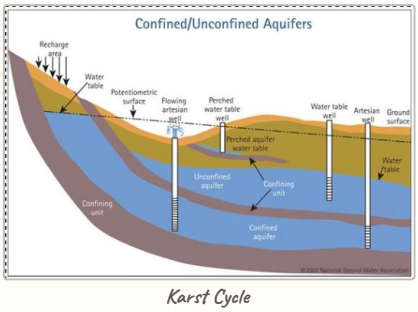
→ Porous (pore spaces, sandstone) · Permeable (water passes; clay porous but not permeable)

→ Aquifer stores and transmits water; water table = top of saturation zone

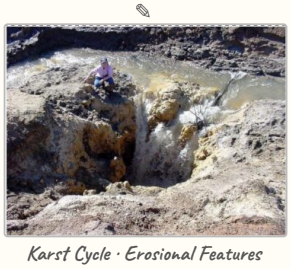
→ Confined (impermeable cover, pressure) vs unconfined

→ Artesian well (rises unpumped); spring; perched water table

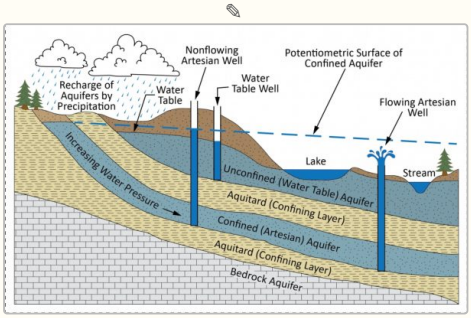
Karst



Karst Cycle



Karst Cycle · Erosional Features



Karst Cycle

→ Named after a province of Yugoslavia (now Slovenia) on Adriatic coast

→ Thick limestone, dolomite or gypsum; rock dense, highly jointed, thickly bedded

Karst Cycle · Erosional Features

Karst Cycle · Erosional Features

Karst Cycle · Blind Valley

Karst Cycle · Blind Valley

Karst Cycle · Blind Valley

Karst Cycle · Blind Valley

Erosional

→ **Sinkholes/swallow holes** (3–9 m; dolines, uvalas, polje)

→ **Lapies** (grooves and ridges)

→ karst window, arch/bridge, caves, blind valley

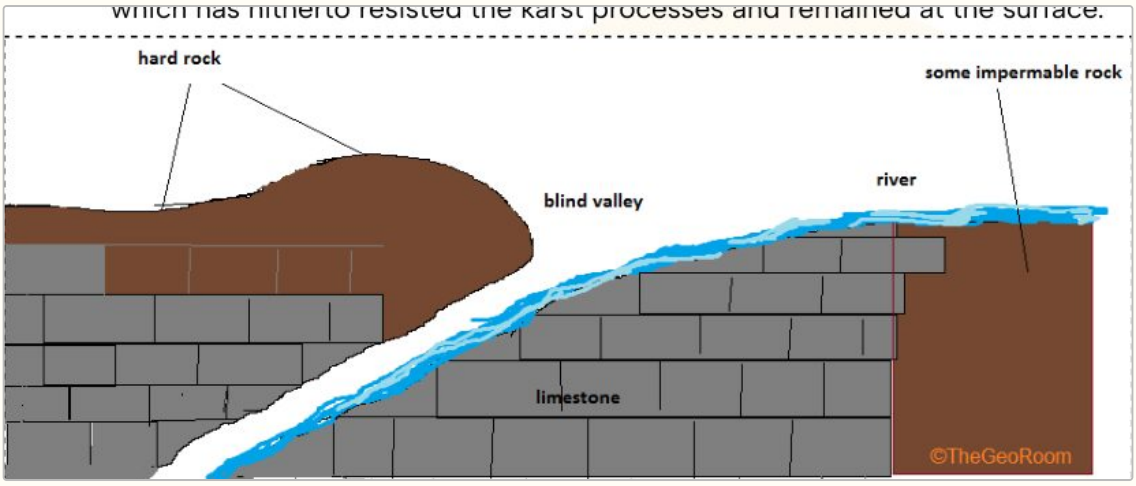
Depositional (speleothems, in caves)

→ **Stalactite** hangs from roof

→ **Stalagmite** rises from floor

→ **Column** when joined

→ flowstone, cave pearls, draperies



Karst Cycle · Blind Valley

Ceiling-Ground – Stalactite = Ceiling (holds tight to the roof). Stalagmite = Ground (might reach the roof).

Karst Cycle

→ **Terminologies associated with Groundwater:**

→ Groundwater: Water that percolates into soil and rock pores.

→ Porous Rocks: Rocks which have a lot of pore spaces between their grains. E.g. sandstone

→ Permeable/Pervious Rocks: Rocks that allow water to pass through them easily.

→ – Most (but not all) porous rocks are permeable. E.g. Clay: highly porous, but the pores are not big enough to allow easy movement of water.

→ Groundwater Springs : Natural discharge points where groundwater flows to the surface, typically occurring where the water table intersects the ground.

Karst Cycle

→ – Sinkholes/Swallow holes: Sinkholes are funnel-shaped depressions having an average depth of three to nine meters. These holes are developed by enlargement of the cracks found in such rocks, as a result of continuous solvent action of the rainwater . The surface streams that sink disappear underground through swallow holes. Larger sinkholes can be called Dolines, Uvalas and Polje (faulting+erosion) .

→ – Lappies: usually formed when the limestones are vertically bedded . Lappies are uneven grooves and ridges that form when the majority of the limestone's surface is removed by the solution process.

→ · Deep grooves separate the etched, fluted, and pitted rock pinnacles that make up the weathered limestone surface seen in the karst region.

→ – Karst Windows: When several adjoining sinkholes collapse , they form an open, broad area called a karst window.

→ – Karst Arch/Bridge: When a part of the cavern collapses the portion which keeps standing forms an arch.

→ – Karst Cave: This is an underground cave formed by water action by various methods in a limestone or chalk area.

→ – Blind Valley:

→ · A stephead valley, or blind valley is a deep, narrow, f lat-bottomed valley with an abrupt ending.

→ · Karst valley abruptly terminated by the passage underground of the watercourse which has hitherto resisted the karst processes and remained at the surface.

→ Depositional Features: Found in the Karst Caves and known as Speleothems .

→ – Stalactites, Stalagmites and Karst Pillars: The water containing limestone in solution, seeps through the roof in the form of a continuous chain of drops.

→ · A portion of the solution hangs on the roof and on evaporation of water, a small deposit of limestone is left behind contributing to the formation of a stalactite, growing downwards from the roof.

→ · The remaining portion of the drop falls to the floor.

→ · This also evaporates, leaving behind a small deposit of limestone aiding the formation of a stalagmite, thicker and flatter, rising upwards from the floor.

→ · Sometimes, stalactite and stalagmite join to form a complete pillar known as the column or Karst pillar .

→ – Flow stones: Flowstone is one of the most common speleothems. It has been described as "melted cake icing" and "frozen waterfalls." Flowstone is usually composed of calcite or other carbonate minerals, and deposition is in layers or bands. Where composed of calcite, individual flowstone layers may be very colourful: yellows, reds, and oranges.

→ – Cave Pearls: Cave pearls are concentrically banded concretions that form in shallow cave pools. Cave pearls vary in shape; they can be spherical, cylindrical, irregular, cubical, or even hexagonal. They can range in size from smaller than a sand grain up to eight inches (20 cm) in diameter.

→ – Banded draperies: When water drops flow down a sloped ceiling before dripping to the floor, calcite can build up in a line. These lines gradually form "draperies" or "cave bacon."

→ · This type of speleothem is found in almost every cave in the world and is universally popular because of its close resemblance to its namesake.

→ · Iron oxide or organic solutions form the baconlike stripes.

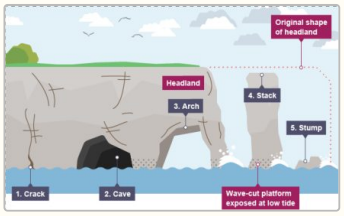
22 · MARINE CYCLE AND COASTS

Understand it first: Waves do the same two jobs as rivers: they erode cliffs and headlands and deposit beaches, bars and spits. Coasts are then classed by whether the land sank (submergence) or rose (emergence).

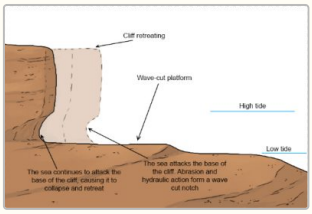
Marine erosion

→ Waves aided by winds, currents, tides, storms; depends on wave size, slope, shore height, coast shape, rock, depth

→ Wave pressure compresses air in fissures → rupture; waves use debris as tools (corrasion); solvent action in limestone and chalk

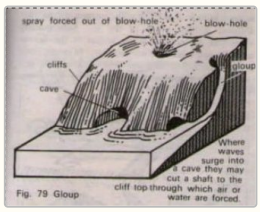


Marine Cycle · Erosional Features



Marine Cycle

Types Of Coastlines



Marine Cycle · Erosional Features

Cats Always Sit Still: C-A-S-S

Cliff → Cave → Arch → Stack → Stump

Erosional features

→ Chasm: narrow, deep, headward erosion along vertical weakness; mouth widens → bay

→ Wave-cut platform: sea cliffs retreat; notch at base (maximum erosion)

→ Sea cave (differential erosion) → sea arch

→ Stack: pillars offshore; further eroded (below low tide) → stumps

→ Blow/spouting hole: air compressed in cave; noise; widen, walls collapse → geo/gloup

Depositional features

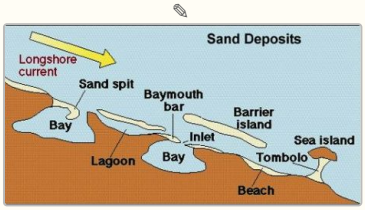
→ Beach: temporary debris cover on wave-cut platform

→ Bar: submerged, offshore; encloses lagoon; barrier = overwater counterpart

→ Spit (one end tied to headland) → hook (curved end)

→ Tombolo: bar joining island

→ Marine dunes: by wind, not waves



Marine Cycle · Depositional Features

Coast type	Cause	Examples
Emergence	land up / sea down	bars, spits, lagoons, salt marshes, beaches, cliffs, arches

Submergence	land sinks / sea up	Ria (river valleys drowned, irregular; SW Ireland) · Fjord (glacial U-trough drowned, long steep inlets; Norway) · Dalmatian (ridges parallel to coast; Yugoslavia) · Drowned lowland (low, no indentations; Baltic coast of E. Germany)
Neutral	new material built out; no net relative change	alluvial fan, delta, volcano, coral-reef coasts
Compound	submergence + emergence (either order)	Norway and Sweden
Fault	down-thrown block submerged, steep fault face to sea	Konkan coast (India)

Coastline of submergence

Coastline of submergence

Marine Cycle

→ The wave pressure compresses the air trapped inside rock fissures, joints, faults, etc . forcing it to expand and rupture the rocks along weak points.

→ Waves also use rock debris as instruments of erosion. These rock fragments carried by waves themselves get worn down by striking against the coast or one another.

→ The solvent or chemical action of waves is another mode of erosion, but it is pronounced only in the case of soluble rocks like limestone and chalk.

→ 1. Chasms: These are narrow, deep indentations (a deep recess or notch on the edge or surface of something) carved due to headward erosion (downcutting) through vertical planes of weakness in the rocks by wave action.

→ 1. With time, further headward erosion is hindered by lateral erosion of the chasm mouth, which itself keeps widening till a bay is formed.

→ 2. Wave Cut Platform: Formed due to the backward retreat of the sea cliffs.

→ 1. The bottom of the cliff suffers the maximum intensive erosion by waves and, as a result, a notch appears at this position.

→ 3. Sea Cave : Differential erosion by sea waves through a rock with varying resistance across its structure produces arched caves in rocks called sea caves. With further erosion, they get converted into sea arches.

→ 4. Stack: When the sea caves get extremely eroded , all that is left is some rock pillars standing in the offshore location called stacks. When these stacks further get eroded (often below the Low Tide limit), we get Stumps.

→ 5. Blow Holes/ Spouting holes : The burst of water through a small hole on a sea cave due to the compression of air in the cave by strong waves. They make a peculiar noise. With further erosion these holes get bigger, the walls collapse, and we get Geos/gloup .

→ 2. Bar: Currents and tidal currents deposit rock debris and sand along the coast at a distance from the shoreline . The resultant landforms which remain submerged are called bars. The enclosed water body so created is called a lagoon . They are of various types:

→ 1. Barrier: It is the overwater counterpart of a bar.

→ 2. Spit and Hook: A spit is a projected deposition joined at one end to the headland, with the other end free in the sea. The mode of formation is like a bar or barrier. A shorter spit with one end curved towards the land is called a hook.

→ 3. Tombolo: Sometimes, islands are connected by a bar called tombolo.

→ 3. Marine Dunes : Deposited in coastal locations, but due to the action of Wind and not Waves.

Types Of Coastlines

→ These are formed either by an uplift of the land or by the lowering of the sea level .

Coastline of submergence

→ Ria Coast: When a region is dissected by streams into a system of valleys and divides, submergence produces a highly irregular shoreline called Ria coastline. The coast of south-west Ireland is a typical example of the ria coastline.

→ 1. After the glaciers have disappeared, a fjord coastline emerges.

→ 2. These coasts have long and narrow inlets with very steep sides.

→ 3. The fjord coasts of Norway are a typical example.

→ These are coastlines formed as a result of new materials being built out into the water. The word 'neutral' implies that there need be no relative change between the level of the sea and the coastal region of the continent.

→ Whatever is taken away by the wave, gets deposited by the agents of erosion, such that there is no net change.

Compound Coastline

→ Such coastlines show the forms of two of the previous classes combined, for example, submergence followed by emergence or vice versa.

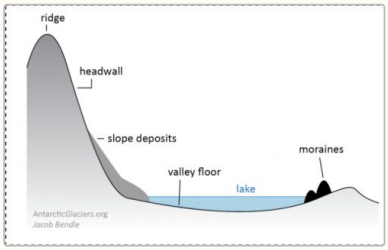
23 · GLACIAL CYCLE

Understand it first: A glacier is a slow river of ice. It plucks and grinds the rock in the mountains (erosional forms, U-shaped valleys) and dumps unsorted debris lower down (moraines and other depositional forms).

Glacier

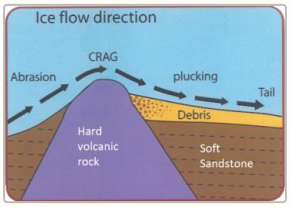
→ Moving ice mass, few metres/day; loaded with rock debris

→ Types: continental, ice cap, piedmont, valley

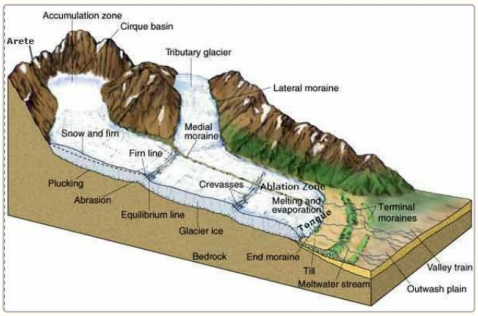


Glacial Cycle

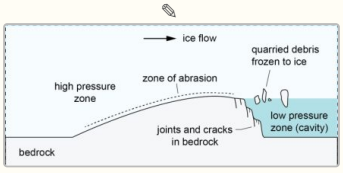
Erosional



Glacial Cycle · Erosional Features



Glacial Cycle · Erosional Features



Glacial Cycle · Erosional Features

→ Cirque/Corrie/Cwm: hollow basin in ridge, steep on 3 sides, open on 1; tarn lake

→ Horn/Pyramidal peak: cut from >2 sides

→ Arête: knife-edge between two cirques

→ Bergschrund: crevasse between moving glacier and stationary ice/rock at upper end

→ Glacial trough: U-shaped valley (uniform horizontal + vertical erosion); flooded = fjord

→ Hanging valley: tributary valley left high, discordant

→ Roche moutonnée ("sheep rock"): smooth side faces ice advance, rough steep side plucked

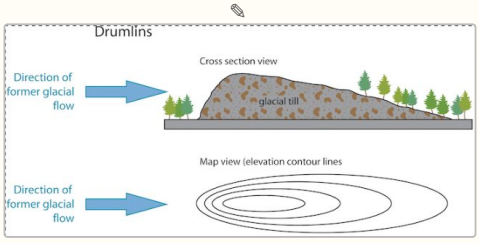
→ Craig and tail: resistant mass + softer ridge downstream

Depositional

→ Outwash/till plain: stratified debris, clay, sand, gravel

→ Esker: winding ridge of unassorted deposits, like a levee, road-making

→ Kame terraces: broken ridges, humps



Glacial Cycle · Depositional Features

→ Drumlin: inverted boat; steep glacier side faces glacier, lee side tapers (shows direction)

→ Kettle holes: depressions in till plain

→ Moraine: ground, lateral, end

Glacial Cycle

→ A glacier is a moving mass of ice at speeds averaging a few meters a day .

→ Erosional Features:

→ 1. Cirque/Corrie/Cwm: Hollow basin cut into a mountain ridge. It has a steep-sided slope on three sides, an open end on one side, and a slightly concave bottom/base. When the ice melts, tarn lakes may develop at the base of the cirques.

→ 3. Arete: It is the knife-edged boundary between two cirques.

→ 4. Bergschrund: Crevasse that forms between a moving glacier and the stationary ice or rock of a mountain, typically at the upper end of a glacier, where it separates the flowing ice from the stagnant ice above.

→ 5. Glacial Trough:

→ Original stream-cut valley, further modified by glacial action. It is a 'U' Shaped Valley . It is at the mature stage of valley formation.

→ Since glacial mass is heavy and slow-moving, erosional activity is uniform – horizontally as well as vertically. A steep-sided and flat-bottomed valley results, which has a 'U' shaped profile. These U-shaped valleys, when filled with water, form Fjords.

→ 6. Hanging Valley: Formed when smaller tributaries are unable to cut as deeply as bigger ones and remain 'hanging' at higher levels than the main valley as discordant tributaries. A valley carved out by a small tributary glacier that joins with a valley carved out by a much larger glacier.

→ 7. Roche Moutonnee: Roche Moutonnée, a term derived from French meaning "sheep rock, " refers to a glacially eroded bedrock formation characterized by a smooth, gently sloping side facing the direction from which the glacier advanced, and a rough, steep side where the glacier plucked and quarried the rock as it moved. These features provide insights into the direction and nature of past glacial movements.

→ 8. Craig & Tail: Landform consisting of a resistant rock mass (the Craig) with a gently sloping ridge of softer material (the Tail) extending downstream, formed by glacial erosion.

→ Depositional Features:

→ 9. Includes both depositional and erosional features .

→ 10. Outwash Plain: When the glacier reaches its lowest point and melts, it leaves behind a stratified deposition material, consisting of rock debris, clay, sand, gravel, etc. This layered surface is called a till plain or an outwash plain.

→ 11. Esker: Winding ridge of un-assorted depositions of rock, gravel, clay, etc. running along a glacier in a till plain. The eskers resemble the features of an embankment and are often used for making roads. They are analogous to levees.

→ 12. Kame Terraces: Broken ridges or un-assorted depositions looking like a hump in a till plain.

→ 13. Drumlin: Inverted boat-shaped deposition in a till plain. They help in determining the direction of movement of the glacier, as the glacier side is always high and steep, while the lee side is smooth and tapers gently in the direction of ice movement.

→ 14. Kettle Holes: Formed when the deposited material in a till plain gets depressed locally and forms a basin.

→ 15. Moraine: General term applied to rock fragments, gravel, sand, etc. carried by a glacier. Depending on its position, the moraine can be ground moraine, lateral moraine, and end moraine.

24 · ARID CYCLE, LAKES, PLATEAUS

Understand it first: In deserts, wind and rare flash floods do the work. Wind removes loose material, sand-blasts rock and piles sand into dunes. Lakes and plateaus are then learnt simply by how they were formed.

H-R-E — Hamada = rocky (large smoothened rocks) · Reg/Serir = stony gravel and pebbles, easier to cross · Erg/Koum = sandy · Cold deserts (high altitude/latitude). H-R-E: Hard → Rubble → Ergs of sand

Water forms (flash floods + wind)	→ Bolson: intermontane basin → Gully (ditch-like, metres–tens of metres); Ravine (narrower than canyon, larger than gully, smaller than valley) → Badland: rills erode weak sediments, no vegetation, no agriculture (Chambal ravines) → Playa: temporary lake; salt-covered = Salina → Pediment: rock-cut surface at mountain foot (erosional) → Bajada: moderately sloping depositional plain
💰 Wind processes	→ Deflation: lifting and carrying away dry dust → Abrasion: sand-loaded wind grinds rock → Attrition: wear of sand particles in transport
Wind erosional	→ Blowouts/deflation hollows (to several km) → Mushroom rock (cap on narrow stem) → Zeugen = Zzz, lies flat (table-shaped, horizontal layers) · Yardang = Y stands up (vertical layers) → Wind window (holes widen) → wind bridge (arch) → Inselberg (= monadnock of deserts) → Pediplain (= peneplain of deserts)
Wind depositional	→ Ripple marks: saltation → Dunes: heaps, few m to 20 m (some hundreds of m high, 5–6 km long); deserts and coasts → Barchan: crescent, windward convex, leeward concave and steep → Longitudinal: parallel to wind (Sahara, Australian, Libyan, S African, Thar) → Transverse: ⊥ wind

→ Parabolic: U-shaped, longer and narrower than barchan, windward concave

→ Star: central peak, 3+ arms, multi-direction wind

→ Loess (ex-situ): fine desert dust carried far (Loess Plateau near Gobi, China)

Lakes

→ Larger, deeper than ponds; most have natural outflow; endorheic = no outflow (lose water by evaporation/seepage)

→ Tectonic: Titicaca, Caspian

→ Rift valley: Tanganyika, Malawi, Dead Sea

→ Glacial: tarns; Scandinavia, Canada

→ Volcanic crater/caldera: Lonar, Krakatoa

→ Erosional: karst, playa, wind-deflated (Great Basin, Utah)

→ Depositional: ox-bow, lagoon (Chilika), moraine-dammed

→ Man-made

Plateaus

→ Flat tablelands, a third of land; mineral-rich (mining); eroded → outliers (old dense rock; iron ore, coal)

Tectonic

→ Volcanism: Deccan

→ Crustal shortening: Tibet (intermontane)

→ Thermal expansion: Ethiopian Highlands

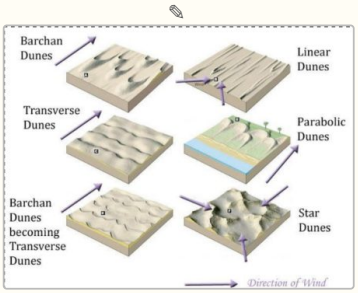
→ Erosional: Allegheny (Appalachians)

Arid Cycle

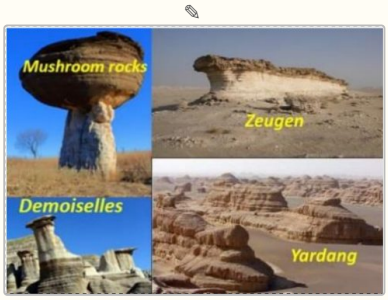
→ Types of Deserts:

→ 1. Hamada (Rocky) : Consist of large rocks that have been smoothened by exogenic forces.

→ 2. Reg (Stony)/Serir : Covered with extensive sheets of gravel and pebbles. More accessible than sandy deserts (easier mobility).



Arid Cycle · 1. Erosional



Arid Cycle · Wind Related landforms

→ 4. Cold Deserts: Either at high altitudes or high latitudes.

→ Water-related landforms:

→ 1. Bolsons: The intermontane basins in dry regions are generally known as bolsons.

→ 2. Erosional:

→ 1. A gully is a landform created by running water. Gullies resemble large ditches or small valleys but are meters to tens of meters in depth and width.

→ 2. Ravine : A ravine is a landform narrower than a canyon and is often the product of stream-cutting erosion. Ravines are typically classified as larger in scale than gullies, although smaller than valleys.

→ 3. Badland Topography: In arid regions, occasional rainstorms produce numerous rills and channels which extensively erode weak sedimentary formations. Lack of any vegetation causes this extreme level of erosion. As the top soil gets eroded, any chance of undertaking agriculture as a source of livelihood also erodes away. This lack of utility gives these regions their name . Ravines and gullies are developed by linear fluvial erosion leading to the formation of badland topography. Example: Chambal Ravines

→ 4. Playas: Small streams flow into bolsons, and this water accumulates in the depressions. These temporary lakes are called playas. After the evaporation of water, salt-covered playas are called Salinas.

→ 1. Bajada: Bajadas are moderately sloping depositional plains in the desert regions.

→ 1. Deflation ⇒ removing, lifting, and carrying away dry, unsorted dust particles by winds.

→ 2. Abrasion ⇒ When wind loaded with sand grains erodes the rock by grinding against its walls.

→ Wind Related landforms:

→ 1. Erosional:

→ 1. Blowouts/ Deflation Hollows: hollows formed by the removal of particles by wind. They are generally small but may be up to several kilometres in diameter.

→ 2. Mushroom rocks: a cap or cap-like structure atop a narrower stem.

→ 3. Zeugen: A table-shaped area of rock found in arid and semi-arid areas formed when more resistant rock is reduced at a slower rate than softer rocks around it. Rocks are in horizontal layers.

→ 4. Yardang: Rocks are in vertical layers . The hard rocks stay resistant while the soft rocks get eroded.

→ 5. Wind bridges and windows: Powerful wind continuously abrades stone lattices, creating holes. Sometimes the holes are gradually widened to reach the other end of the rocks to create the effect of a window—thus forming a wind window.

→ 1. Window bridges, are formed when the holes are further widened to form an arch-like feature

→ 6. Inselbergs: A inselberg is an isolated hill, knob, ridge, outcrop, or small mountain that rises abruptly from a gently sloping or virtually level surrounding plain. Analogous to Monadnocks in the arid regions.

→ 7. Pediplain: equivalent of Peneplain in the arid regions.

→ 2. Depositional:

→ 1. Ripple Marks: These are depositional features on a small scale formed by saltation (the transport of hard particles over an uneven surface in a turbulent flow of air or water).

→ 2. Dunes (in-situ): Dunes are heaps or mounds of sand found in deserts. Generally, their heights vary from a few meters to 20 meters but in some cases, dunes are several hundred meters high and 5 to 6 km long.

→ 1. Formed both in deserts and coastal locations.

→ 2. Longitudinal dunes: Formed parallel to the wind movement . These dunes are commonly found at the heart of trade-wind deserts like the Sahara, Australian, Libyan, South African and Thar deserts.

→ 4. Barchans: Crescent-shaped dunes. The windward side is convex whereas the leeward side is concave and steep.

Lakes

→ Body of water of considerable size, localized in a basin, surrounded by land apart from a river or other outlet that feeds/drains it.

→ Most lakes have at least one natural outflow to maintain its average level by allowing the drainage of excess water.

Plateaus

→ Plateaus occur in every continent and take up a third of the Earth's land

→ Sometimes, a plateau is so eroded that it is broken up into smaller raised sections called outliers.

TRAPS

"Himalayas have volcanoes" ✘ · "P waves can't pass liquids" ✘ (that's S) · "orogeny = continent-building" ✘ (that's epeirogeny) · "solar flares cause tsunamis/forest fires" ✘ · "Blue Moon = eclipse" ✘ · "Sun overhead on equator on 21 June" ✘ (equinoxes).

Connects to: Himalaya and Deccan story → Indian Geography; tilt and 21 June → Climatology; ridges, trenches, coral reefs → Oceanography; Indus/Brahmaputra gorges → Indian drainage. ✅

RECALL DRILL: COVER AND ANSWER

CMB found by whom, when?	Penzias & Wilson, 1965; ~380,000 yrs after Big Bang	White dwarf / neutron star / black hole mass?	<1.44 · 1.44–3 · >3 (Chandrasekhar)
Moons: Jupiter, Saturn, Uranus, Neptune?	101, 274, 28, 16	Earth tilt; sidereal day?	23.5°; 23 h 56 m 4.1 s
Earth volume: crust / mantle / core?	0.5–1% / 83% / 16%	Boundaries top to bottom?	Conrad, Moho, Repetti, Gutenberg, Lehmann
P shadow / S shadow?	103–143° / beyond 103°	Drift / spreading / plates?	Wegener 1912 / Hess 1960 / McKenzie-Parker 1967
Only active lava volcano in South Asia?	Barren Island (India)	Fastest / slowest ridge?	East Pacific Rise / Arctic

🎯 PYQs OF THIS CHAPTER

🎯 Exam lens (from the handout only)

- Priority: HIGH for Prelims. 19 Prelims PYQs are printed in this part of the handout (15 with a year: 2011, 2012×2, 2013×2, 2014, 2015, 2018, 2019, 2022, 2023, 2024×3, 2025; 4 with no year printed). Mains quoted: Juno mission (2017, 10m) and primary rocks (2022, 10m).
- Hot clusters: Earth–Sun geometry (pole star, day length, 21 June, time zones) · solar storm effects · P vs S waves · weathering by rain · drift evidences · volcanic products · chalk vs clay.
- No teacher ink in this PDF, so the priority tag comes only from PYQ counts in the handout.
- Prepare: Prelims – who/when/where of theories, layers and discontinuities, wave properties, landform identification. Mains – drift → plate tectonics reasoning, rock types and economic value, landforms by agent (river, ice, sea, wind, groundwater).

★ *PYQ answers in one line (asked year and the key)*

PYQ 2022 solar storm (ans c): effects = GPS/navigation failure, power-grid damage, intense auroras, satellite orbit disturbance, polar short-wave radio disruption. Not tsunamis (need vertical water shift) and not forest fires (flares don't raise Earth's temperature). May 2024 storm: auroras as far south as Leh. ✅

PYQ 2018 (ans C = 1 and 3): magnetic field reversed every few hundred thousand yrs ✔; life modified early atmosphere ✔; "54% oxygen, no CO_2 at birth" ✘ (early air: water vapour, CO_2, N_2, no O_2). PYQ 2012 (ans B): elements behind origin of life = Carbon, Hydrogen, Nitrogen. ✅

Mains 2017 (10m): "How does NASA's Juno Mission help understand the origin and evolution of the Earth?" Handout prints the question only; only Jupiter notes (gas giant, storms) given. ⭐

PYQ 2023 (ans c): P waves recorded earlier than S; P particles vibrate along direction of travel, S at right angles. Both correct. ✅

PYQ 2024 (ans a): rain weathers rock because rainwater carries dissolved CO_2 and O_2; both statements explain it. ✅

PYQ 2014 (ans c): continental drift and glacial cycles both influenced evolution (unique primates of Australia, S America; genus Homo in last 3 glacial cycles). No-year (ans c, all three): Brazil–W Africa rock belts, Ghana gold from Brazil plateau, Gondwana sediments in six southern landmasses (Africa, S America, Australia, Antarctica, Madagascar, NZ). ✅

PYQ 2013 (ans d): electromagnetic radiation, geothermal energy, gravity, plate movement, rotation, revolution: all cause dynamic surface change (radiation → temperature change, gravity → downslope movement). ✅

✅

PYQ 2024 (ans d): pyroclastic debris, ash and dust, nitrogen compounds, sulphur compounds: all four are volcanic products.

Mains 2022 (10m) "primary rocks": Intro: igneous from magma/lava, hence primary. Body: non-fossiliferous; plutonic vs volcanic; acid vs basic; examples; economic value (ores, building stone, black soil). Close: all other rocks derive via rock cycle. ✅

PYQ 2025 (ans c): chalk = porous and permeable; clay = porous but not permeable (USGS: very high porosity, low permeability). Statement III ("clay not porous at all") wrong; II right and explains I. ✅

✍️ ***The PYQs as printed in the handout, with answers***

Q.1) The term 'Goldilocks Zone' is often seen in the news in the context of (2015)

(a) the limits of habitable zone above the surface of the Earth

(b) regions inside the Earth where shale gas is available

(c) search for the Earth-like planets in outer space

(d) search for meteorites containing precious metals

Answer – C

Explanation:

- Goldilocks zone is the distance from the star, where the temperature is such that water can stay in Liquid form. Earth lies in the Goldilocks Zone of the Sun. If we wish to find life form similar to that found on the Earth, then we need to search for the planets lying in the Goldilocks zone of their respective stars.

Q.2) What is the difference between asteroids and comets? (2011)

1. Asteroids are small rocky planetoids, while comets are formed of frozen gases held together by a rocky and metallic material.

2. Asteroids are found mostly between the orbits of Jupiter and Mars, while comets are found mostly between Venus and mercury.

3. Comets show a perceptible glowing tail, while asteroids do not.

Which of the statements given above is/are correct ?

- 1 and 2 only
- 1 and 3 only
- 3 only
- 1, 2 and 3

Answer. B

Explanation

- The main difference between asteroids and comets is their composition, as in, what they are made of Asteroids are made up of metals and rocky material, while comets are made up of ice, dust and rocky material.
- Both asteroids and comets were formed early in the history of the solar system about 4.5 billion years ago.
- Asteroids formed much closer to the Sun, where it was too warm for ice to remain solid. Comets formed farther from the Sun where ice would not melt. Comets which approach the Sun lose material with each orbit because some of their ice melts and evaporates to form a tail.
- Asteroids are a series of very small planets or fragments of planets lying between the orbits of Mars and Jupiter.
- They revolve around the Sun in the same way as planets, unlike comets.

Q.1) If a major solar storm (solar flare) reaches the Earth, which of the following are the possible effects on the Earth ? (2022)

2. GPS and navigation systems could fail.

3. Tsunamis could occur at equatorial regions.

4. Power grids could be damaged.

5. Intense auroras could occur over much of the Earth.

6. Forest fires could take place over much of the planet.

7. Orbits of the satellites could be disturbed.

8. Shortwave radio communication of the aircraft flying over polar regions could be interrupted.

Select the correct answer using the code given below:

(a) 1, 2, 4 and 5 only

(b) 2, 3, 5, 6 and 7 only

(c) 1, 3, 4, 6 and 7 only

(d) 1, 2, 3, 4, 5, 6 and 7

Answer – C

Explanation: (e) Solar storms can play a havoc on the electrical and electronic systems of the Earth, affecting power systems, satellite communication, nd can cause intense auroras. In May 2024 one such intense solar

storm caused the auroras to be seen as far away from the pole as Leh in India. (f) Tsunamis require vertical shift of water which is not caused by the flares. These flare are not related to increased temperature on earth, so they do not cause forest fires.

Q.2) Which one of the following sets of elements was primarily responsible for the origin of life on the Earth? (2012)

- Hydrogen, Oxygen, Sodium
- Carbon, Hydrogen, Nitrogen
- Oxygen, Calcium, Phosphorus
- Carbon, Hydrogen, Potassium

Answer. B

Explanation The correct answer to this question, as most of you know, is Carbon, Hydrogen and Nitrogen.

- In fact, the early biochemistry was not simply the reactions of Carbon, Hydrogen, Nitrogen and Oxygen was a bit far from it, was CO2, N and water resulting in the organic acids such as oxalic acid.
- The infrared part of solar radiation. The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere because it absorbs the infrared part of the solar radiation. Only water vapour has the ability to absorb both incoming UV and outgoing infrared radiation.

Q.3) A person stood alone in a desert on a dark night and wanted to reach his village which was situated 5 km east of the point where he was standing. He had no instruments to find the direction but he located the polestar. The most convenient way now to reach his village is to walk in the: (2012)

- direction facing the polestar
- a direction opposite to the polestar
- direction keeping the polestar to his left
- direction keeping the polestar to his right

Answer. C

Explanation

- The correct answer is direction, keeping the pole star to his left. In ancient times, people used to determine directions during the night with the help of stars. The North star indicates the north direction. It is also called the Pole Star.

Q.4) Variations in the length of daytime and nighttime from season to season are due to – (2013)

4. The earth's rotation on its axis
5. Revolution of the earth on a tilted axis
6. Latitudinal position of the place
7. The earth's revolution around the sun in an elliptical manner

Answer. B

Explanation Had the axis of the Earth been perfectly perpendicular to the plane of the ecliptic, the sun rays would have been vertical over the equator all year round and the amount of energy received at any place would have remained constant throughout the year; hence no change of seasons.

- Also in that situation, the days and nights all over the earth would have been exactly equal throughout the year as it happens at the time of equinoxes.
- The change between day and night is caused by the rotation of the Earth on its axis. ... The changing length of days and nights depends on where you are on Earth and the time of year. Also, daylight hours are affected by the tilt of the Earth's axis and its path around the sun.
- As the Earth moves around the Sun, the length of the day changes. ... This is all caused by the 23.5-degree tilt of the Earth's axis as it travels around the sun. In the Northern Hemisphere, days are longest at the time of the summer solstice in June, and the shortest days are at the winter solstice in December.

Q.5) Consider the following statements:

- The Earth's magnetic field has reversed every few hundred thousand years.
- When the Earth was created more than 4000 million years ago, there was 54% oxygen and no carbon dioxide.
- When living organisms originated, they modified the early atmosphere of the Earth.

Which of the statements given above is/are correct? (2018)

- 1 only
- 2 and 3 only
- 1 and 3 only
- 1, 2 and 3

Answer. C

Explanation

- A geomagnetic reversal is a change in a planet's magnetic field such that the positions of magnetic north and magnetic south are interchanged. The Earth's field has alternated between periods of normal polarity, in which the predominant direction of the field was the same as the present direction, and reverse polarity, in which it was the opposite. These periods are called chrons.
- Reversal occurrences are statistically random. There have been 183 reversals over the last 83 million years.

- The Earth formed approximately 4.5 billion years ago, along with the other seven planets in the solar system.
- As the Earth cooled, a primitive atmosphere was created by the out-gassing of early volcanoes.
- The early atmosphere contained no oxygen and would have been toxic to human beings, as well as most other life on Earth today.
- Based on an analysis of gases vented by modern volcanoes, it seems likely that this early atmosphere consisted mostly of water vapour (H_2O) and carbon dioxide (CO_2) and nitrogen gas (N_2).
- In the whole process of evolution, the living organism changed the chemical composition of the Atmosphere. For example, oxygen came from photosynthesis.

Q.6) On 21st June, the Sun (2019)

- does not set below the horizon at the Arctic Circle
- does not set below the horizon at Antarctic Circle
- shines vertically overhead at noon on the Equator
- shines vertically overhead at the Tropic of Capricorn

Answer. A

Explanation

- The Sun is directly overhead at "high-noon" on the equator twice per year, at the two equinoxes. On the Arctic Circle, the Sun does not set at all on the Summer Solstice which occurs on 21st June.
- On that one day, the Sun traces a complete circle just above the horizon as the Earth rotates.

Q.7) On June 21 every year, which of the following latitude(s) experience(s) a sunlight of more than 12 hours? (2024)

- Equator
- Tropic of Cancer
- Tropic of Capricorn
- Arctic Circle

Select the correct answer using the code given below:

(a) 1 only
(b) 2 only
(c) 3 and 4
(d) 2 and 4

Answer: (d) 2 and 4

Q.8) Consider the following statements: Statement I: Scientific studies suggest that a shift is taking place in the Earth's rotation and axis. **Statement II:** Solar flares and associated coronal mass ejections bombarded the Earth's outermost atmosphere with tremendous amount of energy. **Statement III:** As the Earth's polar ice melts, the water tends to move towards the equator.

Which one of the following is correct in respect of the above statements?

(a) Both Statement II and Statement III are correct and both of them explain Statement I
(b) Both Statement II and Statement III are correct but only one of them explains Statement I
(c) Only one of the Statements II and III is correct and that explains Statement I
(d) Neither Statement II nor Statement III is correct

Answer: (b) Statement I: Correct Scientific studies, including those from NASA and peer-reviewed journals, confirm that Earth's axis and rotation are gradually shifting due to redistribution of mass (e.g., melting glaciers and sea-level rise). **Statement II:** Correct, but does not explain Statement I Solar flares and coronal mass ejections (CMEs) do indeed impact the Earth's magnetosphere and upper atmosphere. However, they do not significantly affect the Earth's axial tilt or rotation. **Statement III:** Correct and explains Statement I Melting polar ice (especially in Greenland and Antarctica) leads to mass redistribution toward oceans, which contributes to a shift in the Earth's rotational axis (a phenomenon called true polar wander).

Q.9) Consider the following statements: Statement I: Anadyr in Siberia and Nome in Alaska are a few kilometers from each other, but when people are waking up and getting set for breakfast in these cities, it would be different days. **Statement II:** When it is Monday in Anadyr, it is Tuesday in Nome.

Which of the statements given above is/are correct?

(a) I only
(b) II only
(c) Both I and II
(d) Neither I nor II

Answer: (a) I only Statement I: Correct Anadyr is in the Chukotka Autonomous Okrug, Russia (Siberia). Nome is in western Alaska, USA. They are about 800 km apart, separated by the Bering Strait. Time Difference: Anadyr uses UTC+12. Nome uses UTC-9 (Alaska Standard Time). Statement II: Incorrect When it is Monday in Anadyr, it is not Tuesday in Nome : it is actually Sunday.

Q.10) Consider the following countries: 1.United Kingdom 2.Denmark 3.New Zealand 4.Australia 5.Brazil

How many of the above countries have more than four time zones?

(a) All the five
(b) Only four

(c) Only three
(d) Only two
Answer: (b) Only four United Kingdom

- Time zones:
  - Mainland: GMT or BST (1 zone)
  - Overseas territories include Bermuda, Falklands, Gibraltar, etc.
- Total time zones: Up to 9, including overseas territories. Denmark : More than 4 Time Zones
- Mainland: UTC+1 (CET)
- Greenland:
  - Danmarkshavn : UTC+0
  - Ittoqqortoormiit : UTC−1
  - Nuuk (Capital) : UTC−3
  - Thule : UTC−4
- Total: 5 Time Zones New Zealand : More than 4 Time Zones
- Mainland: NZST/NZDT (UTC+12/+13)
- Territories:
  - Tokelau : UTC+13
  - Cook Islands : UTC−10
  - Niue : UTC−11
- Total: 5 Time Zones Australia : More than 4 Time Zones
- Mainland:
  - AWST (UTC+8), ACST (UTC+9:30), AEST (UTC+10)
- Territories:
  - Lord Howe Island : UTC+10:30
  - Norfolk Island : UTC+11
  - Cocos Islands : UTC+6:30
  - Christmas Island : UTC+7
- Total: 6+

Brazil : Only 4 Time Zones

- Historically had 4 zones: UTC−2 to UTC−5
- Now has 3 official zones, but still only 4 recognized historically
- Not more than 4

Q.1) How does the Juno Mission of NASA help to understand the origin and evolution of the Earth? (2017, 10m)

Q1.) Consider the following statements :

- In a seismograph, P waves are recorded earlier than S waves.
- In P waves, the individual particles vibrate to and fro in the direction of Wave propagation whereas. in S waves, the particles vibrate up and down at right angles to the direction of wave propagation.

Which of the statements given above is/are correct? (2023)

- 1 only
- 2 only
- Both 1 & 2
- Neither 1 nor 2

Ans: C

Explanation:

- P or Primary waves are the fastest seismic waves, thats why they are the first to be received. P waves are longitudinal, i.e. direction of propagation of waves and the direction of movement of particles is the same. S waves are transverse i.e. particles move perpendicular to the movement of the wave.

Q2.) Consider the following statements : (2024)

Statement-I: Rainfall is one of the reasons for weathering of rocks.

Statement-II: Rain water contains carbon dioxide in solution.

Statement-III: Rain water contains atmospheric oxygen.

Which one of the following is correct in respect of the above statements?

- Both Statement-II and Statement-III are correct and both of them explain Statement-I
- Both Statement-II and Statement-III are correct, but only one of them explains Statement-I
- Only one of the Statement II and III is correct and that explains Statement-I
- Neither Statement-II nor Statement-III is correct

Answer : (a) Both Statement-II and Statement-III are correct and both of them explain Statement-I.

Q.1) Which of the following phenomena might have influenced the evolution of organisms ?

- Continental drift
- Glacial cycles

Select the correct answer using the code given below. (2014)

- 1 only
- 2 only
- Both 1 and 2
- Neither 1 nor 2.

Answer: (c) Both 1 and 2

9. About 200 million years ago, all the continents on Earth were actually one huge "supercontinent" surrounded by one enormous ocean. This gigantic continent, called Pangaea , slowly broke apart and spread out to form the continents we know today. All Earth's continents were once combined in one supercontinent, Pangaea.

- One of the most prominent examples of the effect of continental drift on the evolution of organisms is the unique primates found in Australia and South America.
- The final stages of evolution of Genus Homo occur in the last 3 glacial cycles.

Q.2) Which of the following are the evidences of the phenomenon of continental drift?

- The belt of ancient rocks from Brazil coast matches with those from Western Africa.
- The gold deposits of Ghana are derived from the Brazil plateau when the two continents lay side by side.
- The Gondwana system of sediments from India is known to have its counterparts in six different landmasses of the Southern Hemisphere.

Select the correct answer using the code given below:

- I and III only
- I and II only
- I, II and III
- II and III only

Answer: (c) I, II and III

- Statement I: Ancient rock belt from Brazil matches with Western Africa Correct. Geological studies show a remarkable similarity between the Precambrian rock formations on the coasts of Brazil and Western Africa.
- Statement II: Gold deposits of Ghana derived from Brazil plateau Correct. Certain mineral deposits such as gold and diamonds found in Ghana and Brazil match in age and type, indicating they were formed in the same geological settings when South America and Africa were joined.

- Statement III: Gondwana sediments in India match those in other Southern Hemisphere landmasses Correct. The Gondwana sedimentary sequences in India show close similarity with those in Africa, South America, Australia, Antarctica, Madagascar, and New Zealand.

Q.1) Consider the following:

- Electromagnetic radiation
- Geothermal energy
- Gravitational force
- Plate Movements
- Rotation of the earth
- Revolution of the earth Which of the above are responsible for bringing dynamic changes on the surface of the earth? (2013)
- 1, 2, 3 and 4 only
- 1, 3, 5 and 6 only
- 2, 4, 5 and 6 only
- 1, 2, 3, 4, 5 and 6

Ans. D All the factors are responsible for bringing dynamic changes on the surface of the earth.

- Wind, water, and ice erode and shape the land. Volcanic activity and earthquakes alter the landscape in a dramatic and often violent manner. And on a much longer timescale, the movement of earth's plates slowly reconfigures oceans and continents. Each one of these processes plays a role in the Arctic and Antarctica.
- Electro-magnetic radiation causes temperature change which induces metamorphism of rocks. Gravity besides being a directional force activating all down slope movements of matter also causes stresses on the earth's materials. Plate movement results into continent building. The energy emanating from within the earth is the main force behind endo-genic geomorphic processes. This energy is mostly generated by radioactivity, rotational and tidal friction and primordial heat from the origin of the earth. This energy due to geothermal gradients and heat flow from within induces diastrophism and volcanism in the lithosphere. Due to variations in geothermal gradients and heat flow from within, crustal thickness and strength, the action of endo-genic forces are not uniform and hence the tectonically controlled original crustal surface is uneven.

Q.1) Consider the following : (2024)

- Pyroclastic debris
- Ash and dust
- Nitrogen compounds
- Sulphur compounds

How many of the above are products of volcanic eruptions?

- Only one
- Only two
- Only three
- All four

Answer : (d) All four

Q.1) Describe the characteristics and types of primary rocks. (2022, 10m)

Q.1) Consider the following statements (2025): Statement I: In the context of effect of water on rocks, chalk is known as a very permeable rock, whereas clay is known as quite an impermeable or least permeable rock. Statement II: Chalk is porous and hence can absorb water. Statement III: Clay is not at all porous.

Which one of the following is correct in respect of the above statements?

(a) Both Statement II and Statement III are correct and both of them explain Statement I

(b) Both Statement II and Statement III are correct but only one of them explains Statement I

(c) Only one of the Statements II and III is correct and that explains Statement I

(d) Neither Statement II nor Statement III is correct

Answer: (c) Only one of the Statements II and III is correct and that explains Statement I

- Chalk is both porous and permeable → absorbs and transmits water easily.
- Clay is porous but not permeable → holds water but does not allow it to flow through.
- USGS Water Science Glossary: Clay has very high porosity, but low permeability.

🔤 **ABBREVIATION COLUMN (full forms as the handout gives them where it does; the one-line meanings are plain-language aids, verify)**

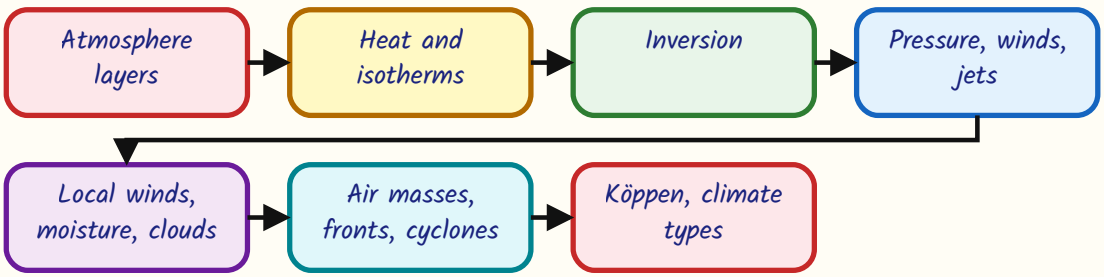
ACST = Australian Central Standard Time
AEST = Australian Eastern Standard Time
AGN = active galactic nucleus — very bright centre of an active galaxy
AWST = Australian Western Standard Time
BST = British Summer Time
CET = Central European Time
CMB = cosmic microwave background — left-over radiation from the early universe
CME = coronal mass ejection — huge burst of charged gas from the Sun
CMEs = coronal mass ejections
CO_2 = carbon dioxide
GICs = geomagnetically induced currents — currents set up in power lines by solar storms
GMT = Greenwich Mean Time — time at the 0° meridian
GPS = Global Positioning System — satellite navigation
ISS = International Space Station
LIGO = Laser Interferometer Gravitational-Wave Observatory — detects gravitational waves
LUCA = last universal common ancestor
NASA = National Aeronautics and Space Administration — US space agency
NiFe = nickel + iron — the core
NW = north-west
NZ = New Zealand
NZDT = New Zealand Daylight Time
NZST = New Zealand Standard Time
PVT = pressure, volume, temperature
PYQ = previous year question
SiAl = silica + aluminium — upper, continental crust
SiMa = silica + magnesium — lower, oceanic crust
SW = south-west
UK = United Kingdom
USA = United States of America
USGS = United States Geological Survey
UTC = Coordinated Universal Time — world time standard
UV = ultraviolet — high-energy part of sunlight

CLIMATOLOGY

PYQ Prelims 2011 ×5, 2012 ×3, 2013, 2014, 2015 ×3, 2018, 2019 ×2, 2020, 2021 ×2, 2022, 2023 ×3, 2024 ×6, 2025 + 2 with no year · Mains 2022 (troposphere, 15m)

✎ *Atmosphere, heat, pressure, winds, jet streams, clouds, air masses, cyclones and climate types: 32 Prelims PYQs make this the most-asked chapter.*

At a glance: the Climatology route



Hot clusters

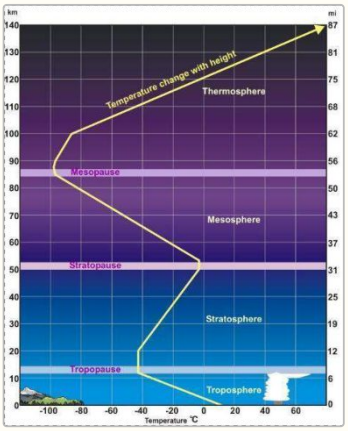
→ heat budget and lapse rate (2012, 2023, 2024)
→ isotherm bending (2024, 2025)
→ westerlies, Coriolis, jet streams (2011, 2015, 2020, 2024)
→ clouds and dew (2019, 2022)
→ cyclone genesis (2015)
→ climate-type identification (2011–2024). **Prepare:** Statement I/II cause-effect pairs (2023, 2024, 2025); Mains: troposphere, jet streams and climate change, tropical vs temperate cyclones. No teacher ink: priority = handout PYQ count.

🎯 1 · ATMOSPHERE: LAYERS AND GASES

PYQ 2011 PYQ 2012 PYQ 2024 PYQ 2022

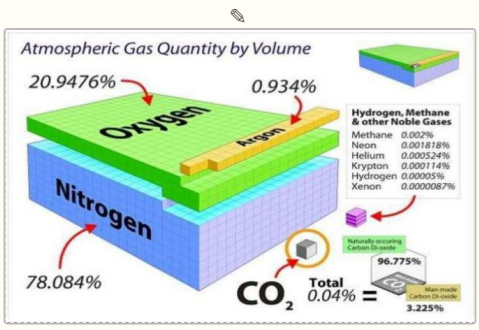
Understand it first: The atmosphere is a stack of layers, sorted by how **temperature changes with height**. Weather stays in the lowest layer; ozone warms the next one. Learn each layer with its height, its temperature trend and the one thing it is known for.

Layer (bottom to top)	Height	Key facts
1 Troposphere (kitchen)	8–18 km (avg 13)	All weather, clouds, jet streams; 18 km at equator (strong convection), 8 km at poles; cools ~1 °C per 165 m
2 Stratosphere (AC floor)	to 50 km	Ozone (lower part) so temp rises with height; almost cloud-free = jets fly here
3 Mesosphere (freezer)	50–80 km	Coldest; meteors burn; noctilucent clouds
4 Thermosphere (antenna)	80–400 km	Temp rises again (UV, X-ray); ISS and satellites; Kármán line 100 km; ionosphere bounces radio
5 Exosphere (terrace)	> 400 km	Particles up to 1,000 °C yet feels cold (too thin to pass heat)



Layers Of The Atmosphere · Troposphere: 8 - 1 8 km (1 3 km avg.)

Gases



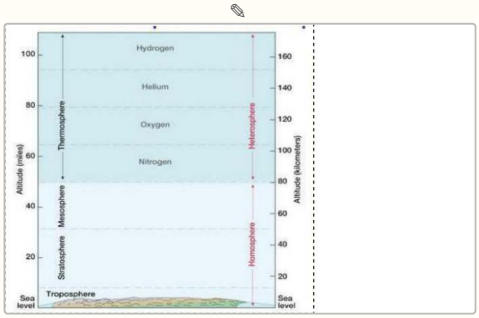
Atmosphere · Composition of the atmosphere

→ N_2 78% · O_2 21% (≈99%) · Argon 0.93% · CO_2 ~0.04%

→ Water vapour 0–5%: >90% below 5 km; falls equator to poles

→ CO_2 and water vapour only up to 90 km; ozone layer 20–25 km

→ By composition: homosphere (<80 km, uniform) · heterosphere (>80 km, layered by molecular mass)



Layers Of The Atmosphere · Based on Composition

→ Aerosols/salt = hygroscopic nuclei; microplastics seed ice 5–10 °C warmer than usual

→ Origin of gases: volcanic eruptions, hot springs, chemical breakdown of solids, biosphere

Source conflicts: lapse rate quoted as 1 °C/165 m and 6.5 °C/1,000 m (about equal). Ionosphere printed 75–1,000 km (PYQ note) vs thermosphere 80–400 km. Water vapour 0–5% (layers) vs "around 1% of air" (PYQ note).

PYQs: 2012 (C) cooler aloft because air is heated from the ground up + thinner air; NOT more moisture. 2012 (D) CO_2 absorbs infrared: transparent to incoming, opaque to outgoing radiation. 2023 (C) water vapour absorbs infrared (also: equator insolation 10× poles ✘; IR two-thirds of insolation ✘; IR not visible spectrum ✘). 2024 (a) troposphere thicker at equator because strong convection lifts heat. 2024 (d) atmosphere heated more by terrestrial (long-wave) than incoming radiation; only Statement II right. Undated (d, II and III): without atmosphere nights freeze but days get extremely hot, so "below freezing everywhere" ✘. Mains 2022 (15m) troposphere: clouds, air masses, jet streams sit here; terrestrial radiation is prime heat source. ✅

Ionosphere 2011 (D): charged particles reflect radio waves; NOT ozone; radio wavelength 1 mm–100 km, not "very long". Ionised by solar and cosmic radiation; D, E, F regions; electron density highest in F (day and night; day by solar radiation, night by cosmic rays); D vanishes at night, E weakens; VLF waves bounce in an Earth–ionosphere waveguide; Earth radius 6,370 km. ✅

CO_2: about 0.038% (now 0.04%) of dry air; product of combustion; greenhouse gas that helps keep the heat budget. **Ionosphere:** ionised by solar and cosmic radiation; lies 75–1000 km above the Earth (thin against the Earth's radius of 6370 km).

Atmosphere

Importance:

→ The Greenhouse Effect by certains gases in the atmosphere helps keep the Earth warm, to allow the survival of life.

→ The atmosphere also takes care of extra-terrestrial objects like meteors that get burnt up while passing through it, due to air friction.

Composition of the atmosphere:

→ The atmosphere is a mixture of many gases. In addition, it contains huge numbers of solid and liquid particles, collectively called aerosols.

→ Nitrogen and oxygen make up nearly 99% of the clean, dry air. The remaining gases are mostly inert and constitute about 1% of the atmosphere.

→ Oxygen constitutes 21% of the total volume of the atmosphere. All living organisms inhale oxygen. Besides, oxygen can combine with other elements to form important compounds, such as oxides. Also, combustion is not possible without oxygen.

→ Nitrogen accounts for 78% of total atmospheric volume. It is a relatively inert gas and is an important constituent of all organic compounds. The main function of nitrogen is to control combustion by diluting oxygen .

→ Carbon Dioxide: constitutes only about 0.038% (now 0.04%) of the dry air and is a product of combustion. Green plants, through photosynthesis, absorb carbon dioxide from the atmosphere and use it to manufacture food and keep other biophysical processes going. It is an important greenhouse gas, and helps maintain the Heat Budget of Earth.

→ Carbon dioxide and water vapour are found only up to 90 km from the surface of the earth.

→ Water vapour depends on temperature and therefore it decreases from the equator poleward in response to changes in temperature (and evaporation).

→ Ozone (O3)

→ – Formed at higher altitudes and transported downwards .

→ – The ozone layer is between 20 km and 25 km altitude where the greatest concentrations of ozone are found.

→ – Protects from harmful UV radiations.

→ Other gases: Neon, helium, hydrogen, xenon, krypton, methane.

→ Particulate Matter: Consists of sand particles (from weathered rocks, and volcanic ash), pollen grains, small organisms, soot, and ocean salts ; the upper layers of the atmosphere may even have fragments of meteors .

→ – Help in the absorbing, reflecting, and scattering of the solar radiation which adds the varied charming colours of red and orange at sunrise and sunset.

→ – Salt particles become hygroscopic nuclei and thus help in the formation of water drops , clouds, and various forms of condensation and precipitation.

→ – Recent research: Microplastics in the atmosphere can act as nuclei for cloud formation, leading to ice crystal formation at temperatures 5 to 10 degrees Celsius warmer than usual.

Layers Of The Atmosphere

👥 Based on Composition:

→ – All weather phenomena like cloud formation, movement of air masses and jet streams, etc , take place in this layer.

→ Stratosphere: Upto 50 kilometres above the earth's surface, beyond the tropopause.

→ – Temperatures are highest over the summer pole and lowest over the winter pole in the stratosphere.

→ – This layer is almost free from clouds and associated weather phenomena, making conditions most ideal for flying aeroplanes.

→ – Sometimes, rarely, cirrus clouds are present at lower levels in this layer.

→ Mesosphere: 50–80 km.

→ – Up to the Mesopause

→ – The air is so cold just below the mesopause that even the scarce water vapour can be sublimated into polar-mesospheric noctilucent clouds .

→ Thermosphere: 80–400 km.

→ – The International Space Station and satellites orbit here.

→ – The Kármán line, located at an altitude of 100 km: is widely used to describe the border between Earth's atmosphere and outer space.

→ – Ionosphere is a part of this layer, where electrically charged currents flow in the air in this sphere. Radio waves are reflected on the earth from this sphere and dve to this radio broadcasting has become possible.

🎯 2 · HEAT, ALBEDO, ISOTHERMS

PYQ 2013 PYQ 2023 PYQ 2024 PYQ 2025

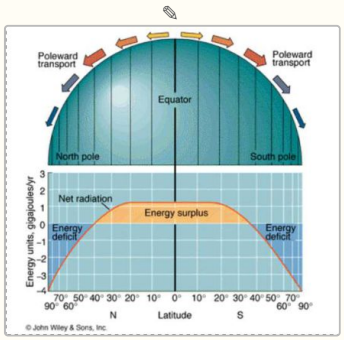
Understand it first: The Earth's surface is heated by the Sun and the air is heated mostly from below by the ground. Over a year the heat coming in equals the heat going out (heat budget). Albedo is simply how much sunlight a surface throws back.

Temperature controls

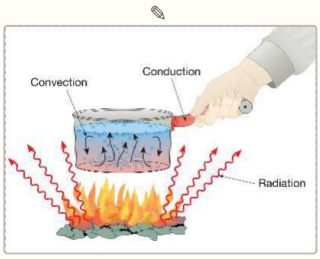
→ Latitude (slant rays) · duration of daylight · sunspot cycle
→ Transparency (cloudy day cooler, cloudy night warmer)
→ Altitude · mountain orientation · vegetation (transpiration cools)
→ Distance from sea · ocean currents · local winds/urban

4 ways heat moves

→ Conduction: through solid, molecule to molecule
→ Convection: fluid movement (liquid/gas)
→ Advection: horizontal movement of air/water
→ Radiation: electromagnetic waves, no medium



Ways of heat transfer

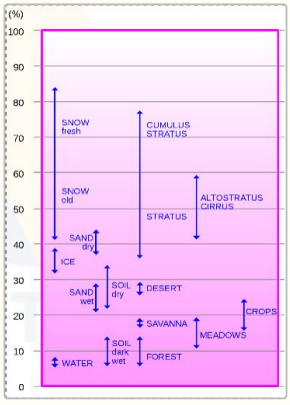


Ways of heat transfer

Con-Con-Ad-Rad – Conduction = Contact (spoon in hot chai) · Convection = Currents (boiling milk) · Advection = Aside, sideways (Loo carries heat in) · Radiation = Rays, no medium (Sun to Earth)

💰 Heat budget

→ Earth gets insolation (short-wave) and returns terrestrial radiation (long-wave) = constant temperature
→ Hotter body = shorter wavelength: Sun short-wave, Earth long-wave



Albedo

Albedo (reflectivity)

→ High: snow/ice = white shirt (reflects, cools)
→ Low: ocean, forest, city = black shirt (absorbs, warms)

→ Energy: 0–40° N/S surplus; beyond 40° deficit; winds and currents carry heat poleward, mostly across 30°–50° (hence storms there)

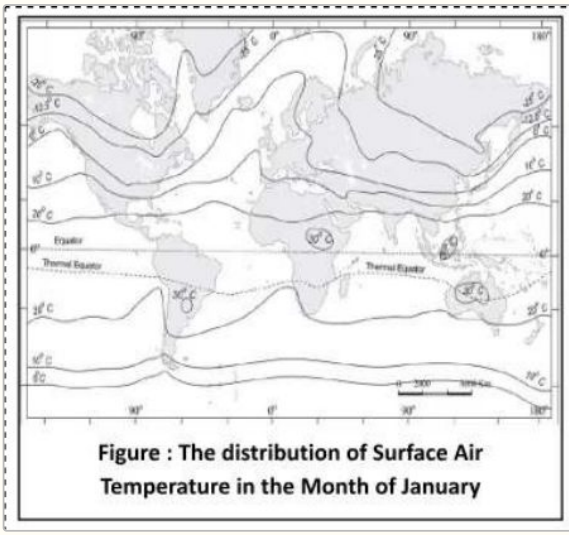


Figure : The distribution of Surface Air Temperature in the Month of January

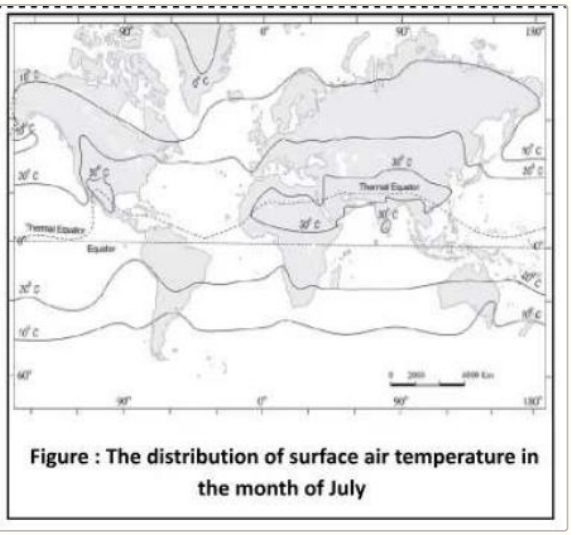


Figure : The distribution of surface air temperature in the month of July

Albedo

Isotherms

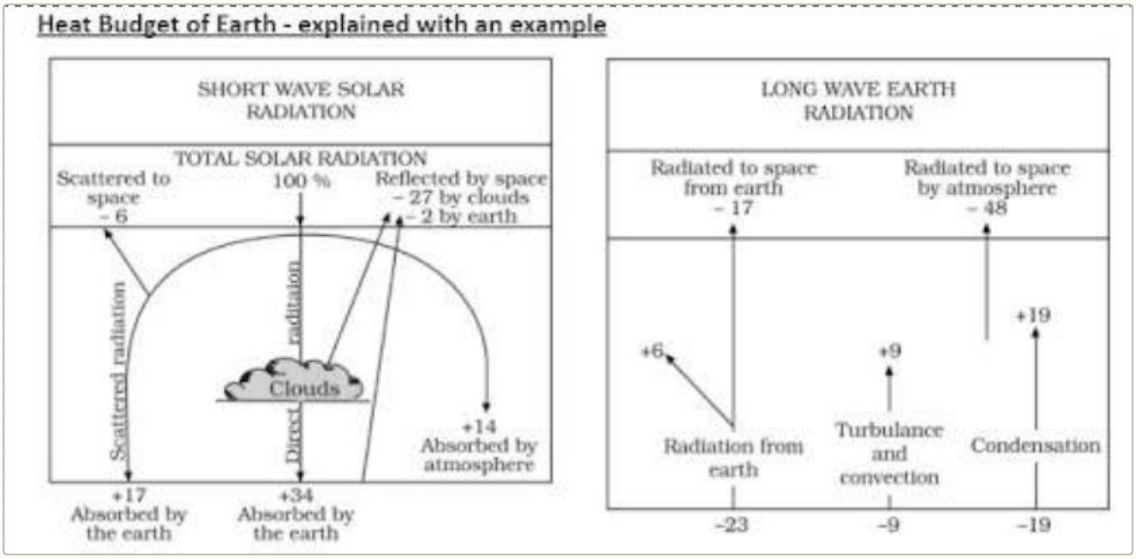
= lines of equal temperature (vertical distribution: see layers)

→ January, NH: bend poleward over oceans (warmer), equatorward over land (cools fast)

→ July: reverse (south over oceans, north over land)

→ SH: shift less evident (little land) and opposite in season: NH January = SH July

→ Cause: water has high specific heat; land heats and cools fast



Horizontal temperature distribution · Shown by "Isotherms"

Handout conflict: isotherm text first says "July: isotherms shift towards equator on land", then summarises "July: north over land, south over oceans". Trust the summary and the January rule.

PYQs: 2013 (A) interiors have bigger annual range due to land–water thermal difference only (not altitude, winds, rain; interiors are dry). Maritime influence = moderating; continental = extreme. 2023 (D) land–sea contrast greater in winter; water has higher specific heat. 2024 (a) January: isotherms north over ocean, south over continent; statement 2 (Gulf Stream, N Atlantic Drift as "cold currents") ✘. 2025 (a) both right and II explains I. ✅

Heat & Temperature

Factors affecting the distribution of temperature:

→ The latitude, which determines the angle of inclination. The higher the angle of inclination, more the distance sun rays have to travel within the atmosphere, which increases the heat lost by the sun rays. So, less

heat is transferred to the Earth. This is why the poles, which get slanting solar radiation, are much cooler than the equator, which gets direct sunlight.

→ Sunspot Cycle (discussed in the Universe part)

→ Transparency of the atmosphere : On a cloudy day, the clouds obstruct the sunlight from reaching the earth's surface, which is why cloudy days are cooler. On the other hand, cloudy nights are warmer as the clouds do not allow terrestrial radiation to escape. The same happens when there are lots of aerosols and particulate matter in the atmosphere.

→ Altitude of the place: as the altitude increases, the temperature reduces per the Normal Lapse rate.

→ Orientation: The places to the North of mountain ranges in the Northern Hemisphere, north of the Tropic of Cancer, and those to the South of the Mountain Ranges, south of the Tropic of Capricorn, are perpetually in the shadow of the mountains (until the mountains are eroded).

→ Vegetation cover: Trees undertake transpiration, which has a cooling effect on the environment.

→ Distance from the sea: The continental locations have more extreme temperatures than the coastal locations, due to the moderating impact of the ocean/sea through the land and sea breezes.

→ Ocean Currents: Warm and cold ocean currents increase or reduce the temperature of the place respectively.

Ways of heat transfer

→ Conduction: Transfer of heat through a solid material from molecule to molecule .

→ – Mechanism: Direct contact between particles , with heat moving from a region of higher temperature to one of lower temperature.

→ – Example: A metal rod heated at one end will gradually transfer heat to the other end.

→ Convection: Transfer of heat by the movement of a fluid (liquid or gas).

→ – Mechanism: Warmer, less dense fluid rises while cooler, denser fluid sinks , creating a circulation pattern.

→ – Mechanism: Large-scale horizontal movement of fluid carrying heat from one place to another.

→ – Example: Warm air moving into a region from another area, such as warm winds bringing heat to a cooler area.

→ Radiation: Transfer of heat through electromagnetic waves without the need for a medium .

→ – Mechanism: Heat is emitted as infrared radiation and can travel through a vacuum .

→ – Example: The heat from the sun warming the Earth.

Horizontal temperature distribution

→ – Shown by "Isotherms"

→ – Generally, from the equator to poleward , the temperature keeps on declining .

→ – In general, the equatorial region is hot , and its temperature is high throughout the year.

→ – The lowest temperature is at and near the poles.

→ – The amount of insolation received varies from latitude to latitude.

→ – The atmosphere (planetary winds) and the oceans (ocean currents) transfer excess heat from the tropics (energy surplus region) towards the poles (energy deficit regions)

Albedo

→ Surfaces with high albedo , such as snow and ice , reflect a large portion of incoming solar radiation, contributing to cooling.

Isotherms are the lines joining locations with same temperature.

→ Bending of isotherms refers to the way lines of equal temperature (isotherms) on weather maps deviate from their expected straight east-west pattern.

→ In Southern Hemisphere (SH), this shift is not very evident due to lack of land. But, though at smaller scale, due to opposite seasons in this hemisphere compared to NH, the shift here is opposite. What NH sees in January, is seen here in July, and vice versa.

→ To summarize, in July, isotherms shift to south over oceans, and north over land. In January the isotherms deviate to the north over the ocean and to the south over the continent.

→ · Topography: Mountains can block or channel air masses , leading to localized temperature variations and bending of isotherms.

3 · TEMPERATURE INVERSION

Understand it first: Normally air gets colder as you go up. In an inversion a warm layer sits above cold air like a lid, so the air below cannot rise: fog and smoke are trapped, but frost-sensitive crops on slopes may be spared.

Inversion: temperature rises with height

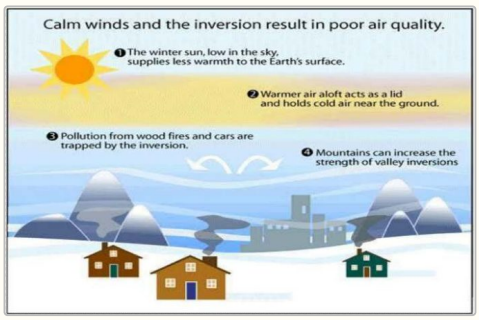
→ Needs clear, calm, long nights, no mixing, valleys/basins
→ Common at fronts, in winter, in mountains and valleys

Good
→ Frost protection for plants

Bad
→ Pollution dome
→ Fog: poor visibility, transport
→ Pests and diseases thrive

Pollution dome

→ Pollutants trapped under warm air; cold ground in cold weather cools air; no vertical mixing
→ Effects: poor air quality; respiratory, cardiovascular problems
→ Mitigation: cut emissions, air-quality monitoring/ forecasting, public awareness



Pollution Dome Effect

→ Beijing: China launched US$100-bn drive in 2013 (vehicle rules, relocate industry, clean energy): PM2.5 down 35% by 2017

Temperature Inversion

→ Inversion of the normal lapse rate , such that as the altitude increases, the temperature starts to increase too.

Conditions for Temperature Inversion:

→ Clear, Calm Conditions: Clear skies and calm winds allow for efficient radiative cooling of the Earth's surface during the night.
→ Long Nights or Extended Darkness: Longer nights or periods of extended darkness allow for more significant radiative cooling of the surface air.
→ Absence of Atmospheric Mixing: A lack of vertical mixing, such as from strong winds or convective activity, prevents the mixing of warmer air aloft with cooler air near the surface, allowing the inversion layer to form and persist.

→ Geographic Features : Certain geographic features, such as valleys or basins, can enhance the likelihood of temperature inversion by trapping cold air near the surface, especially during calm, clear nights.

👫 Most commonly seen at:	→ Winter season

Positive and Negative Impacts

→ Positive:

Negative	→ Reduced Visibility due to fog formation, affects transportation. → Can create conditions where pests and diseases thrive , potentially harming crops if not managed properly.
Pollution Dome Effect	→ During a temperature inversion, a layer of warm air traps cooler air beneath it, along with any pollutants present in the lower atmosphere. This creates a "dome" or "cap" of stagnant air that prevents the vertical dispersion of pollutants. → This usually happens during cold weather, when the air in contact with the cold ground becomes cool, resulting in temperature inversion. → Temperature Inversion's Role: Temperature inversion exacerbates the pollution dome effect by preventing vertical mixing of the atmosphere . Normally, warmer air rises and cooler air sinks, facilitating the dispersal of pollutants. However, during inversion, this vertical movement is suppressed, leading to the accumulation of pollutants near the surface. → Consequences: The pollution dome effect can lead to increased concentrations of pollutants in affected areas , resulting in poor air quality and health hazards for residents. In severe cases, prolonged exposure to high levels of pollutants trapped within the dome can lead to respiratory problems, cardiovascular issues, and other health complications . → Mitigation : Strategies to mitigate the pollution dome effect include reducing emissions of pollutants at their sources, implementing air quality monitoring and forecasting systems , and promoting public awareness and behaviour change to reduce individual contributions to air pollution. → In 2013, China launched a $100-billion initiative to combat air pollution in Beijing, implementing strict regulations on vehicle emissions, relocating polluting industries, and promoting cleaner energy sources. As a result, Beijing's air quality improved substantially, with PM2.5 levels decreasing by 35% between 2013 and 2017.

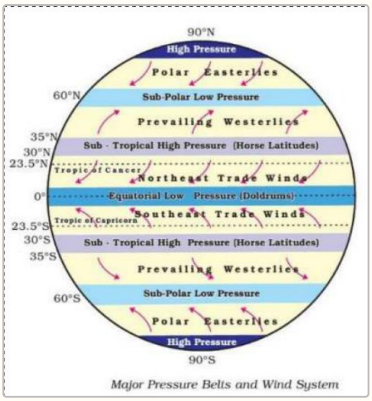
🎯 4 · PRESSURE, PRESSURE BELTS, FORCES PYQ 2024

Understand it first: Air has weight, so it presses. Heated air rises and leaves low pressure; cold or sinking air gives high pressure. Wind is air moving from high to low, turned by the Earth's rotation (Coriolis). That alone gives the pressure belts and the three wind systems.

Air pressure	→ Weight of air column per unit area; maximum at sea level → Heat = expand, rise = low; cool = sink = high → Isobars = lines of equal pressure

→ 7 belts = 3 thermal + 4 dynamic; not uniform (land–water)

→ Equatorial low (10° N–S) = ITCZ = Doldrums (calm, trades converge)



Air Pressure & Pressure Belts

Lazy Hens Lay Happily — Low equator (ITCZ/Doldrums 10° N–S) → High ~30° → Low ~60° → High poles

Belt shift (Earth's 23.5° tilt)

→ 5–10° north (up to 25° over India) when Sun over Tropic of Cancer 21 June

→ 5–10° south when over Tropic of Capricorn 22 December

→ Balanced at equinoxes 21 March, 23 September

→ At 30°–40° = Mediterranean pattern: winter westerlies = rain; summer dry trades offshore

3 forces on wind

→ Pressure gradient: speed + direction (high to low; close isobars = fast)

→ Coriolis: direction only; deflects right in N, left in S; zero at equator, max at poles; grows with wind speed

→ Friction: both; only a few thousand metres up; less over sea so faster

→ Winds are named for where they come from (polar easterlies come from east)

Types of winds

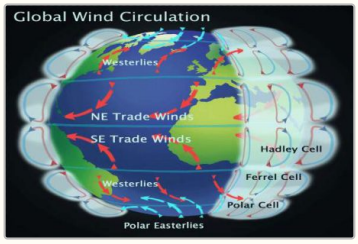
→ Planetary/permanent: all forces, all year (trades, westerlies, polar easterlies)

→ Geostrophic: tropopause level, no friction, parallel to isobars (jet streams)

→ Seasonal: monsoon

→ Diurnal: sea/land breeze, mountain–valley breeze

→ Local: specific place, all year or one season



Planetary Winds And Tri-Cellular Meridional Circulation

Planetary wind	From to	Notes
Trade winds	subtropical high to equator, easterly	helped trade routes; push ocean currents
Westerlies	subtropical high to sub-polar low, W to E	strongest in S. Hemisphere (little land): Roaring 40s, Furious 50s, Shrieking 60s
Polar easterlies	polar high to mid-latitudes	cold, weaker

Hadley Hot — Hadley (equator–30°, biggest, trades) · Ferrel Fill-in (30°–60°, middle, runs opposite) · Polar Puny (60°–pole, smallest, polar easterlies). Tricellular meridional circulation = 3 cells per hemisphere.

PYQs: 2011 (A) southern westerlies stronger because less land (not higher Coriolis). ✅ 2011 (C) jets fly in lower stratosphere: no clouds/vapour and no vertical wind shear. 2015 (B) westerlies blow 30°–60° (not "30°N–60°S"); moist air for NW India's winter rain is westerlies: they start in horse latitudes, steer extratropical cyclones; Mediterranean disturbance dragged by the jet stream; J&K and NW India get the most. 2024 (c) Coriolis rises with wind velocity and is max at poles, absent at equator.

Pressure belts in both hemispheres stay balanced when the Sun is vertical over the Equator: 21 March and 23 September (equinoxes).

Air Pressure & Pressure Belts

→ High-temperature results in Low Pressure and vice versa.

→ When the air gets heated it expands, becomes light and rises vertically. As air rises, the pressure it exerts on the Earth's surface is reduced, causing a low-pressure area. On the other hand, cool air is dense and heavy. As a result, it sinks vertically. It results in additional weight and pressure which causes a high pressure on the ground.

→ Based on temperature distribution, 7 pressure belts are present on the Earth.

→ – 3 of these are thermally induced, created due to the temperature.

→ The Equatorial Low-Pressure belt (10 degrees north and south) is also known as the ITCZ (Inter-Tropical Convergence Zone) or the "Doldrums" (calms), as this region experiences monotonous windless weather, due to the convergence of the trade winds around it.

→ Pressure belts are not homogenous, due to uneven distribution of land and water.

→ Seasonal climate fluctuations are caused by the movement of the pressure belts, especially between the latitudes of 30° and 40° (Mediterranean Climate) in both hemispheres. During winter, Westerlies prevail, resulting in rain. During the summer, dry Trade Winds blow offshore, preventing rain from falling in certain areas.

Winds: Factors Affecting Their Movement

→ Pressure Gradient (Direction and Speed): It denotes the pressure difference between two locations. The winds always move from the High to the Low-pressure region. This determines the direction of the winds.

→ – The steep Pressure Gradient is denoted on the maps via closely spaced isobars. The steeper the gradient is, the higher the speed of movement of the winds.

→ – So overall, this force determines both the speed and the direction of the wind movement i.e. it helps determine the overall velocity of the Winds.

Coriolis Force (Only Direction)	→ This force is generated by virtue of the movement of the Earth and affects only the objects or entities moving on the surface of the Earth, like winds and ocean currents. That is why it is an apparent and not a real force. → It affects only the direction of movement of the winds. → 0 at the equator, maximum at the poles.
Frictional Force (Direction and Speed):	→ It is effective only up to a few thousand meters above the Earth's surface. → The oceans provide much less frictional force than land, as the latter is covered with vegetation, infrastructure, landforms, etc. That is why the wind speeds are higher over water bodies than on the land.

Types Of Winds

→ Diurnal winds: change direction between day and night. E.g. in coastal regions, the sea breeze blows during the day and the land breeze blows during the night. Another example is the mountain valley breeze.

Planetary Winds And Tri-Cellular Meridional Circulation

→ Planetary winds are large-scale wind patterns that circulate the Earth, driven by differences in temperature and pressure between regions.

→ They are present throughout the Year.

→ These winds play a crucial role in redistributing heat and moisture across the planet, influencing weather patterns and climate. There are several key types of planetary winds:

→ – Westerlies : Westerlies are prevailing winds that blow from the subtropical high-pressure belts towards the sub-polar low-pressure region in both hemispheres. They are characterized by their westerly direction (from west to east) and play a crucial role in the movement of weather systems across mid-latitudes. The westerlies are particularly strong and well-defined in the Southern Hemisphere and flow at extremely high speeds. This has given them the names like roaring fourties, furious fifties, and shrieking sixties.

→ Tricellular Meridional Circulation: Tricellular meridional circulation refers to the three-cell circulation pattern in each hemisphere of the Earth's atmosphere, characterized by the movement of air from the equator towards the poles and back again.

→ – Hadley Cell : The Hadley cell is the largest and most prominent cell in the tricellular circulation, extending from the equator to about 30 degrees latitude. Warm, moist air rises at the equator, forming low-pressure zones and generating trade winds. As the air cools and descends around 30 degrees latitude, it creates high-pressure belts and generates the trade winds.

→ – Ferrel Cell : The Ferrel cell is a mid-latitude cell that lies between the Hadley and polar cells. It operates in the opposite direction to the Hadley cell, with air rising at around 60 degrees latitude and descending at around 30 degrees latitude. The Ferrel cell is driven by the interaction between the Hadley and polar cells.

→ – Polar Cell : The polar cell is the smallest and weakest in the tricellular circulation, extending from around 60 degrees latitude to the poles. Cold, dense air sinks at the poles, creating high-pressure zones and generating the polar easterlies. As the air warms and rises around 60 degrees latitude, it creates low-pressure belts and generates westerly winds.

→ – Importance : Tricellular meridional circulation plays a crucial role in regulating global climate and weather patterns by transporting heat, moisture, and energy around the planet. It helps to redistribute heat from the equator towards the poles and vice versa, influencing temperature gradients, atmospheric pressure systems, and wind patterns.

🎯 5 · GEOSTROPHIC WINDS, JET STREAMS PYQ 2020 PYQ 2024

Understand it first: High up, away from friction, the pressure push and Coriolis balance, so wind blows along the isobars: the geostrophic wind. Jet streams are narrow, very fast rivers of such air near the top of the troposphere.

Geostrophic wind: blows parallel to isobars; pressure gradient force balanced by Coriolis (right in N, left in S). 💡

👫 Jet streams: narrow fast bands, 8–15 km

Permanent

→ Polar: 50°–60° N/S, boundary cold polar/warmer subtropical air, strongest in winter

→ Subtropical: 20°–30° N/S, weaker, less consistent

Temporary

→ Tropical Easterly

→ Somali

→ Both hemispheres (2020: "NH only" ✘) · formed by temperature gradients of air masses

→ Wavy jets = Rossby waves

→ Steer storms; carry western disturbances (Mediterranean extra-tropical cyclones, by subtropical westerly jet); control South Asian monsoon; mix troposphere–stratosphere (ozone-depleting substances)

Climate change and jets

→ Arctic warms faster = smaller temperature contrast = weaker, more meandering jets

→ Long heatwaves, coldwaves, droughts or floods

Heat dome

→ Stagnant high pressure; air sinks, compresses, warms

→ Mid/high latitudes; continental climates

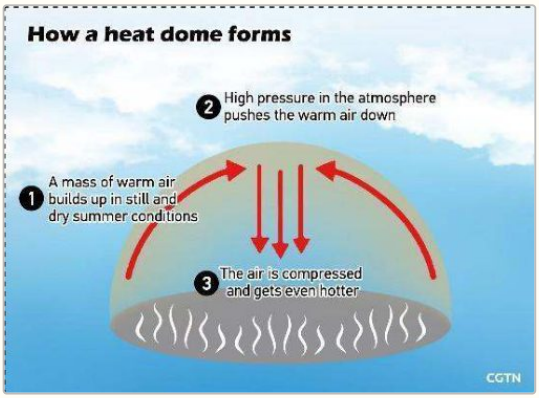
→ Heatwaves, heatstroke; agriculture, energy hit

→ Climate change: more frequent, intense, longer

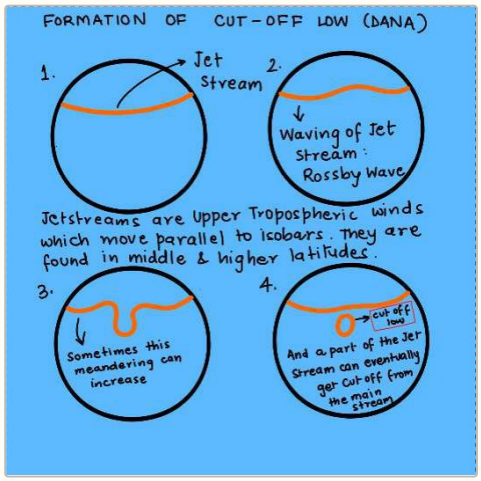
Polar vortex (= DANA, Depresion Aislada en Niveles Altos, isolated depression/cut-off low)

→ Cold air held by polar jet; both poles, N pole matters most

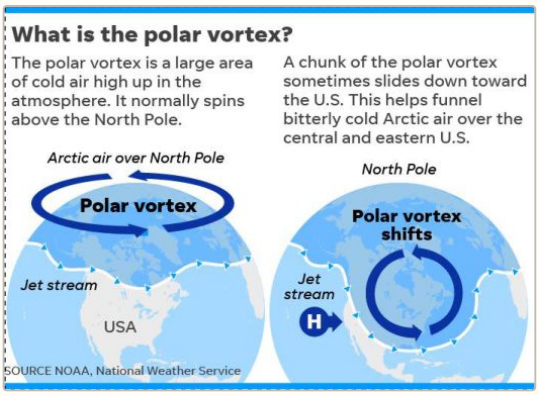
→ Arctic Amplification + meandering jets = cold air spills to low latitudes



Heat Dome Effect



DANA (Depresion Aislada en Niveles Altos)



DANA (Depresion Aislada en Niveles Altos)

Stratospheric cooling

→ CO_2 radiates heat to space in the stratosphere (traps it in troposphere)

→ Ozone loss (CFCs, poles) = less UV absorption = less heating

→ GHGs affected stratosphere by ~1885, five decades before troposphere warming was detected

→ Impact: slows ozone recovery, strengthens polar vortex, disrupts jets and weather, extreme events, affects aviation and satellite orbits, key climate-change indicator ("warm troposphere + cooling stratosphere"); more PSCs

Geostrophic Winds

→ Geostrophic winds are large-scale winds that blow parallel to isobars (lines of constant pressure) in the atmosphere. They are the result of a balance between the pressure gradient force and the Coriolis force , acting perpendicular to the pressure gradient.

→ Formation : Geostrophic winds form when the pressure gradient force , which drives air from areas of high pressure to areas of low pressure, is balanced by the Coriolis force , which

deflects moving air to the right in the Northern Hemisphere and the left in the Southern Hemisphere due to the Earth's rotation.

Jet Streams

→ Jet streams are narrow bands of fast-flowing, high-altitude winds found in the upper atmosphere, typically at altitudes of 8 to 15 kilometres. They are characterized by their high wind speeds and meandering, ribbon-like appearance.

→ Wavy Jet Streams are said to have " Rossby Wave " characteristics.

Impact on Weather & Economy

→ Jet streams play a significant role in shaping weather patterns by steering the movement of weather systems and influencing the development of storms and precipitation.

→ They move air masses from one place to another, leading to the development of fronts and extratropical cyclones.

→ They can also affect aviation routes , with aircraft often taking advantage of the strong tailwinds associated with jet streams to reduce flight times and fuel consumption.

→ Jet Stream may transport ozone-depleting substances higher up in the stratosphere. This vertical air circulation causes a rapid rate of mixing of air between the Troposphere & Stratosphere.

Jet Streams and Climate Change:

→ One potential consequence of climate change on jet streams is their potential weakening in strength. As the Arctic region experiences accelerated warming compared to other parts of the globe, the temperature contrast between the Arctic and lower latitudes diminishes. Since the movement of jet streams is driven by these temperature variations, a less pronounced gradient can lead to jet streams that are weaker and exhibit more meandering paths. This phenomenon can result in prolonged periods of certain weather conditions, such as heatwaves, coldwaves, droughts, or heavy rainfall.

→ Furthermore, the altered temperature differentials may induce shifts in the positioning of jet streams. As they migrate closer to the poles or display more erratic trajectories, they can exert influence on the trajectories of storms and weather systems. Consequently, this can bring about alterations in precipitation distribution, storm intensity, and the frequency of extreme weather occurrences.

Heat Dome Effect

→ A heat dome refers to a weather phenomenon characterized by the trapping of hot air beneath a dome of high pressure in the atmosphere. This leads to prolonged periods of extremely hot temperatures and stifling conditions, particularly during the summer months.

→ Formation: Heat domes typically form when a large, stagnant area of high pressure settles over a region , causing the air beneath it to sink and compress . As the air descends, it warms up due to compression, creating a dome of hot air near the surface.

→ Consequences: The Heat Dome effect can result in dangerously high temperatures, heatwaves, and heat-related illnesses such as heat exhaustion and heatstroke. Prolonged exposure to extreme heat can also impact agriculture, infrastructure, and energy demand .

→ Regional Variability: Most common in regions with continental climates . Also since Jet Streams have a role in creating these Heat Domes, they are mostly found in Middle or higher latitudes.

→ Climate Change: Climate change is expected to increase the frequency, intensity, and duration of heat domes in many regions, exacerbating the risk of heat-related impacts and challenging efforts to adapt to a warming climate.

DANA (Depresion Aislada en Niveles Altos)	→ Can be in both the poles, but impact on human life is mainly due to the North pole vortex. → Due to global warming & Arctic Amplification and resultant meandering of the Jet streams, this parcel of cold air can intrude in the non-polar, lower latitudes, causing extremely cold conditions.
Stratospheric Cooling	→ Stratospheric cooling refers to the gradual decrease in temperature in the stratosphere, primarily due to human-induced changes like greenhouse gas emissions and ozone depletion. → 1. Greenhouse Gas Effect (Counterintuitive Cooling) → Greenhouse gases (especially CO_2) trap heat in the troposphere, warming the lower atmosphere. → However, in the stratosphere, CO_2 acts differently, it radiates heat away into space. → – CO_2 molecules absorb infrared (IR) radiation and then re-emit it in all directions. → – In the dense troposphere, that emitted energy often gets reabsorbed by other molecules, keeping heat trapped (warming effect). → – But in the thin stratosphere, there are fewer molecules to absorb that radiation. → – Because the stratosphere is less dense and lies higher in the atmosphere, much of the IR radiation emitted by CO_2 simply escapes into space. → As more CO_2 accumulates, the stratosphere loses more heat, causing it to cool down. → 2. Ozone Layer Depletion → Slows Ozone Layer Recovery → – Cooling reduces chemical reaction rates needed to rebuild ozone, delaying ozone layer recovery. → – Polar Stratospheric Clouds (PSCs) form more readily in cold conditions, which enhance ozone depletion, especially over Antarctica. → Strengthens the Polar Vortex → – This can trap cold air over polar regions or, when destabilized, lead to cold outbreaks in mid-latitudes (e.g., North America, Europe). → Extreme Weather Events → – This vertical temperature gradient between the stratosphere and troposphere (due to global warming) can enhance convection and influence weather extremes. → Affects Aviation and Satellite Orbits → – Temperature and density changes in the upper atmosphere can alter aircraft flight performance and affect satellite drag, impacting orbital lifetimes. → Key Indicator of Climate Change → – The simultaneous tropospheric warming and stratospheric cooling is a distinct **fi**ngerprint of anthropogenic climate change, not caused by solar activity or natural variability.

🎯 6 · LOCAL AND MONSOONAL WINDS PYQ 2014

Understand it first: Local winds are small-scale and named: they blow because land and sea, or valley and slope, heat and cool at different rates, or because air warms as it comes down a mountain. Monsoon is the same land-sea idea at the scale of a continent and a season.

Wind	Where	Hook
Chinook	E. Rockies (USA, Canada)	warm dry, "snow eater"

Foehn	Alps (leeward)	warm dry descending
Sirocco (Ghibli, Chili, Khamsin)	Sahara to Mediterranean	hot, dry; rainy in S. Europe
Harmattan	W. Africa (Sahara to Guinea)	dry, dusty NE trade; "Doctor" wind
Loo	N. India, Pakistan, May–June	furious, hot, heatstroke
Bora	Adriatic (Italy, Slovenia, Croatia)	extremely cold dry NE, katabatic
Mistral	Rhône valley (France)	cold NW
Santa Ana	S. California	hot dry; worsens wildfires
Khamsin	Egypt	dusty spring wind, up to 50 days
Purga	Siberia, Russian steppes	icy NE blizzard wind
Pamperos	Pampas (Argentina, Uruguay, Brazil)	violent cold SW

Brrr — Hot/warm: Chinook, Foehn, Sirocco, Harmattan, Loo, Santa Ana · Cold: Bora, Mistral, Purga, Pamperos. Mistral and Bora = "Brrr" winds.

Wind Name	Location	Main Characteristics/Effects
Chinook	Eastern slopes of Rockies (USA, Canada)	Warm, dry, descending wind, rapidly melts snow, "snow eater"
Foehn	Leeward side of Alps (Central Europe)	Strong, dry, and warm desceanding wind, rapid temperature rise
Khamsin	Egypt, eastern Mediterranean	Dry, dust-laden spring wind, lasts up to 50 days
Sirocco	Sahara to Mediterranean (Italy, Balkans, North Africa)	Hot, dry winds from Sahara, possibly moist/rainy in southern Europe; also called "Ghibli" (Libya), "Chili" (Tunisia), "Khamsin" (Egypt)
Harmattan	West Africa (Sahara to Gulf of Guinea)	Dry, dust-laden northeast trade wind, lowers humidity & visibility. Aka Guinea doctor/Doctor winds.
Loo	North India, Pakistan (Indo-Gangetic Plain)	Furious, hot and dry summer wind (May-June), causes heatstroke, starts in deserts
Bora	Adriatic Sea coasts (Italy, Slovenia, Croatia)	Extremely cold, dry NE wind, sudden onset, hurricane strength gusts, katabatic
Mistral	Rhône Valley and Provence, France	Cold, strong NW wind, clears skies, accelerates south to Mediterranean
Santa Ana	Southern California (USA)	Hot, dry, gusty desert wind, blows coastal, worsens wildfires.
Blizzard	Tundra/Polar regions, North America, Antarctica	Severe snowstorm with >35 mph wind, <0.25 mile visibility, lasts 3+ hours
Purgas/ Purga	Russia (Siberia, steppes)	Bitterly cold blizzard, icy, strong northeasterly wind, brings snow, reduces visibility
Pamperos	Pampas of Argentina, Uruguay, Brazil	Speedy, violent, cold wind from SW, brings stormy weather, sudden temperature drops

Local Winds

Source note: Khamsin appears twice: as a Sirocco name (Sahara to Mediterranean) and as an Egyptian dusty spring wind.

Monsoon winds: seasonal, tropics, **reverse** with the Sun's apparent position (detail in Indian Geography). **PYQ 2014 (C):** seasonal reversal = **monsoon** climate; "mausim" (Arabic) = season; ~6 months NE, 6 months SW; South Asia, Africa, Australia, Pacific coast of Central America. ✅

Local Winds

→ They affect **only a certain specific part of the world**. These winds **can be both warm and cold**. Here is a list of the major local winds of the world:

Monsoonal Winds

→ These are seasonal winds that are experienced in the tropical regions of the world . The most peculiar part about these winds is the reversal they experience with the switch in the apparent positioning of the sun.

→ This portion will be covered in more detail in the Indian Physical Geography section, as India is highly dependent on these monsoonal winds for its climate, vegetation, economy, as well as society.

🎯 7 · HUMIDITY, CLOUDS, PRECIPITATION PYQ 2019 PYQ 2022 PYQ 2024

Understand it first: Warm air can hold more water vapour. Cool it enough (dew point) and vapour condenses on tiny particles: near the ground it is dew, fog or frost, higher up it is cloud. Clouds are named by height and shape; rain is classed by what lifted the air.

Humidity (condensation = vapour to liquid water)

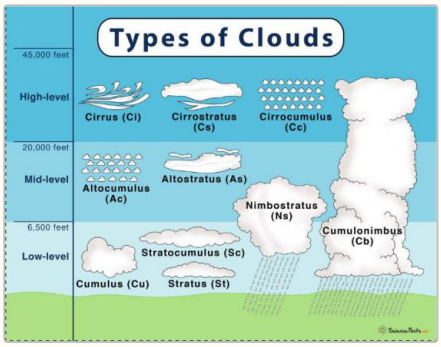
→ Absolute (g/m³) · relative (% of capacity) · specific (g per kg)

→ Dew point = saturation temperature

→ Dew: surface cools below dew point; none on cloudy nights (clouds send radiation back)

→ Frost = deposition (vapour to ice) · Fog = surface cloud · Smog = smoke + fog (sulphurous, photochemical) · Haze = dry particles

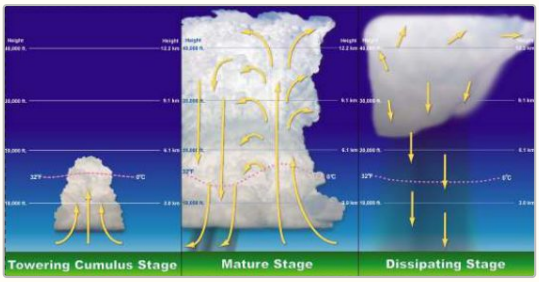
→ Atmospheric rivers: narrow moisture corridors (more water than Amazon), pushed by westerlies to mid-latitudes = heavy rain, floods



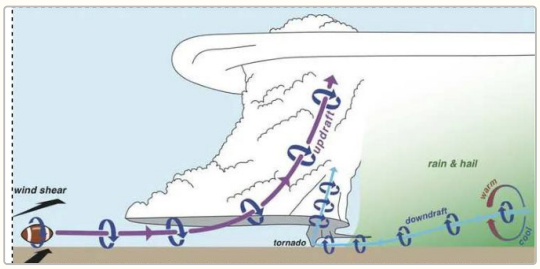
Types Of Clouds

Water Spouts

Clouds	Height	Note
High (Cirrus)	> 6,000 m	wispy ice "chicken feathers"; thin ones trap heat
Mid (Alto-)	2,000–6,000 m	
Low	< 2,000 m	Stratus, stratocumulus, nimbostratus; thick ones reflect
Cumulonimbus	vertical tower	thunder, hail, tornado



Thunderstorms · Formation

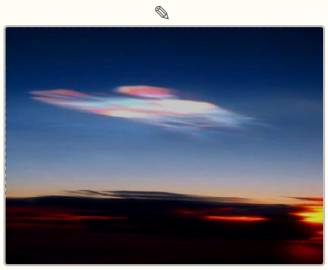


Thunderstorms · Formation

PYQs: 2022 (D) high thin clouds trap heat, low thick clouds reflect/cool: both statements ✘. 2019 (D) cirrus thinning + sulphate aerosols in stratosphere = geo-engineering to reduce global warming (cirrus trap long-wave; aerosols reflect sunlight). 2019 (B) no dew on cloudy night: clouds reflect back Earth's radiation. 2024 (a) water vapour decreases with altitude; not max at poles. ✅

Polar stratospheric clouds (PSCs)

→ 15–25 km, polar winter, below –78 °C



Polar Stratospheric Clouds

Polar Stratospheric Clouds

→ "Nacreous / mother-of-pearl"; bright at twilight

→ Surfaces for chlorine and bromine reactions = ozone destruction, Antarctic ozone hole

Cloud seeding (weather modification)

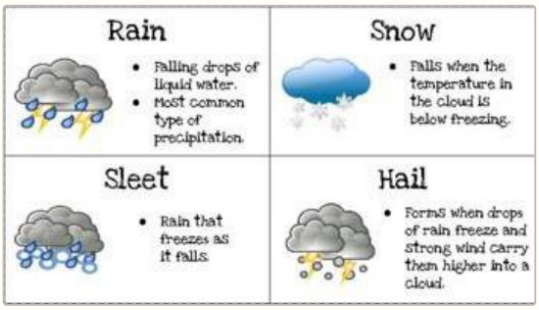
→ Agents: silver iodide, potassium iodide, salt (NaCl), dry ice = hygroscopic nuclei

→ Delivery: aircraft or ground generators (rockets, flares)

→ Uses: rain, snowpack, fog dispersal at airports, hail suppression, pollution control

→ Limits: needs moisture; variable, debated; silver iodide build-up; costly

Precipitation	How	Where
Convectional	heated air rises = cumulonimbus; short, intense, thunder	tropics 0–30°
Orographic	air forced up mountains; rain shadow on lee side	mountains
Frontal	warm over cooler: warm front long steady; cold front short intense	mid-latitudes 30–60°
Cyclonic	convection/frontal on a large scale around a low	cyclones



Types Of Precipitation

Thunderstorm

→ Stages: cumulus, mature, dissipating

→ Collisions of ice/water: lighter + at top, heavier – at bottom = voltage = lightning (excites N_2, O_2), thunder

→ Tall objects hit most: lightning conductors

Tornado

→ Violently rotating column from thunderstorm to ground

→ Over 300 km/h (long text: 300 miles/h)

→ Tornado Alley (Texas to Oklahoma, Kansas, Nebraska): warm moist Gulf air meets cold dry Canadian + warm dry desert air = supercells

→ Also Europe, Asia, Africa, Australia

Waterspout

→ Air + water droplets over water; less intense

→ Fair-weather (calm, little damage) vs tornadic (dangerous)

Humidity & Condensation

→ Humidity refers to the amount of water vapor present in the air. It is a measure of the moisture content of the atmosphere and plays a significant role in determining weather and climate conditions.

There are several different types of humidity

→ Absolute Humidity : Absolute humidity is the actual amount of water vapor present in a unit volume of air , typically expressed in grams per cubic meter (g/m³). It provides a direct measure of the moisture content of the air.

→ Relative Humidity : Relative humidity is the ratio of the amount of water vapor present in the air to the maximum amount of water vapor that the air can hold at a given temperature , expressed as a percentage . It indicates how close the air is to saturation and is commonly used to describe how "humid" or "muggy" the air feels.

→ Specific Humidity : Specific humidity is the ratio of the mass of water vapor to the total mass of air (including dry air and water vapor) in a sample of air. It is expressed in grams of water vapor per kilogram of air and provides a measure of the actual moisture content of the air without regard to temperature.

→ Condensation is the process by which water vapor in the air transforms into liquid water . There are several types of condensation, including:

→ – Dew : Dew forms when the surface temperature of an object, such as grass or car windows, falls below the dew point temperature of the surrounding air. Water vapor in the air condenses onto the cooler surface, forming small droplets of liquid water .

→ · Dew Point: Dew point is the temperature at which air becomes saturated with water vapor and condensation begins to form. It is a measure of how much moisture is present in the air and is often used as an indicator of comfort level, with lower dew points indicating drier air and higher dew points indicating more humid air.

→ – Fog : Fog is a type of condensation that occurs when moist air cools and reaches its dew point, causing water vapor to condense into tiny water droplets suspended in the air near the ground. Fog reduces visibility and is commonly observed in low-lying areas, near bodies of water, or during temperature inversions. Fogs are near-surface clouds.

→ – Clouds : Clouds are large masses of water droplets or ice crystals suspended in the atmosphere. They form when moist air rises, cools, and reaches its dew point, causing water vapor to condense onto tiny particles called condensation nuclei. Clouds play a crucial role in the Earth's climate system and weather patterns.

→ – Frost : Frost forms when water vapor in the air undergoes deposition, transitioning directly from a gas to a solid without passing through the liquid phase. Frost forms on surfaces such as grass, leaves, or windows when the surface temperature falls below the frost point temperature of the surrounding air.

Smog

→ Smog = smoke + fog (smoky fog) caused by the burning of large amounts of coal, vehicular emission, and industrial fumes (Primary pollutants).

→ Smog contains soot particulates like smoke, sulphur dioxide, nitrogen dioxide, and other components.

→ At least two distinct types of smog are recognized: sulphurous smog and photochemical smog.

→ Haze: An atmospheric phenomenon where dust, smoke, and other dry particles obscure the clarity of the sky.

Atmospheric Rivers

→ Creation of Atmospheric Rivers:

→ – Evaporation: Warm ocean waters, particularly in tropical regions, cause large-scale evaporation, adding significant moisture to the atmosphere.

→ – Transport by Wind: Strong winds in the atmosphere, especially within the jet streams, transport this moisture in narrow bands. These winds can extend over thousands of kilometers.

Types Of Clouds

→ Clouds are categorized into different types based on their altitude, appearance, and the processes that form them. Based on altitude, clouds are generally classified into three main categories:

→ – High-Level Clouds (Cirrus Clouds) : These clouds typically form at altitudes above 6,000 meters. They are composed of ice crystals and appear thin, wispy, and feathery. Examples include cirrus, cirrostratus, and cirrocumulus clouds.

→ – Mid-Level Clouds (Alto Clouds) : Mid-level clouds typically form between 2,000 to 6,000 meters above the ground. They are composed of water droplets and sometimes ice crystals . Examples include altocumulus and altostratus clouds.

→ – Low-Level Clouds : These clouds form at altitudes below 2,000 meters . They are composed of water droplets and are often associated with weather phenomena such as rain or drizzle. Examples include stratus, stratocumulus, and nimbostratus clouds.

→ Additionally, some clouds form vertically through the troposphere , extending through multiple altitude levels:

Polar Stratospheric Clouds

→ Polar stratospheric clouds (PSCs) form in the stratosphere at altitudes of 15–25 kilometers in polar regions during the winter months , where temperatures drop below –78°C . Typically, clouds do not form in the stratosphere because it is usually too dry and stable; however, the extreme cold in polar regions during winter allows PSCs to form. Often called nacreous or mother-of-pearl clouds due to their iridescent colors, PSCs appear bright and colorful during twilight. PSCs play a critical role in ozone depletion by providing surfaces for chemical reactions that release ozone-depleting chlorine and bromine, significantly contributing to the formation of the ozone hole over Antarctica.

Cloud Seeding

→ Cloud seeding is a type of weather modification technique where certain substances are introduced into clouds to stimulate rainfall or snowfall. Process

→ Water vapor condenses/freezes around them, forming larger droplets or ice crystals.

→ These eventually grow heavy enough to fall as rain or snow.

Methods of Delivery

→ Aircrafts : release particles directly into target clouds.

Applications

→ Enhancing rainfall in drought-prone regions.

→ Hail suppression in agriculture by reducing hailstone size.

→ Works only if there is moisture in the air .

→ Environmental concerns over silver iodide accumulation .

→ Expensive and logistically complex.

Types Of Precipitation

→ Precipitation occurs when water droplets in clouds coalesce and grow large enough to overcome air resistance and fall to the ground . Raindrops form through the process of collision and coalescence or the freezing of supercooled water droplets at high altitudes.

Precipitation

→ – Intense and localized, often resulting in heavy showers, thunderstorms, and short-lived periods of rainfall.

→ – Precipitation: Rainfall is the most common type of precipitation associated with convective rainfall. It is these clouds that produce thunderstorms and can also produce hail in some cases.

→ Orographic Precipitation :

→ – Forms when moist air is forced to rise over elevated terrain such as mountains or hills.

→ – The leeward side of the mountain may experience a rain shadow effect , with drier conditions and less precipitation.

→ – Latitudes: Orographic rainfall can occur at various latitudes but is most common in regions with mountainous terrain.

→ – Precipitation: Rainfall is the typical outcome of orographic lifting. However, at higher elevations and colder temperatures, precipitation can fall as snow . In some cases it can also fall as hail .

→ Frontal (Cyclonic) Precipitation:

→ – Occurs along the leading edge of a frontal boundary between two contrasting air masses , such as a cold front or warm front.

→ – Can result in prolonged periods of rainfall over a large area in case of warm front, and short periods of intense rainfall in case of cold fronts.

→ – Latitudes: Frontal rainfall is common in mid-latitude regions (30° to 60° latitude), where contrasting air masses meet.

→ – Precipitation: Rainfall is the primary type of precipitation associated with frontal systems. However, in colder regions or during transitional seasons, precipitation can fall as snow or sleet .

→ Cyclonic Precipitation:

→ – Associated with the circulation of air around a low-pressure system, such as a tropical cyclone or extratropical cyclone.

→ – Intense and widespread, with heavy rainfall occurring over large areas for an extended period.

→ – Can result in flooding, landslides, and other severe weather impacts.

→ – Precipitation:

→ · Tropical cyclones typically produce heavy rainfall , sometimes accompanied by thunderstorms and hail .

→ · In colder regions, extratropical cyclones can result in snowfall.

Thunderstorms

→ Thunderstorms are atmospheric disturbances characterized by the presence of thunder and lightning, often accompanied by heavy rain, strong winds, and sometimes hail.

Formation:

→ Thunderstorms typically form in warm, moist air masses where there is instability and upward vertical motion, of air. This upward motion, i.e. convection (updraft) leads to the development of towering cumulonimbus clouds.

→ Stages: Thunderstorms go through three main stages: the cumulus stage, the mature stage, and the dissipating stage.

→ – Finally, in the dissipating stage, the storm weakens as downdrafts dominate and the supply of warm, moist air diminishes.

→ The cumulo nimbus clouds have a large amount of rain drops and ice crystals these particles constantly collide with each other and get charged the lighter positive charges accumulated at the top of clouds while heavier negative charges accumulate at the cloud bottom due to the accumulation of charges at cloud bottom a potential difference (voltage) appears to get created between the ground and the cloud bottom. Initially the air between the cloud and the ground act as an insulator, preventing the passage of any current. However as the potential difference

between the cloud and the grounds keeps on increasing the current starts to move through the air exciting the nitrogen and oxygen molecules thus creating a bluish/ purpulish glow called lightning full stop the sound associated with this lightning discharge is called as thunder. Apart from ground and clouds the lightning discharge can also occur:

→ The taller objects on the ground (building/trees etc) have a high probability of experiencing a lightning event, as they are closer to the cloud bottom. To prevent lightning from destroying the electrical system of a building and endangering the life of the people inside, electric conductors are installed at the top of buildings.

Clouds in Thunderstorms:

→ The main cloud type associated with thunderstorms is the cumulonimbus cloud, which is tall and vertically developed. These clouds can reach heights of up to 10 miles (16 kilometers) or more and are characterized by their anvil-shaped tops. Other cloud types, such as cumulus clouds and stratocumulus clouds, may also be present during the early stages of thunderstorm development.

Hazards:

→ Thunderstorms can pose various hazards, including lightning strikes , which can cause fires and electrical damage, heavy rainfall leading to flash flooding, strong winds capable of causing property damage and power outages, and hail , which can damage crops, vehicles, and buildings.

Tornado

→ It's formed when warm, moist air rises rapidly and meets cooler air, creating a strong updraft. This updraft causes the air to spin, and as it spins faster, it becomes more focused and descends towards the ground. If the rotating air column touches the ground, a tornado is formed.

Formation:

→ Thunderstorm Development: Tornadoes form within powerful thunderstorms, often associated with strong updrafts of warm, moist air.

→ Wind Shear: As the warm air rises, it encounters cooler air at higher altitudes. This creates wind shear, a difference in wind speed and direction at different levels in the atmosphere.

→ Rotation: The wind shear causes the rising air column to begin rotating . This rotation can be further intensified by the Coriolis force .

→ Condensation Funnel: As the air column rotates faster and becomes more focused, it cools and condenses, forming a visible funnel cloud. This funnel can be various colors, including white, gray, or even black depending on the amount of debris it picks up.

Tornado Characteristics:

→ Size: Tornadoes can vary greatly in size, ranging from a few feet wide to over a mile.

→ Windspeed: Wind speeds in tornadoes can reach incredible speeds, exceeding 300 kmph in the strongest ones.

→ Duration: Most tornadoes are short-lived , lasting only minutes to tens of minutes. However, some can persist for longer periods.

→ Damage: Tornadoes can cause significant damage due to their high wind speeds , lifting and throwing debris, and even causing structural damage to buildings.

→ Tornadoes can be incredibly destructive, with wind speeds reaching over 300 miles per hour. They are most common in the central United States , particularly in an area known as Tornado Alley, which stretches from Texas to Oklahoma, Kansas, and Nebraska. However, they can occur in many parts of the world, including Europe, Asia, Africa, and Australia.

→ The majority of tornadoes occur in Tornado Alley in the USA due to the unique convergence of different air masses.

Why the majority in "Tornado Alley"?

→ This convergence creates instability, which, combined with strong wind shear (changes in wind speed and direction with height), leads to the formation of powerful thunderstorms called supercells . These supercells are the primary drivers for the high frequency of tornadoes in Tornado Alley.

→ The geography and flat terrain of the region also help funnel and intensify these weather patterns, making conditions ideal for tornado formation.

Water Spouts

→ A swirling column of air and water droplets that forms over a body of water , typically resembling a tornado.

→ It occurs when warm, moist air rises rapidly and meets cooler air above , causing the air to spin and create a vortex.

→ Usually less intense than tornadoes and are most commonly seen over warm ocean or lake waters.

8 · AIR MASSES AND FRONTS PYQ 2020

Understand it first: An air mass is a large body of air that has taken on the temperature and moisture of the region it sat over. Where two unlike air masses meet, the boundary is a front, and the lighter warm air is forced up, giving cloud and rain.

Air mass: homogeneous temp + humidity, no vertical movement; surface to lower stratosphere; thousands of km; source = high-pressure, calm regions; jets carry them

Types

(Tropical warm, Polar cold, continental dry, maritime moist)

→ mT: Amazon, SE Asia rain; heavy rain and thunderstorms on coasts, cyclones

→ cT: Sahara, central Australia; heatwaves, arid

→ mP: W Europe (UK, Ireland), Japan Sea coast (from Okhotsk) snow; cool, damp

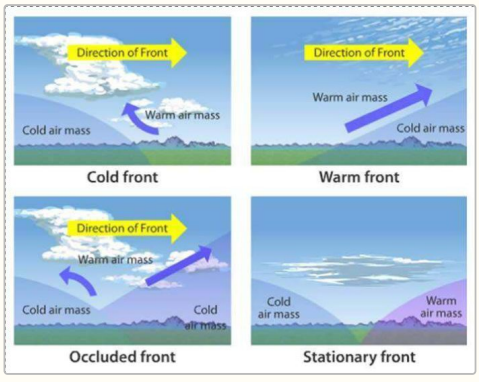
→ cP: Siberia, Canadian Prairies; cold dry winters

→ cA: N Canada, Greenland, Scandinavia; frigid, polar vortex

→ mE (equatorial): Indonesia, Malaysia, W Africa (Gulf of Guinea); heavy rain

Fronts:

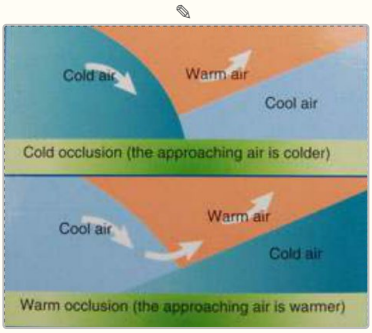
3-D boundary of two contrasting air masses; mid-latitudes, rare in tropics/poles



Fronts · Stationary Front

→ Frontogenesis = formation (face-off); Frontolysis = dissipation

→ Temperate cyclones owe genesis to frontogenesis; bigger contrast = thinner front; cold fronts move faster



Fronts · Warm Occlusion

Occluded: cold front overtakes warm

→ Warm occlusion: air ahead of warm front cooler; clouds cumulus, stratocumulus, nimbostratus; moderate-heavy rain, thunder

→ Cold occlusion: air behind cold front colder; stratocumulus, nimbostratus, altostratus; rain/snow, cooler

→ Stationary = stalemate (light rain for days)

Cold front = Crash

→ (cumulonimbus, thunder, hail)

→ Warm front = Wet blanket (stratus/cirrostratus, steady drizzle, fog)

→ Occluded = cold overtakes warm.

Air Masses

→ When the air remains over a homogenous area for a sufficiently longer time, it acquires the characteristics of the area.

→ This air parcel with distinctive homogeneous characteristics in terms of temperature and humidity is called an air mass.

→ Due to these homogenous conditions, they lack any vertical movement of winds within themselves.

→ The homogenous regions can be the vast ocean surface or vast plains and plateaus.

→ Air masses form an integral part of the global planetary wind system. Therefore, they are associated with one or other wind belts.

→ They extend from the surface to the lower stratosphere and are across thousands of kilometers. This is why the Jet streams can carry them from their source location to other places.

Source Region:

→ They are often the places where the wind movement is absent so that the air can stay at the same place for a longer period of time, resulting in the uniform transfer of the land characteristics to the air.

→ The main source regions are the high-pressure belts in the subtropics (giving rise to tropical air masses) and around the poles (the source for polar air masses):

→ – Warm tropical and subtropical oceans

→ – The subtropical hot deserts

→ – Permanently ice-covered continents in the Arctic and Antarctica

→ When an air mass moves away from a source region, the upper level maintains the physical characteristics for a longer period.

→ This is possible because air masses are stable with stagnant air which does not facilitate convection.

→ Conduction and radiation in such stagnant air are not effective.

Major Air Masses:

→ Continental Arctic (cA)

→ Southeast Asia: The maritime tropical air masses from the South China Sea cause significant rainfall during the wet season, contributing to the region's lush tropical climate.

→ Amazon Basin: The mT air mass from the Atlantic Ocean brings abundant rainfall, supporting the Amazon rainforest ecosystem.

→ Impact: Causes heatwaves and dry conditions in continental interiors, influences summer climates in regions like the southwestern USA.

→ Sahara Desert: The hot, dry cT air mass dominates, leading to extremely high temperatures and arid conditions.

→ Central Australia: The interior experiences hot, dry conditions due to the influence of cT air masses, resulting in arid and semi-arid climates.

→ Brings cool, damp, and cloudy weather with precipitation to coastal and inland regions, particularly in the winter.

→ Western Europe: The mP air mass from the North Atlantic Ocean brings cool, moist conditions, resulting in frequent rain and cloudy weather, especially in the UK and Ireland.

→ Causes cold, dry conditions in winter, can lead to clear, cold weather and influence winter storm tracks in mid-latitudes.

→ Russia (Siberia): cP air masses from the interior lead to extremely cold, dry winters with clear skies and very low temperatures.

→ Brings frigid temperatures and clear skies, influences polar vortex events and can push cold air far into mid-latitudes, causing severe winter weather.

→ Northern Canada and Greenland: cA air masses bring extremely cold, dry conditions, influencing the Arctic climate and contributing to polar vortex events that can push cold air southward.

→ Scandinavia: Periodic invasions of cA air masses can cause severe cold spells and harsh winter weather.

→ Contributes to heavy rainfall and thunderstorms in equatorial regions, influences monsoon climates and enhances tropical cyclone development.

→ Indonesia and Malaysia: The mE air mass brings high humidity and significant rainfall, supporting the region's tropical rainforest climate.

→ West Africa (Gulf of Guinea): The mE air mass contributes to heavy rainfall and the humid tropical climate of the coastal regions.

→ – Overall Impacts on Global Climate:

→ · Temperature Regulation: Air masses redistribute heat globally, balancing temperature differences between the equator and poles.

→ · Precipitation Patterns: Maritime air masses (mT, mP) bring moisture and precipitation, impacting rainfall patterns and water availability.

→ · Weather Systems and Storms: Interactions between different air masses lead to the formation of weather fronts, cyclones, and storms, influencing weather patterns and extremes.

→ · Seasonal Variations: The movement of air masses with seasons contributes to seasonal climate variations, such as monsoons and winter storms.

Fronts

→ These are the 3-dimensional boundary zones, between 2 air masses of contrasting characteristics.

Characteristics of Fronts:

→ The more the temperature contrast between the 2 air masses, the lesser the thickness of the front formed.

→ With a sudden change in temperature through the front, there is a change in pressure also.

→ Sudden wind shifts too.

→ The zone experiences precipitation due to the upliftment (gradual or forced) of the warm air.

Types of Fronts:

→ Stationary Front:

→ – Forms when two air masses meet but neither advances significantly.

→ – Clouds and light precipitation can persist along a stationary front for an extended period.

→ Warm Front:

→ – These clouds often produce steady rain or drizzle and can lead to overcast skies and fog.

→ Cold Front:

→ – Forms when a cold air mass advances and displaces a warm air mass.

→ Occluded Front:

→ – Forms when a fast-moving cold front overtakes a slower-moving warm front.

→ – Can produce a variety of weather conditions , including thunderstorms, rain, and cooler temperatures as warm air is lifted off the surface.

→ Typically associated with moderate to heavy precipitation, including rain and sometimes thunderstorms.

→ The warm sector ahead of the occluded front may experience warmer temperatures compared to the cold sector behind it.

→ Cold air wedges underneath the warm air, lifting it off the surface.

→ Typically associated with widespread precipitation, including rain or snow, and cooler temperatures. A mix of cold & warm front weathers.

9 · CYCLONES PYQ 2015 PYQ 2020

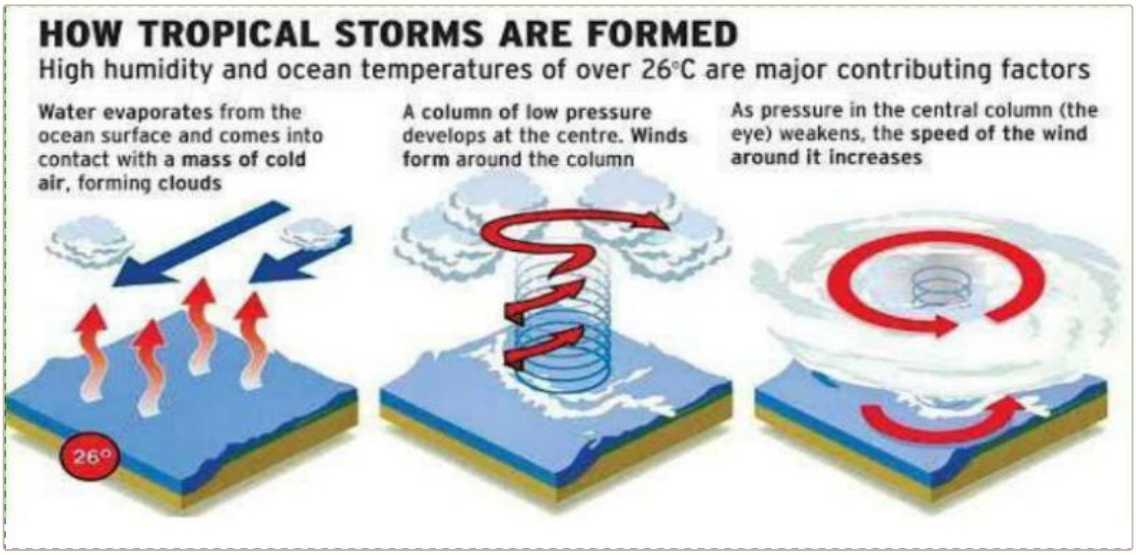
Understand it first: A cyclone is air spiralling into a low-pressure centre. Tropical cyclones are heat engines fed by warm sea water; temperate cyclones are born on fronts where warm and cold air meet. Learn the two side by side.

Cyclone:

low-pressure centre; winds counterclockwise in N, clockwise in S

→ Types: tropical, extratropical, subtropical

→ Names: Cyclone (Indian Ocean) · Hurricane (Atlantic) · Typhoon (W Pacific, S China Sea) · Willy-willy (W Australia)

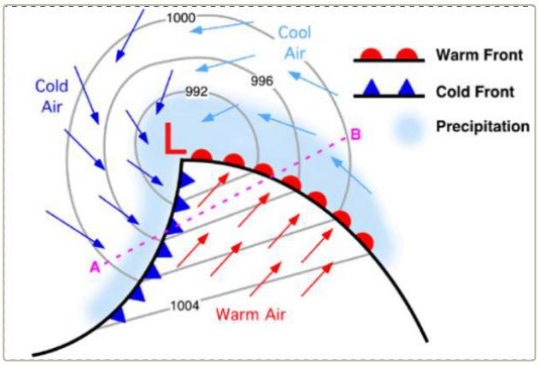


Tropical Cyclones · Formation

	Tropical Cyclone	Temperate Cyclone
Origin	Thermal Origin	Dynamic Origin
Latitude	10 - 30 degree N and S of equator	35 - 65 degrees N and S of equator. More pronounced in Northern hemisphere due to greater temperature contrast
Frontal system	Absent	Cyclone due to Front creation
Formation	Only on warm seas	Can form both on land as well as seas
Season	Seas	Irregular. But few in summers and more in winters
Size	100 – 500 kms in diameter. Varies with the strength of the cyclone	Typical size: 300 – 2000 kms in diameter. Varies from region to region.
Rainfall	Heavy but does not last beyond a few hours.	Rainfall is slow and continues for many days, sometimes even weeks
Wind Velocity and destruction	100 – 250 kmph, and 200 – 1200 kmph in upper troposphere. Greater destruction due to winds, storm surges and torrential rains	Comparatively low. Typical range: 30 – 150 kmph. Less destruction due to winds but more destruction due to flooding.
Lifetime	Doesn't last for more than a week	Last for 2-3 weeks.
Influence on India	Both coasts affected. But east coast is the hot spot	'Western Disturbances'.

TROPICAL v/s TEMPERATE CYCLONES

	Tropical	Temperate (extra-tropical)
🏛️ Origin	Thermal	Dynamic (frontal)
Latitude	10–30°	35–65° (more in N. Hemisphere)
Forms over	warm seas only	land and sea
👥 Size	100–500 km	300–2,000 km
Rain	heavy, a few hours	slow, many days/weeks
Wind	100–250 km/h; wind, surge, rain damage	30–150 km/h; flood damage
Life	not more than a week	2–3 weeks
Fronts	absent	created by fronts
India	both coasts, east coast hot-spot	Western Disturbances

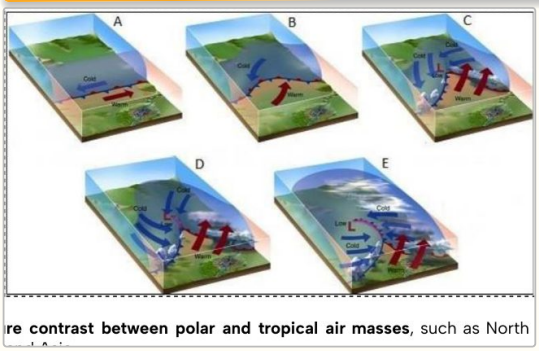


Temperate Cyclones · Warm and Cold Fronts

Tropical cyclone needs

→ Sea-surface ≥ 26.5 °C, moisture, low wind shear, pre-existing disturbance, Coriolis
→ Depression ≤ 62 km/h → cyclone ≥ 64 knots (74 mph)
→ Structure: eye (calm, clear), eyewall (worst), rainbands
→ Impacts: wind, rain, storm surge, tornadoes

Source conflict: cyclone sea temperature given as ≥ 26.5 °C in the notes and "> 27 °C" in the PYQ note.



re contrast between polar and tropical air masses, such as North

Temperate Cyclones · Location

Climate change

Tropical

→ Warmer seas = more intense, more Cat 4–5 (Arabian Sea, N Atlantic), fewer overall (N Atlantic slightly more active)
→ Poleward shift; slower, prolonged rain; rapid intensification; sea-level rise worsens surge

Temperate

→ Tracks move poleward (less cold air at low latitudes)
→ Gradient/jet changes alter intensity; more moisture = heavier rain, floods, landslides

PYQs: 2015 (B) no cyclones in S Atlantic / SE Pacific: ITCZ seldom occurs (also strong wind shear, cooler SST, no pre-existing vorticity). 2020 (C) only some cyclones develop an eye; eye is NOT 10 °C cooler (Erin's eye ~21 °F warmer; above 7.5 miles same as outside); jets in both hemispheres. ✅

Cyclones

→ Cyclones are large-scale atmospheric systems characterized by low-pressure centers and circulating winds that rotate counterclockwise in the Northern Hemisphere and clockwise in the Southern Hemisphere.

Tropical Cyclones

Necessary Conditions:

→ **Moisture:** High levels of atmospheric moisture are essential for sustaining convective activity and cloud formation.

→ **Low Vertical Wind Shear:** Minimal change in wind speed and direction with height allows for vertical development of the storm.

Formation:

→ **A disturbance, such as a low pressure centre** like monsoon trough, **serves as the initial seed** for tropical cyclone development.

→ **If the atmospheric and oceanic conditions are favourable, the disturbance may undergo cyclogenesis**, the process by which a tropical cyclone forms. Warm, moist air rises rapidly, creating an area of low pressure at the surface. As the air converges and rises, it cools and condenses, releasing latent heat and further fuelling the convective activity.

→ As the disturbance intensifies and becomes better organized, it may develop into a tropical depression, characterized by a closed circulation and maximum sustained winds of up to 38 miles per hour (62 kilometres per hour).

→ **If the conditions persist, the depression continues to strengthen into a cyclone.**

→ Once formed, tropical cyclones can track across the ocean and intensify, fuelled by warm ocean waters and favourable atmospheric conditions. They may reach their peak intensity at some point before eventually weakening as they move over cooler waters or make landfall.

Structure:

→ **Eye:** A well-defined center of low pressure characterized by calm winds and clear skies.

→ **Rainbands:** Bands of clouds and thunderstorms extending outward from the center, bringing additional rainfall and gusty winds.

Types of Clouds Associated:

→ **Cumulonimbus Clouds:** Towering clouds with strong updrafts and potential for severe weather, including heavy rainfall, lightning, and hail.

→ **Cirrus Clouds:** High-altitude clouds that may indicate the presence of upper-level outflow associated with the storm.

Impact:

→ **Strong Winds:** Tropical cyclones can produce sustained winds of 64 knots (74 mph) or greater, capable of causing widespread damage to structures, vegetation, and infrastructure.

→ **Heavy Rainfall:** Intense rainfall can lead to flooding, landslides, and mudslides, especially in areas with poor drainage or in mountainous terrain.

→ **Storm Surge:** The onshore flow of water driven by strong winds can result in storm surges, causing coastal flooding and inundation of low-lying areas.

→ **Tornadoes:** Embedded within rainbands, tornadoes can form and cause additional localized damage.

→ **Economic and Humanitarian Impacts:** Tropical cyclones can result in loss of life, displacement of populations, disruption of essential services, and significant economic losses.

Temperate Cyclones

→ They are most common **in regions with a strong temperature contrast between polar and tropical air masses**, such as North America, Europe, and Asia.

Necessary Conditions:

→ **Temperature Contrast:** A significant temperature difference between polar and tropical air masses is essential for the development of temperate cyclones.

→ **Baroclinic Instability:** The interaction between warm and cold air masses along a frontal boundary creates instability, leading to the formation of cyclones.

Formation:

→ **Temperate cyclones form along frontal boundaries,** where warm and cold air masses meet and create a zone of instability.

→ The process **begins with** the **development of a stationary front,** where warm air rises over denser, cooler air.

→ As the temperature gradient steepens and instability increases, a low-pressure system forms and begins to intensify.

→ Temperate cyclones are associated with both warm and cold fronts, which mark the leading edges of advancing air masses.

→ – **Cyclonic Rotation:** In the **Northern Hemisphere** , **temperate cyclones rotate counterclockwise** around their centers, while in the **Southern Hemisphere** , they rotate **clockwise.**

→ – **Occluded Front:** As the cyclone matures, the cold front catches up to the warm front, resulting in an occluded front and the eventual dissipation of the cyclone.

Types of Clouds Associated:

→ **Cumulonimbus Clouds:** Towering clouds associated with thunderstorms and heavy rainfall, especially along the cold front.

→ **Nimbostratus Clouds:** Layered clouds that produce steady rain or snow, often associated with the warm sector of the cyclone.

→ **Altostratus Clouds:** Mid-level clouds that may precede the warm front and bring overcast conditions and light precipitation.

Impact:

→ **Precipitation:** Temperate cyclones can produce a variety of precipitation types, including rain, snow, sleet, and freezing rain, depending on temperature and atmospheric conditions.

→ **Strong Winds:** Gusty winds associated with the cyclone can cause damage to structures, trees, and power lines.

Impact Of Climate Change

→ Rising ocean temperatures due to climate change provide more energy for tropical cyclones, leading to more intense storms with higher wind speeds and heavier rainfall. There has been an increase in the proportion of severe (Category 4 & 5) cyclones, especially in regions like the Arabian Sea and the North Atlantic.

→ Climate models show a decreased overall frequency of tropical cyclones globally, with some regions like the North Atlantic bucking the trend by becoming slightly more active.

→ The cyclones are encroaching even higher latitudes (poleward shift), and their forward motion has slowed down, which can prolong rainfall and increase flood risks.

→ Climate change leads to an increase in rapid intensification events (storms that quickly strengthen), making forecasts and disaster responses more difficult.

Temperate Cyclones

→ Temperate cyclones (also called extratropical cyclones or mid-latitude storms) may experience shifts in their tracks due to climate change, often moving poleward (lack of availability of cold air masses at lower latitudes), which could alter weather patterns in affected regions.

→ Warming impacts the temperature gradients and jet stream patterns that drive temperate cyclones, potentially modifying their intensity, frequency, and seasonal behavior.

→ Increased atmospheric moisture from climate change can lead to more intense precipitation events associated with these cyclones, enhancing the risks of flooding, landslides, and transport disruption.

There is evidence of excess rainfall and unseasonal weather in areas affected by temperate cyclones, which can impact crops and infrastructure.

🎯 10 · KÖPPEN AND CLIMATIC REGIONS PYQ 2011 PYQ 2024

Understand it first: Köppen sorted climates using only **temperature and rainfall**, with letter codes. Climatic-region questions give a description and ask you to name the type, so for each type hold its location, rainfall pattern and natural vegetation.

AllBigCitiesDeserveExtraHalwa — A tropical humid · B dry · C warm temperate · D cold snow-forest · E polar · H highland. Second letters: f no dry season · w winter dry · s dry summer · m monsoon.

Handout conflict: Köppen group table labels Df = humid continental, Dw = subarctic; regional table labels Taiga = Df, Manchurian = Dw. Remember names and places, not D-codes. Rainfall numbers in the regional table (e.g. 150–1000 cm for Af) look scrambled: do not memorise.

Climate (code)	Where	Hooks
Equatorial (Af)	10° N–S: Amazon, Congo, Indonesia, PNG	24–27 °C, tiny range; 4 pm convectional rain; evergreen selvas, mahogany, rosewood, ebony; leached soil; slash-and-burn
Savannah (Aw)	10–30°, interiors: Sudan, Venezuela, Brazil, N Australia	hot rainy summer, dry winter; elephant grass, baobab; Masai; "Big Game country"
Tropical monsoon (Am)	10–30°, E margins: India, Vietnam	3 seasons; 200–250 cm; teak, sal; paddy
Hot desert (BWh)	W margins, subtropical high, cold currents: Sahara, Kalahari, Mojave, Atacama	xerophytes (waxy/thorny leaves, deep roots, succulents), date palms at oases; Bedouin, Tuareg, Bushmen
Cold desert (BWk)	interiors: Gobi, Patagonia	wide range, cold winters; bunchgrasses
📝 Steppe (BS)	45–60°, mid-continental: Prairies, Steppes, Downs	Chernozem; wheat/maize, ranching
Mediterranean (Cs)	30–45°, W margins: California, Chile, SW S Africa, SW Australia	hot dry summer, winter rain; citrus, olives, vineyards; cork oak, eucalyptus
👫 Marine West Coast (Cfb, British type)	45–60°, W margins: NW Europe, SW S America, NZ	mild, rain all year, small range; deciduous forest; dairy
China type (Cw/Cfa; Gulf, Natal)	30–45°, E margins: SE China, Japan	warm moist summer, cool dry winter; typhoons; most productive mid-latitude farming
Taiga/Siberian (Df)	50/60–70°: N Canada, Eurasia	−30 to −40 °C winters; conifers (softwood, pure stands); fur, lumbering
Manchurian/ Laurentian (Dw)	E margins, N Hemisphere only: Manchuria, Japan, NE America	cold dry winters, wet warm summers
Polar (ET, EF)	Greenland, N Canada, Alaska, Antarctica	summer < 10 °C; mosses, lichens; Eskimos, Lapps, Yakuts

Climate PYQ	Answer / rule
👫 2011 rainforest regrows slowly	Soil poor in nutrients: leached, nutrients stored in plants, rapid decay (thin humus)
2011 African–Eurasian desert belt	Subtropical high-pressure cells (not warm currents); rain shadow (Gobi), cold currents (Namib, Atacama) also cause deserts; hot deserts 20°–30°, W coasts
2012 savannah	Definite dry and wet season; 800–1,600 mm rain; less than monsoon

2015 clear morning, thunder, short rain	Equatorial: 26–28 °C, annual range as low as 3 °C, ≥ 2,000 mm, ITCZ two rainfall peaks
2018 desert leaf traits	Hard waxy, tiny, thorns: all three (succulents; cacti spines)
👫 2021 keeps savannah forest in check	Fire, grazing herbivores, seasonal rainfall (2, 3, 4); acacia; maize, millets, sorghum, cassava; groundnuts, cotton, E Africa coffee
👫 2021 fast litter decay, bare soil, epiphytes	Tropical rain forest (canopy 75 m+, 3 layers)
👫 2023 rainforest soil rich?	Statement I ✘ (leached); II ✔
🆕 2024 low annual/daily range, rain all year, 50–250 cm	Marine West Coast
👫 No year dust in subtropical/temperate	I ✔ (deserts: Sahara, Arabia, Central Asia, China; scarce vegetation, high wind); II ✘ (strong dry winds); equatorial humid/ forested, polar ice-covered

Table 12.2 : Climatic Types According to Koeppen

Group	*Type*	*Letter Code*	*Characteristics*
A-Tropical Humid Climate	Tropical wet	Af	No dry season
	Tropical monsoon	Am	Monsoonal, short dry season
	Tropical wet and dry	Aw	Winter dry season
B-Dry Climate	Subtropical steppe	BSh	Low-latitude semi arid or dry
	Subtropical desert	BWh	Low-latitude arid or dry
	Mid-latitude steppe	BSk	Mid-latitude semi arid or dry
	Mid-latitude desert	BWk	Mid-latitude arid or dry
C-Warm temperate (Mid-latitude) Climates	Humid subtropical	Cfa	No dry season, warm summer
	Mediterranean	Cs	Dry hot summer
	Marine west coast	Cfb	No dry season, warm and cool summer
D-Cold Snow-forest Climates	Humid continental	Df	No dry season, severe winter
	Subarctic	Dw	Winter dry and very severe
E-Cold Climates	Tundra	ET	No true summer
	Polar ice cap	EF	Perennial ice
H-Highland	Highland	H	Highland with snow cover

Koppen's Classification

Climate	Koppen Name	Temp.	Rainfall	Location	Biome	Human Adaptation
Hot, Wet Equatorial	Af	24-27 degree. Low Annual And Diurnal Range	150-1000 cm. Avg. annual.Daily Rainfall (4 pm rains).Convectional. Norainlessmonth. Equinox Maxima; Solstice minima. High Relative Humidity (~80%)	10 degree north & South of Equator. E.g.Amazon Basin,Congo Basin,Indonesia, PNG.	Evergreen Forests(Selvas);layered arrangement.Lack of purestands. Hardwood Trees. Mahogany, Rosewood, Ebony; Undergrowth almost absent. Leached Soil. When forests cleared, secondary, less diverse belukar forests crop up. Arboreal animals, insects.	Hunters, Gatherer. Lumbering & Livestock Farming is difficult. Some communities practice slash & burn agriculture. Plantations(due to colonial powers). Diseases Due To Pathogens.
Savannah (Sudan Type)	Aw	High diurnal range (in summers, up to 38 in	Hot rainy summer, dry cold winters. 100-150	10-30 degrees continent interiorsE.g.	Tall grass (6-12 ft; Elephant Grass can be 15ft), little,	Cattle pastoralism (Masai); Subsistence

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		day, 10 in night). About 10 degrees annual range Highest temperatures in April (NH) or October (SH).	cm average annual	South Sudan, Venezuela, Brazil, North Australia, etc.	short trees. Why? Trees either are deciduous or with wide trunks to store water (Baobab). World's largest terrestrial species (Big Game country)	mixed agriculture (Hausa); At professional level in Australia, Brazil. Limited farming due to less rain. (Indigenous cattle are less productive, so temperate varieties are being introduced).
Tropical Monsoon	Am	3 distinct seasons. Summer, Winter, Rainy. High annual range of temperature	Lot of rain in short period of time. 200-250 cm	10-30 degree; East Continental Margins E.g. India, Vietnam, parts of Australia, Central America.	Tropical Monsoon forests, mostly deciduous (Teak, Sal). In wetter regions, tropical evergreen. In drier regions, thorny scrubland. Wide faunal variety.	Intensive subsistence agri. with (wetter) or without (drier) paddy. Now intensive commercial & proliferation of cash crops. Plantation in highlands (coffee, tea). Shifting Agri. Lumbering in accessible forests.
Hot Desert	BWh	Very high temperature. High diurnal and annual range of temperature. No cold season, but frost may occur on winter nights.	Small amount of convective rain. Not more than 25-50 cm	West margins of Tropical landmass. Sub-tropical HP belt, & Impact of Cold Currents. E.g. Sahara, Kalahari, Mojave, Atacama deserts.	Xerophytic vegetation. Waxy/thorny leaves; Deep roots; succulent stems; seeds have tough skins. Nocturnal adaptation of animals; Date palms near oasis. Animals adapted (hump, nocturnal, etc.); reptiles,	Hunters& Gatherers (Kalahari Bushmen; Australian Bindibus), Nomadic Herdsmen (Bedouin Arabs, Tuaregs of Sahara). Caravan travellers; Mining settlements; (E.g. Gold in Kalgoorlie,

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					camels, insects.	Precious Gems in Mojave desert); if water available, agriculture (Nile valley).
Cold Deserts	BWk	Wide range of annual temperature (more than hot deserts). Annual avg <18 degrees C. Severe winters. Flooding during early summers due to ice thawing.	Lesser than hot deserts. 0- 30 cm, due to occasional depressions. (Gurbantunggut desert: 0 rain).	Sub-Tropical Continental interiors E.g. Gobi desert, Siberia, Patagonia	Bunchgrasses. A few thorny trees like Acacia; Cactus are rare. Small burrowing animals Bactrian Camels.	Hunters, Gatherers, Nomadic Herdsmen (Mongols); Mining settlements (REM in Gobi desert).
Steppe Climate	BS	Warm summers (18-20 degree), Cold winters (can be sub-zero). High annual range. However, not that extreme in the Southern Hemisphere (narrow land; impact of oceans)	25-75 cm; decreases with continentality. Summer maxima. Effect of Temperate cyclones too. More rain in Southern Hemisphere due to maritime influence	45-60 degree mid continental regions. Westerly influence E.g. Prairies (impact of Chinook), Steppes, Downs, etc.	Nearly treeless, grasses dominate (pure grasslands). Rain perennials. Chernozem soil Conifers in transitional areas. Wide variety of animals too (Wolves, eagles, owls, snakes)	Wheat/Maize; Ranching; Nomadic Herding in Central Asia. (Kyrghz, Kazakhs Mongols)
Mediterranean	Cs	Long hot summers. (>22 degree C). Cool Winters (0 degree). Impact of many local winds (Sirocco, Mistral, Bora)	Winter rainfall (onshore Westerlies). Summer drought(off shore trade) Shift of pressure	30-45 degree; West margin; Medi. shoreline; Central California; central Chile, SW South Af, SW Aust.	Small broad leaves; Xerophytic; Gardens & vineyards of the World. Can have Mediterranean evergreen forests in wetter margins (cork oak,	Orchards (citrus, olives) Plantations; Wine yards Crop Cultivation (cereals) in some locations. Cattle rearing

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			belts. 50-75 cm		eucalyptus, giant sequoia) & Medi. Conifers in highlands. Few mammals (Iberian Lynx, reptiles)	unimportant due to lack of good quality grass (rain in winters when growth is slow).
Wet West Marine (British Type)	Cf	5-15 degree C; Warmer in Europe (NAD & Norwegian current); cold in rest regions. Low annual range of temperature (cool throughout the year)	Onshore westerlies throughout year. 125cm. Wettest Temperate climate. More rain on western margins.	45-60 degree West margins. North West Europe, South West S.America, parts of New Zealand, Canada	Temperate Deciduous Forests;Pure stands; sparse undergrowth. Small mammals (Bears); Birds like Eagles, Penguins (NZ)	Lumbering; Secondary Industries due to Industrial Revolution (food processing; automobiles etc.); Mixed Farming (cattle/sheep + Crops like potatoes, wheat, barley); Potatoes & Beet; Market gardening; Fishing.
China Type/ Gulf type/ Natal type	Cw (also known as Cfa, but Cw also accepted due to dry winters)	Warm Moist summer; Cool dry winter. Heavy maritime influence. **China:** high annual range **Gulf:** narrow range **Natal**: small annual range	**China type:** SE monsoons in summers; minor snowfall in winters; Typhoons (summer maxima) **Gulf type:** No distinct dry period (summer maxima); Hurricanes & tornadoes. **Natal type:** even rain by	Temperate East margins (30-45 degrees). **China type:** Central & North China; South Japan **Gulf type:** Gulf of Mexico **Natal type:** Southern hemisphere (Uruguay-Paraguay; parts of Brazil & Argentina; East Australia; parts of South Africa.	Luxurious Vegetation due to high rain. Evergreen Broad leaves (in eastern margins) and deciduous forests. Few arboreal and other Mammals and birds.	Most productive mid latitudinal region. Farming main occupation.

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			onshore winds throughout the year.			
Taiga/ Siberian Climate	Df	Exceptionally cold winters (- 30 to -40 deg. C); cool brief summers. V. High temp range	30cm. Mostly convectional and frontal. Summer Maxima	50/60-70 degree latitude. Northern Canada and Eurasia. Continuous strip	Evergreen conifers (softwood & pure stands).Little undergrowth (acidic leached humus-less soils). Greatest single band of conifers in Siberia. Migratory wildlife.	Animal Herding. (Fur Farms: minx, otters, silver fox etc); Lumbering
Manchurian/ Laurentian Climate	Dw	Cold dry winters; wet warm summers	Up to 100 cm (More in N.Am than Asia due to Great lakes); Onshore maritime in summer; During winter off-shore winds.	Temperate East Margin. Only in Northern Hemisphere Manchuria, Japan, Koreas; NE North America	Cool temperate forests. Conifers north of 50 degree; Deciduous to south of 50 degrees Moose, beavers, raccoons, bear, hawk etc.	Lumbering; Fishing (Cold Warm currents meet) Agriculture (Hardy Crops: Potatoes, Oat, barley)
Polar	Sub-Polar (ET); Frosted (EF)	ET: Summer temp. Less than 10 degrees Frost can occur thorough ut the year.	Mostly Snowfall; In summers Rain & sleet; Winter maxima due to Extra-Tropical Cyclones;	Greenland, Northern Canada & Alaska; Antarctica	In ET, mosses & Lichens; in sheltered spots, stunted birches, willows, etc. In brief summer, berry bushes and arctic flowers. Migratory species	Semi-nomadic life; (Eskimos, Lapps, Samoyeds, Yakuts, Chukchis); Mostly hunting; Now mining too (gold, petrol, shipping)

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Name of Shifting Cultivation/	Location/Country
Humah/ Ladang	Java and Indonesia
Tamrai	Thailand
Chena	Sri Lanka
Taungya	Myanmar
Caingin	Philippines
Chetemini	Uganda, Zambia and Zimbabwe
Roka	Brazil
Konuko	Venezuela
Milya	Mexico and Central America
Echalin	Guadeloupe
Milpa	Yucatan and Guatemala
Comile	Mexico
Logan	Western Africa
Fang	Equatorial African Countries
Masole	Congo (Zaire river Valley)
Tavi	Madagascar
Ray	Vietnam

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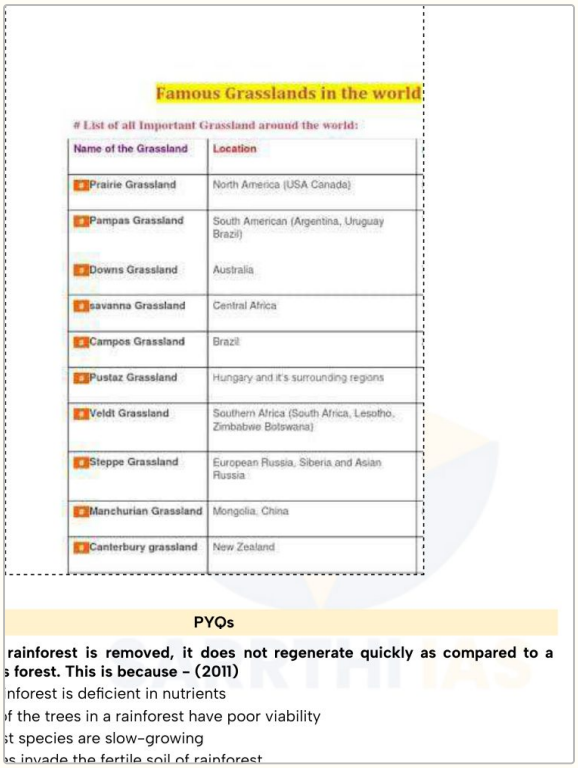


Famous Grasslands in the world

List of all Important Grassland around the world:

Name of the Grassland	Location
Prairie Grassland	North America (USA Canada)
Pampas Grassland	South American (Argentina, Uruguay Brazil)
Downs Grassland	Australia
savanna Grassland	Central Africa
Campos Grassland	Brazil
Pustaz Grassland	Hungary and it's surrounding regions
Veldt Grassland	Southern Africa (South Africa, Lesotho, Zimbabwe Botswana)
Steppe Grassland	European Russia, Siberia and Asian Russia
Manchurian Grassland	Mongolia, China
Canterbury grassland	New Zealand

PYQs

rainforest is removed, it does not regenerate quickly as compared to a
s forest. This is because – (2011)
nforest is deficient in nutrients
f the trees in a rainforest have poor viability
t species are slow-growing
s invade the fertile soil of rainforest

Climatic Regions Of The World

11 · TRAPS AND CONNECTIONS

Traps

- → jet streams only in NH ✘
- → land–sea contrast greater in summer ✘ (winter)
- → westerlies 30°N–60°S ✘
- → ozone reflects radio ✘
- → more moisture aloft = colder ✘
- → Gulf Stream is cold ✘
- → eye 10 °C cooler ✘
- → rainforest soil rich ✘.

Connects to

- → pressure belts and 23.5° tilt / 21 June → Geomorphology
- → Western disturbances, monsoon, Loo → Indian climate
- → ocean currents shape isotherms and deserts → Oceanography
- → climate regions → Economic Geography.

RECALL DRILL: COVER AND ANSWER

Troposphere height; lapse rate?	8–18 km (avg 13); 1 °C per 165 m (≈6.5 °C/km)	Ozone layer height; coldest layer; jet-flight layer?	20–25 km; Mesosphere; lower stratosphere
Kármán line; energy surplus/deficit boundary?	100 km; 40° N/S	Coriolis: zero, max, direction?	Equator; poles; right in N, left in S
Roaring / Furious / Shrieking?	40s / 50s / 60s (S. Hemisphere westerlies: less land)	Polar vs subtropical jet?	50°–60° (strong in winter) / 20°–30°; both hemispheres
Cold front vs warm front clouds?	Cold: cumulonimbus; Warm: stratus, cirrostratus	Cyclone sea temperature; why none in S Atlantic?	≥ 26.5 °C; ITCZ seldom occurs
Chinook: where, nickname?	E. Rockies; snow-eater	PSC altitude and temperature?	15–25 km; below −78 °C
Cloud seeding agents?	Silver iodide, potassium iodide, salt, dry ice	Main cause of desert belts?	Subtropical high-pressure cells (plus rain shadow, cold currents)

🎯 PYQs OF THIS CHAPTER

🎯 Exam lens (from the handout only)

- Priority: VERY HIGH for Prelims. 32 Prelims PYQs are printed in this part, the most of any chapter: 2011×5 · 2012×3 · 2013 · 2014 · 2015×3 · 2018 · 2019×2 · 2020 · 2021×2 · 2022 · 2023×3 · 2024×6 · 2025 · plus 2 with no year printed. Mains quoted: "Troposphere determines weather processes – how?" (2022, 15m).
- Hot clusters: heat budget & lapse rate (2012, 2023, 2024) · isotherm bending (2024, 2025) · westerlies/ Coriolis/jet streams (2011, 2015, 2020, 2024) · clouds & dew (2019, 2022) · cyclone genesis (2015) · climate-type identification (2011–2024: rainforest, savannah, equatorial, Marine West Coast).
- Prepare: Prelims – cause-and-effect statement pairs (the Statement I/II format keeps coming back: 2023, 2024, 2025). Mains – troposphere, jet streams & climate change, tropical vs temperate cyclones.
- No teacher ink in this PDF; priority = PYQ count in the handout.

✍️ ***The PYQs as printed in the handout, with answers***

Q.1) A layer in the Earth's atmosphere called Ionosphere facilitates radio communication. Why?

1. The presence of ozone causes the reflection of radio waves to Earth.
2. Radio waves have a very long wavelength.

Which of the statements given above is/are correct? (2011)

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: D The ionosphere is defined as the layer of the Earth's atmosphere that is ionized by solar and cosmic radiation.

- It lies 75–1000 km (46–621 miles) above the Earth.
- (The Earth's radius is 6370 km, so the thickness of the ionosphere is quite tiny compared with the size of Earth.)
- Because of the high energy from the Sun and from cosmic rays, the atoms in this area have been stripped of one or more of their electrons, or "ionized," and are therefore positively charged. The ionized electrons behave as free particles.
- The Sun's upper atmosphere, the corona, is very hot and produces a constant stream of plasma and UV and X-rays that flow out from the Sun and affect or ionize the Earth's ionosphere.
- Only half the Earth's ionosphere is being ionized by the Sun at any time.
- The ionosphere has major importance to us because, among other functions, it influences radio propagation to distant places on the Earth, and between satellites and Earth.
- For the very low frequency (VLF) waves that the space weather monitors track, the ionosphere and the ground produce a "waveguide" through which radio signals can bounce and make their way around the curved Earth
- The ionosphere is composed of three main parts, named for obscure historical reasons: the D, E, and F regions.
- The electron density is highest in the upper, or F region.
- The F region exists during both daytime and nighttime.
- During the day it is ionized by solar radiation, during the night by cosmic rays.
- The D region disappears during the night compared to the daytime, and the E region becomes weakened.
- Wavelength of radio waves can be 1 milimeter to 100 km, so they are not "very long wavelength waves.

Q.2) Normally, the temperature decreases with the increase in height from the Earth's surface, because

1. the atmosphere can be heated upwards only from the Earth's surface
2. there is more moisture in the upper atmosphere
3. the air is less dense in the upper atmosphere

Select the correct answer using the codes given below : (2012)

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: C The first statement given in this question is correct.

- Temperature falls off with height at a predictable rate because the air near the surface is heated and becomes light, and the air higher up cools to space and becomes heavy.
- The second statement is obviously incorrect. This is because the water vapour, which accounts for around 1% of air, varies greatly in the troposphere and decreases rapidly with altitude.
- As we increase elevation, there is less air above us thus the pressure decreases. As the pressure decreases, air molecules spread out further (i.e. air expands), and the temperature decreases and the atmosphere can be heated upwards only from the Earth's surface.

- Near the Earth's surface, air gets cooler the higher you climb. As you climb a mountain, you can expect the air temperature to decrease by 6.5 degrees C for every 1000 meters you gain. This is called the standard (average) lapse rate.
- The basic answer is that the farther away you get from the earth, the thinner the atmosphere gets. The total heat content of a system is directly related to the amount of matter present, so it is cooler at higher elevations

Q.3) The increasing amount of carbon dioxide in the air is slowly raising the temperature of the atmosphere because it absorbs – (2012)

(a) the water vapour of the air and retains its heat

(b) the ultraviolet part of the solar radiation

(c) all the solar radiations

(d) Infrared part of solar radiation

Answer: D Carbon dioxide is meteorologically a very important gas as it is transparent to the incoming solar radiation but opaque to the outgoing terrestrial radiation.

- It absorbs a part of terrestrial radiation and reflects back some part of it towards the earth's surface. It is largely responsible for the greenhouse effect.
- Its concentration is greater close to the earth's surface as it is denser than air.

Q.4) The annual range of temperature in the interior of the continents is high as compared to coastal areas. What is/are the reason/reasons? (2013)

1. Thermal difference between land and water
2. Variation in altitude between continents and oceans
3. Presence of strong winds in the interior
4. Heavy rains in the interior as compared to coasts

Select the correct answer using the codes given below.

(a) 1 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1, 2, 3 and 4

Answer: A

- The major factor affecting the distribution of the temperature of Earth is distribution of Land and Oceans. Since there is more land in the Northern Hemisphere and more waters in the Southern hemisphere, there is a big difference between the specific heat of land and water; the loss of heat from the continents is bigger than the oceans. The continents get heated faster and get cooled faster in comparison to the Oceans. This is the reason that the temperatures of the Oceans are moderate while that of continents is extreme. The moderating effect on temperature of the land due to proximity of the seas is called Maritime influence. The increasing effect on temperature of the land at the interior of the continents is called Continental Influence.
- Due to this moderating effect of the sea , places near the coast have a low range of temperature and high humidity.The places in the interior of the continent do not experience the moderating effects of the sea. These places have extreme temperatures.
- Statement 4 claims that Heavy rains in the interior as compared to coasts (are responsible for why the annual range of temperature in the interior of the continent is high)
- But the interiors in Central Asia receive very poor rainfall due to the distance from the sea. Annual precipitation is usually low in the interior areas. Hence 4 cannot be the answer

Q.5) Consider the following statements: Statement-I: The temperature contrast between continents and oceans is greater during summer than in winter. Statement-II: The specific heat of water is more than that of land surface.

Which one of the following is correct in respect of the above statements? (2023)

(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I

(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I

(c) Statement-I is correct but Statement-II is incorrect

(d) Statement-I is incorrect but Statement-II is correct

Answer: D

- Water has a very high specific heat, which means that in order to increase or reduce its temperature by 1 degree celsius, it will have to absorb or release huge amount of heat. Compare t that, the land has low specific heat, and can heat up or cool down fast.

Q.6) With reference to the Earth's atmosphere, which one of the following statements is correct? (2023)

(a) The total amount of insolation received at the equator is roughly about 10 times of that received at the poles.

(b) Infrared rays constitute roughly two-thirds of insolation.

(c) Infrared waves are largely absorbed by water vapour that is concentrated in the lower atmosphere.

(d) Infrared waves are a part of the visible spectrum of electromagnetic waves of solar radiation.

Answer: C

- Water vapour is an important Greenhouse Gas, which absorbs the heat in the form of Infrared Radiations.
- The troposphere is a very significant atmospheric layer that determines weather processes. How ? (2022, 15m)

Q.7) Consider the following statements: (2024) Statement-I: The atmosphere is heated more by incoming solar radiation than by terrestrial radiation.

Statement-II: Carbon dioxide and other greenhouse gases in the atmosphere are good absorbers of long wave radiation.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

(b) Both Statement-I and Statement-II are correct but Statement-II does not explains Statement-I

(c) Statement-I is correct, but Statement-II is incorrect

(d) Statement-I is incorrect, but Statement-II is correct

Answer:

(d) Statement-I is incorrect, but Statement-II is correct

Q.8) Consider the following statements: (2024) Statement-I: Thickness of the troposphere at the equator is much greater as compared to poles. Statement-II: At the equator, heat is transported to great heights by strong convectional currents.

Which one of the following is correct in respect of the above statements?

(a) Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

(b) Both Statement-I and Statement-II are correct, but Statement-II does not explains Statement-I

(c) Statement-I is correct, but Statement-II is incorrect

(d) Statement-I is incorrect, but Statement-II is correct.

Answer:

- Both Statement-I and Statement-II are correct and Statement-II explains Statement-I

Q.9) Which of the following is/are correct inference/inferences from isothermal maps in the month of January? (2024)

1. The isotherms deviate to the north over the ocean and to the south over the continent.

2. The presence of cold ocean currents, Gulf Stream and North Atlantic Drift make the North Atlantic Ocean colder and the isotherms bend towards the north.

Select the answer using the code given below:

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Answer:

(a) 1 only

Q.10) Consider the following statements: Statement I: In January, in the Northern Hemisphere, the isotherms bend equatorward while crossing the landmasses, and poleward while crossing the oceans. Statement II: In January, the air over the oceans is warmer than that over the landmasses in the Northern Hemisphere.

Which one of the following is correct in respect of the above statements? (2025)

(a) Both Statement I and Statement II are correct and Statement II explains Statement I

(b) Both Statement I and Statement II are correct but Statement II does not explain Statement I

(c) Statement I is correct but Statement II is not correct

(d) Statement I is not correct but Statement II is correct

Answer:

(a) Both Statement I and Statement II are correct and Statement II explains Statement I Statement I: Correct

- In January (winter in the Northern Hemisphere), isotherms (lines of equal temperature) bend equatorward over landmasses because land cools faster than water.
- Over oceans, where temperatures remain relatively warm, isotherms bend poleward.
- This distortion of isotherms is a well-known climatological feature seen on global isotherm maps for January.

Statement II: Correct

- Due to higher specific heat and slower cooling, oceans retain more heat than land during winter. Hence, in January, air over oceans is warmer than over land.
- This temperature differential explains why maritime climates are milder than continental ones in winter.

Q.11) Consider the following statements:

1. Without the atmosphere, temperature would be well below freezing point everywhere on the Earth's surface.

2. Heat absorbed and trapped by the atmosphere maintains our planet's average temperature.

3. Atmosphere's gases, like carbon dioxide, are particularly good at absorbing and trapping radiation.

Which of the statements given above are correct?

(a) I and III only
(b) I and II only
(c) I, II and III
(d) II and III only

Answer: (d) II and III only **Statement I:** It's overgeneralized and incorrect to say temperatures would be "well below freezing everywhere." While nights would be freezing, daytime temperatures could become extremely hot. **Statement II:** Correct The atmosphere traps infrared radiation, regulating Earth's temperature. **Statement III:** Correct Greenhouse gases like CO_2, CH_4, and H_2O vapor absorb and re-radiate heat.

Q.1) Westerlies in the southern hemisphere are stronger and persistent than in the northern hemisphere. Why?

1. The Southern hemisphere has less landmass as compared to the Northern hemisphere.
2. Coriolis force is higher in Southern hemisphere as compared to the Northern hemisphere

Which of the statements given above is/are correct? (2011)

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: A

- The Westerlies are strongest in the Southern hemisphere and at times when the pressure is lower over the poles, while they are weakest in the Southern hemisphere and when pressures are higher over the poles.
- The Westerlies are particularly strong in areas where land is absent because land mass obstructs the movement of westerlies.
- Westerlies wind of the Southern hemisphere are known by different names like at 40 degrees South latitude they are called the Roaring forties, at 50 degrees latitude, they are called Furious fifties.
- The Westerlies play an important role in carrying the warm, equatorial waters and winds to the western coasts of continents, especially in the southern hemisphere because of its vast oceanic expanse.
- Wind never proceeds or moves in a straight line manner, during the movement of the wind deflection is caused due to Coriolis Force.
- Coriolis Force is highest at the Poles and zero at Equator.

Q.2) The jet aircraft fly very easily and smoothly in the lower stratosphere. What could be the appropriate explanation?

1. There are no clouds or water vapour in the lower stratosphere.
2. There are no vertical winds in the lower stratosphere.

Which of the statements given above is/are correct in this context? (2011)

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: C

- Commercial jet aircraft fly in the lower stratosphere to avoid the turbulence.
- The Stratosphere is very dry; air there contains very little water vapour. Because of this, few clouds are found in this layer.
- Vertical wind shear is a change in wind speed or direction with change in altitude.
- Wind shear has significant effects on control of an aircraft, and it has been a sole or contributing cause of many aircraft accidents.
- There is no vertical wind shear in the lower stratosphere.
- That's plane fly in that region to avoid turbulence or an accident

Q.3) With reference to "Coriolis force", which of the following statements is/are correct? (2024)

1. It increases with increase in wind velocity.
2. It is maximum at the poles and is absent at the equator.

Select the answer using the code given below:

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: (c) Both 1 and 2

Q.1) The seasonal reversal of winds is the typical characteristic of – (2014)

(a) Equatorial climate
(b) Mediterranean climate
(c) Monsoon climate
(d) All of the above climates

Answer: C

- Monsoon is traditionally defined as a seasonal reversing wind accompanied by corresponding changes in precipitation but is now used to describe seasonal changes in atmospheric circulation and precipitation associated with the asymmetric heating of land and sea.
- A monsoon is a seasonal reversal in wind patterns over a region. The word "monsoon" comes from the Arabic word mausim, meaning "season." The seasonal wind shift is usually accompanied by a dramatic change in precipitation.
- Winds that blow for approximately six months from the northeast and six months from the southwest. The most prominent monsoons occur in South Asia, Africa, Australia, and the Pacific coast of Central America.

Q.2) Consider the following statements

1. The winds that blow between 30 degrees N and 60 degrees S latitudes throughout the year are known as westerlies.
2. The moist air masses that cause winter rains in the North-Western region of India are part of westerlies.

Which of the statements given above is/are correct? (2015)

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: B

- The westerlies, anti-trades, or prevailing westerlies, are prevailing winds from the west toward the east in the middle latitudes between 30 and 60 degrees latitude.
- They originate from the high-pressure areas in the horse latitudes and trend towards the poles and steer extratropical cyclones in this general manner.
- Tropical cyclones which cross the subtropical ridge axis into the westerlies recurve due to the increased westerly flow.
- Westerly disturbances originate in the Mediterranean sea, this sea is surrounded on all sides by high mountains and plateau, as the result high pressure is developed there in the month of September- October, while low pressure still developed over the Mediterranean water body.
- This disturbance is dragged to India or brought to India by the Jet stream (westerly winds (flowing west to east)).
- This is the reason that J&K and NW part of India receives highest rainfall from westerly disturbances.

Q.3) In the context of which of the following do some scientists suggest the use of cirrus cloud thinning technique and the injection of sulphate aerosols into the stratosphere? (2019)

(a) Creating the artificial rains in some regions
(b) Reducing the frequency and intensity of tropical cyclones
(c) Reducing the adverse effects of solar wind on the Earth
(d) Reducing global warming

Answer: D

- Cirrus clouds do not reflect a lot of solar radiation back into space, but because they form at high altitudes and cold temperatures, they trap long-wave radiation and have a climate impact similar to greenhouse gases.
- Thus Thinning cirrus clouds would reduce the absorption of infrared radiation and proposed a form of climate engineering to reduce global warming.
- Another proposed Geo-engineering technique involves injecting reflective sulphate aerosol particles into Earth's lower stratosphere to cast a small proportion of the inbound sunlight back into space and cool the planet off.
- Thus both are climate engineering/ Geo-engineering techniques to reduce global warming.

Q.4) Why are dewdrops not formed on a cloudy night? (2019)

(a) Clouds absorb the radiation released from the Earth's surface.
(b) Clouds reflect back the Earth's radiation.
(c) The Earth's surface would have low temperature on cloudy nights.
(d) Clouds deflect the blowing wind to ground level.

Answer: B

- Less dew is formed on cloudy nights because radiation during cloudy nights occurs at a very slow pace.
- The radiation occurs very slowly because the clouds are thick during cloudy nights and trap the heat and water vapour inside the clouds.
- All this indicates why the level of dew is reduced during nights that are heavy with clouds.
- Dew is a natural form of water, formed as water vapour condenses. Dew, like the glistening drop on this grassy field in Anaconda, Montana, forms as water near the surface of the ground is cooled to its dew point, the temperature at which water vapour condenses. When the sky is cloudy, the trees and plants do not get cool in the night and hence there is less dew formation because on a cloudy night, the clouds send the heat (earth's radiation) back to the ground so the ground never gets cold enough for the dew to be formed.

Q.5) Consider the following statements:

1. Jet streams occur in the Northern Hemisphere only.
2. Only some cyclones develop an eye.
3. The temperature inside the eye of a cyclone is nearly 10°C lesser than that of the surroundings.

Which of the statements given above is/are correct? (2020)

(a) 1 only
(b) 2 and 3 only
(c) 2 only
(d) 1 and 3 only

Answer: C

- Jet streams occur in both the Northern and Southern Hemispheres. The actual appearance of jet streams result from the complex interaction between many variables – such as the location of high and low pressure systems, warm and cold air, and seasonal changes. Statement 1 is incorrect.
- The warmest part of Erin's eye was almost 21 degrees (Fahrenheit) warmer than the surrounding air, a dramatic difference from the air around it. Above 7.5 miles high, the eye's temperature dropped quickly to the same temperature as the air outside the eye. Statement 3 is incorrect.

Q.6) Consider the following statements:

1. High clouds primarily reflect solar radiation and cool the surface of the Earth.
2. Low clouds have a high absorption of infrared radiation emanating from the Earth's surface and thus cause warming effect.

Which of the statements given above is/are correct? (2022)

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer: D

- High clouds are often thin and do not reflect very much. They let lots of the Sun's warmth in. They radiate less energy into space than the lower, warmer clouds.

Therefore, high clouds work to "trap" more energy than the low clouds.

- Low clouds are often quite thick and reflect lots of sunlight back to space. Low clouds are excellent reflectors. But, they don't stop the longwave energy from escaping to space. Therefore, low clouds help to cool the Earth.

Q.7) With reference to "water vapour", which of the following statements is/are correct? (2024)

1. It is a gas, the amount of which decreases with altitude.
2. Its percentage is maximum at the poles.

Select the answer using the code given below:

(a) 1 only
(b) 2 only
(c) Both 1 and 2
(d) Neither 1 nor 2

Answer :

(a) 1 only

Q.1) In the South Atlantic and South Eastern Pacific regions in tropical latitudes, cyclones do not originate. What is the reason? (2015)

(a) Sea Surface temperature are low
(b) Inter-Tropical Convergence Zone seldom occurs
(c) Coriolis force is too weak
(d) The absence of land in those regions

Answer: B The most proximate reasons for the lack of cyclone formation in the South Atlantic are sea surface temperatures that tend to run a shade cooler than ideal for tropical cyclone formation even in the southern summer, climatologically high values of vertical wind shear across that basin throughout the year, and a lack of pre-existing centres of rotation (vorticity) in that area.

- Tropical cyclones are the zones of low pressure systems around which there is closed air circulation in counterclockwise direction(Northern hemisphere) and circular direction(Southern hemisphere).
- Important condition for the origin of tropical cyclones:
  - Sea Surface temperature should be greater than 27 degree C.
  - Pre existing weak tropical disturbances
  - Continuous supply of energy.
  - Coriolis force.
  - Weak wind shear
- Strong Wind shear present at South Atlantic removes the heat and moisture from the place where it is required to drive the heat engine of cyclone i.e near the centre.

- Without an Inter-tropical Convergence Zone (ITCZ) to provide synoptic vorticity and convergence (i.e large scale spin and thunderstorm activity) as well as having strong wind shear, it becomes very difficult to nearly impossible to have genesis of tropical cyclones.

Q.1) If a tropical rainforest is removed, it does not regenerate quickly as compared to a tropical deciduous forest. This is because – (2011)

(a) the soil of rainforest is deficient in nutrients

(b) propagules of the trees in a rainforest have poor viability

(c) the rain forest species are slow-growing

(d) exotic species invade the fertile soil of rainforest

Answer: A Due to heavy rainfall and weathering, most of the nutrients have been washed out of the soil.

- Only a thin layer of decaying organic matter is found, unlike in temperate deciduous forests.That's why the soil of most of the tropical rainforest is poor in nutrients.
- The high temperature and moisture of tropical rainforests cause dead organic matter in the soil to decompose more quickly than in other climates, thus releasing and losing its nutrients rapidly. The high volume of rain in tropical rainforests washes nutrients out of the soil more quickly than in other climates.
- Why are rainforest soils low in nutrients? One reason the rain forest soil is so poor is that most of the nutrients are stored in the plants themselves. In any forest, dead organic matter falls to the ground, providing valuable nutrients for new growth. In cooler or drier climates, the nutrients build up in the soil.

Q.2) What could be the main reason/reasons for the formation of the African and Eurasian desert belt?

1. It is located in the sub-tropical high-pressure cells.
2. It is under the influence of warm ocean currents.

Which of the statements given above is/are correct in this context? (2011)

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Answer: A

- Hot deserts are located between 20°30° latitude in both the hemisphere on the western coast of the continents.
- They are also called Tropical Deserts.
- They are different from the midlatitude deserts and cold deserts, hot deserts are found in High-pressure belts along the Tropics in the trade winds region.
- Distribution Of Hot Deserts:
  - Asia Thar desert and the Arabian Desert;
  - Africa The Sahara and the Kalahari;
  - N. America Lower California and Arizona states of the USA;
  - S. AmericaAtacama desert;
  - Australia The Great Australian desert.
  - The Sahara desert of Africa is the biggest hot desert in the world.
  - The Hot desert regions are located in the subtropical high-pressure belts, which are not favourable regions for rainfall.
  - There are three main causes of deserts:
    - Most of the world's deserts are made by the equatorial wind belts.
    - At the equator, warm air is always rising.
    - This creates wind belts that blow north and south away from the equator.
  - When air descends at the tropics of cancer and Capricorn it is too dry for clouds to form.
  - No clouds mean no rain producing Deserts. E.g. the Sahara desert
  - Rainshadow also produces Deserts.
  - Any moisture in the air will fall as air passes high mountain ranges, so land beyond the mountains receives little or no rain.
  - The Gobi desert is produced by Rainshadow.
  - Cold Currents also produce deserts.
  - The southwestern coast of Africa and South America are swept by cold currents upwelling from the ocean floor. These currents cool the air that passes over them, causing the water in the air to fall as rain before it reaches the land. The Namib and Atacama desert have formed because of these cold currents.
  - The Eurasian Belt is related to cold currents NOT warm current, so statement 2 is incorrect.

Q.3) Which one of the following is the characteristic climate of the Tropical Savannah Region? (2012)

(a) Rainfall throughout the year

(b) Rainfall in winter only

(c) An extremely short dry season

(d) A definite dry and wet season

Answer: D The tropical savanna climate is also called the tropical wet and dry climate. It is the climate experienced in savanna or tropical grassland regions of the world.

- These places are located near the equator, and they lie between the Southern and the Northern Tropics.
- The climate dominates many parts of the African continent, the northern region of South America, and parts of Asia such as India.
- The tropical savanna climate has alternating dry and wet seasons, hence its name.
- It shares some similar characteristics with the tropical monsoon climate, but it receives less annual rainfall as compared to the tropical monsoon climate.
- Tropical savanna climates are relatively hot since they lie within the tropical latitudes.
- Precipitation in the tropical savanna climate is mainly in the form of rainfall. On average, this climatic zone receives between 800 and 1,600 millimetres of rain annually.

Q.4) "Each day is more or less the same, the morning is dear and bright with a sea breeze; as the Sun climbs high in the sky, heat mounts up, dark clouds form, then rain comes with thunder and lightning. But rain is soon over".
Which of the following regions is described in the above passage? (2015)
(a) Savannah
(b) Equatorial
(c) Monsoon
(d) Mediterranean

Answer: B Regions with this climate experience high temperatures all year round.

- The average monthly temperatures are about 26 – 28 degrees Celsius.
- The annual temperature range (the difference between the average temperature of the hottest and coldest months) is very small.
- The annual temperature range may be as low as 3 degrees Celsius.
- The diurnal or daily temperature range (the difference between the highest temperature in the day and the lowest temperature at night) is usually greater. Humidity is usually very high.
- Another major characteristic of this climate is high rainfall.
- These regions usually experience 2000 mm of rainfall or more in a year.
- Rainfall is high for most of the year.
- Many equatorial regions are affected by the ITCZ.
- As the ITCZ passes over these areas it brings heavy rainfall and thunderstorms.
- In some areas, the ITCZ causes two periods of very heavy rainfall every year.
- One occurs when the ITCZ crosses these areas on its way north and another occurs when it crosses these areas again on its way south.
- The climograph below shows the rainfall and temperature pattern of an area which experiences an equatorial climate.

Q.5) Which of the following leaf modifications occur(s) in the desert areas to inhibit water loss?
1. Hard and waxy leaves
2. Tiny leaves
3. Thorns instead of leaves
Select the correct answer using the code given below: (2018)
(a) 2 and 3 only
(b) 2 only
(c) 3 only
(d) 1, 2 and 3

Answer: D Leaf modifications of Desert Plant to inhibit water loss for adaptations and survival of Desert Plant-

- Wax coatings on leaves prevent water loss through transpiration, which in the hot desert can cause loss of water from both the surface and the inside of the leaves. Leaves are also smaller on desert plants, further reducing the possibility for water loss.
- Deciduous plants in desert ecosystems have adapted through the activity of their leaves. Leaves on these plants are typically smaller and coated with wax to prevent evapo-transpiration.
- Plants are equipped with fleshy leaves that contain much of their water supply. Because of their moist inner bodies, these plants are called succulents. They typically feel spongy and when cut open are filled with a pulpy flesh, protected by a waxy outer layer.
- Many plants in the desert conserve water by not having any leaves at all. Cacti are the most prolific of this plant type. Many cacti have spines instead of leaves, which conduct photosynthesis and catch dew when the climate is right. These small structures also reflect light, further reducing water loss. During heavy rains, cacti will grow temporary root systems and absorb water. They will then shed the roots when the ground has dried.

Q.6) The vegetation of savannah consists of grassland with scattered small trees. The forest development in such areas is generally kept in check by one or more or a combination of some conditions.
Which of the following are such conditions?

1. Burrowing animals and termites
2. Fire
3. Grazing herbivores
4. Seasonal rainfall
5. Soil properties

Select the correct answer using the code given below (2021)

(a) 1 and2
(b) 4 and 5
(c) 2, 3 and 4
(d) 1, 3 and 5

Answer: C

- Savanna vegetation includes scrub , grasses and occasional trees, which grow near water holes , seasonal rivers or aquifers . Plants and animals have to adapt to the long dry periods. Many plants are xerophytic – for example, the acacia tree with its small, waxy leaves and thorns.
- The crops grown in the savanna are mainly annuals, the staple food crops being maize, millets and sorghum, but in many regions cassava is planted as a 'hedge' against crop failure. Cash crops include groundnuts, cotton and, at higher altitudes in East Mrica, coffee.
- So, the Vegetation is deliberately reduced in size by keeping in mind of the various factors which can play an important role in increasing the growth of the disasters.

Q.7) "Leaf litter decomposes faster than in any other biome and as a result, the soil surface is often almost bare. Apart from trees, the vegetation is largely composed of plant forms that reach up into the canopy vicariously, by climbing the trees or growing as epiphytes, rooted on the upper branches of trees." This is the most likely description of (2021)

(a) Coniferous forest
(b) Dry deciduous forest
(c) Mangrove forest
(d) Tropical rain forest

Answer: D

- Option d is correct. In tropical rainforest biome, hot, moist atmosphere and all the dead plant material create the perfect conditions in which bacteria and other microorganisms can thrive. This leads to rapid decomposition of leaf litter material. The tropical rainforest is known for its dense canopies of vegetation that form three different layers. The top layer or canopy contains giant trees that grow to heights of 75 m (about 250 ft) or more. Thick, woody vines are also found in the canopy. They climb trees in the canopy to reach for sunlight.

Q.8) Consider the following statements: Statement-I : The soil in tropical rain forests is rich in nutrients. Statement-II : The high temperature and moisture of tropical rain forests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the above statements? (2023)

(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
(c) Statement-I is correct but Statement-II is incorrect
(d) Statement-I is incorrect but Statement-II is correct

Answer: D

- The soil here is not rich in nutrients. It is in fact leached, as the fast growing vegetation constantly keep on absorbing the nutrients being created in the soil.

Q.9) Consider the following description: (2024)

1. Annual and daily range of temperatures is low.
2. Precipitation occurs throughout the year.
3. Precipitation varies between 50 cm – 250 cm.

What is this type of climate?

(a) Equatorial climate
(b) China type climate
(c) Humid subtropical climate
(d) Marine West coast climate

Answer: (d) Marine West coast climate

Q.10) Consider the following statements: Statement I: The amount of dust particles in the atmosphere is more in subtropical and temperate areas than in equatorial and polar regions. Statement II: Subtropical and temperate areas have less dry winds.

Which one of the following is correct in respect of the above statements?

(a) Both Statement I and Statement II are correct and Statement II explains Statement I
(b) Both Statement I and Statement II are correct but Statement II does not explain Statement I

(c) Statement I is correct but Statement II is not correct
(d) Statement I is not correct but Statement II is correct

Answer: (c) Statement I is correct but Statement II is not correct Statement I: Correct

- Dust concentration is higher in subtropical and temperate regions, particularly in arid and semi-arid zones like the Sahara Desert, Arabian Peninsula, Central Asia, and parts of China.
- These areas have scarce vegetation, high wind activity, and dry conditions, ideal for dust uplift.
- In contrast, equatorial regions are humid and forested (e.g., Amazon, Congo), limiting dust production. Polar regions are ice-covered, also limiting dust.

Statement II: Incorrect

- Subtropical regions, especially deserts (e.g., Sahara, Thar), experience strong dry winds like:
  - Trade winds
  - Harmattan (West Africa)
  - Santa Ana (California)
- These dry winds are key agents for transporting dust particles across continents and oceans.
- Therefore, the statement that these areas have "less dry winds" is factually incorrect.

🔤 ***ABBREVIATION COLUMN (full forms as the handout gives them where it does; the one-line meanings are plain-language aids, verify)***

AC = air conditioner
BS = steppe climate (Köppen)
BWh = hot desert climate (Köppen)
BWk = cold desert climate (Köppen)
CFCs = chlorofluorocarbons — ozone-destroying gases
CH = CH_4 = methane
DANA = Depresión Aislada en Niveles Altos — isolated high-level depression (cut-off low)
GHGs = greenhouse gases
ISS = International Space Station
ITCZ = Inter-Tropical Convergence Zone — low-pressure belt where the trade winds meet
NaCl = sodium chloride — common salt
NH = Northern Hemisphere (in highway lists: National Highway)
NW = north-west
NZ = New Zealand
PM2 = PM2.5 — fine particles 2.5 micrometres or smaller
PNG = Papua New Guinea
PSC = polar stratospheric cloud
PSCs = polar stratospheric clouds
PYQ = previous year question
SST = sea surface temperature
SW = south-west
UK = United Kingdom
US = United States
USA = United States of America
UV = ultraviolet — high-energy part of sunlight
VLF = very low frequency — radio waves

OCEANOGRAPHY

PYQ Prelims 2013 x2, 2015 x2, 2021 x2 · Mains 2017 (oceanic salinity, 15m)

✎ *Ocean floor, temperature-salinity-density, currents, tides, waves, corals, deposits, resources and UNCLOS: 6 Prelims PYQs, heavy on current-affairs themes.*

The chapter in 60 seconds

- The sea floor steps down from the **shelf** to the **slope**, the **rise**, the **abyssal plain** and the **trench**.
- Three properties of sea water, **temperature**, **salinity** and **density**, decide where water sinks and where it flows.
- Almost all water is salty. Of the fresh water, most is **ice**, then **groundwater**; rivers and lakes hold very little.
- **Currents** are one-way rivers in the sea (wind + Coriolis + continents); **tides** come from Moon, Sun and centrifugal force; **waves** carry energy, not water.
- **Corals**, **dead zones** and the melting **cryosphere** show how warming and pollution hurt the ocean; **UNCLOS** sets who may use which belt of sea.

Words you need first

Word	Meaning
Continental shelf	The gently sloping sea floor next to land (1–2°).
Abyssal plain	The wide, flat deep-ocean floor.
Salinity	Dissolved salt in sea water, about 35 ppt on average.
Thermohaline circulation	Slow deep circulation driven by temperature and salt differences.
Gyre	A big loop of ocean current.
Upwelling	Cold, nutrient-rich water rising to the surface.
Spring tide	Highest high tide, when Sun, Moon and Earth are in one line.
Neap tide	Weak tide when Sun and Moon pull at right angles.
Bleaching	A stressed coral expels its alga and turns white.
Hypoxia	Oxygen so low that most life cannot survive (a dead zone).
EEZ	Exclusive Economic Zone: coastal State's resource rights up to 200 nm.
AMOC	Atlantic Meridional Overturning Circulation, a deep-water conveyor.

One line to carry

Coast to deep, warm meets cold, 12–24–200.

What the exam asks

Prelims hooks

- why the equatorial counter-current flows east (convergence, not rotation)
- what causes tides: Sun, Moon and centrifugal force together
- freshwater order: ice, then groundwater, then rivers and lakes
- west warmer in the tropics, east warmer in the temperate zone
- where fishing grounds sit: warm and cold currents meeting
- UNCLOS belts: 12, 24 and 200 nm

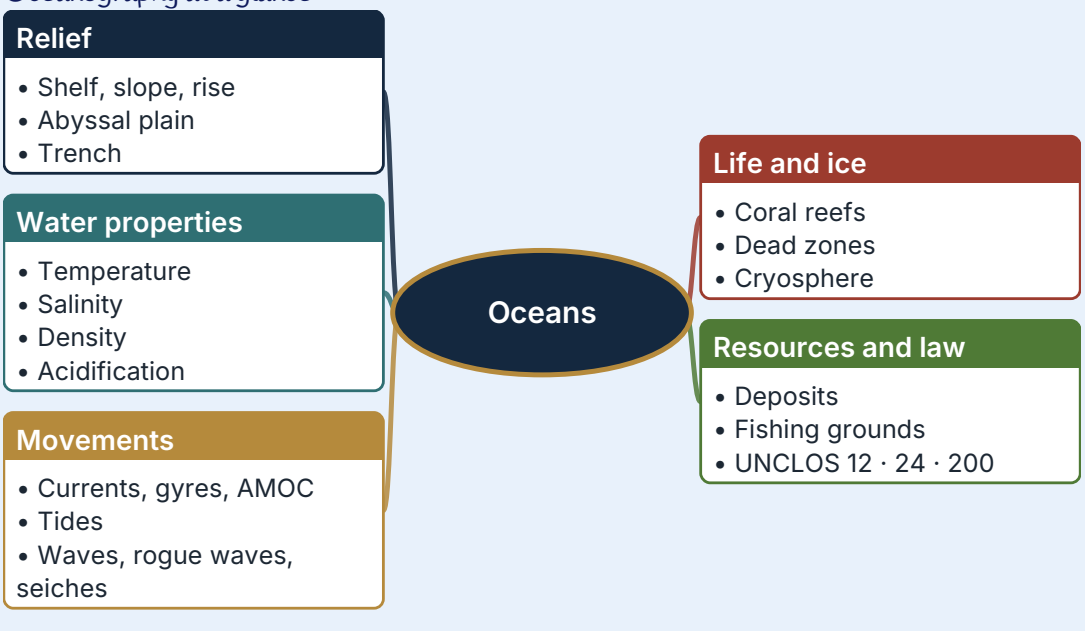
Mains hooks

- variations in oceanic salinity and their multi-dimensional effects (2017, 15 marks)
- AMOC weakening and what it does to climate
- coral bleaching, dead zones, marine heatwaves: causes, effects, remedies
- cryosphere melt and its effect on oceans and cities

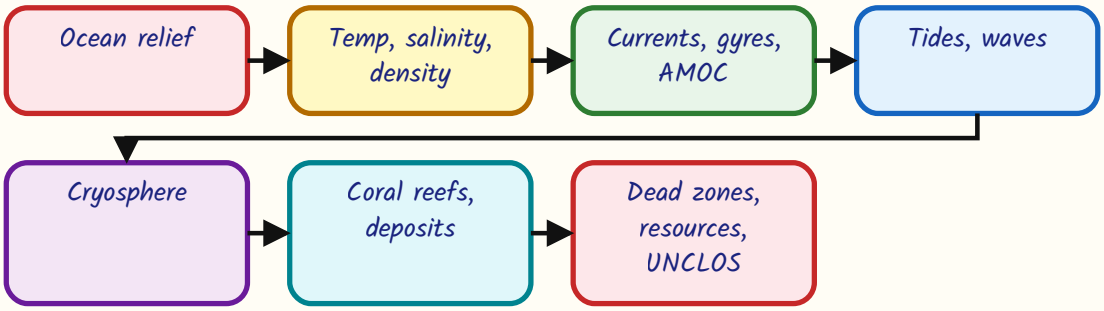
The exam asks why, not what: why the counter-current flows east, why tides occur, why fishing grounds sit where warm and cold currents meet. Learn the cause chain in each topic and the current-affairs themes (AMOC, corals, dead zones) for Mains.

Chapter map

Oceanography at a glance



At a glance: the Oceanography route



Pattern: the exam asks why: counter-current flows east, tides occur, fishing grounds sit where warm and cold currents meet. **Current-affairs heavy:** AMOC weakening, coral bleaching, dead zones, marine heatwaves, Southern Ocean carbon sink (good Mains material). Priority printed as MEDIUM-HIGH (6 Prelims, 1 Mains). No teacher ink. ⭐

1 · OCEANS AND RELIEF

Understand it first: Think of walking into the sea from the beach. First a gentle, shallow step (the **shelf**), then a steep staircase (the **slope**), a soft landing of sediment (the **rise**), a wide flat floor (the **abyssal plain**) and, where one plate dives under another, a deep cellar (the **trench**). The house picture is only a memory aid. Learn each part with its gradient, depth and one example.

Basics

→ Hydrosphere ~71% of Earth's surface

→ 5 oceans: Pacific, Atlantic, Indian, Arctic, Southern

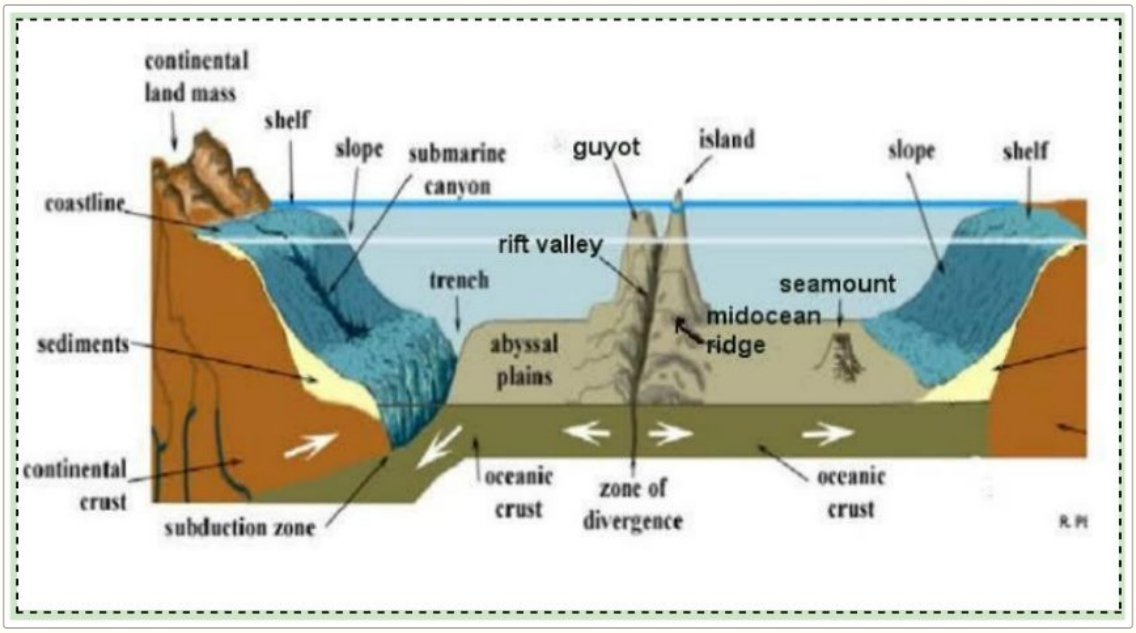
→ Average depth 3,800 m vs lithosphere average height 840 m

→ Relief from tectonic, volcanic, erosional, depositional processes

Oceans	:	97.3	Saline Water
Ice-caps	:	02.0	
Ground water	:	0.68	
Fresh water lakes	:	0.009	
Inland seas and salt lakes	:	0.009	Fresh Water
Atmosphere	:	0.0019	
Rivers	:	0.0001	
		100.00	

Water on Earth

Part (house)	Facts
Continental shelf (front porch)	gentle 1–2°; to ~200 m (up to 500 m); continental crust + sediment (sand, silt, clay); oil and gas, fisheries; North Sea, Gulf of Mexico; Great Barrier Reef (>2,300 km); width few km to hundreds of km
Continental slope (steep stairs)	3–6° (up to 25°); ~200–3,000 m; submarine canyons (Monterey, Hudson); Blake Plateau; deep-sea ecosystems (pressure, dark, cold)
Continental rise (landing)	gentle sediment apron at slope base; deep-sea fans, turbidity currents; Amazon rise
Abyssal/deep-sea plain (floor)	>2,000 m (2,000–6,000+); flat, fine sediment (clay, silt, sand); polymetallic nodules: Clarion–Clipperton Zone (Pacific, Hawaii–Mexico)
Trench (cellar)	>6,000 m; convergent boundaries, subduction, quakes; Mariana



Continental Shelf

Continental Shelf · Economic Significance

Source overlap: slope runs to ~3,000 m but abyssal plain starts >2,000 m, and the notes also print "deep ocean 2–4 °C": treat depths as approximate.

Other features

→ Seamounts: ≥ 1,000 m underwater volcanoes (Davidson)

→ Guyots (table mounts): flat-topped seamounts, wave-eroded then submerged (Emperor Seamounts)

→ Hydrothermal vents: mineral-rich water; black smokers; life may have begun here

Term	Handout example
Strait (narrow waterway joining two waters)	Palk (India–Sri Lanka)
Isthmus (land strip joining landmasses)	Kra (Malay Peninsula–Asia)
Archipelago	Japanese (W Pacific)
Lake	Baikal (Russia), Wular (India)
Bay	Bengal
Cape	Comorin = Kanyakumari (southernmost tip of Peninsula); Good Hope (Africa)
Lagoon (behind barrier island/reef)	Chilika (brackish, Odisha)
Island	Lakshadweep (Arabian Sea)
Gulf (constricted entrance vs bay)	Kutch (Gujarat)
Peninsula (3 sides water)	Indian (Arabian Sea, Bay of Bengal, Indian Ocean)

Oceans And Their Relief

→ The hydrosphere covers nearly 71% of the total surface area of the earth.

→ Divided into Oceans, Inland Seas, Small enclosed seas, Bays, etc. based on size and location.

Continental Shelf

→ The width of the continental shelf varies widely , ranging from a few kilometers to hundreds of kilometers.

→ The depth of the continental shelf is relatively shallow , typically extending to depths of about 200 meters (656 feet) but can reach depths of up to 500 meters (1,640 feet) in some areas.

→ It may feature submerged ridges, valleys, and canyons formed by geological processes such as erosion, sedimentation, and tectonic activity.

💡 Economic Significance:

→ The continental shelf is often rich in hydrocarbon resources such as oil and natural gas, which accumulate in sedimentary layers over millions of years. Offshore drilling and exploration are conducted on continental shelves to extract these resources.

→ The continental shelf supports diverse marine ecosystems , including coral reefs, kelp forests, and commercially important fish species.

→ It provides habitat and feeding grounds for marine organisms and supports commercial fisheries .

→ This biodiversity can also attract tourism .

🎯 Examples:

→ The North Sea : Located off the coasts of Norway, Denmark, Germany, the Netherlands, and the United Kingdom, the North Sea has a wide continental shelf and is known for its oil and gas reserves.

→ The Gulf of Mexico : Bordered by the southern United States and Mexico, the Gulf of Mexico has an extensive continental shelf and is a major oil-producing region.

→ The Great Barrier Reef : Off the coast of Australia, the Great Barrier Reef is the world's largest coral reef system, extending over 2,300 kilometers along the continental shelf of Queensland.

Continental Slope

→ The depth of the continental slope varies widely, ranging from depths of about 200 meters (656 feet) to 3,000 meters (9,843 feet) or more.

→ It supports deep-sea ecosystems adapted to high pressure, low light, and cold temperatures.

Resource Exploration

→ The continental slope is of interest for resource exploration, including oil and gas deposits, mineral resources, and deep-sea mining opportunities. But we are yet to develop proper technologies for resource extraction.

→ Deep-sea research and exploration expeditions study the geological, biological, and environmental characteristics of the continental slope.

🎯 Examples

→ The Blake Plateau: Located off the southeastern coast of the United States, the Blake Plateau features a prominent continental slope and is known for its rich biodiversity and geological formations.

→ The Antarctic Slope Front: Surrounding the Antarctic continent, the Antarctic Slope Front marks the boundary between the continental shelf and the deep ocean and plays a key role in ocean circulation and climate dynamics.

Deep Sea Plains

→ Deep-sea plains are found in the abyssal zones of the ocean, typically beyond the continental slope and rise, covering large areas of the ocean basin.

→ Deep-sea plains are characterized by depths ranging from 2,000 meters (6,560 feet) to 6,000 meters (19,685 feet) or more, with relatively gentle slopes and low relief.

→ Deep-sea plains have relatively flat topography compared to the rugged terrain of the continental slope and rise.

→ They are composed of fine-grained sediments, including clay, silt, and sand, deposited over millions of years by marine currents and processes.

Biological Activity:

→ Deep-sea plains support diverse ecosystems adapted to extreme conditions, including high pressure, low light, and cold temperatures.

→ Benthic organisms such as deep-sea corals, sponges, and echinoderms inhabit the sediments and substrates of deep-sea plains.

Mineral Resources:

→ These resources contain valuable metals and minerals used in various industries, including technology, manufacturing, and construction.

→ However, their extraction is a challenge. We are yet to develop technologies for that.

Geological Processes

→ Sediments accumulate slowly over time, forming flat layers that may preserve geological and environmental records.

Trenches:

→ They are the deepest parts of the ocean, with depths exceeding 6,000 meters (19,685 feet) in some cases.

Submarine Canyons

→ They are formed by erosional processes such as turbidity currents, underwater landslides, and river discharge.

→ Submarine canyons serve as conduits for sediment transport from the continental shelf to the deep ocean, and they provide habitat for diverse marine life.

→ E.g. Hudson Canyon, off the coast of New York.

Continental Rise:

→ It consists of accumulated sediments transported from the continental shelf and slope by gravity, turbidity currents, and sediment gravity flows.

→ The continental rise marks the transition between the continental margin and the deep ocean basin.

→ Trenches are associated with intense geological activity, including subduction zones, earthquakes, and volcanic eruptions.

Seamounts:

→ Seamounts are underwater mountains rising from the ocean floor, typically with heights of 1,000 meters (3,280 feet) or more above the surrounding seabed.

→ They can be either active or extinct volcanoes, formed by hotspot activity or tectonic processes.

→ Seamounts provide habitat for diverse marine life, including deep-sea corals, sponges, and fish, and they serve as important fishing grounds and navigation hazards.

→ E.g. Davidson Seamount Off the coast of central California, USA

Guyots: → Guyots provide valuable insights into past sea level changes and geological processes, and they support unique ecosystems adapted to their deep-sea environments.

General Terminologies Associated With Oceans

→ **Archipelago** : A group or chain of islands clustered together in a body of water. Example:Japanese Archipelago, in Western Pacific Ocean.

→ **Bay:** A recessed, coastal body of water that is partly surrounded by land. Example: Bay of Bengal.

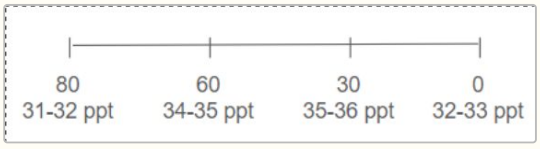
2 · TEMPERATURE, SALINITY, DENSITY PYQ 2017

Understand it first: Three properties decide how sea water behaves: how **warm** it is, how **salty** it is and so how **heavy** (dense) it is. Sun and rain set the first two at the surface: hot, dry belts give warm and very salty water; rain, rivers and melting ice dilute it. Cold and salty water is heavier and sinks, and that sinking is what drives the deep currents of the next topic.

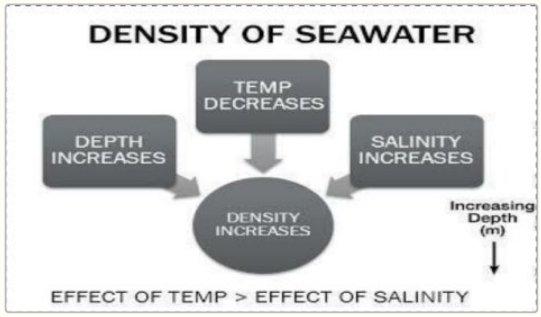
	Temperature	Salinity	Density
Key numbers	26.7 °C equatorial mean; 12.7 °C at 45° N/S; poles freezing; deep 2–4 °C; mixed layer to 200–400 m; thermocline = rapid fall	deep ocean ≈ 34.7 ppt (avg ~35); Great Salt Lake 220 · Dead Sea 240 · Lake Van 330	cold > warm; high salinity > low; deep > surface (pressure)
Pattern	warmest at equator; Indian Ocean warmest; N Hemisphere warmer (more land)	highest at ~20°–30° (high evaporation, little rain); lower at equator and poles	temperature matters more than salinity
Factors	latitude; currents (Gulf Stream, Kuroshio warm); atmospheric circulation; continentality; depth; season	evaporation/precipitation; rivers; sea-ice (freezing leaves salt); currents (Gulf Stream saline, Labrador cold fresher); winds	temperature, salinity, pressure

Salts	Percentage
1. Sodium chloride	77.8%
2. Magnesium chloride	10.9%
3. Magnesium sulphate	4.7%
4. Calcium sulphate	3.6%
5. Potassium sulphate	2.5%

Properties Of The Ocean Water: Salinity



Properties Of The Ocean Water: Salinity · Latitudinal Variations



Properties Of Ocean Water: Density

Sun + No Rain = Salty – Subtropics (hot, dry) saltiest; equator and poles diluted. Enclosed seas keep getting saltier: 220 – 240 – 330 = Great Salt Lake → Dead Sea → Lake Van. Continentality: sea ringed by land is warmer.

Ocean acidification

→ pH 8.2 (pre-industrial) → 8.11 (1985) → 8.04 (2024); ~30% more acidic in 200 yrs

→ More in Arctic than tropics (cold water absorbs more CO_2)

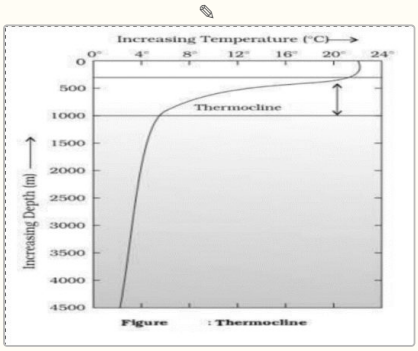
→ Causes: CO_2 dissolving, volcanoes, agricultural runoff

→ Effects: coral bleaching, weaker shells/exoskeletons, phytoplankton changes

Deep ocean (abyssal zone, below the thermocline): uniform and cold, about 2°C to 4°C; unaffected by solar heating (volcanic hot spots are the exception).

Properties Of The Oceanic Water: Temperature

→ Temperature is a vital property of ocean water that influences a wide range of physical, chemical, and biological processes.



Properties Of The Oceanic Water: Temperature · Vertical Distribution of Temperature

Factors Affecting Temperature Distribution

→ **Solar Radiation (Latitude)** : Solar radiation is the primary driver of ocean temperature distribution. Near the equator, where solar radiation is most intense, ocean temperatures tend to be warmer. At higher latitudes, solar radiation is less direct, resulting in cooler surface temperatures.

→ **Atmospheric Circulation** : Atmospheric circulation patterns, such as the Hadley, Ferrel, and Polar cells, affect ocean temperature distribution by influencing wind patterns, evaporation rates, and heat exchange between the atmosphere and the ocean.

→ **Continentality:** The water body surrounded by land on its majority of shores, will be warmer than a body surrounded by land on less of its sides. This is because just like oceans have a moderating impact on the land's temperature, similarly land has the opposite i.e. **"extreming" effect** on the sea's temperature.

→ **Ocean Depth** : Temperature distribution in the ocean varies vertically with depth. Surface waters are generally warmer due to solar heating and mixing by wind and waves, while deeper waters tend to be colder.

Horizontal Distribution of Temperature:

→ **Near the equator, ocean temperatures are generally warm, with surface temperatures often exceeding 25°C (77°F) due to direct exposure to intense solar radiation.**

→ **As you move away from the equator toward higher latitudes, ocean temperatures tend to decrease.** In polar regions, surface temperatures can drop below freezing due to the influence of cold polar air masses.

→ Ocean currents play a crucial role in redistributing heat horizontally across the ocean surface. Warm ocean currents, such as the Gulf Stream in the North Atlantic and the Kuroshio Current in the North Pacific, transport heat from the equator toward higher latitudes, influencing regional climates and ecosystems.

Vertical Distribution of Temperature:

→ The vertical distribution of ocean temperature **varies with depth** , forming distinct layers known as water masses.

→ In the deep ocean, temperatures are relatively uniform and cold, typically ranging from 2°C to 4°C (35.6°F to 39.2°F). This deep layer, known as the abyssal zone, extends from the base of the thermocline to the ocean floor and is largely unaffected by solar heating or surface processes. Here sporadically certain locations are heated due to volcanic activities, but that is an exception and not a rule.

Significance of Temperature:

→ Ocean temperature influences the distribution and behaviour of marine organisms, including fish, plankton, and marine mammals.

→ Temperature gradients in the ocean drive ocean currents , which play a crucial role in regulating Earth's climate and transporting heat around the globe.

→ Changes in ocean temperature can impact weather patterns, sea level rise, and the frequency and intensity of extreme events such as hurricanes and droughts.

→ The Northern Hemisphere is warmer than the Southern Hemisphere, due to more presence of land.

Properties Of The Ocean Water: Salinity

→ Salinity, the concentration of dissolved salts in ocean water , is a crucial property that influences various oceanographic processes.

Factors Affecting Salinity Distribution:

→ Riverine Input: Rivers transport freshwater and dissolved minerals from land to the ocean, contributing to variations in salinity along coastlines and estuaries. Regions near river mouths tend to have lower salinity due to freshwater influx, while areas further offshore may exhibit higher salinity.

→ Sea Ice Formation and Melting: During sea ice formation, salt is excluded from the ice crystals , increasing the salinity of surrounding seawater. Conversely, when sea ice melts, freshwater is added to the ocean, reducing salinity . These processes influence salinity distribution in polar regions and areas affected by seasonal ice cover.

Horizontal Distribution of Salinity:

→ Latitudinal Variations:

→ – Salinity levels tend to be higher in subtropical regions and lower in equatorial and polar regions. Subtropical regions, around 20–30 degrees latitude, experience high evaporation rates and limited precipitation, leading to elevated salinity levels.

→ – Equatorial regions receive high levels of precipitation but also experience significant evaporation, resulting in relatively lower salinity.

→ Coastal Influence: Salinity near coastlines can vary significantly due to factors such as river discharge, freshwater input, and local weather patterns.

→ – Riverine input from streams and rivers can reduce salinity near coastlines, especially during periods of high runoff.

→ Inland Seas:

→ – The salinity of the inland Seas and lakes is very high because of the regular supply of salt by 'the rivers falling into them.

→ – The oceans and salt lakes are becoming saltier as time goes on because the rivers dump more salt into them, while freshwater is lost due to evaporation.

Vertical Distribution of Salinity:

→ Halocline: Below the surface layer lies the thermocline, a transition zone characterized by a rapid change in temperature with depth, creating a layer of Halocline. Salinity levels may also change within this zone but generally remain relatively stable compared to temperature changes.

💡 Significance of Salinity	→ Salinity influences seawater density, which drives ocean circulation patterns, mixing processes, and vertical stratification . → Changes in salinity can affect ocean density gradients, leading to alterations in thermohaline circulation, nutrient distribution, and marine habitats. → Salinity variations play a crucial role in shaping marine ecosystems, species distributions, and biogeochemical cycles , influencing the productivity and biodiversity of marine environments.

Properties Of Ocean Water: Density

→ Density is a critical property of ocean water that influences various physical processes, including ocean circulation, stratification, and mixing.

Factors affecting Density Distribution	→ Temperature : Cold water is denser than warm water , so temperature variations play a significant role in determining ocean density. → – Vertical variations in temperature also contribute to density differences within the ocean. → Salinity : Salinity affects seawater density, with higher salinity generally increasing water density. → – Evaporation and precipitation influence salinity levels, leading to variations in density in regions with different water balances.
Horizontal Distribution:	→ Latitudinal Variations: Density varies horizontally across latitudes due to temperature and salinity gradients. → Coastal Influence: Coastal areas may exhibit density variations due to factors such as river runoff, freshwater input, and mixing with terrestrial sources. → – Estuaries and coastal upwelling zones often experience density gradients due to interactions between freshwater and seawater.

→ Vertical Distribution : Vertical variation of density is not constant and is different for different latitudinal areas. However, a general trend is that it increases with depth due to an increase in pressure.

💡 Significance:	→ Density variations drive vertical and horizontal ocean circulation , influencing heat transport, nutrient distribution, and marine habitats. → Ocean density gradients contribute to the formation of ocean currents, upwelling and downwelling zones , and the global conveyor belt circulation system.

Ocean Acidification

→ Seen more in arctic region compared to tropics, as colder water have a higher capacity to absorb CO2.

Causes	→ Dissolution of Carbon Dioxide in Water.

🎯 3 · FRESHWATER PYQ 2013 PYQ 2021

Understand it first: Two PYQs test one idea: almost all of Earth's water is salty, and most of the small fresh share is locked in ice. Of what is left, groundwater is far bigger than all rivers and lakes together. Read each statement against that order: ice, then groundwater, then surface water.

Freshwater share	→ Only ~3% of Earth's water is fresh (of freshwater, other ~70% after ice: groundwater ~30%, rivers/lakes ~1%); 2.5% unavailable (ice, atmosphere, soil, polluted, too deep); 0.5% available

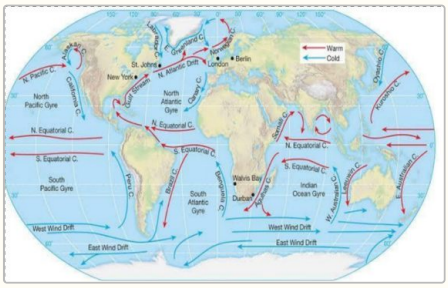
→ Ice caps/glaciers ~68–69% · groundwater ~30% (largest active resource; equals a lake 180 m deep over the Earth) · rivers/lakes ~1%

Handout conflict: surface water (lakes, rivers, swamps) given as 0.3% in 2013 note and ~1% in 2021 note. 2013 (C): after ice, most freshwater is groundwater. 2021 (B): ice caps/glaciers hold more than groundwater; rivers/lakes not more.

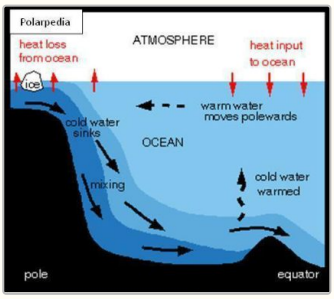
4 · OCEAN CURRENTS PYQ 2015 PYQ 2021

Understand it first: A current is a river inside the ocean that flows in one direction all year. Wind pushes the top layer, the Earth's spin (Coriolis) turns it and the continents block it, so the water goes round in big loops called gyres. Below, cold salty water sinks at the poles and creeps along the bottom (thermohaline). Western sides of oceans get warm, fast currents; eastern sides get cold, slow ones with upwelling.

NEW **Currents:** continuous flow of large water volumes in a definite direction, all year; move heat tropics to poles



Types of currents · Factors affecting Ocean Currents



Types of currents · Deep Water Currents

→ **Surface:** top ~400 m, only ~10% of ocean water; wind-driven

→ **Deep (thermohaline):** density-driven; down-welling at poles, up-welling in tropics

Circumpolar

→ Continuous loop round poles, isolate polar waters

→ Antarctic Circumpolar Current, West Wind Drift

→ Factors: latitude, wind, Coriolis, gravity, temperature, salinity, continental margins

Equatorial system

Trades push water E to W (+ Earth's spin) → *Coriolis: N and S Equatorial Currents* → *Water piles up in the west* → *Gravity: flows back east* → *Equatorial Counter Current (3–10° N)*

West = Warm & Wild – Western boundary currents (Gulf Stream, Kuroshio, Brazil, East Australian, Agulhas): warm, narrow, fast, strong. East = cold & Easygoing: Eastern boundary (Canary, Benguela, California, Peru, West Australian): cold, broad, slow, nutrient upwelling.

Gyres (wind + Coriolis + coastlines): N Hemisphere clockwise, S anticlockwise

→ N Atlantic (clockwise): Gulf Stream, N Atlantic Current, Canary, N Equatorial; Sargasso Sea in middle = only sea without a coastline (Sargassum weed)

→ S Atlantic (anti): Brazil, Benguela, S Equatorial

→ N Pacific (clockwise): Kuroshio, N Pacific, California

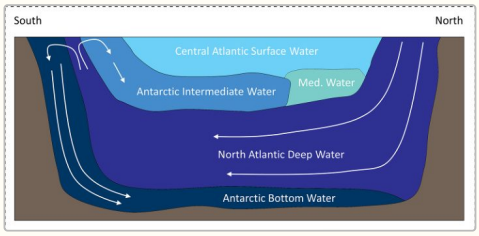
→ S Pacific (anti): East Australian, Peru, S Equatorial

→ Indian (anti): Agulhas, West Australian, S Equatorial

Water masses: big homogeneous bodies with own temp, salinity, density from a source region; keep properties far

→ Move by gravity, density, wind stress, Earth's rotation (3-D); backbone of global circulation

→ Set habitat: warm clear = coral; cold nutrient-rich = kelp, high phytoplankton and productivity; deep = specialised life



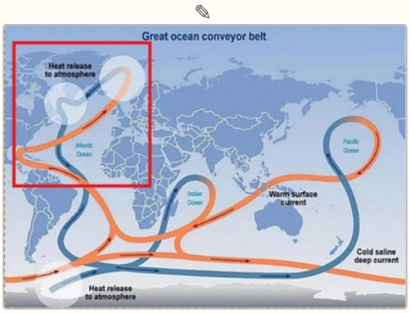
Water Masses

AMOC (Atlantic Meridional Overturning Circulation)

→ Warm salty surface water north (Gulf Stream) → cools, sinks in N Atlantic = NADW → flows south at depth

→ Regulates Atlantic circulation of heat, climate, weather, rain (also written AOC)

→ Weaker than any time in past millennium (warming; melting-ice and rain freshwater disrupt sinking)



Atlantic Meridional Overturning Circulation: (AOC)

→ If weak: cooler W Europe, more extreme events (heat, cold, storms), stronger US east-coast hurricanes, less CO_2 uptake, sea-level rise, fisheries shifts

Southern Ocean carbon anomaly

→ Absorbs ~40% of CO_2 taken up by oceans (stronger sink than models)

→ Glacier-melt + rain freshwater = light "lid", blocks mixing, traps carbon-rich water 100–200 m down

→ "Fleeting reprieve": lid weakens = CO_2 release, faster warming

NEW Ekman current

→ Wind-driven; water ~45° right (N) / left (S) of wind

→ Layers turn = Ekman spiral

→ Gives coastal upwelling (nutrients, fisheries), heat redistribution, El Niño/La Niña links

NEW Climate change on currents

→ Weaker thermal/salinity gradients = thermohaline slowdown

→ Less upwelling nutrients; less oxygen in downwelling zones; shifted winds; ice melting from below; biodiversity decline

Marine heatwaves: prolonged unusually warm sea

→ Episodes doubled since 1982; warming rate nearly doubled since 2005 (Copernicus)

→ Causes: climate change, high pressure, shifting currents, El Niño, feedback loops

→ Effects: coral bleaching, ecosystem disruption, current changes

Ocean Movements: Ocean Currents

→ Unlike waves, ocean currents are continuous in nature, flowing all year round .

NEW Types of currents

→ Surface Currents :

→ – Surface currents are driven primarily by wind patterns and occur in the upper layer of the ocean (typically the top 400 meters).

→ – They flow horizontally and can extend over large distances, influencing weather patterns, climate, and marine ecosystems.

→ These currents converge near the equator and form the eastward-flowing Equatorial Counter Current.

→ Eastern boundary currents are slower, broader, and cooler currents that flow along the eastern edges of ocean basins.

→ – They are created due to the impact of the Westerlies , causing the water to accumulate at the Eastern boundaries of the oceans in the upper latitudes.

→ – Deep Water Currents :

→ · These currents result from variations in temperature and salinity , with cold, dense water sinking and warmer, less dense water rising.

→ Factors affecting Ocean Currents:

→ – Wind : Surface currents are primarily driven by wind patterns, especially in the upper layer of the ocean. Prevailing winds, such as the trade winds and westerlies, play a significant role in generating ocean currents.

→ – Temperature : Temperature differences between water masses contribute to the formation of density gradients, which drive thermohaline circulation. Cold, dense water sinks, while warmer, less dense water rises, creating vertical movement and influencing ocean currents.

→ – Continental Margins : Ocean currents are influenced by the shape and topography of the ocean floor, particularly around continental margins. Submarine features such as underwater ridges, seamounts, and canyons can deflect or channel ocean currents.

→ Sargasso Sea , the only sea without a coastline lies in the middle of this Gyre . It is named after the Sargassum weed , which grows here, sustaining a huge biodiversity.

Water Masses

→ Water masses primarily define the physical and chemical environment for marine life. Their temperature, salinity, and nutrient content determine the types of organisms that can thrive in a particular region.

→ Water masses, once formed, move and circulate through the ocean due to a combination of gravity, density differences, wind stress, and Earth's rotation . This movement happens in three dimensions – vertically and horizontally – and forms the backbone of the global ocean circulation system .

📜 Impact Of Water Masses

→ Defining the Physical and Chemical Environment: Water masses primarily determine the "habitat" for marine life by influencing characteristics like temperature, salinity, and nutrients.

→ Supporting Specific Ecosystems: Different water masses create unique environments that favor specific types of ecosystems:

→ – Coral Reefs: Warm, nutrient-poor water masses with high clarity often support vibrant coral reef ecosystems.

→ Impact on Marine Productivity: The temperature, salinity, and nutrient content of a water mass directly influence the overall productivity of an area . Nutrient-rich, cooler water masses can support larger populations of phytoplankton (the base of the marine food chain), leading to higher overall productivity.

Atlantic Meridional Overturning Circulation: (AOC)

→ The AMOC plays a vital role in regulating climate patterns, and influencing weather, and precipitation in regions bordering the Atlantic.

→ Changes in the AMOC can lead to shifts in regional climate patterns, affecting temperature, rainfall, and storm frequency.

→ Climate change is affecting the AMOC through various mechanisms, including increased freshwater input from melting ice caps and increased precipitation.

→ Weakening of the AMOC could lead to altered weather patterns, more frequent extreme events (such as heatwaves, coldwaves and storms), and changes in ocean currents and ecosystems.

→ – Increased frequency and intensity of hurricanes along the east coast of North America due to changes in ocean temperature gradients and atmospheric dynamics. Human-induced climate change has elevated Atlantic hurricane wind speeds by 18 miles per hour (29 kilometers per hour) over the past six years.

→ – It could also contribute to sea level rise and impact global climate stability.

→ Impact of Climate Change on Ocean Currents:

→ – Warmer waters beneath the Arctic and Antarctic ice caps can affect the melting of ice from below.

→ – Changes in ocean currents can disrupt marine ecosystems, affecting species distribution, migration patterns, and breeding grounds. This can lead to a decline in biodiversity and changes in the availability of marine resources.

→ – Coastal communities may experience changes in weather patterns, increased frequency of extreme weather events, and impacts on fisheries, tourism, and shipping routes. Altered ocean currents can also affect regional climates, agriculture, and water resources.

NEW Ekman Currents

→ As you go deeper, each layer of water moves slightly differently, forming a spiral pattern called the Ekman spiral (in the image).

→ Significance:

→ – Wind Stress and Transport : Ekman transport, the net movement of surface water, occurs at an angle to the wind, influencing coastal circulation and the formation of important features like gyres.

→ – CO_2 Absorption : The upwelling associated with Ekman currents aids in the exchange of carbon dioxide between the atmosphere and ocean, impacting the carbon cycle.

Marine Heatwaves

→ Causes:

→ – Atmospheric Conditions : High-pressure systems can reduce cloud cover and wind, allowing more sunlight to warm the ocean surface.

🎯 5. TIDES AND WAVES PYQ 2015

Understand it first: Tides are the sea rising and falling twice a day because the Moon (mainly) and the Sun pull the water, with the Earth's centrifugal force making the bulge on the far side. When Sun, Moon and Earth are in one line the pull adds up (spring); at right angles it partly cancels (neap). Waves are different: wind passes energy through the water while the water itself only circles in place.

Tide: rise (high/flood) and fall (low/ebb) of seawater

→ Cause: gravity of Moon (main) and Sun + centrifugal force of Earth = bulges near and far side

Timing

→ Most places semi-diurnal: 2 highs, 2 lows a day; same tide every 12 h 26 min

→ Moon takes 24 h 52 min to return over a tide centre

→ Diurnal: Gulf of Mexico, Gulf of Thailand

→ Height depends on: Sun-Moon-Earth distances, funnel-shaped bays/estuaries, local winds, pressure (low pressure raises sea)

→ Use: nutrient circulation, temperature moderation, flushing pollution

SPrung Straight – Spring tide = syzygy: new and full moon, Sun–Moon–Earth aligned; highest highs, lowest lows, ~20% higher; twice a month; nothing to do with the season.

Nearly right-angle – Neap tide = quadrature: quarter moons, forces oppose; lower highs, higher lows.

Tidal bore

→ Rare: incoming tide makes wave travelling upstream in narrow estuary/shallow river; absent at neaps

→ India: Hooghly; world's biggest: Qiantang (China)

Climate change on tides

→ Higher sea level = larger areas tidal

→ May amplify tidal ranges; threatens mangroves

Waves: energy moves, not water; particles move in small circles

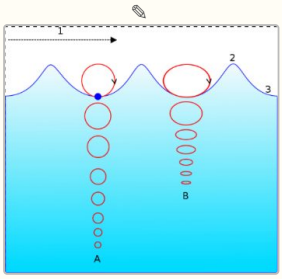
→ Wind gives energy; gravity pulls crests down; falling water lifts troughs

Tsunami: very long seismic sea waves

→ Causes: earthquakes at subduction zones (size ∝ vertical shift), volcanoes, landslides, meteorite impact, nuclear tests

→ 500–1,000 km/h in deep water, < 1 m high (invisible); slows and grows near coast; many waves

→ Warning: sea-level change, odd ocean behaviour



Ocean Movements: Waves

Rogue (freak) waves

→ Unusually large, spontaneous, even in calm seas

→ Constructive interference: wave systems in phase (rare, random)

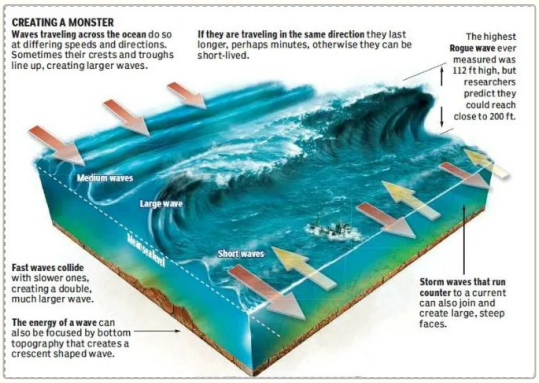
→ Focusing by currents (Agulhas Current opposing waves) and topography (ridges, shelves)

Seiche: standing wave in lake/bay/reservoir

→ Causes: wind, pressure change, quakes/landslides

→ Resonance with natural rhythm grows waves

→ Basin size, shape, depth decide size and duration



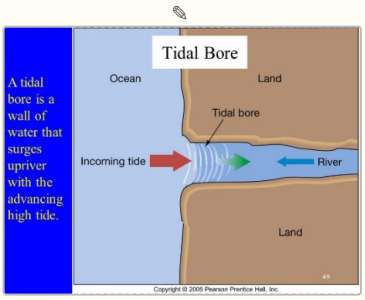
2 . Focusing by Currents and Topography

Tsunami, human cause: nuclear explosions, e.g. US nuclear testing at the Marshall Islands (1940s and 1950s).

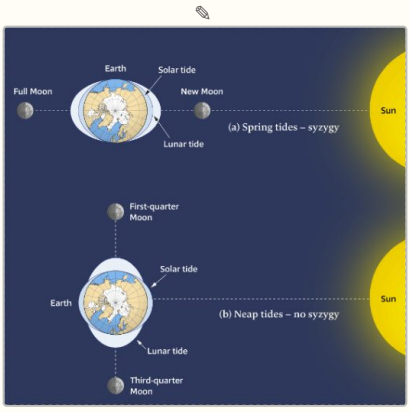
Ocean Movements: Tides

→ Period: 12 hr 26 minutes.

→ So, a place experiences the same tide after 12 hours 26 minutes.



Ocean Movements: Tides · Removal of pollutants



Ocean Movements: Tides

→ Diurnal tides occur in specific regions due to the unique geography and resonance effects, resulting in one dominant high tide and one dominant low tide each day. (Gulf of Mexico, Gulf of Thailand);

→ Spring & Neap Tides: There is a lot of temporal and spatial variation in the tide-producing forces due to different positions of the sun and the moon.

→ – Spring Tides : Very high tides are caused when the sun, the moon, and the earth are almost in the same line.

→ · This is the syzygy position.

Factors affecting tides:

→ Shape of Shoreline:

→ – Irregular coastlines with headlands and bays can cause tidal currents to intensify and create areas of convergence and divergence, affecting local tidal patterns.

→ Local winds:

→ – Wind can influence tidal currents by creating surface water movement and altering the distribution of water masses.

→ – Strong onshore winds can push water toward the shore, causing higher high tides, while offshore winds can result in lower high tides.

→ – Wind-driven currents can interact with tidal currents, enhancing or opposing their direction and intensity, especially in coastal areas with complex bathymetry.

📜 Impact of tides:

→ Sustenance of life (nutrient circulation):

→ – Tidal movements play a crucial role in the circulation of nutrients within coastal ecosystems, including estuaries, salt marshes, and intertidal zones.

→ – Tidal currents transport nutrients such as nitrogen, phosphorus, and organic matter from the open ocean to coastal areas , where they support the growth of phytoplankton and other primary producers.

→ – As tides ebb and flow, they mix and distribute nutrients throughout coastal waters, enhancing productivity and biodiversity.

→ – Nutrient-rich tidal flats and marshes serve as important feeding and breeding grounds for a variety of marine organisms, including fish, birds, and invertebrates.

→ Temperature moderation:

→ – Tidal movements help regulate water temperatures in coastal areas by mixing surface waters with cooler, deeper waters .

→ – During incoming tides (flood tide), cooler water from deeper layers is brought to the surface, lowering surface temperatures.

→ – Conversely, during outgoing tides (ebb tide), warmer surface waters are mixed with cooler, deeper waters, moderating temperatures.

→ – This temperature moderation effect is particularly important in coastal regions with extreme temperature fluctuations, helping to create more stable and hospitable environments for marine life.

→ Removal of pollutants:

→ – Tidal movements contribute to the removal and dispersion of pollutants in coastal waters through the process of flushing.

→ – As tides ebb, outgoing currents transport pollutants back out to sea, diluting their concentration and reducing their impact on coastal ecosystems.

→ – Tidal flushing helps mitigate the accumulation of pollutants in estuaries, harbors, and coastal zones, improving water quality and reducing the risk of contamination to marine organisms and human populations.

→ Most prominent in the world:

📜 Impact of Climate Change on Tides:

→ In some regions, the rising sea level may amplify the existing tidal range, increasing the difference between high and low tides. This can exacerbate coastal flooding during storm surges and extreme weather events.

→ Many marine species depend on predictable tidal patterns for nesting and breeding. Mangroves too stay in such tidal regions. Alterations in tidal ranges and flooding patterns can disrupt these critical habitats.

Ocean Movements: Waves

→ The falling water pushes the former troughs upward, and the wave moves to a new position.

→ The actual motion of the water beneath the waves is circular.

📜 Impact of Climate Change:

→ Global warming can lead to stronger and more frequent storms, particularly hurricanes and typhoons, which generate larger and more powerful waves. This increased wave energy can contribute to coastal erosion and damage infrastructure.

→ Higher sea levels mean that waves, including those from storm surges, can travel further inland, increasing the potential for coastal flooding and erosion.

→ Changes in wind patterns and ocean currents due to global warming can alter wave patterns. This can impact coastal erosion rates, sediment transport, and the stability of coastal structures.

→ Increased wave energy and frequency can damage coral reefs, which are already stressed by warmer water temperatures and ocean acidification. The physical destruction of reefs reduces their ability to act as natural barriers against waves.

Tsunamis: Extremely long waves,

→ When they reach the coast, they can cause dangerous coastal flooding and powerful currents that can last for several hours or days.

→ Causes:

→ – Just like winds are the source of energy in the case of normal surface waves, for Tsunamis, this energy comes from some other sources.

→ – Earthquake – Most tsunamis are generated by earthquakes that occur along the subduction boundaries of plates along the ocean trenches. The size of the Tsunami is related to the size of the earthquake and the vertical shift.

→ – Volcanic eruption

→ – Meteorite Impacts – Though no historical example of meteorite impact has caused tsunamis, the apparent impact of a meteorite about 5 million years ago produced tsunamis leaving deposits along the Gulf Coast of Mexico and the United States.

→ Characteristics:

→ – **Long Wavelengths** : Tsunamis have long wavelengths, often hundreds of kilometers, which allow them to travel rapidly across ocean basins.

→ – **High Speeds** : Tsunamis travel at high speeds in deep water, typically around 500 to 1000 kilometers per hour (300 to 600 miles per hour).

→ – **Amplification near Coast** : As tsunamis approach shallow coastal areas, their speed decreases while their wave height increases, leading to amplification and potential devastation upon landfall.

→ – **Destructive Power** : Tsunamis can cause catastrophic flooding, inundation of coastal areas, destruction of infrastructure, and loss of life.

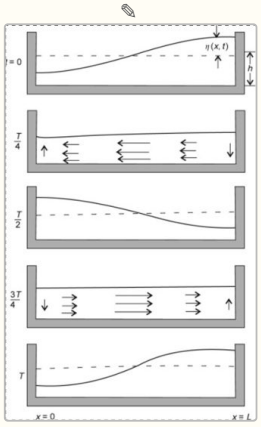
→ – **Multiple Waves** : A single tsunami event may consist of multiple waves, with successive waves arriving minutes to hours apart.

→ – **Warning Signs** : Before the arrival of a tsunami, warning signs may include rapid sea level changes, unusual ocean behavior, and the sound of approaching waves.

Rogue Waves

Creation

→ **Rogue waves** , also called **freak waves** , **monster waves** , or **killer waves** , are **unusually large, spontaneous, and unpredictable ocean surface waves** that are **far larger** than the surrounding waves. They can occur even in **relatively calm seas** and pose significant danger to ships and offshore structures. **Creation:**



Creation

1. Constructive Interference

→ When several wave systems **coincide in phase** , their energies **combine** to form a single **giant wave** .

NEW 2. Focusing by Currents and Topography

→ **Strong ocean currents** (e.g., Agulhas Current) flowing opposite to wave direction can **compress wave fronts** , increasing wave height. **Seafloor features** like ridges or continental shelves can also focus energy.

Seiches

Creation

→ A **seiche** is a **standing wave** oscillation of water in an **enclosed or semi-enclosed water body** , such as a **lake, bay, reservoir, or even a swimming pool** , caused by disturbances like **strong winds, atmospheric pressure changes** , **earthquakes** , etc. **Creation:**

→ **Wind Action:** Strong, persistent winds push water toward one end of a lake or basin. When the wind stops suddenly, the water sloshes back, creating oscillations.

→ **Seismic Activity:** Underwater earthquakes or landslides can displace water, initiating back-and-forth wave motion.

→ **Resonance Effect:** If the disturbance (like wind or pressure) matches the natural rhythm of the lake, the waves grow stronger and keep going longer—like pushing a swing at just the right time.

6 · CRYOSPHERE

Understand it first: The cryosphere is all the ice on Earth. It acts as a freshwater store, a mirror (albedo) and a lid over trapped gases. When it melts, each of those jobs fails: water supply changes, more heat is absorbed, gases escape and ocean circulation shifts.

Cryosphere = Earth's ice

→ 69% of freshwater; albedo; stores methane/CO_2; feeds rivers; affects circulation; GLOFs in Himalaya, Alps, Andes

Melting effects

→ Thermohaline change; winds/jet streams
→ Freshwater (short-term up, long-term down)
→ Lower albedo; GHG release
→ Heavy elements and pathogens released
→ Isostatic rebound and volcanic activity

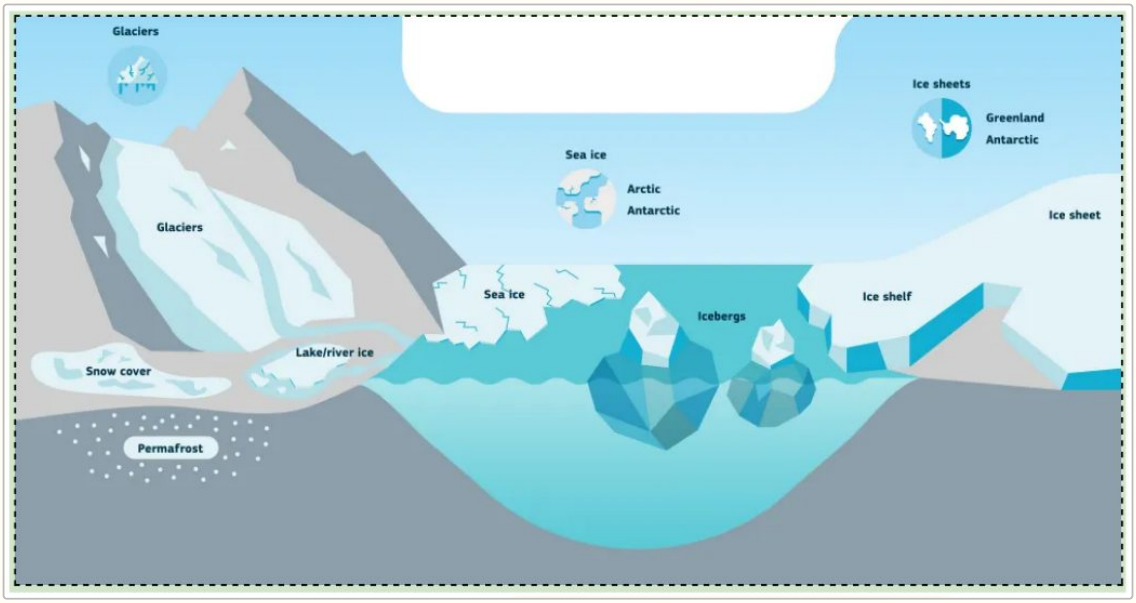
Arctic

→ Weaker polar vortex = cold spells in N America/Europe; more temperate storms
→ New shipping routes (thick ice drifts into Northwest Passage); resource access
→ Hundreds of Greenland lakes burst since 2008; trapped mercury leaks (reaches Arctic by air and ocean circulation)

Antarctica

→ Jet-stream weakening; winter heatwaves (mid-July −25 to −30 °C instead of −50 to −60 °C); wildlife and research

→ Oceans: acidification; SST rise 0.06 °C/decade (1980s) → 0.27 °C/decade now; circulation and nutrient changes



Earth's ice in all its forms

Climate change and cities: a quarter of the world's urban population faces a 0.5°C rise by 2040: health risks, lower labour productivity, more cooling demand, more extreme weather.

Cryosphere

Earth's ice in all its forms

→ Significance:
→ – During the warm season, these ice reservoirs feed the rivers of the world.
→ – Albedo , which prevents heating of the earth via insolation.

→ – These ice sheets/permafrost have trapped within themselves many greenhouse gasses when they were freezing. E.g. Methane, CO2

→ – Their melting due to Global warming will affect the global oceanic salinity, further affecting the oceanic circulations.

→ – In hilly regions, like the Himalayas, Alps, Andes, etc, the melting of the cryosphere will cause increased GLOF events.

→ Impacts of melting of Cryosphere:

→ – Impact on prevailing winds, jet streams

→ – Release of GHG, thus global warming

→ – Increase GLOF events

→ – Release of pathogens and micro-organisms

→ – Impact on climate, flora, fauna & economy.

How Arctic and Antarctic melting affect us?

→ Arctic:

→ – Climatic Impacts

→ · Due to the southward movement of cold air masses, more frequent Temperate storms , as these cold air masses meet the warmer air masses at lower latitudes.

→ – Impact on Human Activities

→ · Opening up of new navigation routes. However, recent studies on the Northwest passenger of Canada have shown that global warming is causing thick ice blocks to flow into this channel from Greenland. This can pose a danger to shipping vessels, reducing the navigability of the shipping passages, at least in the short term.

→ · Easing resource exploration in these regions.

→ · Change in fish and wildlife patterns , that may affect the local communities and their traditional livelihoods.

→ · According to a latest study, toxic Mercury trapped in the Arctic is now leaking into the environment.

→ Antarctica:

→ – Climate

→ · Jet stream weakening can cause changes in weather in South America, southern Africa, and Australia.

→ · The weakening of the polar vortex is causing winter season heatwaves here. Some parts of Antarctica, in mid-July had temperatures as high as -25 to -30 degrees Celcius compared to -50 and -60 degrees Celcius.

→ – On Humans

→ · Affect the wildlife in the region (Penguins, seals, microorganisms, etc)

📜 Impact of Climate Change on Climate & Oceans

→ Climate

→ – Urban Areas: Quarter of global urban population to face 0.5°C temperature increase by 2040. The temperature rise will disproportionately impact vulnerable populations, leading to increased health risks, reduced labor productivity, and greater energy demands for cooling.

→ – Increase instances of extreme weather events.

→ Biodiversity

→ – 2023 report by the Intergovernmental Panel on Climate Change had warned that if the carbon dioxide emission and global warming continued unabated, the tropical rainforest community may altogether collapse much before the end of this century.

→ – According to a BHU study, while increased CO_2 can initially enhance crop growth through the "CO_2 fertilization effect," surpassing certain temperature thresholds due to climate change leads to significant reductions in crop yields.

→ Oceans

→ – Increased acidification

7 · CORAL REEFS

Understand it first: A coral is a tiny animal (polyp) living with an alga, **zooxanthellae**, inside it. The alga makes food from sunlight, so reefs need warm, clear, shallow, low-nutrient water. Under stress, mainly heat, the coral throws the alga out and turns white: **bleaching**.

Corals: marine invertebrates, colonies of polyps; secrete calcium carbonate; "rainforests of the sea"

→ Symbiotic algae zooxanthellae give energy by photosynthesis

Kinds

→ Hard (stony, reef-builders): Acropora, Porites

→ Soft: octocorals, sea fans

→ Cold-water: deep, polar/temperate

→ Black corals: 30–300+ m, Gulf of Mannar

Warm, Clear, Shallow, Salt-stable, Light-snack, Mild-waves – 20–29 °C · clear sunlit water · <50 m deep · salinity 32–40 ppt · low nutrients (oligotrophic; runoff feeds algae that choke them) · moderate waves

Fringing = Friend – Fringing = Friend, hugs shore (boat channels) · Barrier = Bodyguard, stands off, lagoon between · Atoll = Wedding ring around lagoon (sinking volcanic island/seamount)

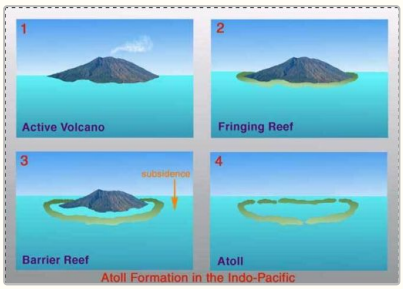
Bleaching: corals expel zooxanthellae (white; recover or die); more disease

→ Causes: heat, acidification, pollution, UV, tectonic exposure/drowning, overfishing (fish keep reefs clean), disease, tourism

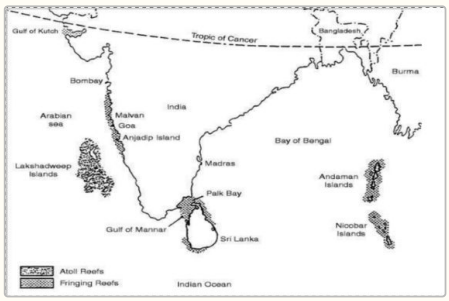
→ Great Barrier Reef: 4th mass bleaching ongoing since early 2023

→ India: bleaching in Gulf of Mannar and Andamans; Lakshadweep, March 2024

→ Climate: acidification (harder to build skeletons); stronger storms (damage, smothering sediment)



Coral Reefs · Barrier Reefs



Coral Reefs · Factors Responsible for this

Source conflict: 4th mass bleaching start printed as January 2023 once and February 2023 once.

Coral Reefs

→ Corals are marine invertebrates, which exist in colonies composed of numerous individual coral polyps, which are tiny, soft-bodied organisms.

→ Corals secrete calcium carbonate to form hard skeletons , which accumulate over time to create large, underwater structures known as coral reefs.

→ These reefs provide habitat and shelter for a diverse range of marine life and are often referred to as the "rainforests of the sea" due to their high biodiversity.

Necessary conditions:

→ Shallow Depths: Most coral reefs develop in shallow water environments, typically less than 50 meters deep. Shallow depths allow corals to access sunlight for photosynthesis and promote the growth of reef-building corals.

→ Stable Salinity Levels: Corals prefer stable salinity levels within the range of 32 to 40 parts per thousand (ppt). Fluctuations in salinity can stress corals and affect their ability to maintain osmotic balance.

→ Low Nutrient Levels: Coral reefs thrive in oligotrophic (nutrient-poor) waters . Excessive nutrient runoff from land-based sources can lead to eutrophication, which promotes the growth of algae that can outcompete corals for space and resources.

→ Moderate Wave Action: Moderate wave action helps to oxygenate the water and remove sediment , promoting coral growth and reef development. However, excessive wave energy from storms or strong currents can damage coral reefs.

Other type of Corals:

→ Hard Corals:

→ – Examples include branching corals like Acropora and massive corals like Porites.

→ Soft Corals:

→ – Lack a hard skeleton and have flexible structures.

→ – Often referred to as gorgonians or octocorals.

→ – Include sea fans, sea whips, and sea pens.

→ Cold-water Corals:

→ – Found in deep, cold waters, including polar regions and temperate seas.

→ – Adapted to low light levels and cooler temperatures compared to tropical corals.

→ – Examples include species of Lophelia, Paragorgia, and Primnoa.

→ Black Corals:

→ – Typically found in deeper waters, ranging from 30 meters to over 300 meters deep.

→ – Have a dark appearance due to the presence of a protein called melanin.

→ – Often prized for their use in jewelry and ornamental pieces.

→ – Found in India in Gulf of Mannar region, along with other corals.

Types of Coral Reefs:

→ Fringing Reefs:

→ – Found close to the shoreline, directly adjacent to land.

→ – Develop along the edges of continental coastlines or islands.

→ – Characterized by their shallow depths and proximity to land, with the reef crest often exposed during low tide.

→ – They might create small lagoons, known as boat channels .

→ – Parallel to the shoreline but separated from land by a lagoon.

→ – Form farther offshore compared to fringing reefs and are separated from the mainland by a deep lagoon.

→ – Barrier reefs can stretch for long distances and are often associated with large oceanic islands or continental landmasses.

→ Atolls:

→ – Circular or horseshoe-shaped reefs surrounding a central lagoon.

→ – Formed from the gradual subsidence of volcanic islands or seamounts over time, leaving behind a ring-shaped coral reef.

→ – Atolls are often found in the open ocean, far from continental landmasses, and may contain several small islands or islets within the lagoon.

Coral Bleaching:

→ Bleached corals appear white or pale due to the loss of pigmented algae.

→ **This may be temporary or permanent.**

→ Bleaching weakens corals and can lead to their death if the stress is prolonged.

→ **Factors Responsible for this:**

→ – **Elevated Water Temperatures** : High water temperatures, particularly during periods of ocean warming associated with climate change, can trigger coral bleaching.

→ – **Ocean Acidification:** Increased carbon dioxide levels in the atmosphere lead to ocean acidification, which can weaken coral skeletons and make them more susceptible to bleaching.

→ – **Pollution:** Pollution from agricultural runoff, sewage, and coastal development can increase stress on corals and contribute to bleaching events.

→ – **UV Radiation:** Excessive exposure to ultraviolet (UV) radiation, often exacerbated by clear skies and calm seas, can also induce coral bleaching.

→ – **Tectonic Activity:** Sudden exposure of the corals above the sea level due to land upliftment. Or drowning of corals to non-photic zones due to subsidence.

📜 **Impact Of Climate Change**

→ **Coral Bleaching:** Rising sea temperatures due to global warming are the primary culprit. Bleached corals are weakened and more susceptible to diseases.

→ – Example: The Great

→ Barrier Reef in Australia has experienced large-scale bleaching events in recent years, jeopardizing its vast coral ecosystems. Currently, 4th mass bleaching event going on since Feb 2023.

→ **Ocean Acidification:** Increased absorption of carbon dioxide from the atmosphere by the oceans leads to acidification. This makes it harder for corals to build their calcium carbonate skeletons, hindering their growth and survival.

→ – Example: Coral reefs in Florida, USA, have shown signs of reduced growth and weakened skeletons due to ocean acidification.

→ **Increased Storm Intensity:** Warmer ocean temperatures contribute to stronger storms and hurricanes. These storms can physically damage coral reefs through strong waves and generate large amounts of sediment that can smother corals.

→ – Example: Coral reefs in the Caribbean have suffered extensive damage from recent hurricanes, hindering their recovery from other stressors like bleaching.

→ **Corals in India:**

→ – Indian reefs have also **experienced several bleaching events** in recent decades, with significant damage reported in the Gulf of Mannar and the Andamans.

→ – These events threaten the vibrant coral ecosystems and the associated biodiversity.

→ – Pollution from coastal development, destructive fishing practices, and sedimentation from rivers can exacerbate the impacts of global warming on Indian coral reefs.

8 · OCEANIC DEPOSITS

Understand it first: Whatever settles on the sea floor is sorted by where it came from: land, living things, chemical precipitation from sea water, or space. Four sources, four names.

Land, Life, Liquid, Cosmos – Terrigenous (land: rivers, glaciers, wind; margins) · Biogenous (organisms): calcareous ooze $CaCO_3$ (warmer, shallower), siliceous ooze (diatoms, radiolarians; colder, deeper) · Hydrogenous (chemical): manganese nodules, phosphorite · Cosmogenous (meteorite dust)

Oceanic Deposits

Terrigenous Deposits:

→ Found primarily on continental margins , including continental shelves and slopes.
→ Examples include sand, silt, and clay, as well as larger particles like gravel and boulders.

Biogenous Deposits

→ Composed of remains of marine organisms , such as shells, skeletons, and coral fragments.
→ Accumulate on the seafloor over long periods.
→ Found in both shallow and deep ocean environments , including continental shelves, slopes, and abyssal plains.
→ Biogenous deposits include :
→ – Calcareous Ooze : Composed primarily of calcium carbonate shells and skeletons of foraminifera, coccolithophores, and other calcareous organisms. Found in warmer, shallower waters.

Hydrogenous Deposits:

→ Formed through chemical precipitation from seawater .
→ Found in various oceanic environments , including continental shelves, slopes, and deep-sea hydrothermal vent areas.
→ Examples include:
→ – Manganese Nodules : Formed of manganese, iron, and other metals precipitated from seawater onto the seafloor.
→ – Phosphorite Deposits : Consist of phosphate minerals precipitated from marine sediments.

Cosmogenous Deposits:

→ Derived from extraterrestrial sources , such as meteorites and interplanetary dust.
→ Found distributed throughout the oceans , with higher concentrations in deep-sea environments.
→ Examples include microscopic meteorite particles and cosmic dust grains.

9 · DEAD ZONES PYQ 2013

Understand it first: A dead zone is water with too little oxygen for most life. The chain is always the same: nutrients run off the land, algae bloom, the algae die and sink, and bacteria use up the oxygen while decomposing them.

→ Cause: nutrient pollution (N, P: farm runoff, sewage, industry, even wind-blown dust: Madagascar bloom)
→ Chain: algal bloom → dies, sinks → bacteria use up oxygen

👫 Dead zones (hypoxic): oxygen too low for most life

→ Effects: marine life suffocates, fisheries/tourism lose, toxins and bacteria contaminate seafood

🎯 Examples

→ N Gulf of Mexico (Mississippi runoff, annual)
→ Baltic Sea (chronic, nutrient pollution)
→ Black Sea (natural, poor circulation)

→ Fix: cut nutrient pollution (farming rules, wastewater treatment), monitoring, public awareness

Dead Zones

→ Dead zones, also known as hypoxic zones , are areas in oceans and large lakes where oxygen levels become so low that most marine life cannot survive.

→ Causes: The primary culprit behind dead zones is nutrient pollution , particularly excess nitrogen and phosphorus. This pollution often comes from:

→ – Agricultural runoff: Fertilizers used in farming can be washed away by rain, carrying these nutrients into rivers and ultimately into coastal waters.

→ – Sewage discharge: Untreated or inadequately treated sewage can also contribute excess nutrients.

→ – Industrial waste: Certain industries may release nitrogen and phosphorus into waterways.

→ – Wind Deposits: A massive algal bloom off the coast of Madagascar was triggered by dust blown from drought-stricken regions of southern Africa. The dust contained nutrients like iron and phosphorus that acted like fertilizers.

Formation

→ When these nutrients reach coastal waters, they trigger a rapid growth of algae. This phenomenon is called an algal bloom . As the algae die and decompose, they sink to the bottom, where they are consumed by bacteria. This bacterial decomposition process consumes a large amount of oxygen dissolved in the water.

Consequences

→ The spread of dead zones poses a serious threat to marine ecosystems, with cascading effects: Loss of Marine Life: When oxygen levels become depleted, fish, shrimp, and other marine organisms suffocate and die. This disrupts the food chain and can lead to ecosystem collapse. Dead zones can become "ecological graveyards" devoid of most marine life. Economic Damage: Dead zones can have a significant economic impact on fisheries, tourism, and other industries that rely on healthy marine ecosystems. As fish populations decline, fishing communities suffer economic losses, and tourism to affected areas can decrease. Public Health Risks: Decomposition of organic matter in dead zones can release harmful toxins and bacteria. This can contaminate seafood and pose a risk to human health if affected fish are consumed.

🎯 Examples

→ Northern Gulf of Mexico: One of the world's largest dead zones occurs annually in the Gulf of Mexico, primarily due to agricultural runoff from the Mississippi River basin.

→ Baltic Sea: This semi-enclosed sea suffers from chronic oxygen depletion due to nutrient pollution from surrounding countries.

→ Black Sea: The Black Sea has a naturally occurring dead zone in its deep waters due to limited circulation and high organic matter content.

How to Combat?

→ Reducing Nutrient Pollution: Implementing stricter regulations on agricultural practices, improving wastewater treatment, and promoting sustainable farming methods are essential to curb nutrient runoff.

→ Monitoring and Management: Regular monitoring of water quality and implementing measures to manage nutrient loads in vulnerable areas can help prevent dead zone formation.

→ Public Awareness: Raising public awareness about the causes and impacts of dead zones is essential for promoting behavioral changes and supporting policies that address this environmental threat.

10 · OCEAN RESOURCES AND FISHING GROUNDS PYQ 2013

Understand it first: The ocean gives living resources (fish, seaweed, pearls) and non-living ones (oil and gas, energy, salts, nodules). The best fishing grounds sit where a warm and a cold current meet, because the mixing feeds plankton.

Resources

Biotic

→ Fish: tuna, salmon (Pacific); cod (N Atlantic); sardines (Mediterranean, Peru)

→ Shellfish; seaweed (Japan nori); marine pharma; pearls

Abiotic

→ Offshore oil/gas: North Sea, Gulf of Mexico, Persian Gulf

→ Tidal, OTEC, wave, offshore wind

→ Dissolved salts (NaCl); manganese nodules (source of "dark oxygen"); phosphorite (Morocco, Peru, USA)

Services

→ Absorb ~90% of excess heat, 25–30% of human CO_2 (blue carbon: mangroves, seagrass)

→ Phytoplankton make 50–60% of Earth's oxygen

Ocean Resources

Biotic Resources:

→ Edible Resources:

→ – Shellfish: Shrimp and crabs are harvested in the Gulf of Mexico and along the coast of Alaska. Lobsters are abundant in the waters of Maine and the North Atlantic, while oysters are farmed in the Chesapeake Bay and along the Gulf Coast.

→ – Seaweed: Japan is a major producer of seaweed, particularly nori, used in sushi. Other countries like China, Korea, and Indonesia also cultivate seaweed for culinary and industrial purposes.

→ Non-Edible Resources:

→ – Marine Pharmaceuticals: Marine organisms from coral reefs in the Caribbean and the Great Barrier Reef in Australia have been studied for their potential pharmaceutical properties.

→ – Coral Reefs: The Great Barrier Reef in Australia, the Coral Triangle in Southeast Asia, and the Caribbean Sea are renowned for their rich coral reef ecosystems.

→ – Pearls: Pearl farming occurs in regions like the South Pacific, particularly in countries like Japan, China, and Indonesia, where pearl oysters are cultivated for their pearls.

→ – Seal and Whale Products: Seal fur and whale blubber have historically been harvested in regions such as the Arctic and Antarctic, though regulations have limited commercial harvesting in recent years.

Non-Biotic Resources:

→ Energy Resources:

→ – Renewable Energy: Tidal energy, Ocean Thermal Energy Conversion, Wave energy; Offshore wind farms are located in the North Sea, the Baltic Sea, and along the coastlines of countries like

the United Kingdom, Germany, and Denmark. Wave energy converters are being tested in regions with strong wave action, such as the Pacific Northwest and the Atlantic coast of the United States.

→ Mineral Resources:

→ – Manganese Nodules (Poly-Metallic Nodules): Found on the abyssal plains of the Pacific Ocean, particularly in the Clarion-Clipperton Zone. Recent discovery states that these nodules, also known as Polymetallic nodules, are an important source of "dark oxygen", sustaining the life in the ocean bottom regions.

→ – Phosphorite Deposits: Located off the coast of countries like Morocco, Peru, and the United States, where upwelling currents bring nutrient-rich waters to the surface.

Intangible Resources

→ Regulating Services (Environmental Balance)

→ – Climate regulation : Oceans absorb ~90% of the excess heat trapped by greenhouse gases.

→ – Carbon sink : Absorb ~25–30% of anthropogenic CO_2; "blue carbon" ecosystems like mangroves and seagrass sequester it long-term. In fact, recent research has shown that the ocean's surface absorbs significantly more carbon dioxide than previously believed, especially the upper layers.

→ – Heat & circulation regulation : Ocean currents redistribute heat, influencing monsoons and global climate.

→ – Weather & water cycle regulation : Evaporation drives rainfall, storms, and global hydrological balance.

→ – Coastal protection : Coral reefs, mangroves, wetlands reduce storm surges, erosion, and flooding.

→ Cultural & Recreational Services

→ – Tourism & recreation : Beaches, diving, surfing, cruises fuel coastal economies.

→ – Cultural & spiritual values : Sacred seascapes, heritage of coastal communities.

→ – Education & research : Source of marine knowledge, climate science, biodiversity studies.

→ – Aesthetic & inspiration : Oceans inspire literature, art, and human imagination.

→ Supporting Services (Underlying Systems)

→ – Biodiversity hotspots : Coral reefs, estuaries, deep-sea vents harbor immense species diversity.

→ – Nutrient cycling : Recycle nitrogen, phosphorus, and carbon essential for global food webs.

→ – Habitat provision : Support breeding, nursery, and migratory grounds for marine life.

📜 11 · LAW OF THE SEA: UNCLOS 1982

Understand it first: UNCLOS 1982 divides the sea into belts measured outward from the coast: the closer the belt, the stronger the coastal State's rights. Remember the three distances, 12, 24 and 200 nautical miles, and what the State may do in each.

12 - 24 - 200 — Territorial sea 12 nm (22.2 km) · Contiguous zone 24 nm (44.4 km; customs, fiscal, immigration, sanitary laws) · EEZ 200 nm (370.4 km; sovereign rights over resources) · High seas beyond (navigation, overflight, fishing, research). A dozen → a day → a double century.

UNCLOS (United Nations Convention on the Law of the Sea), adopted 1982: main legal framework on maritime boundaries, navigation, resources, protection of marine environment.

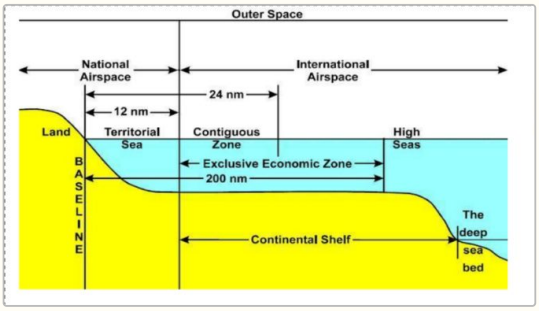
Law Of Sea

→ The Law of the Sea refers to a body of international law that governs the use of the world's oceans and seas. It encompasses rules and principles regarding maritime boundaries, navigation, exploitation of marine resources, conservation, and protection of the marine environment.

Components:

→ Territorial Sea: A belt of water adjacent to a coastal state's land territory, usually extending up to 12 nautical miles (22.2 kilometres) from the baseline.

→ Exclusive Economic Zone (EEZ): An area extending up to 200 nautical miles (370.4 kilometres) from a coastal state's baseline, within which the state has sovereign rights over the exploration and exploitation of natural resources.



Law Of Sea

12 · TRAPS AND CONNECTIONS

Traps

→ spring tide = in spring ✘
→ counter-current due to Earth's rotation ✘
→ salinity highest at equator ✘
→ density driven more by salinity ✘ (temperature)
→ tsunamis tall in deep ocean ✘
→ rivers/lakes hold most freshwater after ice ✘ (groundwater)
→ atolls form on continents ✘ (sinking volcanic islands).

Connects to

→ trenches, guyots, hotspots → Geomorphology
→ Coriolis, trade winds, westerlies, cyclones, El Niño → Climatology/ Indian climate
→ resources, ports, fisheries → Economic Geography
→ Palk Strait, Chilika, Gulf of Kutch, Cape Comorin → Indian Geography.

RECALL DRILL: COVER AND ANSWER

Average ocean depth; shelf and slope gradient?	3,800 m; 1–2° / 3–6°	Trench depth; example?	> 6,000 m; Mariana
Mean temperature equator vs 45°?	26.7 °C / 12.7 °C	Saltiest listed water bodies?	Lake Van 330 > Dead Sea 240 > Great Salt Lake 220
Ocean pH now?	8.04 (2024); pre-industrial 8.2	Why does the counter-current flow east?	Convergence/pile-up of the two equatorial currents
Only sea with no coastline?	Sargasso Sea (N Atlantic gyre)	Tide repeats after; spring tide when?	12 h 26 min; new and full moon (syzygy)
Tidal bore in India; world's biggest?	Hooghly; Qiantang (China)	Tsunami speed in deep ocean?	500–1,000 km/h, under 1 m high
Coral temperature / depth / salinity?	20–29 °C / <50 m / 32–40 ppt	UNCLOS zones?	12 · 24 · 200 nm

🎯 PYQs OF THIS CHAPTER

🎯 Exam lens (from the handout only)

- Priority: MEDIUM–HIGH. 6 Prelims PYQs printed: 2013×2 (freshwater share; fishing grounds) · 2015×2 (equatorial counter-current; tides) · 2021×2 (freshwater; warm/cold sides of oceans). Mains quoted: "Account for variations in oceanic salinity and discuss its multi-dimensional effects" (2017, 15m).
- Pattern: the exam asks why — why the counter-current flows east, why tides occur, why fishing grounds sit where warm and cold currents meet.
- Current-affairs heavy handout: AMOC weakening, coral bleaching, dead zones, marine heatwaves, Southern Ocean carbon sink — good Mains material.
- No teacher ink in this PDF; priority = PYQ count in the handout.

★ *PYQ answers in one line (asked year and the key)*

Mains 2017 (15m) salinity: Intro: dissolved salts, avg ~35 ppt, deep 34.7. Variation: latitude (subtropical high), evaporation vs rain, rivers, sea ice, currents, inland seas (220/240/330). Effects: density → thermohaline circulation, stratification (halocline), nutrients, habitats and productivity. Close: melting-ice freshwater now alters salinity and circulation. ✅

2015 (B): counter-current flows east due to convergence/pile-up of the two equatorial currents, not Earth's rotation directly. NECC 100–150 m deep, in Atlantic, Indian, Pacific. Counter Equatorial Current (CEC). Not the Equatorial Undercurrent (EUC) (eastward along equator, ~200 m deep in W Pacific, 100 m in E Pacific). ✅

2021 (C), both right: tropics: west of ocean warmer (trade winds carry warm water west); temperate: east warmer (westerlies carry warm equatorial water to continents' west coasts). ✅

2015 (D): tides = Sun's gravity + Moon's gravity + Earth's centrifugal force (all three). ✅

2013 (C) fishing grounds: best where warm and cold oceanic currents meet (plankton mixing): Newfoundland banks, Sea of Japan. Helpers: coastal upwelling, indented coasts, shelf food from erosion + sunlight, nearby forests for boats, rugged land pushing people to fishing, temperate water below 20 °C. ✅

✍️ *The PYQs as printed in the handout, with answers*

Q.1) On the planet earth, most of the freshwater exists as ice caps and glaciers. Out of the remaining freshwater, the largest proportion (2013)

(a) is found in the atmosphere as moisture and clouds

(b) is found in freshwater lakes and rivers\

(c) exists as groundwater

(d) exists as soil moisture

Ans: C According to the U.S. Geological Survey, over 68 percent of the freshwater on Earth is found in ice caps and glaciers, and just over 30 per cent is found in groundwater.

- Only about 0.3 percent of our freshwater is found in the surface water of lakes, rivers, and swamps.
- Over 68 percent of the freshwater on Earth is found in icecaps and glaciers, and just over 30 percent is found in groundwater. Only about 0.3 percent of our freshwater is found in the surface water of lakes, rivers, and swamps.
- US and Canadian researchers recently calculated the total amount of the world's groundwater and estimated that it is equivalent to a lake 180 metres deep covering the entire Earth. This makes groundwater the largest active freshwater resource on the planet.

Q.2) With reference to the water on the planet Earth, consider the following statements: (2021)

1. The amount of water in the rivers and lakes is more than the amount of groundwater.

2. The amount of water in polar ice caps and glaciers is more than the amount of groundwater.

Which of the statements given above is/are correct?

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Ans.- B

- Groundwater represents about 30% of world's fresh water. From the other 70%, nearly 69% is captured in the ice caps and mountain snow/glaciers and merely 1% is found in river and lakes.

- Of the total freshwater, 69% resides in glaciers and 30% underground. Polar ice caps and glaciers has more water than groundwater.
- 3% of the earth's water is fresh. 2.5% of the earth's fresh water is unavailable: locked up in glaciers, polar ice caps, atmosphere, and soil; highly polluted; or lies too far under the earth's surface to be extracted at an affordable cost. 0.5% of the earth's water is available fresh water.

Q.3) Account for variations in oceanic salinity and discuss its multi-dimensional effects. (2017, 15m)

Q.1) What explains the eastward flow of the equatorial counter-current? (2015)

(a) The Earth's rotation on its axis

(b) Convergence of the two equatorial currents

(c) Difference in salinity of the water

(d) Occurrence of the belt of calm near the equator

Ans: B The Equatorial Counter Current is an eastward flowing, wind-driven current which extends to depths of 100-150m in the Atlantic, Indian, and Pacific Oceans.

- As the earth rotates from west to east, water near the equator tends to move from east to west (due to the prevailing winds). This leads to the formation of Equatorial currents, both North and South of the equator.
- Now, as the North Equatorial Current and South Equatorial Current take the water from east to the western part of the ocean basins, there is a piling up water in the western parts of ocean basins (or eastern boundaries of land masses near the equator). This piled up water then flows from west to east, under the effect of gravity, resulting in Counter Equatorial Current (CEC).
- So, while the rotation of the earth is the primary causative factor for Equatorial Currents, it is piling up of water and subsequent backflow (convergence of both equatorial currents) under gravity which causes the eastward flow of CEC.
- More often called the North Equatorial Countercurrent (NECC), this current flows west-to-east at about 3-10°N in the Atlantic, Indian Ocean and Pacific basins, between the North Equatorial Current (NEC) and the South Equatorial Current (SEC).
- The NECC is not to be confused with the Equatorial Undercurrent (EUC) that flows eastward along the equator at depths around 200m in the western Pacific rising to 100m in the eastern Pacific.

Q.2) Tides occur in oceans and seas due to which among the following? (2015)

1. Gravitational Force of the Sun
2. Gravitational Force of the Moon
3. Centrifugal force of the Earth

Select the correct answer using the code given below

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2, and 3

Ans: D Tides are the rise and fall of sea levels caused by the combined effects of the gravitational forces exerted by the Moon and the Sun, and the rotation of the Earth.

- The periodical rise and fall of the sea level is known as tide. The Moon's gravitational pull is the major cause of a tide. Sun's gravitational pull and centrifugal force due to the gravitational pull of the earth to a lesser extent so causes types.
- Tides are very long waves that move across the oceans. They are caused by the gravitational forces exerted on the earth by the moon, and to a lesser extent, the sun. Gravity pulls the ocean towards the moon and high tide occurs.
- High tides and low tides are caused by the moon. The moon's gravitational pull generates something called the tidal force. The tidal force causes Earth—and its water—to bulge out on the side closest to the moon and the side farthest from the moon.

Q.3) Consider the following statements:

1. In the tropical zone, the western sections of the oceans are warmer than the eastern sections owing to the influence of trade winds.
2. In the temperate Zone, westerlies make the eastern sections of oceans warmer than the western sections

Which of the statements given above is/are correct? (2021)

(a) 1 only

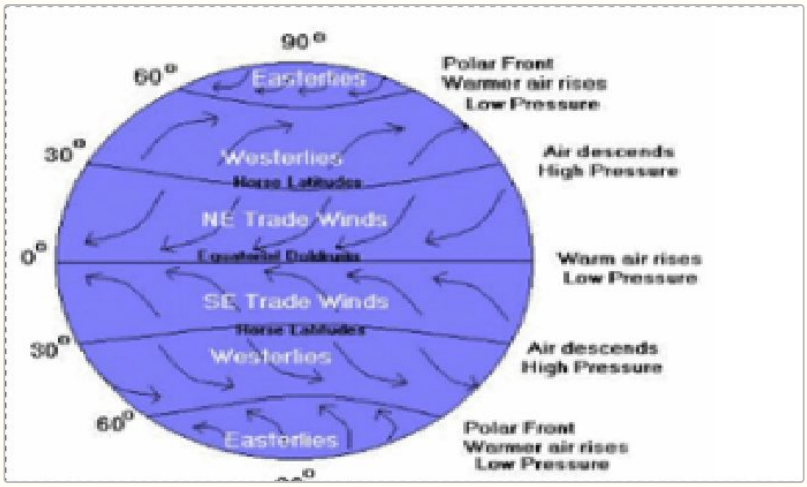
(b) 2 only

(c) Both 1 and 2

(d) Neither 1 and 2

Ans.- C

- Statement 1 is correct. Warmer water is transported westward in the ocean by the Northeast trade winds in the Northern hemisphere of the tropical zone (as the figure given below shows). So, in tropical zones, the western section of ocean is warmer than eastern sections due to trade winds.



- Statement 2 is correct. Similarly, the Westerlies play an important role in carrying the warm, equatorial waters and winds to the western coasts of continents (that is eastern section of the Oceans in temperate zone). Thus, in temperate zones, westerlies make the eastern section of the ocean warmer than the western sections.

Q.1) The most important fishing grounds of the world are found in the regions where: (2013)

(a) Warm and cold atmospheric currents meet

(b) Rivers drain out large amounts of fresh water into the sea

(c) Warm and cold oceanic currents meet

(d) Continental shelf is undulating

Ans.- C Reasons that determining Fishing grounds

- The location of major world fishing grounds is governed by many factors:
  - Abundance of plankton
- coastal waters are beneficial to fish in more than one way—
- There is plentiful food supply from the erosion of continents, which provide mineral and organic matter.
- There is enough sunlight penetration, which is critical to the photosynthesis process required by plants.
- Areas where warm and cold currents meet tend to have high biological productivity, because plankton growth is encouraged by the mixing of cold currents and warm currents.
- Some of the world's most productive fishing grounds are located here, where warm and cold currents converge.
- Examples of these phenomena are the Newfoundland 'banks' and the Sea of Japan.
  - In areas of coastal upwelling high rates of primary production takes place. Conventional mixing provides plenty of oxygen to sustain life in coastal zones as well as assuring a favourable dispersion of nutrients.
  - Broken coastline with numerous indentations provides excellent berthing facility.
  - Accessible forest resources, which favour the manufacture of fishing boats, jetties, etc.
  - The character of land: Rugged land, which hinders the development of other activities like agriculture and forces the people to go for fishing.
  - Climatic condition: Temperate climate with a temperature of less than 20°C is the best region for marine life.

NUMBER CARD, TRAPS AND SELF-TEST

Number card

- 3,800 m — average ocean depth
- 1–2° / 3–6° — shelf / slope gradient
- > 6,000 m — trench (Mariana)
- 35 ppt — average salinity
- 26.7 °C / 12.7 °C — mean temperature, equator / 45°
- 8.04 — ocean pH (2024); 8.2 pre-industrial
- 12 h 26 min — tide repeats after
- 20–29 °C · <50 m · 32–40 ppt — coral reef limits
- 12 · 24 · 200 nm — UNCLOS belts

Trap: Counter-current and Earth's rotation

The equatorial counter-current is not caused by the Earth's rotation directly. Exam-hall rule: pick "convergence (pile-up) of the two equatorial currents".

Trap: Salinity at the equator
Salinity is not highest at the equator: the subtropics are saltiest; equator and poles are diluted. Exam-hall rule: high salinity = hot, dry, low-rain belts.

Trap: What drives density
Density is driven more by temperature than by salinity. Exam-hall rule: temperature first, salinity second.

Trap: Freshwater after ice
Rivers and lakes do not hold most fresh water after ice; groundwater does. Exam-hall rule: ice, groundwater, then surface water.

Trap: Spring tide and atolls
Spring tide is not a tide in spring (it is Sun-Moon-Earth in line); atolls form on sinking volcanic islands, not on continents. Exam-hall rule: read the word by its cause, not its name.

Handout conflict: Surface water share
Surface water is printed as 0.3% in one note and about 1% in another; the 4th mass bleaching event starts in January 2023 once and February 2023 once. Exam-hall rule: the ranking (ice, groundwater, surface) decides the answer, not the exact percentage.

Self-test (answers at the end of the box)

Q1. Which belt gives the coastal State sovereign rights over resources up to 200 nautical miles?
(a) Territorial sea (b) Contiguous zone (c) Exclusive Economic Zone (d) High seas

Q2. Western boundary currents of oceans are generally:
(a) cold, broad and slow (b) warm, narrow and fast (c) cold, narrow and fast (d) warm, broad and slow

Q3. Spring tides occur when:
(a) Sun, Moon and Earth are in one line (b) the Moon is at right angles to the Sun (c) it is the spring season (d) the Earth is farthest from the Sun

Q4. A dead zone is a region of:
(a) very high salinity (b) oxygen depletion after an algal bloom (c) extreme cold (d) coral growth

Q5. Which is a cold eastern boundary current?
(a) Gulf Stream (b) Kuroshio (c) Canary (d) Agulhas

Answers: Q1 c, Q2 b, Q3 a, Q4 b, Q5 c

🔤 **ABBREVIATION COLUMN (full forms as the handout gives them where it does; the one-line meanings are plain-language aids, verify)**

AMOC = Atlantic Meridional Overturning Circulation — the Atlantic conveyor: warm surface water north, cold deep water south
AOC = Atlantic (Meridional) Overturning Circulation — the handout's other short form for AMOC
CaCO = $CaCO_3$ = calcium carbonate — what coral skeletons and shells are made of
CEC = Counter Equatorial Current — eastward counter-current
CO_2 = carbon dioxide
EEZ = Exclusive Economic Zone — up to 200 nautical miles: sovereign rights over resources
EUC = Equatorial Undercurrent — eastward flow below the surface along the equator
GHG = greenhouse gas — heat-trapping gas such as carbon dioxide or methane
GLOF = glacial lake outburst flood — sudden flood when a glacier-dammed lake breaks
NaCl = sodium chloride — common salt

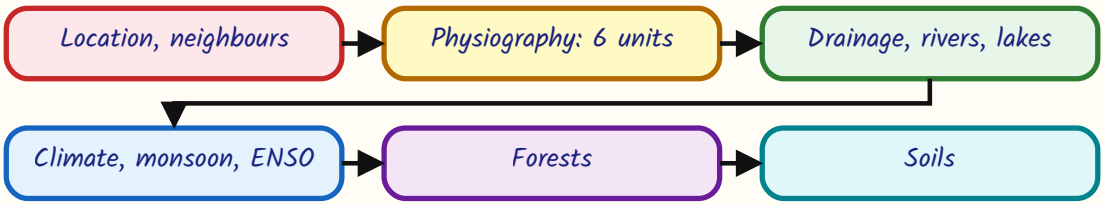
NADW = North Atlantic Deep Water — cold, salty water that sinks in the North Atlantic and flows south at depth
NEC = North Equatorial Current — westward surface current north of the equator
NECC = North Equatorial Counter Current — eastward return flow between the two equatorial currents
nm = nautical mile — sea distance unit; the handout gives 12 nm = 22.2 km
OTEC = ocean thermal energy conversion — power from the temperature gap between warm surface and cold deep water
pH = potential of hydrogen — acidity scale; lower = more acidic
ppt = parts per thousand — unit of salinity
PYQ = previous year question
SST = sea surface temperature
UNCLOS = United Nations Convention on the Law of the Sea — the 1982 treaty that sets maritime zones
US = United States
USA = United States of America
UV = ultraviolet — high-energy part of sunlight

INDIAN GEOGRAPHY

PYQ Prelims 40 PYQs: physiography 15 (2012, 2014 x2, 2015, 2016, 2017 x2, 2018 x2, 2020, 2022, 2023 + 3 undated) · drainage 12 (2011, 2013, 2014, 2015, 2016 x2, 2017, 2021 x2, 2022, 2023 + 1 undated) · climate 3 (2011, 2017 + 1 undated) · forests and soils 10 (2011, 2013, 2015 x3, 2016, 2021, 2023 x2 + 1 undated) · Mains none quoted

✎ *Location, six physiographic units, rivers, climate and monsoon, forests and soils: the largest PYQ bank of the six chapters (40 Prelims), all map-and-list questions.*

At a glance: the Indian Geography route



Pattern: "which pairs / rivers / peaks / states?": tributaries of Godavari and Brahmaputra, hills of which range, which state has what, which forest has which tree. Place every fact on a map. Mains hooks (none quoted): Western Disturbances, cloudbursts, urban flooding, ELLA, river-linking, mangroves, social forestry. No teacher ink: priority = PYQ count. Image-only pages (peaks, glaciers, passes, waterfalls, lakes) are kept below as tables.

1 · LOCATION OF INDIA

PYQ 2015 · PYQ 2018 · PYQ 2017

Understand it first: Fix India on the globe first: its latitudes and longitudes, the Tropic of Cancer cutting it in half, the standard meridian and its size and neighbours. Most other facts (climate, time, coastline) follow from this position.

Fact	Number
Area	32,87,263 km² · 7th largest · 2.4% of world area
Extent	N–S (Indira Col, Kashmir → Kanyakumari) 3,214 km · E–W (Rann of Kutch → Arunachal) 2,933 km
Latitude / longitude	8°4′N (6°45′ for islands) to 37°6′N · 68°7′E to 97°25′E · wholly Northern Hemisphere
Southernmost point	Indira Point (Pygmalion Point), 6°45′N
Time	1° longitude = 4 min · E–W gap ≈ 2 hrs (~30 × 4) · Chai Bagan time = 1 hr ahead of IST (British, NE India, to raise productivity)
Tropic of Cancer	splits India in two; area north ≈ 2× south · 8 states: Gujarat, Rajasthan, MP, CG, Jharkhand, West Bengal, Tripura, Mizoram
Borders	land 15,106.7 km through 17 states · coastline 11,098 km, 13 states/UTs · longest = Bangladesh · shortest = Afghanistan (Wakhan)

🧍🧍 No border / no coast	Telangana, MP, CG, Jharkhand, Delhi, Chandigarh, Haryana; all others = frontline states (border management)

U C My Old Aunt — Standard Meridian 82°30'E passes through 5 states: Uttar Pradesh · Chhattisgarh · Madhya Pradesh · Odisha · Andhra Pradesh (handout mnemonic).

Neighbours (clockwise) → Pakistan · Afghanistan · China · Nepal · Bhutan · Bangladesh · Myanmar · Sri Lanka · Maldives · Indonesia

Neighbour	Bordering Indian states/UTs
Pakistan	J&K, Ladakh, Punjab, Rajasthan, Gujarat
Afghanistan	Ladakh
China	Ladakh, HP, Uttarakhand, Sikkim, Arunachal
Nepal	UP, Uttarakhand, Bihar, Sikkim, West Bengal
Bhutan	Assam, Sikkim, Arunachal, West Bengal
Bangladesh	West Bengal, Assam, Meghalaya, Mizoram, Tripura
Myanmar	Arunachal, Nagaland, Manipur, Mizoram

State / UT	New Length	Old length
1. Gujarat	2340.62	1217.7 ↑
2. Tamil Nadu	1068.69	906.9 ↑
3. Andhra Pradesh	1053.07	973.7 ↑
4. Maharashtra	877.97	652.6 ↑
5. West Bengal	727.02	157.5 ↑
6. Kerala	600.15	569.7 ↑
7. Odisha	574.71	464.4 ↑
8. Karnataka	343.3	280 ↑
9. Goa	193.95	101 ↑

India And Its Neighbours

State / UT	New Length	Old length
10. Andaman & Nicobar	3083.5	1962 ↑
11. Lakshdweep	144.8	132 ↑
12. Daman & Diu Dadra & Nagar Haveli	54.38	42.5 ↑
13. Puducherry (includes all 4 parts)	42.65	47.6 ↓

India And Its Neighbours

India And Its Neighbours

Source doubt: coastline is 11,098 km here, but the cyclone note says "over 7,500 km" (as printed). Use 11,098 km for coastline.

Location Of India

→ India accounts for about 2.4 percent of the total surface area of the world .

→ East-West time difference is nearly 2 hrs . (A difference of 1° longitude will make a difference of 4 minutes in time. ~30 x 4 = ~120 minutes or ~2 hours)

→ The Tropic of Cancer passes through the middle of the country dividing it into two latitudinal halves. The area to the north of the Tropic of Cancer is nearly twice the area that lies to the south of it.

2 · PHYSIOGRAPHY: SIX UNITS PYQ 2012 PYQ 2014 PYQ 2023

Understand it first: India has six physical units. Start with the Himalayas: young fold mountains in parallel ranges from north to south, and also divided west to east by river gaps. Learn the ranges, then the regional divisions, passes and peaks on top of that frame.

My Parents Do Play Cricket Indoors — handout's six units top to bottom: Mountains · Plains · Desert · Plateau · Coasts · Islands (Northern and North-eastern Mountains · Northern Plains · Indian Desert · Peninsular Plateau · Coastal Plains · Islands).

Northern and NE mountains

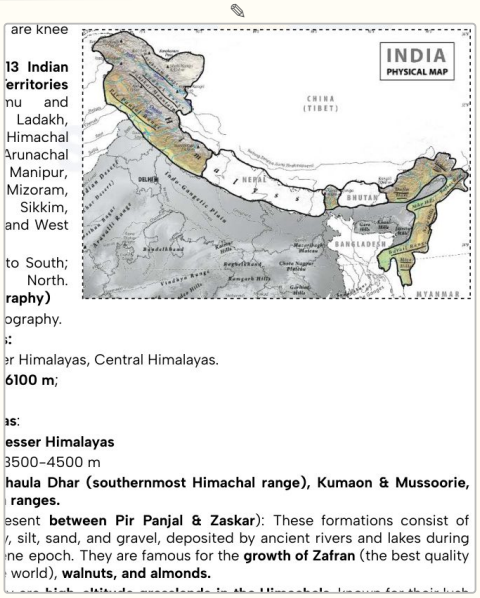
→ Origin: Indian + Eurasian plates converge (Tethys closes); Tethys sediments fold

→ Multiple upliftments = parallel ranges; steep south, gentle north (hogback); ridge-valley

→ 2,500 km long; width 500 km (Kashmir) → 200 km (Arunachal); 13 states/UTs

→ Main Himalaya: Nanga Parbat (Indus gorge) to Namche Barwa (Brahmaputra gorge) = two syntaxial (knee-joint) bends; then Purvanchals

→ Transhumance: seasonal herding to bugyals: Bakarwals, Gaddis, Bhotiyas, Drokpas



The Northern And North-Eastern Mountains · Syntaxial Bends are knee joint bends.

Belt	Also called	Notes
Trans-Himalaya	Karakoram, Ladakh, Zaskar	highest, cold dry · most glaciers in Karakoram · K2 (Godwin-Austen) · Ladakh = highest Indian plateau · Zaskar south of Indus · Kailash entirely outside India
Great Himalaya	Himadri, inner	avg 6,100 m; tallest peaks
Middle Himalaya	Himachal, Lesser	3,500–4,500 m · Pir Panjal, Dhauladhar (southernmost HP range), Kumaon, Mussoorie, Mahabharat · Karewas (clay-silt-sand-gravel of Pleistocene lakes, between Pir Panjal and Zaskar) → saffron (zafran), walnuts, almonds · Bugyals
Outer Himalaya	Shiwaliks	lowest; nearly unbroken, gap of 80–90 km at Teesta valley

Foothills (top to bottom)

→ Bhabar (stones, rivers vanish) → Terai (swamp, rivers reappear) → plains

Duns (W)

→ valleys between Shiwaliks and Himachals (HP, Uttarakhand); longitudinal

→ Dehradun, Patli Dun, Kota Dun

Duars (NE)

→ floodplain foothills north of Brahmaputra valley (Assam, Arunachal)

→ alluvial, fertile, dense vegetation

Chos / Choes / Raos

→ seasonal Shiwalik streams, NW Punjab foothills; sandy beds

→ less fertile than Duars; need irrigation

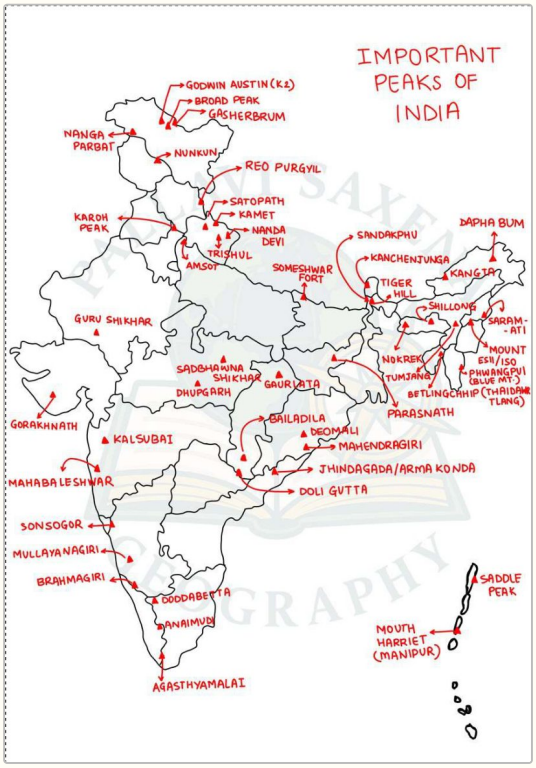
Purvanchals

→ crescent, convex to west; perpendicular to Himalaya; rough, forested = hard to develop

→ Mishmi, Patkai Bum, Naga, Kohima, Manipur hills (Loktak), Barail, Mizo (Lushai)

→ Barail: separates Manipur and Naga hills; divides Brahmaputra and Barak basins

Regional Himalaya	Limits	Peaks / valleys (as printed)
Punjab / Kashmir / Himachal (west)	Indus → Sutlej	Nanga Parbat 8,126 m · K2 8,611 m · Dhaulagiri I 8,167 m · Kashmir, Kullu, Kangra valleys
Kumaon (central)	Sutlej → Kali	Nanda Devi 7,816 m (highest entirely in India) · Nanda Kot 7,818 m · Trishul 7,120 m · Valley of Flowers (UNESCO) · Ganga source
Nepal (eastern)	Kali → Teesta	Everest 8,848 m · Kanchenjunga 8,586 m (3rd highest) · Lhotse 8,516 m · Annapurna I 8,091 m
Assam (easternmost)	Teesta → Arunachal / Purvanchals	Namche Barwa 7,756 m · Manas NP (UNESCO)



Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

	Peak	Characteristics
1.	Nanga Parbat	In Himadris, in Ladakh; Himalayas make a Syntaxial bend here. (> 8000 m high)
2.	Godwin Austin (K2)	In Karakoram Range; 2nd highest peak of the world.
3.	Broad Peak, Gasherbrum 1, Gasherbrum 2.	All three are part of Gasherbrum group of peaks; all are higher than 8000 m.
4.	Nun Kun	Nun in J&K; Kun in Ladakh; Part of Himadris
5.	Reo Purgyil	Highest Peak of H.P. Part of Zanskar Range
6.	Satopath	In Uttarakhand. Satopath Lake at its Foot In Greater Himalayas
7.	Kamet	Highest peak of Uttarakhand & of Zaskar Range
8.	Trishul	Group of 3 peaks (look like a Trishul / Trident)
9.	Nanda Devi	In Greater Himalayas. It is a UNESCO World Heritage Site. Also Nanda Devi National Park
10.	Karoh Peak	Part of Shivaliks. Highest peak of Haryana
11.	Amsot Peak	Part of Shivaliks. Highest peak of U.P.
12.	Someshwar Peak	Part of Shivaliks. Highest peak of Bihar
13.	Kunchenjunga	Highest Himalayan Peak in India. Part of Greater Himalayas. In Sikkim. Presence of Biosphere Reserve, National Park, India's First UNESCO Mixed Heritage site
14	Sandakphu	Highest Peak of W.B. Part of Himachals
15	Tiger Hill	In Darjeeling. Known for Tea Cultivation Part of Himachal Range.
16	Kangto	Part of Greater Himalayas. Highest peak of Arunachal Pradesh.

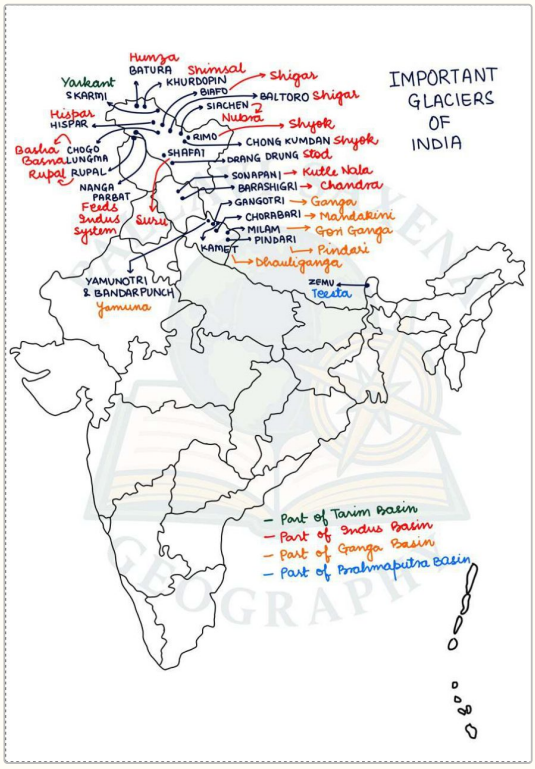
Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

	Peak	Characteristics
17	Dapha Bum	Highest Peak of Purvanchals. Located in Mishmi hills.
18.	Saramati	Highest peak of Nagaland.
19.	Mt. Iso / Iso	Highest peak of Manipur
20.	Phawngpui (Blue Mountain)	Highest peak of of Mizoram
21.	Betlingchhip (Thaidalor Tlang)	Highest peak of Tripura
22.	Nokrek	Highest peak of Garo Range. Also a National Park & a Biosphere Reserve.
23.	Shillong Peak	In Khasi hills. Highest peak of Meghalaya
24.	Tumjang / Sielkal / Thingkubung	In Barail Range. Highest peak of Assam.
25.	Parasnath	Highest peak of Jharkhand. Important Jain pilgrimage site. Part of Chotanagpur Plateau
26.	Guru Shikhar	Highest peak of Aravallis, and of Rajasthan
27.	Gorakhnath	Highest peak of Gujarat. Also an imp. pilgrimage site for Hindus. Located in Girnar hills.
28	Dhupgarh	Highest peak of Satpuras and of M.P.
29.	Sadbhawna Shikhar (Goodwill Peak)	Highest peak of Vindhyas.
30.	Kalsubai	In Western Ghats. Highest peak of Maharashtra
31.	Mahabaleshwar	Part of Western Ghats. Origin region of Krishna river
32.	Sonsogor	In Western Ghats. Highest peak of Goa. Near trijunction of Maharashtra, Goa & Karnataka

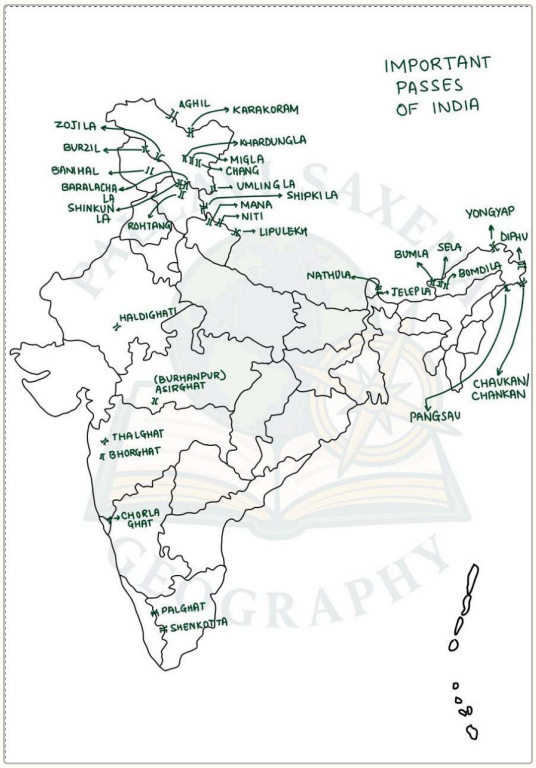
Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

Peak	Characteristics
33. Mullayanagari	Part of Western Ghats. Highest peak of Karnataka.
34. Kudremukh	Important peak for mineral resources. Also a Tiger Reserve & at National Park.
35. Brahmagiri	In Karnataka. Source of Kaveri
36. Doddabetta.	Highest peak of Tamil Nadu and of Nilgiris.
37. Anaimudi	Highest peak of Western ghats. Also highest peak of Kerala.
38. Agastyamalai	Part of Western Ghats. At Tamil Nadu-Kerala border.
39. Saddle Peak	Highest peak of Andaman & Nicobar
40. Mt. Hasiret	Now known as Mt. Manipur
41. Bailadila	Important peak of Chattisgarh. Part of Deccan Plateau
42 Gaurlata	Highest peak of Chattisgarh. Part of Central Highlands.
43. Doliguta	Highest peak of Telangana. Part of Deccan Plateau
44. Mahendragiri	Imp. peak of Eastern Ghats.
45.	Highest peak of Odisha. Part of Eastern Ghats.
46. Arma Konda/ Jhindagada	Highest peak of Eastern Ghats & of Andhra Pradesh

Regional Division of Himalayas · Assam Himalayas (Easternmost Section)



Regional Division of Himalayas · Assam Himalayas (Easternmost Section)



Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

	Mountain Pass	State/UT	Characteristic
1	Agzil	Ladakh	In Karakoram Range Historically used for trade. Military importance
2.	Karakoram	Ladakh	In Karakoram pass. Was part of ancient silk route
3.	Khardung la	Ladakh	connects Leh & Siachen
4	Mig la	Ladakh	world's highest motorable pass.
5.	Chang la	Ladakh	connects Pangong TSO with rest of Ladakh
6.	Umling La	Ladakh	2nd highest motorable road in the world.
7.	Zoji La	Ladakh & JnK	Connects Ladakh and Kashmir valley. Often obstructed by snow
8.	Burzil La	JnK	Currently in PoK. Used for trading activities
9.	Banihal	JnK	Known for Jawahar Tunnel (NH 44)
10.	Shinkun La	Himachal Pradesh	Known for NPD Route (Nimmu - Padum - Darcha) All weather connectivity to Ladakh.
11.	Baralacha La	Himachal Pradesh	Known for Leh-Manali Highway
12.	Rohtang	Himachal Pradesh	Important for Leh connectivity. Atal Tunnel is under Rohtang Pass
13.	Shipki La	Himachal Pradesh	At India - China border. Sutlej enters India from here.

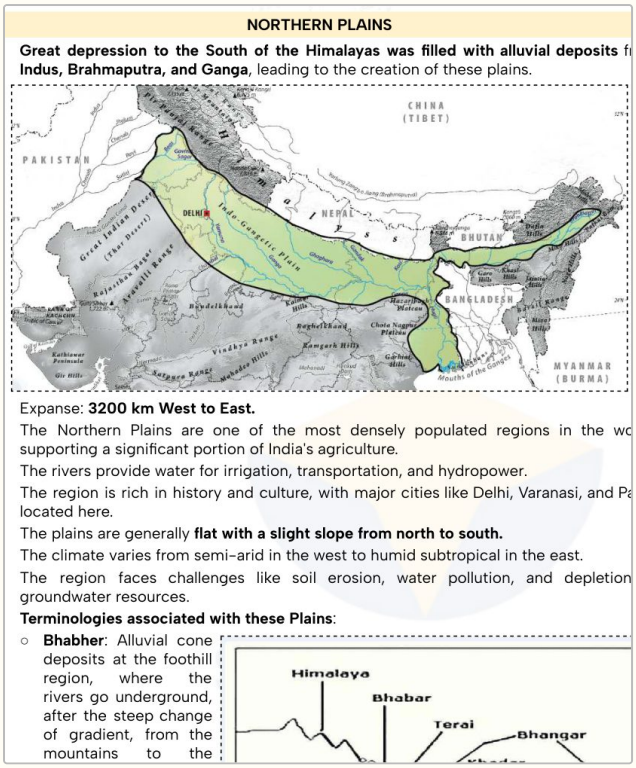
Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

Mountain Pass	State/UT	Characteristic
14. Mana	Uttarakhand	Highest motorable pass of Uttarakhand
15. Niti	Uttarakhand	Northernmost pass of Uttarakhand
16. Lipulekh	Uttarakhand	Nepal claims it, but belongs to India. Used for Mansarovar Yatra.
17. Nathu La	Sikkim	Important for Indo-China trade. Used for Mansarovar Yatra.
18. Jelep la	Sikkim	Connects India & Bhutan
19. Bum la	Arunachal Pradesh	Used by Dalai Lama to enter India in 1959.
20. Sela	Arunachal Pradesh	Sela Tunnel is located here. Important for connecting Tawang with rest of Arunachal Pradesh.
21 Bomdila	Arunachal Pradesh	Helpful for Tawang connectivity. Used by India for the 1962 Indo-China War.
22. Yongyap	Arunachal Pradesh	Near the entry point of Dihang in India. British built the Sitwell Road to connect Myanmar & China through here.
23. Diphu	Arunachal Pradesh	At trijunction of India-China-Myanmar
24. Chaukan	Arunachal Pradesh	Important for Indo-Myanmar connectivity.
25 Pangsau	Arunachal Pradesh	Sitwell road also passes through here.

Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

Mountain Pass	State/UT	Characteristic
26. Haldighati	Rajasthan	In Aravallis. Historically important for Battle between Maharana Pratap & Mughals.
27. Asirgrat / Burhanpur Darra	Madhya Pradesh	In Satpuras. Known for Asirpur fort. Key to Deccan. Connects Northern & Southern India.
28. Thalghat	Maharashtra	Known for Mumbai-Nashik Highway. In Western ghats
29. Bhorghat	Maharashtra	Known for Mumbai-Pune Highway. In Western Ghats
30. Chorla Ghat	Goa	Near Goa-Maharashtra-Karnataka trijunction. In Western Ghats.
31. Palghat / Palakkad	Kerala-Tamil Nadu	Widest gap in Western ghats. At Kerala-Tamil Nadu Border. Gateway of Monsoon in Tamil Nadu. Between Nilgiris in North & Annamalai in South.
32. Shenkotta	Kerala-Tamil Nadu	At Kerala-Tamil Nadu border. In Western Ghats.

Regional Division of Himalayas · Assam Himalayas (Easternmost Section)



Regional Division of Himalayas · Assam Himalayas (Easternmost Section)

Source conflicts: Nanda Devi 7,816 m vs Nanda Kot 7,818 m (so Nanda Devi is "highest entirely in India" only by that label; the 2022 PYQ note says Nanda Devi 7,817 m). Kamet is printed as "highest of Uttarakhand" though Nanda Devi (Uttarakhand) is higher. Learn ~7,816 and rely on the label "highest entirely in India".

Region	Peak: what the handout says
Trans / Himadri	Nanga Parbat (Ladakh Himadri, >8,000 m, syntaxial bend) · K2/Godwin-Austen (Karakoram, 2nd highest in world) · Broad Peak, Gasherbrum I and II (Gasherbrum group, all >8,000 m) · Nun (J&K), Kun (Ladakh): Himadri
HP	Reo Purgyil: highest of HP, Zaskar Range
Uttarakhand	Satopanth (lake at foot) · Kamet: highest of Uttarakhand and of Zaskar · Trishul: 3 peaks like a trident · Nanda Devi: Greater Himalaya, UNESCO World Heritage (with its NP)
Shiwalik peaks	Karoh: highest of Haryana · Amsot: highest of UP · Someshwar: highest of Bihar
👫 East Himalaya	Kanchenjunga: highest Himalayan peak in India, Sikkim, biosphere reserve + NP, India's first UNESCO mixed heritage site · Sandakphu: highest of West Bengal (Darjeeling, tea) · Tiger Hill (Himachal range) · Kangto: highest of Arunachal
Purvanchal / NE	Dapha Bum: highest of Purvanchals (Mishmi) · Saramati: Nagaland · Mt Esii/Iso: Manipur · Phawngpui (Blue Mountain): Mizoram · Betlingchhip (Thaidawr Tlang): Tripura · Nokrek: Garo, NP + biosphere reserve · Shillong Peak: Khasi, highest of Meghalaya · Tumjang/Sielkal/Thingtuibung: Barail, highest of Assam
Peninsula (north → south)	Parasnath: highest of Jharkhand, Jain pilgrimage, Chotanagpur · Guru Shikhar: highest of Aravallis and Rajasthan · Gorakhnath: highest of Gujarat, Girnar hills · Dhupgarh: highest of Satpuras and MP · Sadbhawna Shikhar (Goodwill Peak): highest of Vindhyas · Kalsubai: highest of Maharashtra (W. Ghats) · Mahabaleshwar: origin region of Krishna · Sonsogor: highest of Goa (Maharashtra-Goa-Karnataka trijunction)
Western Ghats south	Mullayanagiri: highest of Karnataka · Kudremukh (minerals, tiger reserve, NP) · Brahmagiri: source of Kaveri · Doddabetta: highest of Tamil Nadu and Nilgiris · Anaimudi: highest of Western Ghats and Kerala · Agasthyamalai: TN-Kerala border
Others	Saddle Peak: highest of Andaman and Nicobar · Mt Harriet (now Mt Manipur) · Bailadila: Chhattisgarh, Deccan · Gavrlata: highest of Chhattisgarh, Central Highlands · Doligutta: highest of Telangana · Mahendragiri: Eastern Ghats · peak No. 45: highest of Odisha, Eastern Ghats (name blank in handout) · Arma Konda / Jindhagada: highest of Eastern Ghats and Andhra Pradesh

Peak pegs — Trishul = trident (3 peaks) · Kamet = the "Kamet-ition winner" of Uttarakhand · Doddabetta = "Dodo-big-bat" of the Nilgiris · Anaimudi = "Anna" is tallest in the Ghats' family (Kerala) · Kalsubai = Maharashtra's "kal-subah" (tomorrow morning) climb.

Glaciers (colour = basin on handout map)

- **Tarim** → Yarkant, Skarmi
- **Indus system**
  - → Hispar, Chogo, Basha, Lungma, Rupal (Nanga Parbat)
  - → Hunza, Batura, Khurdopin, Biafo, Baltoro → Shigar
  - → Siachen, Rimo → Nubra/Shyok; Chong Kumdan, Drang Drung → Shyok/Stod
  - → Shafat · Sonapani → Kutle Nala · Barashigri → Chandra · Suru (Nanga Parbat side)
- **Ganga basin**
  - → Gangotri → Ganga · Chorabari → Mandakini · Milam → Gori Ganga · Pindari → Pindari · Kamet · Dhauliganga
  - → Yamunotri and Bandarpunch → Yamuna
- **Brahmaputra basin** → Zemu → Teesta

Pass	State/UT	Hook
Aghil	Ladakh	Karakoram; historic trade; military
Karakoram	Ladakh	ancient silk route
Khardung La	Ladakh	connects Leh and Siachen
Mig La	Ladakh	printed as "world's highest motorable pass"
Chang La	Ladakh	connects Pangong Tso to rest of Ladakh
Umling La	Ladakh	printed as 2nd highest motorable road
Zoji La	Ladakh–J&K	connects Ladakh and Kashmir valley; snow-blocked
Burzil La	J&K (now PoK)	trade
Banihal	J&K	Jawahar Tunnel (NH44)
Shinku La	HP	NPD route (Nimmu–Padum–Darcha), all-weather to Ladakh
Baralacha La	HP	Leh–Manali highway
Rohtang	HP	Atal Tunnel under it; Leh connectivity
Shipki La	HP	India–China border; Sutlej enters India
Mana	Uttarakhand	highest motorable pass of Uttarakhand
Niti	Uttarakhand	northernmost pass of Uttarakhand
Lipulekh	Uttarakhand	Nepal claims it, handout says India's; Mansarovar Yatra
Nathu La	Sikkim	Indo-China trade; Mansarovar Yatra
Jelep La	Sikkim	connects India and Bhutan
Bum La	Arunachal	Dalai Lama entered India 1959
Sela	Arunachal	Sela Tunnel; connects Tawang
Bomdila	Arunachal	Tawang connectivity; used in 1962 war
Yongyap	Arunachal	near entry of Dihang; British Stilwell Road
Diphu	Arunachal	tri-junction India–China–Myanmar
Chaukan, Pangsau	Arunachal	Indo-Myanmar; Stilwell Road passes Pangsau
Haldighati	Rajasthan	Aravallis; Maharana Pratap vs Mughals
Asirgarh / Burhanpur	MP	Satpuras; Asirgarh fort; key to the Deccan (north-south link)
Thalghat	Maharashtra	Mumbai–Nashik highway (Western Ghats)
Bhorghat	Maharashtra	Mumbai–Pune highway
Chorla Ghat	Goa	near Goa–Maharashtra–Karnataka junction
Palghat (Palakkad)	Kerala–TN	widest gap in Western Ghats; monsoon gateway to TN; between Nilgiris (north) and Annamalai (south)

Shenkotta	Kerala-TN	Western Ghats gap

The Northern and Northeastern Mountains

The Northern And North-Eastern Mountains

→ (hogback topography)
→ – Great Himalayas:
→ · Avg. height: 6100 m ;
→ – Middle Himalayas :
→ · Avg. height: 3500-4500 m
→ · Bugyals : They are high-altitude grasslands in the Himachals , known for their lush greenery, colourful flowers, and grazing pastures.
→ The transhumance activity takes place here , which is the seasonal movement of herders and their livestock between different elevations in search of grazing pastures.
→ During the summer months, herders from lower valleys bring their livestock, such as sheep, goats, and cattle, to graze on the lush grasslands of bugyals situated at higher altitudes. These bugyals provide abundant fodder and grazing opportunities during the warmer months when lower elevations may experience heat and scarcity of forage. In the winter, as bugyals become snow-covered and inaccessible, herders return with their livestock to lower elevations where they can find shelter and fodder.
→ Important tribes involved in this: Bakarwals, Gaddis, Bhotiyas and Drokpas.
→ – Outer Himalayas: Shivaliks
→ · Jammu Hills (Jammu), Dafla, Miri, Abor (Arunachal); Dhang and Dudhwa in Uttarakhand; Churia Ghat in Nepal
→ Soil: Alluvial soil, fertile and good for agriculture.
→ · Duns : (Sometimes Doons)
→ Seasonal streams or rivers found primarily in the foothills of the Shivalik (Siwalik) range. Sandy riverbeds are characteristic.
→ Alternate names: Choes/Raos.
→ Soil: These regions are less fertile compared to Duars , may require irrigation for agriculture.
→ Trans Himalayas:
→ – Comprises rugged and high-altitude terrain with stark landscapes, characterized by barren plateaus, deep valleys, and towering peaks.
→ – They have high altitudes with a cold, dry climate. Vegetation is sparse, and permanent human habitation is limited.
→ – Along with the Himalayas, they act as a climatic barrier , influencing weather patterns in Asia. They are also a source of freshwater through glaciers and rivers, and a region of cultural and spiritual importance for many communities.
→ – Karakoram :
→ · Located in the UT of Ladakh, extending into Afghanistan, Pakistan and China.

Ladakh (highest Indian Plateau)

→ · Situated in the eastern part of Ladakh Union territory , adjacent to the Karakoram Range.
→ · Known for its high-altitude deserts, deep gorges, and Buddhist monasteries.
→ – Zaskar :

→ · Features rugged terrain with deep canyons, glaciers, and remote valleys inhabited by the Zanskar people.

→ – Kailash (entirely outside India):

→ · Located in the western part of the Tibetan Plateau, straddling the border between China and the Tibet Autonomous Region.

→ · Houses Mount Kailash, a sacred peak revered by Hindus, Buddhists, Jains, and Bonos.

→ Purvanchals:

→ – They have very rough terrain, and are full of dense forests, making the region challenging from a developmental perspective.

Regional Division of Himalayas

Kashmir/ Punjab/Himachal Himalayas (Westernmost Section):

→ Significance: This section is known for its scenic beauty, and diverse landscapes with valleys, meadows, and glaciers. It's also a region of cultural and historical significance, with Kashmir Valley being a major tourist destination.

Kumaon Himalayas (Central Section):

→ Significance: This section is known for its high peaks, including Nanda Devi, and for housing pilgrimage sites like the source of the Ganges River. The Valley of Flowers National Park is a UNESCO World Heritage Site known for its vibrant alpine meadows.

Nepal Himalayas (Eastern Section):

→ Significance: This section boasts the highest peaks in the Himalayas, including Everest. It's a major center for mountaineering expeditions and adventure tourism.

→ Nepal is a landlocked country, and the Himalayas play a vital role in its geography and culture.

Assam Himalayas (Easternmost Section):

→ Valleys: Manas National Park

→ Significance: This section is known for its lush green valleys, diverse ecosystems, and abundant rainfall. Manas National Park is a UNESCO World Heritage Site known for its rich biodiversity.

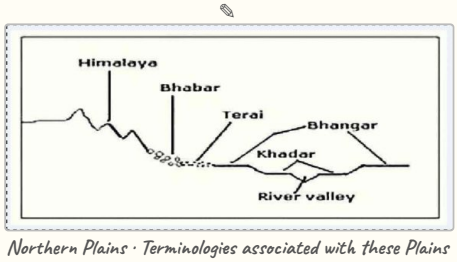
NORTHERN PLAINS

Understand it first: The Northern Plains are the fill: sediment brought by the Indus, Ganga and Brahmaputra systems into the trough south of the Himalayas. From the foothills outward the belts change with the size of the deposit: coarse first, fine later.

Northern plains

→ Great depression south of Himalaya filled by Indus, Ganga, Brahmaputra alluvium; 3,200 km W to E

→ Flat, slight slope north to south; one of the most densely populated regions; rivers give irrigation, transport, hydropower; Delhi, Varanasi, Patna



Northern Plains · Terminologies associated with these Plains

→ Climate: semi-arid (west) → humid subtropical (east)

→ Problems: soil erosion, water pollution, groundwater depletion

Four sections

→ Rajasthan plain: bagar, saline lakes

→ Punjab–Haryana: Bari and Bist doab, canals

→ Ganga plain: Yamuna eastward; Rohilkhand, Awadh, Mithila, Magadha, Bengal; delta

→ Brahmaputra plain: chars / point bars, floods, tea, Sundarbans delta

Term	Meaning
Bhabar	alluvial cones at foothills; rivers vanish
Terai	swampy; rivers re-emerge
Bhangar	older coarse alluvium away from channel; kankar nodules; less fertile
Khadar	newer alluvium near rivers; renewed yearly; very fertile
Reh / Kallar	barren saline patches (UP, Haryana); spread by over-irrigation
Bhur	elevated land along the Ganga (Ganga-Yamuna doab)
Bagar	semi-arid plain from Aravalli foothills, Rajasthan; Rohi = fertile patches in it
Doab	between two rivers: Bari, Bist, Rechna, Chaj
Betlands	Khadar plains of Punjab

Old Bhangar, New Khadar – Bhang-ar = bhang, old and coarse, kankar. Khad = manure: renewed every year, super fertile.

Northern Plains

→ Expanse: 3200 km West to East.

→ The Northern Plains are one of the most densely populated regions in the world, supporting a significant portion of India's agriculture.

→ The region is rich in history and culture, with major cities like Delhi, Varanasi, and Patna located here.

Terminologies associated with these Plains

→ Bhabher : Alluvial cone deposits at the foothill region, where the rivers go underground, after the steep change of gradient, from the mountains to the plains.

→ Reh/Kallar : Barren saline drier areas found in UP, Haryana. They are extending beyond their natural zone due to bad agricultural practices, like excessive irrigation.

→ Doab : Land between 2 rivers. Examples: Bari, Bist, Rechna, Chaj

Rajasthan Plains:

→ Location: Westernmost part of the Northern Plains, including the Rajasthan bagar, which extends into the Thar Desert.

Punjab Haryana Plains:

→ Characteristics: Fertile land, ideal for agriculture (wheat, rice, cotton), network of canals for irrigation.

Ganga Plains:

→ Location: The most extensive part, stretching from Yamuna River eastwards to Bangladesh.

→ Characteristics: Highly fertile land due to Ganga-Yamuna Doab, densely populated, diverse agriculture (wheat, rice, sugarcane, pulses), home to the Ganges River Delta.

Brahmaputra Plains:

→ Location: Easternmost part , bordering Bangladesh.

→ Characteristics: Riverine islands (Point bars), also called chars , prone to flooding, lush green due to high rainfall, Brahmaputra River Delta Sundarbans. These plains, in the foothills, are also known for known for tea plantations.

INDIAN DESERT: THAR

Understand it first: The Thar lies in the rain shadow of the Aravallis. Learn its dry landforms and its one inland drainage story.

→ Cause: sub-tropical high pressure + rain shadow of Aravallis; extension of Rajasthan plains

Thar (Great Indian Desert; local Marusthali)	→ ~2,00,000 km²; 85% India, 15% Pakistan; Rajasthan, Gujarat, Punjab, Haryana → Borders: Indus plain (west) · Punjab plain (north, NE) · Aravalli (SE) · Rann of Kutch (south); natural India–Pakistan border → Features: dunes (some >150 m), sandy plains (Indira Gandhi Canal), low barren hills bhakars, seasonal "rivulets", saline lakes Sambhar, Didwana → Source of salt, limestone, gypsum

The Indian Desert: Thar

→ The Thar Desert, also known as the Great Indian Desert, is a vast arid region straddling the northwestern part of the Indian subcontinent.

→ Created due to a combination of the Sub-Tropical High Pressure belt, and this location being in the rainshadow zone of the Aravallis.

→ Covers an area of approximately 200,000 square kilometers .

→ The Aravalli Range acts as a barrier to the southeast.

The desert landscape is dominated by	→ Sand dunes : These are constantly shifting due to wind action, with some reaching heights of over 150 meters (500 ft). → Sandy plains : These plains separate the dune fields. Here the Indira Gandhi Canal Project has provided the irrigational facilities, providing people with a source of employment. → The landscape is punctuated by intermittent rivers, seasonal streams (known as "Rivulets"), and saline lakes (such as the Sambhar Lake, Didwana, etc.).

PENINSULAR PLATEAU PYQ 2023

Understand it first: The Peninsular Plateau is the old, stable block of India: a tilted tableland of hard rock, bounded by hill ranges, higher in the west and sloping east, which is why most of its big rivers flow to the Bay of Bengal.

Peninsular plateau	→ Oldest and largest unit; irregular triangle; broken from Gondwanaland; rich in metallic minerals; includes Deccan Traps → Meghalaya plateau cut off by Rajmahal–Garo (Malda) Gap / Ganga Suture Zone during northward drift → Two halves: Central Highlands and Deccan Plateau, split by the Narmada rift

Parts of the Peninsular Plateau

Part	Where / what
Mewar / Marwar	NW, Rajasthan, near Aravallis; scrub
Malwa	Aravallis (NE) to Bundelkhand (E) to Vindhyas (S); Chambal badlands; black soil, cotton
Bundelkhand	between Vindhyas and Ganga plain (MP, UP); sal, teak, kheria
Baghelkhand	north Deccan, MP/CG; forests; coal
Chotanagpur (Eastern Highlands)	Hazaribagh, Ranchi, Rajmahal Hills; coal, mica; sal
Meghalaya (NE tip)	Garo, Khasi, Jaintia, Mikir, Rengma; Mawsynram wettest on Earth
Deccan	largest, central; avg 600 m; Deccan Traps; Chhattisgarh plateau and plains (Dandakaranya, Bastar; Mahanadi rice bowl)

Range	Facts (as printed)
Aravalli	relict of world's oldest mountains · NE edge = Delhi Ridge · highest Guru Shikhar · Gujarat, Rajasthan, Haryana, Delhi
Vindhyas	Jobat (Gujarat) → Sasaram (Bihar); east as Bharner and Kaimur · Gujarat, MP, UP, CG, Bihar
Satpuras	higher than Vindhyas · Rajpipla, Mahadev, Maikala · Dhupgarh highest · Gujarat, Maharashtra, MP, CG
Western Ghats (Sahyadris)	Satmala, Harishchandra, Balaghat, Ajanta, Baba Budan, Nilgiri, Annamalai, Cardamom, Palani · climatic AND drainage divide · 6 states: Gujarat, Maharashtra, Goa, Karnataka, Kerala, TN
Eastern Ghats	Garjhat, Mahendragiri, Eramalla, Nallamalla, Seshachalam, Velikonda, Palkonda, Javadi, Shevroy, Kalrayan, Biligiri Rangan · highly dissected by rivers · highest Jindhagada (Araku) · Odisha, Telangana, AP, Karnataka, TN
Meeting point	Eastern and Western Ghats meet at the Nilgiris

Thal-Bhor-Chorla-Pal-Shenkotta — Western Ghats gaps north → south: Thalghat · Bhorghat · Chorla · Palghat · Shenkotta. Ghats compare: Western = Wall (continuous, high, wet windward, divide); Eastern = Entry-gate pieces (broken by rivers, low, drier, scrub).

Peninsular Plateau Of India

→ An ancient landmass formed by the breakup of Gondwanaland.

Parts of the Peninsular Plateau

Mewar/ Marwar Uplands:

→ Characteristics: Arid and semi-arid climate with scarce rainfall, supports scrub vegetation and hardy desert plants. The Thar Desert borders its western flank.

Malwa Plateau:

→ Characteristics: Known for its fertile black soil (regur), ideal for cotton cultivation. The network of rivers like Narmada and Chambal provides water resources for agriculture.vegetation.

Bundelkhand Uplands:

→ Location: Between the Vindhya Range and the Ganga Plains, East of Malwa Plateau, covering parts of Madhya Pradesh and Uttar Pradesh.

→ Characteristics: Rugged terrain with a mix of hills, valleys, and plateaus. The climate supports mixed deciduous forests with trees like sal, teak, and kheria.

Baghelkhand:

→ Characteristics: Hilly terrain with dense forests of sal, teak, and bamboo. The region is rich in mineral resources, particularly coal.

Eastern Highlands:

→ Characteristics: Rolling hills with a network of rivers. Dense forests with sal trees dominate the landscape. The region is known for its mineral wealth, including coal and mica.

→ North-Eastern Tip: Meghalaya (Garo, Khasi, Jaintia, Mikir and Rengma Hills):

→ – Location: Northeastern tip of the plateau, bordering Bangladesh, in Meghalaya and Assam.

→ – Characteristics: Home to Mawsynram, the wettest place on Earth. The region receives heavy rainfall, supporting lush green forests with diverse flora and fauna. The region is known for its unique tribal cultures.

Deccan Plateau:

→ It's called Regur in Maharashtra and Malnads in Karnataka.

→ Characteristics: A vast, relatively flat tableland with an average elevation of 600 meters. The landscape varies from forests and grasslands to rocky outcrops. The Deccan Traps, lava formations from volcanic eruptions millions of years ago, are a distinctive feature.

→ Chhattisgarh Plateau & Plains:

→ – Also has fertile plains formed by the Mahanadi River and its tributaries. The fertile soil supports rice cultivation, making it a vital agricultural region.

→ Western Ghats:

→ – Location: Along the western coast of India, running parallel to the Arabian Sea.

→ – Important gaps include Thalghat (aka Kasara ghat; Mumbai-Nasik Highway), Bhorghat (Mumbai-Pune highway), Chorla Ghat (Goa), Palghat (Palakkad) & Shenkotta gaps (Gateway of monsoon to Tamil Nadu).

→ – Tallest peak Anaimudi

→ – Characteristics: The mountain range acts as a barrier, causing heavy rainfall on the windward side due to the southwest monsoon. The region is known for its lush green forests, rich biodiversity, and numerous waterfalls. It's a source of several major rivers flowing eastward, and many minor west flowing rivers. So it is both a climatic and a drainage divide.

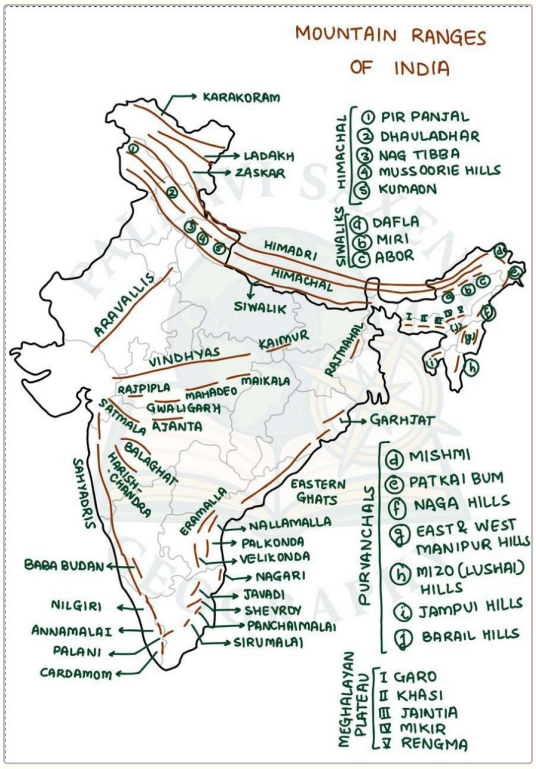
→ Eastern Ghats:

→ – Location: Along the eastern coast of India, running parallel to the Bay of Bengal.

→ – The highest peak is Jindhagada (Araku Valley, Andhra Pradesh).

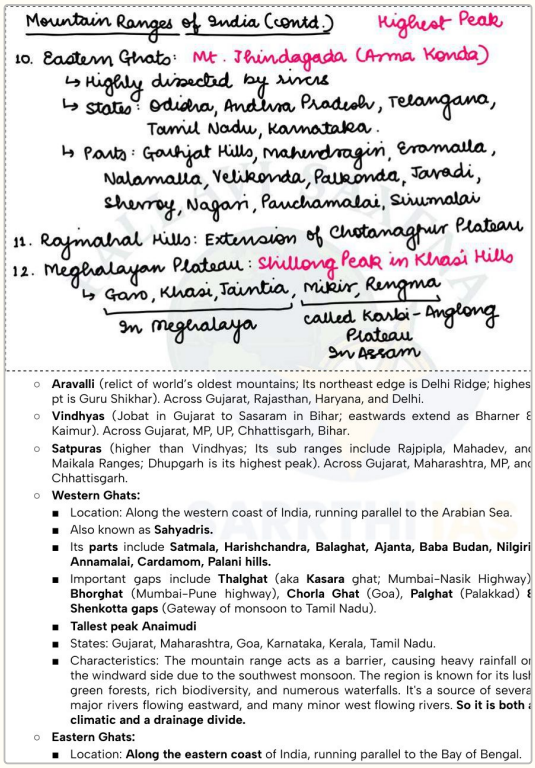
→ – States: Odisha, Telangana, Andhra Pradesh, Karnataka, and Tamil Nadu.

→ – Characteristics: Less continuous and lower in elevation compared to the Western Ghats. The rainfall is lower, resulting in drier ecosystems with deciduous forests and scrub vegetation. The Eastern Ghats play a crucial role in draining many major rivers of the Indian Peninsula.



Parts of the Peninsular Plateau · Important Hills Ranges

Parts of the Peninsular Plateau · Important Hills Ranges

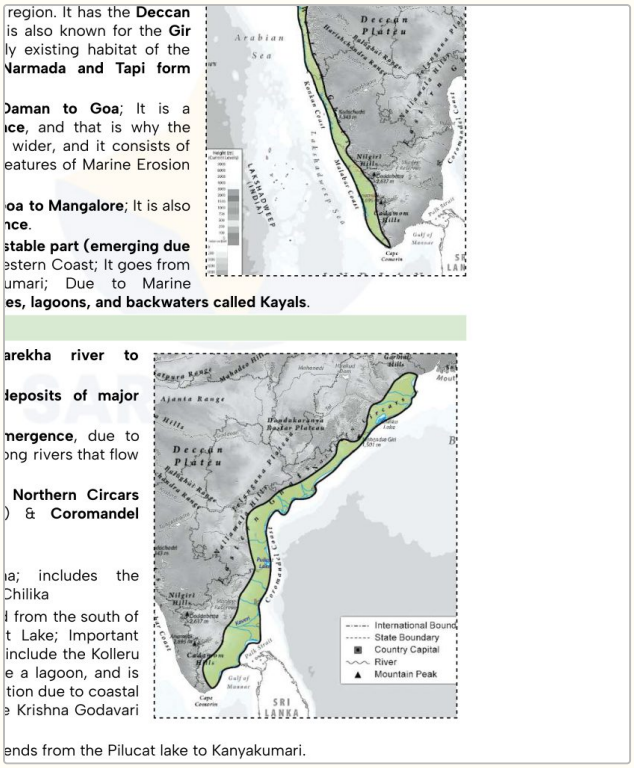


Parts of the Peninsular Plateau · Important Hills Ranges

COASTAL PLAINS PYQ 2022

Understand it first: Compare the two coasts as a pair: the **western** one is narrow, with estuaries and backwaters; the **eastern** one is broad, with deltas. Learn the local names section by section.

Coast	Facts
Western plain	mixed shoreline (emergence, submergence, stable)
Kachchh	emergence (tectonic + depositional) → Rann of Kachchh salt marsh; Banas and Luni inland drainage; Banni grassland (one of India's largest)
Kathiawar (Saurashtra)	Deccan Trap black soil; Gir: only Asiatic lions; Narmada, Tapi form estuaries
Konkan (Daman–Goa)	submergence; wider shelf; sea cliffs, marine erosion
Karnataka plain (Goa–Mangalore)	submergence
Malabar (Mangalore–Kanyakumari)	stable (deposition); kayals = lakes, lagoons, backwaters
Eastern plain	Subarnarekha → Kanyakumari; emergence by deposition (long rivers into Bay of Bengal); Northern Circars (Mahanadi–Krishna) + Coromandel (south of Krishna)
Utkal / Andhra / TN	Chilika (biggest lake, Utkal) · Kolleru (once lagoon, now inland) + Krishna–Godavari delta · plain ends at Pulicat → Kanyakumari



Western Coastal Plains

Eastern and western coasts meet at Kanyakumari. ⭐

Eastern Coastal Plains

→ They include alluvial deposits of major rivers .

→ 2 major parts include Northern Circars (Mahanadi to Krishna) & Coromandel Coast (South of Krishna)

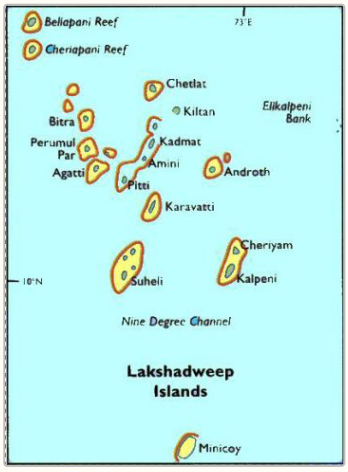
Regionally divided into → Tamil Nadu Plain : Extends from the Pilucat lake to Kanyakumari.

🎯 ISLANDS PYQ 2014 PYQ 2018

Understand it first: Two island groups with different origins: the Andaman and Nicobar group is the top of a submerged mountain chain (with volcanoes), the Lakshadweep group is built of coral. Channels and the named points are the usual exam targets.

Islands	Facts
Andaman and Nicobar	500+ islands, 6°–13°N, 92°–93°E, Bay of Bengal · extension of Arakan Yoma (plate collision; a few volcanic) · Saddle Peak highest · 10° Channel separates Andaman and Nicobar · 6° Channel: Great Nicobar–Sumatra · Barren Island = only active lava volcano in South Asia (Indian plate under Burmese); Narcondam extinct · Baratang = only active mud volcano · Havelock (Swaraj Dweep) largest
Lakshadweep	36 coral islands and reefs, ~400 km off Kerala, Arabian Sea · coral atolls on submerged volcanic Lakshadweep–Chagos Ridge; low, sandy, no rivers or permanent freshwater · Andrott largest · Minicoy (Maliku) southernmost, split by 9° Channel

Andaman & Nicobar Islands



Andaman & Nicobar Islands

10 - 9 - 6 — Channels = own number: 10° (Andaman | Nicobar) · 9° (Lakshadweep | Minicoy) · 6° (Great Nicobar | Sumatra): count down from the north.

Andaman & Nicobar Islands

→ Emerged due to the collision of tectonic plates millions of years ago. They are an extension of the Burmese mountain range Arakan Yoma, including a few volcanic islands as well.

Lakshadweep Islands

→ Landforms: Flat and sandy islands with surrounding coral reefs.

3 · DRAINAGE

PYQ 2011 · PYQ 2013 · PYQ 2014 · PYQ 2015 · PYQ 2016 · PYQ 2017 · PYQ 2021 · PYQ 2022 · PYQ 2023

Understand it first: Drainage is the pattern rivers make. Two families: **Himalayan** rivers (snow-fed, perennial, long, still cutting) and **Peninsular** rivers (rain-fed, seasonal, old and settled). Learn the vocabulary first, then each system with source, course and tributaries.

Factors governing drainage

→ Topography (slope, relief)
→ Nature of rocks (basalt low permeability; granite impermeable)
→ Amount of water (rain + snow)
→ Periodicity (perennial / seasonal)
→ Geological time
→ Anthropogenic: deforestation, urbanisation, dams

Terms

→ Catchment (drainage basin / watershed): whole land area draining to one point
→ Drainage divide: elevated landform separating catchments
→ Basin size: major >20,000 km² · medium 2,000–20,000 · minor <2,000
→ Inland drainage: does not reach sea (Kachchh, shoreline of emergence)

Type	Meaning	Example
Consequent	flows down original slope	Himalayan streams
Subsequent	tributary on softer rock	Appalachian
Antecedent	river older than the uplift; cuts gorges	Indus, Brahmaputra (through Himalaya); Ghaghara
Superimposed	pattern from older landscape laid onto new rocks	Subarnarekha
Concordant / Discordant	follows structure (stable regions) / does not (uplift, erosion past)	

Big numbers

→ 77% drains to Bay of Bengal · 23% to Arabian Sea
→ Divide: Delhi Ridge – Aravallis – Vindhyas – Satpuras – Western Ghats
→ Himalayan rivers 40–60 mn yrs; peninsular 200 mn yrs
→ Basin areas: Indus 11.78, Ganga 8.61, Brahmaputra 5.8 lakh km²; Godavari 3.12 lakh km² = largest peninsular basin

	HIMALAYAN rivers = teenagers	PENINSULAR rivers = grandparents
Age, valley	young, V-valleys; shift channels (avulsion, e.g. Kosi); origin: Shiwalik river divided into Indus and Brahmaputra	mature (~200 mn yrs), broad shallow graded valleys
Flow	perennial (snow + rain); meanders; big deltas; flooding	seasonal (except Kaveri); straight courses; small basins; Narmada and Tapi make estuaries, others deltas

Factors Governing Drainage Of A Location

Topography:

→ Slope and Relief: Steeper slopes and higher relief lead to faster drainage. Water flows rapidly downhill, creating well-defined channels and rivers. (Example: The Himalayas with their steep slopes have fast-flowing rivers like the Indus and Brahmaputra)

Nature of Rocks:

→ Permeability : Permeable rocks like sandstone allow water to seep through, reducing surface runoff and contributing to groundwater recharge. (Example: The Deccan Plateau with its basalt rocks has low permeability, leading to seasonal streams and some areas with poor drainage). Impermeable rocks like granite don't allow water to pass through, resulting in higher surface runoff and potentially flash floods. (Example: The Aravalli Range with its granitic rocks has a limited capacity for water infiltration, leading to seasonal rivers)

Amount of Water:

→ Rainfall : Areas with high rainfall will naturally have more water to drain. Heavy precipitation can overwhelm drainage systems, leading to flooding. (Example: The Western Ghats receive heavy rainfall, leading to numerous perennial rivers)

→ Snowfall: Melting snow in mountainous regions contributes significantly to river flow, impacting seasonal drainage patterns. (Example: The Himalayan rivers like the Ganga and Yamuna experience increased flow during the spring thaw)

Periodicity of Flow:

→ Perennial Rivers: These rivers flow continuously throughout the year, fed by constant sources like glaciers, springs, or consistent rainfall. (Example: The major rivers of India like the Ganga, Brahmaputra, and Indus are perennial)

→ Seasonal Rivers: These rivers flow only during specific seasons, typically during the rainy season. (Example: Many rivers in the Deccan Plateau are seasonal, flowing only during the monsoon months)

Geological Time:

→ Long-term processes like tectonic activity, erosion, and deposition can shape the landscape and influence drainage patterns. (Example: The uplift of the Himalayas millions of years ago created the current drainage pattern of major rivers flowing southwards)

Anthropogenic Activities:

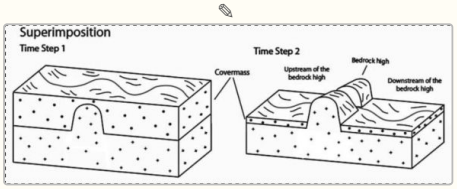
→ Deforestation : Cutting down trees reduces water infiltration and increases surface runoff, leading to flash floods and soil erosion.

→ Urbanization : Paving over land with concrete reduces natural drainage and increases the burden on storm drains. (Example: Rapid urbanization in Indian cities like Delhi and Mumbai has placed stress on drainage systems during heavy rainfall)

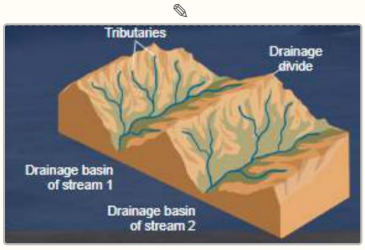
→ **Dams and Reservoirs** : These structures can regulate water flow, reduce flooding, and provide irrigation water, but can also disrupt natural drainage patterns and ecosystems downstream. (Example: The Bhakra Nangal Dam on the Sutlej River in Punjab helps control floods and provides water for irrigation)

Terminologies Related To Drainage

→ (Drainage patterns already covered in Geomorphology).



Terminologies Related To Drainage · Drainage Types



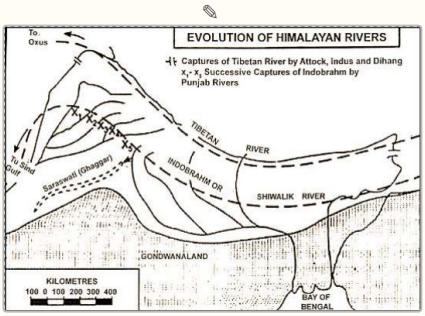
Terminologies Related To Drainage

Drainage Types:

→ **Concordant drainage systems suggest a long-term evolution** where the rivers have carved their paths in harmony with the landscape. It is found in regions with tectonic stability.

River Systems In India

→ **River basins are divided based on their size** into: Major River basins: >20,000 sq. km; Medium: 2000-20000 sq. km; Minor: <2000 sq. km.



River Systems In India

HIMALAYAN RIVERS: INDUS PYQ 2021

Understand it first: For every river hold the same five things: source, course, right-bank and left-bank tributaries, and mouth. The Indus system is tested through its five Punjab tributaries and their order.

Indus

→ Origin **Bokhar Chu**, Kailash; **3,180 km**, rises W Tibet; in India only **Ladakh**
→ Riparian: **Ladakh, J&K, HP, Punjab**; gorge near **Nanga Parbat**
→ **Panjnad = I-J-K-R-B-S** (left tributaries, as printed)
→ Right bank: **Shyok** (Rimo glacier, Siachen tongue; joins at **Keris**, PoK), Gilgit · Others: Zanskar, Suru

River	Origin / hook
Jhelum	Verinag; meanders; Wular Lake; joins Chenab in Pakistan
Chenab	near Baralacha La (Lahaul-Spiti); joins Sutlej in Pakistan
Ravi	west of Rohtang, Kullu hills; joins Chenab in Pakistan
Beas	Beas Kund near Rohtang; gorges at Kati and Largi (Dhauladhar); entirely in India; meets Sutlej in Punjab
Sutlej	Rakshasthal near Mansarovar; enters India at Shipki La; joins Indus in Pakistan as Panjnad

GANGA SYSTEM

Understand it first: The Ganga is built from headstreams that join one by one at the prayags, then collects Himalayan tributaries from the left and both Himalayan and Peninsular ones from the right. Bank-wise lists are the exam favourite.

Ganga

→ Origin: Gangotri (Bhagirathi) + Satopanth (Alaknanda); Uttarakhand, UP, Bihar, Jharkhand, WB; Padma in Bangladesh

→ NW-1: Haldia → Prayagraj

Panch Prayag (up → down) → Vishnu · Nand · Karna · Rudra · Dev

Feeders → Dhauli (Ganga) · Nandakini · Pindar · Mandakini · Bhagirathi

Vasudev Ne Karna ko Rasta Dikhaya — Panch Prayag: Vishnuprayag · Nandprayag · Karnaprayag · Rudraprayag · Devprayag. Feeders: Dhul Nandakini ke Pair Mein Bikhri = Dhauli(Ganga) · Nandakini · Pindar · Mandakini · Bhagirathi.

Rama Ghoomte hue Ghaghara Ganda Kar Mat — LEFT-bank tributaries: Ramganga · Gomti · Ghaghara · Gandak · Kosi · Mahananda (+ Burhi Gandak between Gandak and Kosi).

Left bank	Facts
Ramganga	Chamoli; meets Ganga at Kannauj
Gomti	Pilibhit; meets Ganga at Ghazipur
Ghaghara	Tibet; antecedent; joins in Bihar; tribs Sarda, Saryu, Rapti, Choti Gandak
Gandak	Tibet–Nepal border; Kali Gandak, Mayangadi, Trishuli; joins near Patna
Burhi Gandak	West Champaran; Kanwar Tal (India's biggest oxbow; Sikrahna)
Kosi	Sapt Kosi (7 tribs: Arun, Tumar, Sun Kosi); Sorrow of Bihar (shifting course); Nepal; joins near Kursela
Mahananda	near Tiger Hill; "last major right-bank tributary" to join Ganga in India (handout wording)
Right bank	Yamuna (Yamunotri; joins at Prayagraj; Chambal, Sind, Ken, Betwa) · Tons/Tamsa · Son (Amarkantak, radial drainage with Narmada, Johila; joins near Patna) · Phalgu (Gayaji)
Distributary	Hooghly (Giria → Sagar Island, estuary) · Damodar (Chotanagpur → Hooghly near Kolkata) · Mayurakshi, Ajay

Source conflict: Mahananda sits in the left-bank list but is described as the "last major right bank tributary". Treat it as printed (left list).

→ Brahmaputra And Barak: The Brahmaputra changes its name along its course and brings a very heavy load of silt, which makes it braided and flood-prone. The Barak is the separate system to its south.

BRAHMAPUTRA AND BARAK

PYQ 2011 PYQ 2014 PYQ 2016 PYQ 2017

Understand it first: The Brahmaputra changes its name along its course and brings a very heavy load of silt, which makes it braided and flood-prone. The Barak is the separate system to its south.

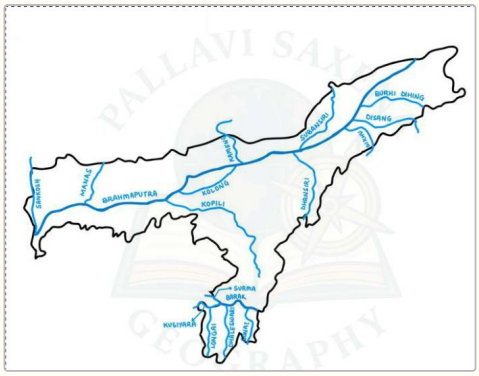
Brahmaputra ("son of Brahma"; Luit, Dilao)

→ Source Chemayungdung glacier (also Angsi), SW Tibet, Kailash; near sources of Indus and Sutlej; Mariam La

separates it from Manasarovar; Indus–Tsangpo Structure Zone between Great Himalaya and Kailas Range

→ U-turn at Namcha Barwa → enters as Dihang (Siang); Lohit and Dibang join; Sadiya → Dhubri

→ Name chain: Tsangpo – Dihang – Brahmaputra – Jamuna – Padma – Meghna



Forms Majuli Island where Subansiri joins.

→ Majuli island where Subansiri joins; Assam floods

Tributaries

→ Right: Subansiri · Kameng · Manas · Beki · Raidak · Jaldhaka · Teesta (+ Sankosh)

→ Left: Lhasa · Nyang · Parlung Zangbo · Lohit · Dibang · Burhi Dihing · Dhansiri · Kolong

SKM-ST – Brahmaputra right-bank tributaries: Subansiri · Kameng · Manas · Sankosh · Teesta.

Teesta (2017)

→ 315 km; Pahunri (Teesta Kangse) glacier, above 7,068 m; Sikkim, West Bengal, Bangladesh, Bay of Bengal

→ Fed by Thangu, Yumthang, Donkha ranges; Rangpo joins; border of Sikkim–WB to Teesta Bazaar

→ Main tributary Rangeet (originates in Sikkim)

→ Source is not the same as Brahmaputra (originates in Sikkim)

Barak

→ 900 km; Manipur hills; enters plains near Lakhipur; Manipur, Nagaland, Mizoram, Assam

→ Splits into Surma + Kushiyara → Meghna (after meeting Padma, Ganga + Brahmaputra flow)

→ Separated from Brahmaputra by Barail Hills; inland waterway; rich flora and fauna

→ States it moves through: Arunachal Pradesh, Assam

→ Experiences a shifting river course and causes flooding in Assam, especially during the monsoon season.

4. Barak River:

→ Called Meghana in Bangladesh after it merges with Padma.

→ States: Manipur & Assam. Other riparian states: Meghalaya, Nagaland, Mizoram, Tripura.

PENINSULAR DRAINAGE

PYQ 2013 PYQ 2015 PYQ 2021 PYQ 2022 PYQ 2023

Understand it first: Peninsular rivers split by direction: most flow **east** into the Bay of Bengal and build deltas; the Narmada and Tapi flow **west** through rift valleys and form estuaries. Then inland drainage, river-linking and lakes.

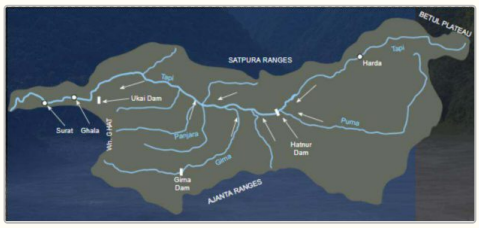
Peninsular rivers

→ By direction: East-flowing Mahanadi, Godavari, Krishna, Kaveri · West-flowing Narmada, Tapi · NE-flowing Chambal, Ken

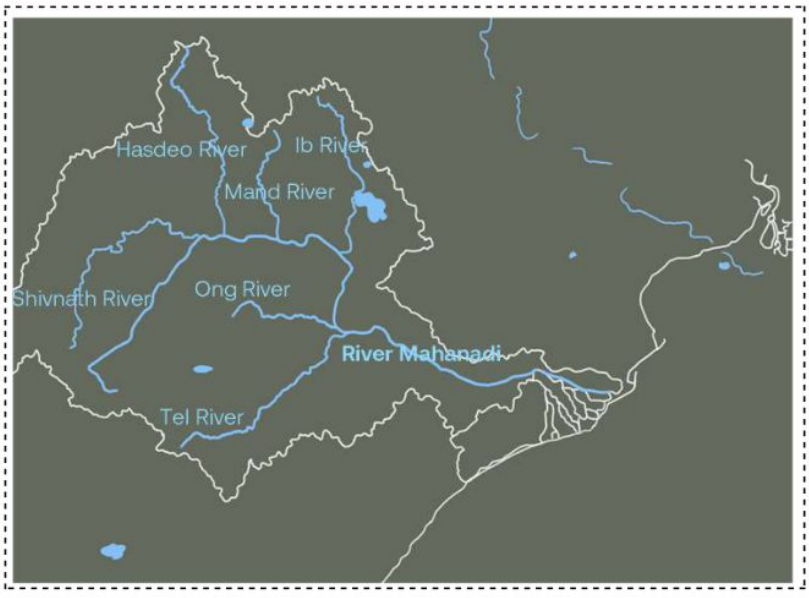
→ ~200 mn yrs, mature, broad shallow graded valleys

South-Eastward flowing rivers · Godavari River

West Flowing Rivers

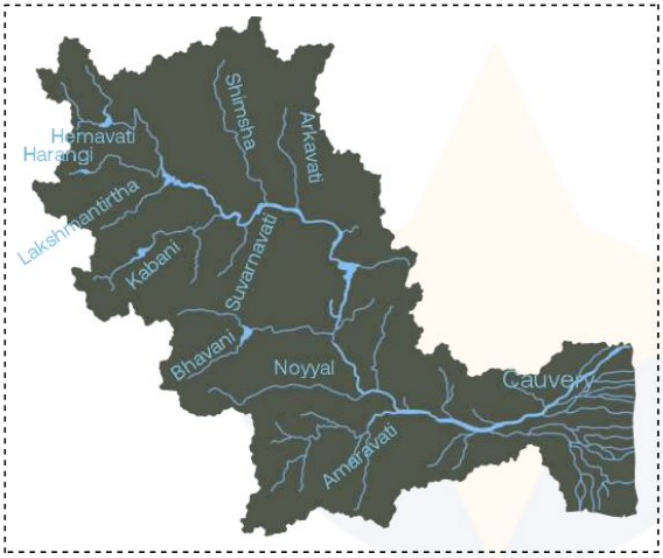


West Flowing Rivers · Tapi



South-Eastward flowing rivers

- Kaveri River:



  - Origin: Brahmagiri Hills, Kodagu, Karnataka.
  - **Perennial** river which gets fed by both southwest monsoon (in its upper course) and

South-Eastward flowing rivers · Kaveri River

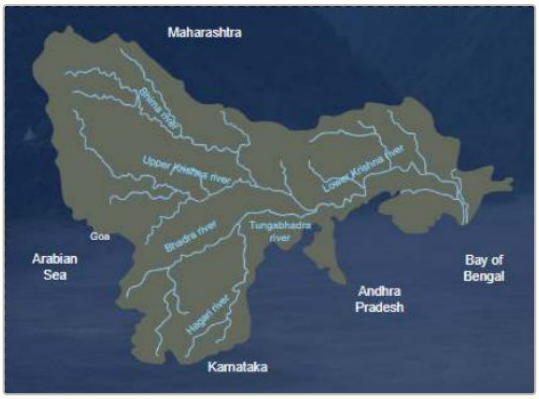
Right Bank	Left Bank
• Lakshmana Tirtha • Kabini • Suvarnavathi • Bhavani • **Noyall** • **Amaravati**	• Harangi • Hemavati • **Shimsha** • Arkavati

South-Eastward flowing rivers · Kaveri River

Seonath Man Hasdo On the Joke I Told — Mahanadi: left Seonath · Mand · Hasdo · right Ong · Jonk · Tel · and Ib (left). ("I" = Ib, "Told" = Tel.)

River	Facts
Mahanadi	Sihawa, Raipur (CG); CG + Odisha (53% / 47% as printed); Hirakud Dam; gorge near Tikkarpara; delta
Godavari	Longest peninsular; Vridha / Dakshin Ganga; 2nd longest river of India after Ganga; Trimbakeshwar (Nashik); Maharashtra, Telangana, CG, AP, Yanam · right: Manjara, Pravara, Manair (16.1% of basin) · left: Purna, Pranhita (Wainganga + Penganga), Indravati, Sabari (59.7%); below Rajahmundry splits into Gautami + Vashishtha; KG delta
Krishna	Mahabaleshwar; Maharashtra, Karnataka, Telangana, AP · right: Koyna, Panchganga, Dudhganga, Ghataprabha, Malaprabha, Tungabhadra · left: Bhima, Dindi, Halia, Musi
Kaveri	Brahmagiri (Kodagu); perennial (SW monsoon upper + NE monsoon lower); Kerala 3%, Karnataka 41%, TN 56% (as printed); delta; splits at Tiruchirapalli into Kollidam + Kaveri, rejoin at Srirangam island · right: Lakshmana Tirtha, Kabini, Suvarnavathi, Bhavani, Noyyal, Amaravati · left: Harangi, Hemavati, Shimsha, Arkavati

Others (east)	Brahmani (S. Koel + Sankh near Rourkela; rises near Nagri, Ranchi, ~600 m; Eastern Ghats start south of it) · Subarnarekha (Ranchi plateau, Nagri ~600 m; Odisha–WB boundary; estuary) · Pennеru (Nandi Durg, Chikballapur; Gandikota canyon, Kadapa, "Grand Canyon of India", Erramala hills; estuary) · Ponniyar, Vaigai (short) · Vamsadhara (Kalahandi, Eastern Ghats; between Rushikulya and Godavari; Kalingapatnam) · Nagavali (Lakhbahal, Kalahandi, 1,300 m; Langulya)
Narmada	Amarkantak (Maikala, ~1,080 m); rift valley between Vindhya and Satpura, few tributaries (short, trellis); Marble Rocks Gorge; estuary ~21 km wide below Bharuch, Gulf of Khambhat; Hiran, Kolar, Hatni, Choti Tawa; MP, Maharashtra, Gujarat; meander near Bharuch
Tapi	"Narmada's twin"; Multai (MP); Purna, Girna, Panjara; Satpura north, Mahadev east, Ajanta/Satmala south
Luni	west of Ajmer; Sagarmati → Luni after Sarsuti joins; ephemeral; best inland-drainage example in the world → Rann of Kachchh; Rajasthan's biggest system
Sabarmati	Aravallis → Gulf of Khambhat; gorge at Dharoi
Mahi	Vindhyas (MP) → Gulf of Khambhat; only river to cross the Tropic of Cancer twice
Short west rivers	parallel drainage off W. Ghats, no deltas: Gujarat (Auranga, Par, Daman Ganga) · Goa (Mandovi, Zuari, Rachol) · Maharashtra (Ulhas, Kalindi) · Karnataka (Kalinadi, Sharavati – Jog Falls, Tadri, Netravathi) · Kerala (Bharathapuzha, Periyar, Pamba)
Inland	Ghaggar-Hakra (Ghaggar in India, Hakra in Pakistan after Ottu barrage; ends in Thar) · Luni, West Banas (Rajasthan → Gujarat → Rann of Kachchh)



South-Eastward flowing rivers · Also called Vridha Ganga and Dakshin Ganga

Right Bank	Left Bank
• **Koyna** • Panchganga • Dudhganga • **Ghataprabha** • Malaprabha • **Tungabhadra**	• **Bhima** • Dindi • Halia • **Musi**

South-Eastward flowing rivers · Makes a delta near its mouth. (KG delta)

Inland Drainage In India

		to perform. Rarely do these rivers form gorges.
Drainage Type	Both antecedent and consequent drainage found	These are examples of consequent drainage. Subarnarekha is an example of superimposed drainage.
Water Flow	The Himalayan rivers are perennial in nature, i.e., water flows throughout the year in these rivers. These rivers receive water from the S-W Monsoons as well as Snowmelt.	The Peninsular rivers receive water only from rainfall and water flows in these rivers in the rainy season only. Therefore, these rivers are seasonal or perennial (except Kaveri).
Stage	These rivers flow across the Young Fold mountains and are still in a youthful stage.	These rivers have been flowing in one of the oldest plateaus of the world and have reached maturity.
Meanders	The upper reaches of the Himalayan rivers are highly tortuous. When they enter the plains, there is a sudden reduction in the speed of flow of water. Under these circumstances these rivers form meanders and often shift their beds.	The hard rock surface and non-alluvial character of the plateau permits little scope for the formation of meanders. As such, the rivers of the Peninsular Plateau follow more or less straight courses.
Deltas and Estuaries	The Himalayan rivers form big deltas at their mouths. The Ganga-Brahmaputra delta is the largest in the world.	Some of the Peninsular rivers, such as the Narmada and the Tapti form estuaries. Most of the rest make deltas.

Difference Between Peninsular And Himalayan Drainage

River-linking project

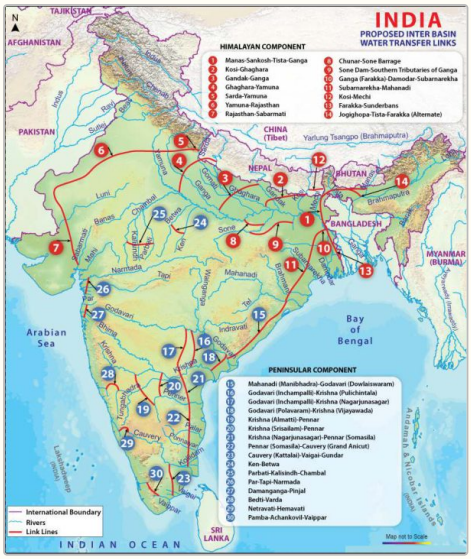
→ Idea: Sir Arthur Cotton (19th c.); revived 1960s, momentum 1980s (scarcity, floods)

→ NWDA set up 1982 to study feasibility; Supreme Court intervened several times

→ Himalayan component 14 links, peninsular 16 links; some states propose intra-state links; under debate

→ Ken–Betwa pilot: cabinet approval 2014, hurdles (environment, inter-state)

→ Pattiseema lift scheme (Godavari → Krishna; Polavaram right canal to Krishna at Vijayawada), inaugurated 2016

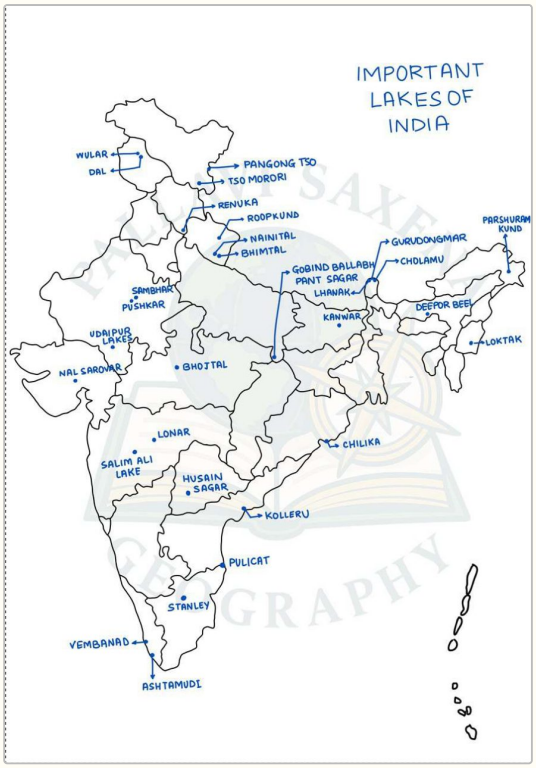


Indian River Linking Project

State	Waterfall (height m) – river
Karnataka	Kunchikal 455 (Varahi) · Barkana 259 (Sita) · Jog 253 (Sharavati) · Magod 198 (Bedti) · Shivanasamudra 98 (Kaveri) · Hebbe 168 (Thunga) · Shimsha 46
Odisha	Barehipani 399 (Budhabalanga) · Khandadhar 244 (Korapani Nala) · Koilighughar 61 (Ahiraj)
Kerala	Meenmutty 300 (Kallar) · Soochipara 200 (Chaliyar) · Athirapally 25 (Chalakuddy → Periyar)
Chhattisgarh	Amritdhara 90 (Hasdeo) · Teerathgarh 91 (Munga) · Chitrakote 29 (Indravati)
MP	Chachai 130 (Bihad → Tons) · Dhuandhar 30 and Kapildhara 30 (Narmada) · Sonemuda 15 (Son)
Jharkhand	Lodh 143 (Budha) · Hundru 99 (Subarnarekha) · Panchghagh 45 (Banai)
Others	Nohsngithiang (Seven Sisters) 315, Meghalaya · Vantawng 229, Mizoram · Kempty 40 (Yamuna, Uttarakhand) · Dudhsagar 320 (Mandovi, Goa) · Thalaiyar (Rat Tail) 297 (Manjalar/Vaigai, TN) · Maharashtra: Thoseghar, Someshwar 10 (Godavari)

Tallest first – Kunchikal 455 (Karnataka) > Barehipani 399 (Odisha) > Dudhsagar 320 > Nohsngithiang 315 > Meenmutty 300 > Thalaiyar 297 > Jog 253.

Lake	State/UT	Hook
Wular	J&K	freshwater, part of Jhelum
Dal	J&K	Srinagar
Pangong Tso	Ladakh	high-altitude saline; Indo-China border
Tso Moriri	Ladakh	high-altitude saline
Renuka	HP	largest natural lake of HP
Roopkund	Uttarakhand	near Trishul; skeletons at bottom
Nainital / Bhimtal	Uttarakhand	natural freshwater / artificial FW
Gobind Ballabh Pant Sagar	UP	Rihand dam; largest artificial lake of India
Sambhar	Rajasthan	playa lake
Pushkar, Udaipur lakes	Rajasthan	artificial; Pushkar religious
Kanwar	Bihar	largest oxbow lake of Asia; Burhi Gandak
Lhonak, Cholamu	Sikkim	high-altitude glacial
Gurudongmar	Sikkim	highest lake of India
Deepor Beel	Assam	Brahmaputra basin
Loktak	Manipur	between west and east Manipur hills; phumdis; Keibul Lamjao NP, sangai deer



Important Lakes in India

	Lake	State/UT	Characteristics
1	Wular	J&K	FW lake. Part of Jhelum
2	Dal	J&K	In Srinagar
3.	Pangong Tso	Ladakh	High altitudinal saline lake. Indo-China border
4.	Tso Morori	Ladakh	High altitudinal saline lake
5.	Renuka	HP	Largest natural lake of HP.
6.	Roopkund	Uttarakhand	Near Trishul peak. Skeletons at its bottom
7.	Nainital	Uttarakhand	Natural freshwater lake
8.	Bhimtal	Uttarakhand	Artificial FW lake
9.	Gobind Ballabhpant sagar	Uttar Pradesh	Rehand dam. Largest artificial lake of India
10.	Sambhar	Rajasthan	Playa lake
11.	Pushkar	Rajasthan	Artificial lake. Religious importance
12.	Udaipur lakes	Rajasthan	Artificial lakes
13.	Kanwar	Bihar	Largest oxbow lake of Asia. By Burhi Gandak
14.	Lhonak, Cholamu	Sikkim	High altitudinal glacial lakes
15.	Gurudongmar	Sikkim	Highest lake of India
16.	Deepor Beel	Assam	In Brahmaputra basin
17.	Loktak	Manipur	Between West & East Manipur hills. Known for Phumdis. Keibul Lamjao NP. Sangai deer.

Important Lakes in India

	Lake	State/UT	Characteristics
18	Nal Sarovar	Gujarat	Freshwater. Largest lake of Gujarat.
19.	Bhojtal	M.P	Artificial lake
20.	Lonar	Maharashtra	Meteorite crater lake
21	Salim Ali	Maharashtra	Chhatrapati Sambaji Nagar (earlier called Aurangabad)
22	Chilika	Odisha	Lagoon lake. Largest lake of India
23.	Husain Sagar	Telangana	Artificial lake in Hyderabad. Gets water from musi river
24.	Kolleru	Andhra Pradesh	Was a lagoon lake. But now it is completely inland
25.	Pulicat	AP-TN	At the border of Andhra Pradesh and Tamil-Nadu. Lagoon lake
26.	Vembanad	Kerala	Longest lake of India. Backwater 6 rivers have a mouth here. Pamba, Periyar
27	Ashtamudi	Kerala	Octopus lake. It is also a Kayal/backwater.

PYQs:

.) The Brahmaputra, Irrawaddy and Mekong rivers originate in Tibet and flow throug
arrow and parallel mountain ranges in their upper reaches. Of these rivers, Brahmaputr
akes a "U" turn in its course to flow into India. This "U" turn is due to – (2011)

(a) Uplift of folded Himalayan series
(b) Syntaxial bending of geologically young Himalayas
(c) Geo-tectonic disturbance in the tertiary folded mountain chains
(d) Both (a) and (b) above

ns: B

bout 70 million years ago, there was a sea called Tethys sea where the Himalayas stan
day. The sea started to recede during the early Eocene which was marked by the comin
gether of Indian and Asian plates.

- This trend of marine regression continued until the final emergence of the Himalaya

Important Lakes in India

Named on map, no notes: Parshuram Kund, Nal Sarovar, Bhojtal, Lonar, Salim Ali, Husain Sagar, Kolleru, Pulicat, Stanley, Vembanad, Ashtamudi, Chilika. ⭐

Peninsular Drainage

→ Classification based on the direction of flow:

South-Eastward flowing rivers

Mahanadi River:

→ Hirakud Dam is built on it.

→ States it moves through: Maharashtra, Telangana, Chhattisgarh, Andhra Pradesh, and Yanam (Puducherry). Riparian states include Odisha and Karnataka apart from the above.

→ Major Tributaries:

→ Makes a delta at its mouth (KG delta)

Krishna River:

→ Originates in Mahabaleshwar .

→ Moves through: Maharashtra, Karnataka, Telangana, and Andhra Pradesh. Riparian states also same.

→ Important Tributaries:

→ Moves through Karnataka and Tamil Nadi. Riparian States: Kerala (3%), (41%) Karnataka and 56% Tamil Nadu and Puducherry.

→ Makes a delta.

→ Important Tributaries:

Other East flowing Peninsular rivers:

→ Penneru: Originates from Nandi Durg in Karnataka ; Initially flows to the north, then attains an eastward course; forms an estuary before joining the Bay of Bengal. It makes a canyon at Gandikota.

→ Ponniyar (Karnataka & Tamil Nadu) & Vaigai (Tamil Nadu) : Have a short course with a near coast existence only.

West Flowing Rivers

Narmada:

→ Originates at the Amarkantak Plateau, M.P. forming a radial drainage pattern with the river Son.
→ Makes an estuary.
→ Origin: Multai in M.P.

Luni River:
→ Also known as Salt (Lun) River
→ Ephemeral river, i.e. it flows only when there is rain.

Sabarmati:
→ Culturally important river of India.

Mahi:
→ Origin: Vindhyas, M.P.

Other short West flowing rivers:
→ They are a prime example of parallel drainage, where these short channels originate in the Western Ghats and end in the Arabian Sea, while moving parallel to each other from West to East.

Inland Drainage In India

→ Refers to the drainage systems that do not drain into seas/oceans.

Indian River Linking Project

→ History of the Project:
→ The concept originated in the 19th century by British engineer Sir Arthur Cotton.
→ Post-independence, the idea was revisited in the 1960s and gained momentum in the 1980s due to concerns about water scarcity and floods.
→ The Supreme Court of India has intervened multiple times, directing the government to finalize plans and initiate implementation.

About the Project
→ The project aims to link numerous Indian rivers through a network of reservoirs and canals.

Chhattisgarh

Waterfall	Height (m)	River (including Tributary Info)
Amritdhara Falls	90	Hasdeo (Tributary of Mahanadi)
Teerathgarh	91	Munga (Tributary of Kanger which joins Mahanadi)
Chitrakote Waterfalls	29	Indravati (Tributary of Godavari)

Madhya Pradesh

Waterfall	Height (m)	River (including Tributary Info)
Dhuandhar Waterfall	30	Narmada (Independent)
Chachai	130	Bihad (Tributary of Tons, which is a tributary of Ganga)
Kapildhara Falls	30	Narmada (Independent)
Sonemuda Falls	15	Son (Tributary of Ganga)

Important Dams of India

Jharkhand

Waterfall	Height (m)	River (including Tributary Info)
Panchghagh Falls	45	Banai (Tributary of Subarnarekha)
Lodh Falls	143	Budha (Tributary of North Koel)
Hundru Falls	99	Subarnarekha (Independent)

Odisha

Waterfall	Height (m)	River (including Tributary Info)
Barehipani	399	Budhabalanga (Tributary of Subarnarekha)
Khandadhar Falls	244	Korapani Nala (Tributary of Brahmani)
Koilighughar Falls	61	Ahiraj (Tributary of Mahanadi)

Important Dams of India

Karnataka

Waterfall	Height (m)	River (including Tributary Info)
Kunchikal Falls	455	Varahi (Tributary of Krishna)
Barkana Falls	259	Sita (Tributary of Krishna)
Jog Falls	253	Sharavati
Magod Falls	198	Bedti
Shimsha Falls	46	Shimsha (Tributary of Cauvery)
Shivanasamudra Falls	98	Cauvery
Hebbe Falls	168	Thunga (Tributary of Krishna)

Important Dams of India

Other States

Waterfall	Height (m)	River (including Tributary Info)
Nohsngithiang (Seven-Sisters) Waterfall	315	Cherrapunji (Rain-fed), Meghalaya
Vantawng Falls	229	Lau (Tributary of Tlawng which joins Barak river), Mizoram
Kempty Falls	40	Yamuna, Uttarakhand
Dudhsagar	320	Mandovi, Goa
Thalaiyar Falls (Rat Tail Falls)	297	Manjalar (Tributary of Vaigai), Tamil Nadu

Important Dams of India

Difference Between Peninsular And Himalayan Drainage

Differences	Himalayan Drainage	Peninsular Drainage
Origin	These rivers originate from the lofty Himalayan ranges	These rivers originate in the Peninsular Plateau
Catchment Area	Himalayan rivers have large basins and catchment areas. E.g. The total basin area of the Indus, the Ganga and the Brahmaputra is 11.78, 8.61 and 5.8 lakh square km, respectively.	These rivers have small basins and catchment areas. The Godavari has the largest basin area of 3.12 lakh square km.
Valleys	The Himalayan rivers flow through deep V-shaped valleys and sometimes form gorges.	The Peninsular rivers flow in comparatively shallow valleys. These are more or less completely graded valleys. The rivers have little erosional activity

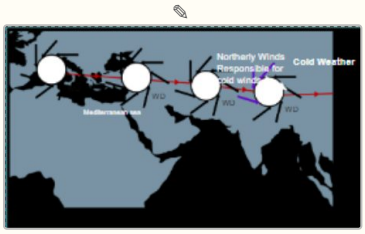
Important Dams of India

4 · INDIAN CLIMATE PYQ 2011 PYQ 2017

Understand it first: India's climate is the monsoon rhythm: winds reverse with the season. Learn the controls first, then the four seasons in order, and then the ocean and atmosphere patterns (El Niño, IOD and others) that strengthen or weaken the monsoon.

Monsoon climate

→ ~75% rain in monsoon months; tropical south to very cold north, desert west to heavy rain east



Western Disturbances

Controls

→ Latitude ~6°N–35°N (tropical)

→ Himalayas: block Central Asian cold; shape monsoon

→ Pressure and winds: summer low pulls SW winds; winter high → dry NE winds; sub-tropical westerly jet → Western Disturbances

→ Ocean currents (Indian Ocean Equatorial Current: warm, humid coasts)

→ Continentality (inland extremes)

→ Relief: W. Ghats barrier, rain shadows

WiSAR — Four seasons: Winter (mid-Nov–Feb) · Summer / pre-monsoon · Advancing monsoon (Jun–Sep) · Retreating monsoon (Oct–Nov).

Winter (mid-Nov–Feb)

→ Fog (can turn to smog), inversion, pollution domes, frost, cold waves in north

→ Offshore NE trade winds

→ WD rain = "Mawath / Mahawat" (Mewat) vital for Rabi

→ Fewest tropical cyclones (ITCZ shifts, high pressure on land and ocean)

Western Disturbances

→ Extratropical storms from the Mediterranean; steered east by the sub-tropical westerly jet; lose moisture on the Himalaya

→ Winter and pre-monsoon rain in Punjab, Haryana, UP, Delhi; Himalayan snow (feeds Ganga / Indus perennial flow); moderate winter cold; hail, thunderstorms

→ Climate change: possibly more intense (warmer, moister air), shifted timing/path, now seen in summers, interacting with cyclones and monsoon → extreme rain

Summer

→ Sun north → ITCZ shifts (~25°N)

→ Loo, dust storms (Aandhi; WD can bring showers); heat waves

→ Pre-monsoon showers (warm continental + maritime air meet): 1% of annual rain; lots of cyclonic activity

Pre-monsoon showers: what they help — Mango showers (Karnataka/Kerala) · Coffee / Cherry-blossom showers (Karnataka, Kerala) · Norwester / Kaal Baisakhi (WB, Bihar, Jharkhand, Odisha: rice, tea, jute) · Bardoli Chheerha / Bordoisila = tea showers (Assam).

Cyclones in India

→ Coast >7,500 km (as printed); season April–Dec, peaks May–June and Oct–Nov

→ Bay : Arabian Sea ≈ 4 : 1 (bigger warm-water area, lower wind shear); 5–6 per year, 2–3 severe

→ Climate change: intensity up, not necessarily frequency; Arabian Sea activity rising; 2022 study: Bay numbers may fall, intensity rise

Naming

→ WMO/ESCAP Panel on Tropical Cyclones, since September 2004

→ 13 countries × 13 names: Bangladesh, India, Iran, Pakistan, Oman, Maldives, Myanmar, Sri Lanka, Thailand, UAE, Yemen, Qatar, Saudi Arabia

→ RSMC New Delhi assigns; names retired after use; a South China Sea cyclone entering the Bay keeps its name

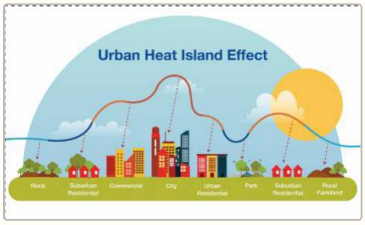
IMD colour warning	Lead time
Pre-cyclone watch (no colour)	72 h
Cyclone Alert: Yellow	≥ 48 h
Cyclone Warning: Orange	≥ 24 h (landfall point forecast)
👫 Post-landfall outlook: Red	≥ 12 h

Mnemonic (handout): 72 – 48 – 24 – 12, halving each step: Watch → Yellow → Orange → Red. ⭐

Urban heat island

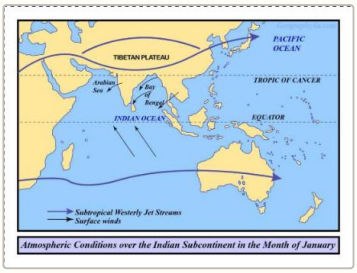
→ Urban areas much hotter than rural surroundings

→ Causes: concrete/asphalt, less vegetation, vehicle/industrial/AC heat, anthropogenic energy, urban geometry ("heat canyons")

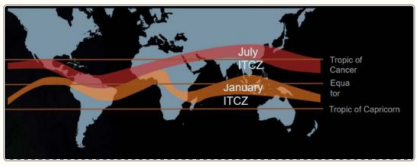


Urban Heat Island Effect

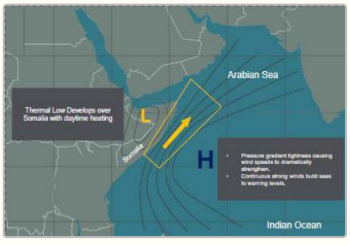
Advancing monsoon (Jun–Sep; "mausim"; seasonal wind reversal; 75% rain)



Seasons Of India: Advancing Monsoon · Subtropical Westerly Jet Stream

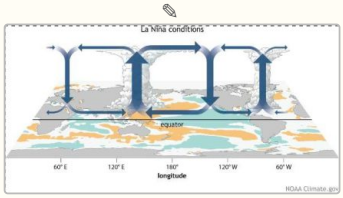


Seasons Of India: Advancing Monsoon · Differential Heating of Land and Water

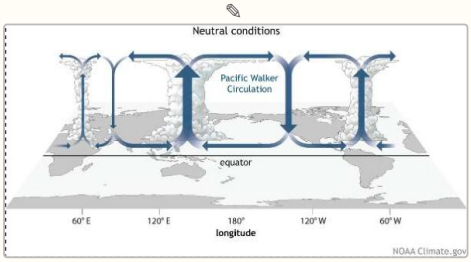


Seasons Of India: Advancing Monsoon · Somali Jet Stream

Seasons Of India: Advancing Monsoon · Subtropical Westerly Jet Stream



Seasons Of India: Advancing Monsoon · El Niño Southern Oscillation (ENSO)



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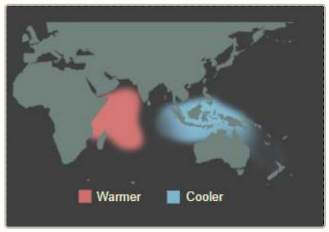
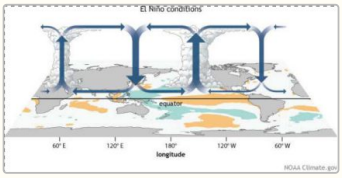
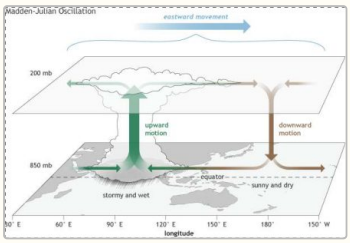


Seasons Of India: Advancing Monsoon · Indian Ocean Dipole (IOD)



Seasons Of India: Advancing Monsoon · El Niño Southern Oscillation (ENSO)



Seasons Of India: Advancing Monsoon · Madden-Julian Oscillation (MJO)

Drivers

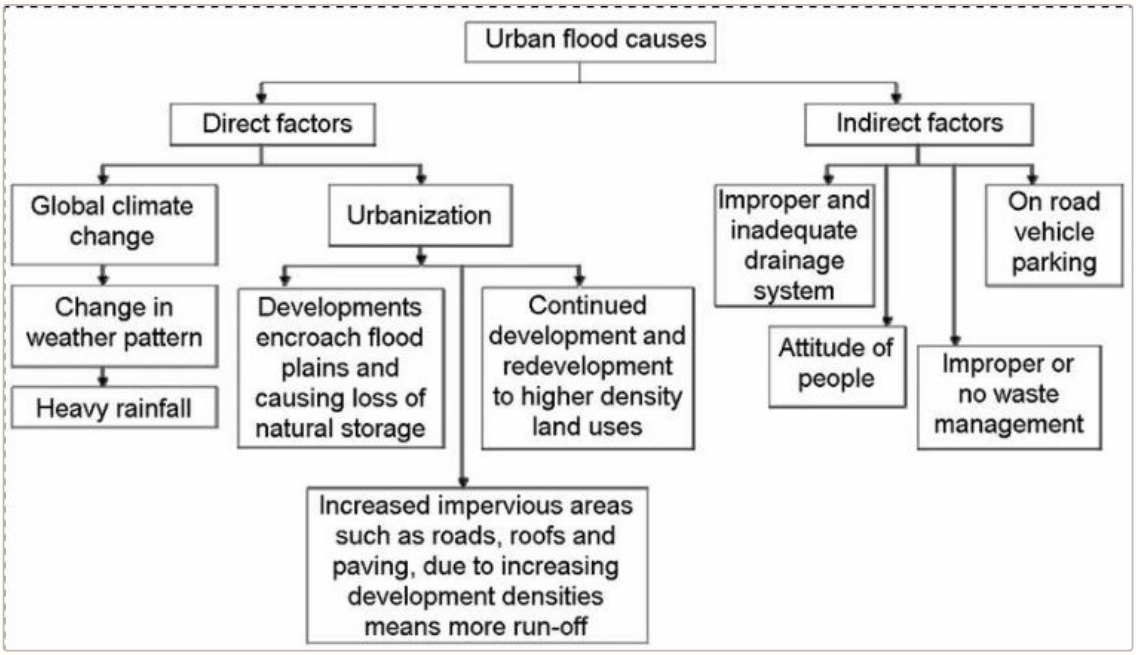
→ Land–sea differential heating (primary)
→ ITCZ shifts north
→ Sub-tropical westerly jet (weaker/northward = stronger monsoon)
→ Tibetan Plateau heating → Tropical Easterly Jet (TEJ)
→ Somali jet (moisture across Arabian Sea)
→ ENSO · IOD · MJO
→ Arctic sea ice (less ice → more rain in central and NE India, less in west and south peninsula; mechanism unknown)

→ Monsoon trough = low over India attracting SW winds
→ Burst = sudden onset with thunder, lightning, downpour; ~1 June, Kerala

Two branches

→ Arabian Sea (stronger): Malabar, Konkan, Kathiawar
→ Bay of Bengal (blocked by Arakan): Utkal and North-Eastern
→ Meet around Agra–Ferozepur

→ Tamil Nadu dry: rain shadow of W. Ghats; branches parallel to its coast
→ Break: usually August: weak Tibetan Low + winds parallel to west coast; flooding common



Urban Flooding In India

Cloudbursts

→ ≥ 10 cm rain in an hour over a small area; flash floods, landslides

→ Orographic lift + rapid condensation + cold-air downdraft; more in mountains, also plains

→ Himalaya, Western Ghats; monsoon; hard to predict (dense radar, high-res models); more with warming (warmer air holds more moisture)

Urban flooding

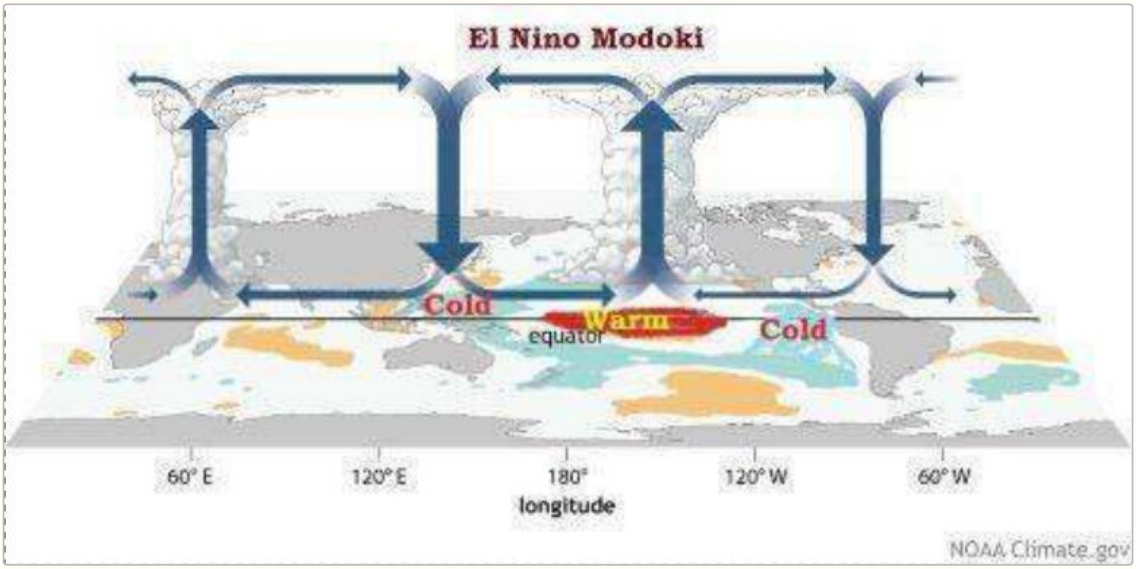
→ Impervious surfaces (concrete, asphalt) + overwhelmed drains

→ Mumbai 2005 (944 mm in a day) · Chennai 2015 (drainage, water-body encroachment) · Bengaluru (unplanned growth) · Hyderabad (lake encroachment) · Mumbai mangrove loss · heavy rain in Mumbai/Kolkata linked to climate

El Nasty / La Nice – El Niño = El Nasty (warm east Pacific → weak SW monsoon, drought in India/Australia; NE monsoon stronger). La Niña = La Nice (cold east Pacific → good monsoon, Australian floods; NE monsoon weaker).

Phenomenon	What it is	India link
El Niño	trades weaken; warm water sloshes east; Humboldt (Peruvian) current replaced, less upwelling	weak SW monsoon; Australia drought/fire; Peru floods; California storms
La Niña	trades strengthen; cold east Pacific; more upwelling	better monsoon; floods in Australia, E. Africa, SE Asia
ELLA years	transition between El Niño and La Niña (IMD-IITM Pune study)	severe, prolonged summer heat; more heatwaves in NW and central India; early identification helps forecasting
El Niño Modoki	"similar but different" (Japanese); warmth in central Pacific (dateline El Niño), cooler east and west	different global pattern
PDO	North Pacific; each phase 20–30 yrs (ENSO = 2–7 yrs); positive = warm east	modulates ENSO links to monsoon; amplifies or weakens El Niño/La Niña
IOD	west–east SST gap in the Indian Ocean	Positive IOD (warm west near Arabian Sea, cool east) → stronger monsoon, even in El Niño years; negative → weaker, more break days
Atlantic Niño / Niña (Atlantic Ocean Dipole / Zonal Mode)	Niño: east equatorial Atlantic warms, west cools; Niña = reverse	not as wide-ranging as Pacific
Arctic Ocean Dipole	Arctic sea-level pressure pattern (Arctic Dipole Anomaly)	–

MJO	eastward cloud-rain pulse, 30–60 days; active (wet) vs suppressed (dry)	active phase enhances monsoon; modulates cyclone formation; shorter cycle = better monsoon impact



El Nino Modoki

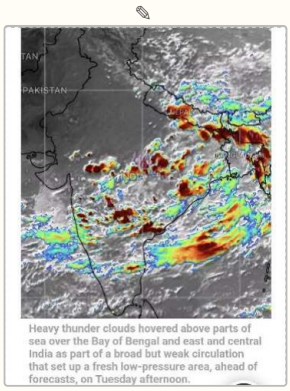
Arctic Ocean Dipole

Retreating monsoon (Oct–Nov)

→ ITCZ shifts; pressure rises over north India; "October heat" (humid, clear)

→ NE monsoon rain: coastal Andhra + most of Tamil Nadu (NE trades over Bay); El Niño: NE stronger; La Niña: NE weaker, SW stronger

→ Season of most severe cyclones in Bay (hot sea, trough moves south, wind change)



Seasons Of India: Retreating Monsoon Season

Rainfall map

→ >200 cm: windward W. Ghats (S. Gujarat → Kerala), NE (Assam, Meghalaya, Arunachal; funnelling: Cherrapunji, Mawsynram), Andaman and Nicobar

→ 100–200: Himalayan foothills, east India, east TN

→ 50–100: parts of Deccan, NW India

→ <50: west Rajasthan, W. Ghats rain-shadow, Ladakh

Year	Asked	Answer
2011	La Niña vs El Niño: (1) La Niña = cold equatorial Indian Ocean? (2) La Niña no effect on monsoon?	D neither: La Niña = cooling of tropical west coast of S. America (Peru/Chile cold current returns); helps monsoon; Australia floods

2017	IOD: (1) W Indian Ocean vs Eastern Pacific? (2) can influence El Niño's effect	B only 2: IOD = W (near Arabian Sea) vs E (near Bay of Bengal) Indian Ocean
–	Andaman and Nicobar climate: humid tropical coastal ✔; SW + NE monsoon ✔; max rain Dec–May ✘	B (1 and 2); max rain May–Sept (SW monsoon); ocean winds onshore both ways

Factors affecting Indian Climate

→ Latitude : India's latitudinal extent stretches from roughly 6°N to 35°N , placing it primarily in the tropical zone. This means it receives a high amount of solar radiation throughout the year, contributing to generally warm temperatures. However, there are variations. As you move north towards higher latitudes, temperatures tend to decrease as the climate gets more temperate in nature.

→ Distance from the Sea (Continentality) : As we move inland from the coast, the influence of the sea on the climate diminishes. Locations further away from the sea experience more extreme temperatures, with hotter summers and colder winters compared to coastal areas.

Seasons Of India: Winter Season

→ From mid-Nov till Feb

Western Disturbances

→ As these disturbances move towards the Indian subcontinent, they encounter the Himalayas, which cause them to lose moisture, resulting in precipitation over northern India.

Impact:

→ Temperature Regulation : These disturbances help regulate winter temperatures, preventing extreme cold conditions by bringing cloud cover and moisture.

→ Water Resources : The snowfall in the Himalayas contributes to the perennial flow of rivers such as the Ganges and Indus, which are vital for irrigation and drinking water supply. Important for Rabi crops.

→ Weather Events : These disturbances can also lead to extreme weather events like heavy rainfall, hailstorms, and thunderstorms, which can cause damage to crops, infrastructure, and even loss of life.

Possible Impact of Climate Change:

→ Frequency and Intensity : Climate change may alter the frequency and intensity of western disturbances. Some studies suggest that there could be an increase in the intensity of these disturbances due to warmer air holding more moisture.

→ – Increased intensity could lead to more severe weather events, including heavier rainfall and more extensive snowfall.

→ Shift in Patterns : There is a possibility of a shift in the typical pathways and timing of western disturbances due to changes in global atmospheric circulation patterns.

→ – Changes in the timing and pathway can affect the predictability and reliability of rainfall and snowfall during the winter months, impacting agriculture and water resources.

Cyclones In India

→ India has a tropical location, with a vast coastline of over 7500 km, making it susceptible to Tropical Cyclones.

→ The period between April and December experiences the most significant cyclone activity in India, with the peak months being May–June and October–November .

→ Favorable Conditions : This period coincides with:

📜 Impact of Climate Change:	→ Warmer Seas, Stronger Storms: → – Warmer water acts as fuel for cyclones, providing the energy they need to form, intensify, and sustain themselves. This can lead to cyclones reaching higher wind speeds and causing more extensive damage. → Increased Intensity, Not Necessarily Frequency: → – While the overall number of cyclones might not significantly increase, climate change is projected to make existing cyclones more severe . → – Stronger winds, heavier rainfall, and higher storm surges are all potential consequences of climate change-induced intensification. → Impact on the Arabian Sea: → – Historically, the Arabian Sea has been less prone to cyclones compared to the Bay of Bengal. However, recent studies suggest an increase in cyclone activity in the Arabian Sea, which some experts attribute to climate change. → Impact on the Bay of Bengal: → – The Bay of Bengal is already a hotspot for cyclones. With rising sea surface temperatures, cyclones here could become even more powerful . → – Increased storm surges due to rising sea levels can cause greater coastal inundation, displacing communities and damaging infrastructure. → – According to a 2022 study, numbers might decrease and intensities will increase here. → Adaptation and Mitigation: → – Adaptation : Building seawalls, improving early warning systems , and promoting disaster preparedness measures are crucial for adapting to the changing cyclone landscape. → – Mitigation : Addressing climate change at its source by reducing greenhouse gas emissions is essential for long-term mitigation of cyclone intensity.

→ Colour-Coded warnings: The Indian government uses a well-defined color-coded cyclone warning system to disseminate information and prepare communities:

→ – Cyclone Alert (Yellow): issued at least 48 hrs in advance of the expected commencement of adverse weather over the coastal areas. It contains information on the location and intensity of the storm likely direction of its movement, intensification, coastal districts likely to experience adverse weather, and advice to fishermen, the general public, media, and disaster managers.

→ – Cyclone Warning (Orange): issued at least 24 hours in advance of the expected commencement of adverse weather over the coastal areas. The landfall point is forecast at this stage.

→ – Post Landfall outlook (Red): issued at least 12 hours in advance of the expected time of landfall. It gives the likely direction of movement of the cyclone after its landfall and adverse weather likely to be experienced in the interior areas.

→ Naming of the Cyclones in the North Indian Ocean region: The World Meteorological Organization/Economic and Social Commission for Asia and the Pacific (WMO/ESCAP) Panel on Tropical Cyclones plays a crucial role in naming cyclones that form over the Bay of Bengal and the Arabian Sea, a practice that began in September 2004.

→ – Names are chosen from a predefined list.

→ – Names are new and not used elsewhere globally to avoid confusion.

→ – Once used, a name is retired from the list for future storms .

▶ Urban Heat Island Effect

Causes:	→ Concrete and Asphalt Surfaces : Buildings, roads, and other infrastructure absorb and retain heat, leading to higher temperatures.

→ Urban Geometry : The configuration of buildings and streets can trap heat and restrict airflow, creating "heat canyons."

Seasons Of India: Advancing Monsoon

Factors affecting Indian Monsoons:

→ Subtropical Westerly Jet Stream:

→ – High up in the atmosphere, the Subtropical Westerly Jet Stream is a fast-moving band of westerly winds. While its direct influence on monsoon rainfall is minimal, its position plays a role.

→ Tibetan Plateau and Tropical Easterly Jet Stream (TEJ):

→ – The massive Tibetan Plateau acts like a giant elevated heat source during summer. This heating strengthens the TEJ , an easterly jet stream blowing over the subcontinent.

→ El Niño Southern Oscillation (ENSO):

→ – This large-scale climate-ocean phenomenon occurring in the Pacific Ocean has a significant impact on the Indian monsoon.

→ – During El Niño events (warmer than average sea surface temperatures in the eastern Pacific compared to Western Pacific), monsoon rainfall over India tends to be weaker.

→ – Conversely, La Niña events (cooler than average sea surface temperatures in the Eastern Pacific) are often associated with stronger monsoons.

→ Madden-Julian Oscillation (MJO):

→ Flooding can be common during this season, especially in the regions where rivers move.

Cloudbursts

→ Cloudbursts are intense and sudden rain events where a significant amount of rain falls over a small area in a very short period, often leading to flash floods.

→ Cloudbursts are typically defined as instances where 10 cm of rain or more falls within an hour at a particular location.

→ Cloudburst Formation Process

→ – Rapid Upward Movement of Moist Air: Warm, moisture-laden air rises quickly, often due to geographical factors like mountain ranges. This ascent can also be triggered by unstable atmospheric conditions.

→ – Sudden Rainfall Release: Once the cloud becomes too heavy, it releases the moisture all at once, resulting in an intense and concentrated downpour. The rate of rainfall is so high that the ground cannot absorb it quickly enough, causing flash floods.

→ – The release of moisture is also aided by the cold air influx into the clouds at high altitudes (lapse rate), which can force a downdraft (like a piston), causing the raindrops to be pushed down.

→ These events are difficult to predict because they occur on such a small scale , both in time and space. For effective prediction, we need either a high-resolution radar network or advanced weather models capable of capturing small-scale weather dynamics.

Urban Flooding In India

Causes:

→ Poor Drainage Systems : Inadequate, poorly designed, or poorly maintained drainage infrastructure can lead to blockages and insufficient water flow. E.g. Chennai floods of 2015 highlighted the city's inadequate drainage systems and encroachments on water bodies.

→ Unplanned Urbanization : Rapid and unplanned urban growth reduces the amount of open land that can absorb rainwater, increasing surface runoff. E.g. Bengaluru has seen significant urban flooding due to rapid, unplanned urbanization and reduced green spaces.

→ Improper Waste management can choke the natural and artificial drains, exacerbating the problem.

→ Encroachment on Natural Water Bodies : Construction and encroachment on lakes, wetlands, and rivers disrupt natural drainage and water absorption. E.g. Hyderabad's flooding issues are partly due to encroachment on lakes and water bodies.

→ Deforestation : Reduction of green cover in and around cities decreases natural absorption of rainwater, leading to increased runoff. E.g. The loss of mangroves in coastal cities like Mumbai reduces natural flood buffers.

El Nino Southern Oscillations

→ El Niño Southern Oscillation (ENSO) is a coupled ocean-atmosphere phenomenon that disrupts typical weather patterns across the globe. It's essentially a large-scale climate dance between the Pacific Ocean and the atmosphere above it, with two distinct phases: El Niño and La Niña.

Climatological and Oceanographic Impacts:

→ El Niño: During El Niño events, trade winds weaken in the tropical Pacific . Warm surface water typically confined to the western Pacific sloshes eastward , leading to:

→ – Reduced upwelling of cold, nutrient-rich water from the deep ocean.

→ La Niña : La Niña represents the opposite phase. Trade winds intensify, pushing warm surface water westward. This results in:

Global Climate Impacts of El Niño:

→ Warmer and Drier Conditions : El Niño is often associated with drier-than-average conditions in many parts of the world, including:

→ – Australia : Drier conditions can contribute to wildfires and heatwaves.

→ Increased Rainfall and Flooding : Conversely, El Niño can bring wetter-than-average conditions to other regions:

→ – California (USA): Increased storminess and precipitation are common during El Niño events.

Global Climate Impacts of La Niña:

→ Cooler and Wetter Conditions : La Niña often brings:

→ – Colder-than-average temperatures in parts of North America and Europe.

→ Droughts : Conversely, La Niña can cause drier conditions in some regions:

→ – Peruvian Coast : La Niña can contribute to drier-than-average conditions and more supressed rainfall.

Ella

→ Extended duration of extreme heat , making summers notably harsher. Early identification of ELLA years could improve heatwave forecasting and preparedness in India. Recognizing this pattern adds nuance beyond standard ENSO impacts, highlighting critical transitional phases that affect summer temperatures profoundly.

El Nino Modoki

→ El Niño Modoki, also known as "Central Pacific El Niño" or "dateline El Niño," is a variation of the traditional El Niño phenomenon.

→ The term "Modoki" comes from a Japanese word meaning "similar but different."

→ Unlike the conventional El Niño, where the warmest sea surface temperatures (SSTs) are found in the eastern Pacific Ocean, El Niño Modoki features warm SST anomalies centered in the central Pacific, with cooler temperatures in both the eastern and western Pacific.

Characteristics of El Nino Modoki:

→ Location of Warm SSTs : In El Niño Modoki events, the central Pacific (around the dateline) experiences significant warming, while the eastern Pacific may not see as much warming and can even experience cooling.

Pacific Decadal Oscillation (PDO)

→ The Pacific Decadal Oscillation (PDO) is a long-term fluctuation in sea surface temperatures (SSTs) and atmospheric circulation over the North Pacific Ocean , with each phase typically lasting 20–30 years . It has two phases: positive (warm) , when the eastern Pacific is warmer and the western Pacific cooler than average, and negative (cool) , with the opposite pattern. Unlike ENSO , which operates on a 2–7 year timescale, the PDO influences climate over decadal scales and can amplify or weaken El Niño and La Niña impacts. It affects temperature, rainfall, marine ecosystems, fisheries, and storm tracks across the Pacific basin. In India, the PDO can modulate monsoon variability by influencing ENSO teleconnections , thereby affecting droughts, floods, and long-term climatic trends.

Indian Ocean Dipole

Climatological Impacts of Indian Ocean Dipole

→ Indian Subcontinent:

→ – Monsoon Variability : A positive IOD often enhances the Indian monsoon , leading to above-average rainfall in the Indian subcontinent.

→ – Conversely, a negative IOD can suppress the monsoon , causing drought conditions in the region.

→ Australia : Positive IOD events are typically associated with drier conditions in Australia, particularly in the southeastern parts, leading to droughts and increased bushfire risk. Negative IOD events bring wetter conditions , increasing the likelihood of floods.

→ East Africa : Positive IOD events can lead to increased rainfall in East Africa, resulting in wetter conditions and sometimes causing floods. Negative IOD can lead to drier conditions and droughts in the region.

→ Indonesia and Southeast Asia : During a positive IOD , Indonesia and parts of Southeast Asia can experience drier conditions , affecting agriculture and water resources. Negative IOD usually brings wetter conditions to these areas.

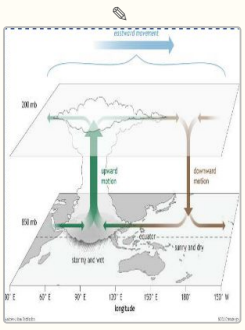
Atlantic Ocean Dipole

Atlantic Nino & Nina

→ Do not have as wide climatic impacts as Pacific El Nino and la nina.

Madden Julian Oscillation

→ The Madden-Julian Oscillation (MJO) is a major atmospheric phenomenon that significantly influences global weather patterns and climate. It is characterized by a large-scale coupling between atmospheric circulation and tropical deep convection, primarily affecting the Indian and Pacific Oceans.



Characteristics of the MJO

→ Phases: The MJO consists of two primary phases:

→ – **Active (Convective) Phase:** Characterized by **enhanced cloudiness and precipitation.** This phase is associated with increased convection and storm activity.

→ – **Inactive (Suppressed) Phase:** Marked by **reduced cloudiness and precipitation**, leading to drier and more stable conditions.

Seasons Of India: Retreating Monsoon Season

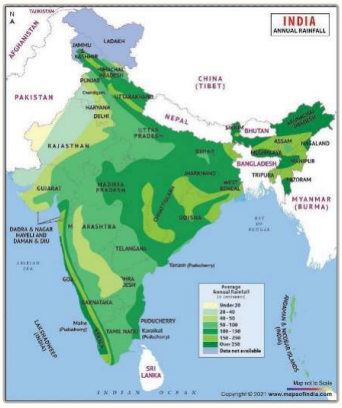
→ **October Heat : High Temperature and Humidity.** Its causes include:

→ – With the **reduction in cloud cover** post-monsoon, there is **increased solar radiation**, which raises daytime temperatures.

Spatial Distribution Of Rainfall In India

High Rainfall Regions (>200 cm annually)

→ **Northeastern India:** Primarily Assam, Meghalaya, and parts of Arunachal Pradesh. Notable places include Cherrapunji and Mawsynram, which receive some of the highest rainfall in the world. **Funneling effect:** Clouds get channeled into constricted space between mountains. Happens in NE India.



Spatial Distribution Of Rainfall In India · High Rainfall Regions (>200 cm annually)

Medium Rainfall Regions (100–200 cm annually)

→ **Foothills of the Himalayas:** This includes regions in Uttarakhand, Himachal Pradesh, and northern West Bengal (like Darjeeling).

Very Low Rainfall Regions (<50 cm annually)

→ **Rajsthan:** This includes the semi-arid regions of Rajasthan, particularly the Western part.

5 · FORESTS PYQ 2015 PYQ 2016 PYQ 2023 PYQ 2011

Understand it first: Forest type follows rainfall (and height in the mountains): wettest areas carry evergreen forest, moderate rain gives deciduous, dry areas thorn, coasts mangrove, and high altitudes alpine. Learn each with rainfall, region and typical trees.

Forest	Climate	Where	Trees / traits
Wet evergreen	>200 cm; 25–27 °C	W. Ghats (south of Mumbai), NE, A&N	no undergrowth (dense canopy); jackfruit, rosewood, mahogany, aini, ebony, agarwood
Semi-evergreen	175–225; 24–27 °C	W. coast, Assam, lower E. Ghats, Odisha, Andamans	wet evergreen + moist deciduous mix; evergreen due to undergrowth; kadam, cashew, cedar, chestnut, thorny bamboo
Dry evergreen	~100 cm all year	TN coast	short; tamarind, neem, toddy palm

Moist deciduous	100–200; ~27 °C	around W. Ghats, NE, Shiwalik (Terai-Bhabar), E. MP, CG, Chotanagpur, Odisha	shrubby undergrowth; teak, sal, mahua, tendu, bamboo (also shisham, sandalwood, khair, kusum, arjun, mulberry); leaf-fall 6–8 weeks; humidity 60–75%
Dry deciduous	75–100	rainier peninsula, plains of UP, Bihar, MP, Rajasthan	transition to thorn; teak, tendu, red sanders, sal, khair
Thorn	<50; 25–30 °C	Rajasthan, SW Punjab, W. Haryana, Kachchh, Saurashtra, some UP, lee of W. Ghats (MH, Karnataka, Telangana, AP, TN)	low scattered; acacia, babul, neem, khair, cacti, wild date
Littoral and swamp (mangroves)	brackish water	Sundarbans (Sundri), A&N, Mahanadi-Godavari-Krishna deltas	viviparity, pneumatophores; sundri, keora, agar, rhizophora; ~7% of world mangroves; fresh + brackish

Rainfall ladder (cm) — >200 wet evergreen → 175–225 semi-evergreen → 100–200 moist deciduous → 75–100 dry deciduous → ~100 spread through year dry evergreen (TN coast) → <50 thorn. Wetter = evergreener, taller; drier = shorter, thornier.

Mangroves protect coast

→ Erosion, storm surge, tsunami (2004 PYQ); trap sediment (protect coral, seagrass)

→ Fisheries nurseries; rot-resistant timber, fodder, medicine; tourism

→ Handout alert: capacity to adapt to rising seas is being compromised; roots salt-tolerant (not salt-sensitive), do not store freshwater

Mountain forests

Himalaya (north)

→ base moist deciduous → 1,000–2,000 m wet temperate with pine (British replaced oak with pine for railway sleepers)

→ up to ~3,000 m blue pine, spruce, grassland

→ 3,000–4,000 m silver fir, juniper, rhododendron + bugyals → tree line → alpine tundra → snow line; northern slopes little vegetation

Western Ghats (south)

→ evergreen base → temperate evergreen "sholas" (Nilgiris, Anamalai, Palani) both slopes; no alpine tundra (height limit)

Andaman and Nicobar forests (2015 PYQ)

→ ~2,200 plant varieties, 200 endemic, 1,300 not on mainland; forest cover 86.2%

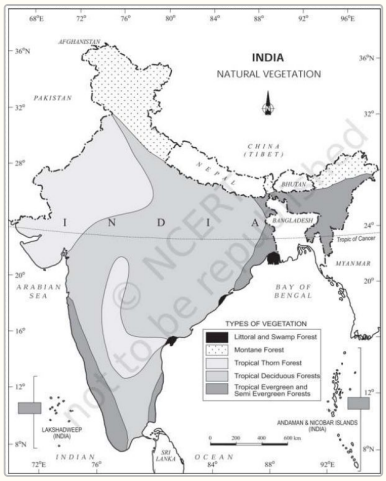
→ South Andaman: epiphytes (ferns, orchids) · Middle: moist deciduous · North: wet evergreen + woody climbers

→ North Nicobar: no evergreen; central/south Nicobars mostly evergreen; grasslands only in Nicobars

→ Mangroves 1,136 km² (Andaman) + 35 km² (Nicobar)

Forest status	Figure (handout)
Forest + tree cover	8,27,357 km² = 25.17% (target 33% for stability)
Forest / tree cover	7,15,343 km² = 21.76% / 1,12,014 km² = 3.41%
Change vs 2021	+1,445 km² (forest +156, tree +1,289); lower than +1,540 in 2021
Mangrove	4,992 km² (−7.43 km²) · bamboo area +5,227 km²
States	>33% forest: 19 states/UTs · >75%: 8 (Mizoram, Lakshadweep, A&N, Arunachal, Nagaland, Meghalaya, Tripura, Manipur)

🧑‍🤝‍🧑 Highest % cover	Lakshadweep 91.33 · Mizoram 85.34 · A&N 81.62
🧑‍🤝‍🧑 Largest area	forest + tree: MP, Arunachal, Maharashtra · forest: MP, Arunachal, Chhattisgarh · bamboo: MP, Arunachal, Maharashtra
Max increase	total: Chhattisgarh, UP, Odisha · forest: Mizoram, Gujarat, Odisha · tree: Gujarat, Bihar, Kerala
Megacities	Delhi has the highest forest cover among the six
Worries	decline in Andhra Pradesh, Maharashtra · Western Ghats ESA −58.22 km² in 10 years · NE −327.30 km²
🆕 New in 2023 report	agroforestry chapter · "forest characteristics" chapter back after 10 yrs · bamboo in tree cover · first estimate of 5–10 cm DBH small trees · 751 districts (636 in 2021)
FAO	India +2,66,000 ha/yr 2010–2020, 3rd (after China, Australia) · GFRA 2025: 9th in total forest area, 3rd in annual gain · Asia the only continent with forest gain since 1990



Forests Of India

Category	Details
Largest Total Forest and Tree Cover	Madhya Pradesh, Arunachal Pradesh, Maharashtra
Largest Forest Cover	Madhya Pradesh, Arunachal Pradesh, Chhattisgarh
Highest Percentage of Forest Cover	Lakshadweep (91.33%), Mizoram (85.34%), Andaman & Nicobar Islands (81.62%)
Maximum Increase in Total Cover	Chhattisgarh, Uttar Pradesh, Odisha
Maximum Increase in Forest Cover	Mizoram, Gujarat, Odisha
Maximum Increase in Tree Cover	Gujarat, Bihar, Kerala
Maximum Bamboo-Bearing Area	Madhya Pradesh, Arunachal Pradesh, Maharashtra

Status of Indian Forests · Key Points

Social forestry

→ Increase tree cover + rural livelihoods through local people; term first used in India in 1976 by the National Commission on Agriculture

→ Types: community, urban (Miyawaki = dense native planting, Japanese), extension, agroforestry (trees with crops), farm forestry

→ Benefits: jobs, fuel/fodder, NTFPs, carbon, erosion control, habitat, less pressure on natural forests

Forests Of India

→ India boasts of a variety of vegetation , owing to the diverse climatic conditions . Important forest types in India include:

→ Montane forests.

Tropical Evergreen Forests

→ The trees do not shed their leaves due to optimal conditions all year round.

→ Depending upon the precipitation levels, they can be further divided into the following types:

Semi-Evergreen: → Bordering wet evergreen

Dry Evergreen → Found on the Tamil Nadu coast .

Tropical Deciduous Forests

→ The trees in these forests shed their leaves once a year.

→ Based on the amount of precipitation received, can be classified into:

Moist Deciduous: → Found in parts surrounding western ghats, NE India, Shiwalik range (along Terai & Bhabher belt); hills of Eastern MP, CG, Chotanagpur plateau; most of Odisha, parts of WB & Andaman and Nicobar.

Dry Deciduous: → Some parts of Rajasthan, also show these type of forests, but with scantier growth.

Tropical Thorny Forests

→ Found in Arid locations.

Littoral & Swamp

→ In and around deltas, estuaries, and creeks.

→ Highly developed in the Andaman and Nicobar Islands and Sunderbans (Sundri tree) of West Bengal. Also seen in Mahanadi, the Godavari and the Krishna deltas.

Alpine

→ In the Northern Mountains: The base region has moist deciduous forests , as the height increases, the forests become temperate. At further heights, beyond the tree line, up to the snow line, there is alpine tundra vegetation .

→ – At 1000-2000m we get wet temperate forests with Pine trees. British replaced Oaks with Pine in these forests, and used it for laying Railways.

→ – Above this upto 3000 m (approx) are Blue Pine, Spruce and some Grasslands.

→ – 3000-4000m: Alpine forests (Silver firs, Junipers, Rhodohenderon), and Grasslands (Bugyals)

Status of Indian Forests

Key Points: → However, this is still below the recommended 33% for ecological stability.

→ The bamboo-bearing area has increased compared to the 2021 assessment by 5227 sq. km.

→ Forest cover in hill districts increased by 234.14 sq.km.

⚠ Concerns:

→ Total cover grew only by 1445 sq.km, compared to 1540 sq.km in 2021. Notable decrease in Andhra Pradesh and Maharashtra.

Difference between 2021 & 2023 reports:

→ Re-introduction (after 10yrs) of chapter on "Forest characteristics" which talks about soil health, grazing instances etc.

FAO Report:

→ According to the latest report (State of World's Forests Report) by FAO (released July 22, 2024), India gained 2,66,000 hectares of forest area annually from 2010 to 2020 , securing the third spot (after China and Australia) among the top 10 countries with the most significant forest area gains during this period.

→ FAO appreciated Indian efforts in Agroforestry and reclamation of degraded lands for forestry.

Social Forestry

→ The emphasis is on planting trees that meet the social and economic needs of local communities, while also providing environmental benefits.

Its types include

→ Community Forestry:

→ – This involves local communities actively participating in the management of forests on degraded or wasteland.

→ Urban Forestry:

→ – This type of social forestry involves planting and managing trees in urban areas , including parks, streetscapes, and institutional lands.

→ Extension Forestry:

→ – Extension of forests in previously non-forested regions. This approach involves educating and empowering rural communities about the benefits of trees and providing technical assistance for tree planting and management.

→ Agroforestry:

→ – This combines trees and shrubs with crop production on the same land. (Not necessarily by a single farmer).

Farm Forestry:

→ Focus: This approach encourages farmers to integrate trees into their existing agricultural lands for commercial as well as protection purposes.

→ Benefits: Trees planted along field borders or in small woodlots provide wind protection for crops, reduce soil erosion, and offer additional income from timber, fruits, or other non-timber forest products (NTFPs).

Advantages:

→ Social Benefits:

→ – Creates rural employment opportunities through tree planting and maintenance.

→ – Provides fuelwood, fodder, and other forest products for local communities , improving livelihoods.

→ – Empowers local communities through participation in decision-making and management of forest resources.

→ Economic Benefits:

→ – Provides a source of income for rural communities through the sale of timber, fruits, and other non-timber forest products (NTFPs) like medicinal plants and honey.

→ Environmental Benefits:
→ – Reduces soil erosion and promotes land reclamation .
→ – Provides habitat for diverse species, promoting biodiversity .
→ – Reduces dependence on traditional forests , potentially leading to less deforestation.
→ – Supports sustainable agriculture by improving soil quality and providing wind protection.

🎯 6 · SOILS PYQ 2013 PYQ 2021 PYQ 2023

Understand it first: A soil is its parent rock plus climate plus time. Each Indian soil type is learnt the same way: where it occurs, how it formed, its colour and texture, what it is rich or poor in, and the crops it suits.

Soil formation factors → Relief · parent rock · climate · biota · time · humans

A Red Black Arid Laterite Saline Peaty Forest – ICAR's 8 soils: Alluvial · Red & yellow · Black · Arid desert · Laterite · Saline · Peaty/marshy · Forest & mountain. Pairs: Black = cotton + cracks (self-ploughing); Red = iron; Laterite = tea-coffee-cashew (acidic, leached); Alluvial = the breadbasket.

Soil	Where	Rich / poor	Crops and hooks
Alluvial	N. plains, coastal plains; >40% of India, most common	rich: potash, lime, phosphoric acid; poor: nitrogen, humus	sugarcane, paddy, wheat, pulses; Bhangar / Khadar
Black (Regur)	Maharashtra, Saurashtra, MP, CG; Godavari-Krishna; 3rd most common	rich: Mg, potash, iron, lime, alumina; poor: phosphoric, N, organic	cotton, sugarcane, pulses; cracks when dry; high water retention; volcanic (Deccan / Rajmahal trap lava, fissure volcanic rock)
Red and yellow	low-rain E and S Deccan; W. Ghat piedmont; yellow in parts of Odisha/CG	rich: iron, potash; poor: N, phosphorus, humus	crystalline igneous rocks; yellow = hydrated iron; wheat, cotton, pulses, tobacco, oilseeds, potato
Laterite	wetter margins; south, W. Ghats of Maharashtra, Odisha, WB, NE; also Karnataka, TN, MP, Bihar, Assam, Meghalaya	iron oxide + potash high (notes); poor: organic, N, phosphate, calcium	acidic; tea, coffee, cashew (tapioca: PYQ); rusty red
Arid	Rajasthan / Thar	little humus, N; phosphorus adequate; kankar	millets, fodder, guar; Indira Gandhi Canal
Saline (Usara)	arid/semi-arid Rajasthan, Punjab, UP, Haryana, Kachchh, Maharashtra	salts of Na, K, Mg	some coconut, salt-resistant fodder (White Revolution)
Peaty / marshy	heavy rain: Odisha, Sundarbans, Bihar, UP, Kerala	humus/organic 40–50%	rice if drained
👫 Forest and mountain	Himalaya, W and E Ghats	varies	valley sides loamy-silty (rice, wheat); upper slopes coarse (horticulture)

Source conflict: laterite is "rich in iron oxide and potash" in the soil table but "poor in nitrogen, potash, lime, organic matter; rich in iron and aluminium" in the 2013 explanation. Exam rule: "laterite rich in N?" = no; laterite is red, cashew + tapioca ✔.

Soil degradation
→ Erosion (deforestation, bad tillage, overgrazing)
→ Chemical (fertiliser/pesticide, salinisation, irrigation acidity)

→ Physical (mining, construction, machinery)
→ UNCCD: affects 15 million km² (larger than Antarctica), growing ~1 million km²/yr; excessive degradation → desertification

India: a sub-continent

→ Himalaya = natural border
→ Own plate: India = Gondwanaland; rest of Asia on Eurasian plate (Laurasia)
→ Almost all major climate and forest types (desert to tundra)
→ Distinct history and culture

Multiple factors affect the formation of soils

Relief (Topography):

→ Slope: Steeper slopes experience faster erosion, leading to thinner and less developed soils. Gentle slopes allow for better weathering and accumulation of organic matter, promoting richer soil development.
→ Drainage: Well-drained areas allow for water to percolate through the soil, influencing nutrient leaching and mineral composition. Poorly drained areas can lead to waterlogging, affecting oxygen availability and plant growth.
→ Aspect: South-facing slopes in the Northern Hemisphere receive more sunlight, leading to faster decomposition of organic matter and potentially drier soils. North-facing slopes may have slower decomposition and cooler, moister soils.

Multiple factors affect the formation of soils · Anthropogenic Factors

Parent Material (Bedrock):

→ Texture: The type of rock the soil forms from determines its initial mineral content. For example, granite weathers into sandier soils, while basalt weathers into clay-rich soils.

Climate:

→ Temperature: Warmer temperatures accelerate the rate of chemical weathering of rocks, releasing minerals for soil formation. Cold temperatures slow down the process.
→ Precipitation: Rainfall patterns directly impact soil development. High rainfall can leach nutrients from the upper layers, while low rainfall can lead to salinization in arid regions.

Vegetation and Other Forms of Life (Biota):

→ Plants: Plant roots help break down rocks, contributing to physical weathering. Decomposing plant matter adds organic material to the soil, enhancing fertility and soil structure.
→ Microorganisms: Bacteria and fungi in the soil break down organic matter and rock minerals, releasing nutrients for plants and influencing soil structure.
→ Animals: Earthworms and burrowing animals improve aeration and drainage by creating tunnels in the soil.

Time:

→ Soil formation is a slow and continuous process. Over time, the interaction of the above factors leads to the development of distinct soil horizons with varying characteristics. The older the soil, the more developed and mature its profile will be .
→ Time helps in the further disintegration of the rocks into finer material and the addition of minerals to the soil via various biotic and abiotic activities.

→ Anthropogenic Factors:

Descriptions of the soils

→ Black (Regur):

→ – Plateaus of Maharashtra, Saurashtra, Madhya Pradesh, and Chhattisgarh; also extend in the Southeast direction along the Godavari and the Krishna valleys.

→ – When dry, it develops cracks which gives it self-ploughing ability.

→ Red (& Yellow) soil

→ – Red soil is found in areas of low rainfall in the eastern and southern parts of the Deccan Plateau and also along the Piedmont zone of the Western Ghats.

→ – Fertile if **fi**ne-grained.

→ Laterite Soil

→ – It is an acidic soil, making it unsuitable for cultivation.

Descriptions of the soils

→ Found in Southern states, the Western Ghats region of Maharashtra, Odisha, some parts of West Bengal, and the North-east regions.

Arid Soils

→ Include Calcium Kankar nodules which reach the topsoil layers by the capillary action.

→ Can be cultivated if irrigation is available . E.g. The Indira Gandhi Canal project has helped cultivate these soils.

Saline Soils

→ Infertile, with a large proportion of salts of Sodium, Potassium, Magnesium, etc.

Peaty/Marshy

→ In the areas of heavy rainfall and high humidity.

Forest & Mountain Soil

→ In Forest areas, with adequate rainfall .

→ In Forest regions of Himalayas, Western and Eastern Ghats.

→ They vary in their structure and texture depending upon the mountain environment where they are formed.

Soil Degradation

→ Soil degradation is the decline in quality and health of soil . It's a gradual process caused by various factors that diminish the soil's ability to support life. Excessive degradation can cause desertification.

→ According to a UNCCD Report, deforestation, urbanization, and unsustainable farming practices are causing land degradation at an unprecedented scale, affecting 15 million square kilometers—an area larger than Antarctica. The report emphasizes that the global area impacted by land degradation is expanding by about 1 million square kilometers each year, posing significant threats to biodiversity, climate stability, and human livelihoods.

Causes of Soil Degradation:

→ Soil Erosion: Wind and water can wear away topsoil, the most fertile layer. This happens due to:

→ – Unsustainable farming practices: Tilling practices and leaving fields bare leave soil vulnerable.

→ – Overgrazing: Excessive grazing by livestock removes vegetation and compacts soil.

→ Chemical Degradation:

→ – Improper use of chemicals can lead to salinization, where salts accumulate in the soil, making it unsuitable for plant growth.

→ – Improper irrigation practices can lead to an increase in the acidity of soils.

→ Sits on a separate plate which has a geological history separate from the rest of Asia (rest of Asis in Eurasian plate, which was part of Laurasia; India was a part of Gondwanaland).

- Human activity also plays a role in soil formation through **practices like agriculture, deforestation, and urbanization, which can accelerate erosion, deplete nutrients, and alter soil composition.**

Types of Soils in India

Multiple factors affect the formation of soils · Anthropogenic Factors

RECALL DRILL: COVER AND ANSWER

Standard meridian and 5 states?	82°30'E: UP, CG, MP, Odisha, AP (U C My Old Aunt)	Area, rank, coastline, land border?	32,87,263 km², 7th; 11,098 km; 15,106.7 km, 17 states
Channels 10, 9, 6?	Andaman|Nicobar · Lakshadweep|Minicoy · Great Nicobar|Sumatra	Highest: India Himalayan peak; entirely in India; Maharashtra; Kerala; TN?	Kanchenjunga; Nanda Devi; Kalsubai; Anaimudi; Doddabetta
Panch Prayag in order?	Vishnu, Nand, Karna, Rudra, Dev	Left-bank Ganga tributaries?	Ramganga, Gomti, Ghaghara, Gandak, Kosi, Mahananda
Why does Narmada flow west? Brahmaputra U-turn?	Linear rift valley; syntaxial bend at Namcha Barwa	Drainage split; river that crosses Tropic of Cancer twice?	77% Bay : 23% Arabian Sea; Mahi
Cyclone alert lead times; peak months?	72 – 48 – 24 – 12 h; May–June, Oct–Nov	IOD positive; El Niño; ELLA?	good monsoon; weak SW, stronger NE; heatwave years
Soils: alluvial share, black, laterite crops?	>40%; cotton, volcanic; tea, coffee, cashew	Forest + tree cover; target?	21.76% + 3.41% = 25.17%; 33%

🎯 PYQs OF THIS CHAPTER

🎯 Exam lens (from the handout only)

- Priority: VERY HIGH (largest PYQ bank of the six). The handout prints 40 Prelims PYQs here: physiography 15 (2012, 2014×2, 2015, 2016, 2017×2, 2018×2, 2020, 2022, 2023; three carry no year) · drainage 12 (2011, 2013, 2014, 2015, 2016×2, 2017, 2021×2, 2022, 2023; one undated) · climate 3 (2011, 2017, one undated) · forests & soils 10 (2011, 2013, 2015×3, 2016, 2021, 2023×2, one undated). Counted from the handout text.
- Pattern: "Which of these pairs/rivers/peaks/states?" Map-and-list questions: tributaries of Godavari/Brahmaputra, which hills belong to which range, which state has what, which forest has which tree. Facts you must be able to place on a map.
- Mains quoted: none quoted in the material. Hooks: Western Disturbances, cloudbursts, urban flooding, ELLA, river-linking, mangroves, social forestry – all written as current-affairs notes.
- No teacher ink in this PDF; priority = PYQ count in the handout. The handout also has image-only pages (peaks, glaciers, passes, waterfalls, lakes) – I read them and kept them below as compact tables.

★ ***PYQ answers in one line (asked year and the key)***

PYQ BANK: PHYSIOGRAPHY

Year	Asked	Answer / trap
2012	Young fold mountains: deep gorges, U-turn courses, parallel ranges, steep gradients	D, all four (folding of crust; highest ranges; steep slopes = recent; old fold mountains are lower)
2014	Pairs: Cardamom-Coromandel? Kaimur-Konkan? Mahadeo-Central India? Mikir-NE?	C: Mahadeo and Mikir only. Kaimur = eastern Vindhya (~483 km, Katangi near Jabalpur → Sasaram, Rohtas; Rewa, Mirzapur). Cardamom (Yela Mala) = south W. Ghats, SE Kerala + SW TN, UNESCO
2014	Ten Degree Channel separates	A Andaman and Nicobar
2015	Easternmost and westernmost states	D Arunachal and Gujarat

2016	Place-region: Khajuraho-Bundelkhand ✔, Tirupati-Rayalaseema ✔; Bodhgaya-Baghelkhand ✘, Shirdi-Vidarbha ✘, Nashik-Malwa ✘	C (2 and 5). Malwa = W MP + SE Rajasthan; Baghelkhand = NE MP + small W UP; Vidarbha = E Maharashtra (Nagpur, Amravati divisions)
2017	Kohima → Kottayam: minimum states	B = 7 (Nagaland, Assam, WB, Odisha, AP, TN, Kerala)
2017	Himalaya in 5 states only? W. Ghats in 5 only? Pulicat in 2 only?	B: only 3 right. Himalaya >5 states (J&K, HP, Uttarakhand, Sikkim, Arunachal + hill parts of Assam, WB); W. Ghats 6 (30–50 km inland); Pulicat = AP (40%) + TN (60%), 2nd largest brackish lagoon after Chilika, Sriharikota barrier island (Satish Dhawan Space Centre)
2018	Barren Island: active ✔; ~140 km east of Great Nicobar ✘; last erupted 1991, inactive since ✘	D (1 only). Erupting again since 1991, last as recent as 2022
2018	City on longitude nearest Delhi	A Bengaluru (both 77°E)
2020	Siachen Glacier is…	D north of Nubra Valley (eastern Karakoram, ~70 km / 44 mi)
2022	Namcha Barwa-Garhwal ✘ · Nanda Devi-Kumaon ✔ · Nokrek-Sikkim ✘	B (2 only). Namcha Barwa in SE Tibet inside Great Bend of Yarlung Tsangpo; Nokrek biosphere on Tura Range, Meghalaya Plateau
–	Repeated falls in sea level → marshland	D Rann of Kutch (shoreline of emergence)
2023	Amarkantak = Vindhya + Sahyadri? Biligiri = easternmost Satpura? Seshachalam = southernmost W. Ghats?	D none: Amarkantak = Vindhya + Satpura; Biligiri Rangan in SE Karnataka; Seshachalam = Eastern Ghats
–	Peninsular Block: W-coast submergence ✔; residual hills (Velikonda, Mahendragiri) ✔; deep V-valleys ✘	B: ancient stable craton; Eastern Ghats denuded; peninsular rivers mature
–	UP has most neighbouring states · Rajasthan longest int'l border · Sikkim only one-neighbour state	A as printed (see conflict)

Source conflict: last PYQ prints answer **A** (statement 1 not correct), but its explanation says **West Bengal** has the longest international border and **Meghalaya** also has only one neighbouring state, which would make statements 2 and 3 wrong. Check atlas / class notes.

PYQ BANK: DRAINAGE

Year	Asked	Answer / rule
2011	Why Brahmaputra makes a U-turn	B syntaxial bending of young Himalaya. Tethys ~70 mn yrs ago, receded from early Eocene; maximum push from Aravalli (W) and Shillong Plateau (E); Arakan Yoma bends near Namcha Barwa, Sulaiman near Nanga Parbat (hairpin)
2013	Why Narmada flows west	A linear rift valley only (not "slope west")
2014	Through Arunachal: Barak, Lohit, Subansiri	B Lohit + Subansiri; Barak = Manipur, Nagaland, Mizoram, Assam
2015	Godavari tributaries: Vamsadhara, Indravati, Pranahita, Pennar	D Indravati + Pranahita
2016	Brahmaputra tributaries: Dibang, Kameng, Lohit	D all three
2016	Recent river linking	B Godavari-Krishna (Pattiseema)
2017	Teesta: same source as Brahmaputra? Rangeet a tributary? Flows into Bay on the border?	B only 2 right
2021	Which joins the Indus directly: Chenab, Jhelum, Ravi, Sutlej	D Sutlej (others join the Sutlej system first)

2021	Rise from Eastern Ghats: Brahmani, Nagavali, Subarnarekha, Vamsadhara	B Nagavali + Vamsadhara (Brahmani, Subarnarekha rise on Ranchi plateau)
2022	Gandikota canyon	C Pennar
2023	Jhelum through Wular ✔; Krishna feeds Kolleru ✘; Gandak meander → Kanwar ✘	A only one. Kolleru lies between Krishna and Godavari; Kanwar = Burhi Gandak (Sikrahna)
–	River: antecedent, 3 countries, Tibetan origin, irrigation, no distributaries	C Sutlej (Brahmaputra and Indus have distributaries; Teesta rises in Sikkim)

PYQ BANK: FORESTS AND SOILS

Year	Asked	Answer / rule
2023	Rainforest soil nutrient-rich because of fast decomposition	D: I false, II true; soil leached and poor, trees take nutrients as fast as produced
2011	Mangroves as safety hedge (2004 tsunami)	D: extensive roots hold against storms and tides
2013	Laterite: red ✔; rich in N and potash ✘; Rajasthan/UP ✘; tapioca + cashew ✔	C (1 and 4)
2015	Region with mangrove + evergreen + deciduous	D Andaman and Nicobar
2015	Tropical wet evergreen in: Arunachal, HP, Mizoram	C Arunachal + Mizoram (also Assam, Meghalaya, Nagaland, Tripura, WB, A&N, W. Ghats slopes in TN, Karnataka, Kerala, Maharashtra)
2015	Teak dominant in	A tropical moist deciduous
2016	Red Sanders: found in a part of S. India ✔; important in tropical rainforest ✘	A (1 only): Pterocarpus santalinus, endemic to southern Eastern Ghats; wood not aromatic, not Santalum sandalwood; dry deciduous
2021	Black cotton soil forms from weathering of	B fissure volcanic rock
2023	Jackfruit, Mahua, Teak: how many deciduous?	B two (jackfruit evergreen)
–	Mangrove rationale for climate resilience	D bio-shield + livelihood; roots salt-tolerant, not salt-sensitive; do not store freshwater

Traps

→ "highest of..." lists are state-specific (Karoh Haryana, Amsot UP, Someshwar Bihar are Shiwalik; Kangto Arunachal; Shillong Peak Meghalaya)
→ Narmada flows west because of the rift, not slope (2013)
→ Nokrek = Meghalaya, not Sikkim; Namcha Barwa ≠ Garhwal (2022)
→ Amarkantak = Vindhya + Satpura (2023)
→ IOD is Indian Ocean W-E, not Pacific (2017)
→ La Niña helps the monsoon (2011)
→ Rainforest soil is poor (2023)
→ Red Sanders is not aromatic sandalwood. Connects to: Human Geography, Geomorphology, Climatology.

✍ ***The PYQs as printed in the handout, with answers***

Q1.) When you travel to the Himalayas, you will see the following :

- Deep gorges
- U-turn river courses
- Parallel mountain ranges
- Steep gradients causing land-sliding Which of the above can be said to be the evidence for the Himalayas being young fold mountains? (2012)
- 1. 1 and 2 only
- 2. 1, 2 and 4 only
- 3. 3 and 4 only
- 4. 1, 2, 3 and 4

Answer: D The Himalayas are known as the young fold mountains because they were formed only a few million years ago.

- They were formed because of the folding of the earth's crust due to tectonic activities.
- Gorge is a deep cleft resulting from weathering and the erosive activity of a river over geologic timescales.
- These deep gorges are irregular and don't have steep slopes which shows that these are formed recently.
- It shows that the Himalayas are young folded mountains.
- Similarly, other features like U-turn river courses, Parallel mountain ranges, Steep gradients causing landslides etc can be seen generally in young folded mountains only.
- These all features constitute that the Himalayas are young folded mountains.
- The old folded mountain is lower than a young folded mountain.
- The Himalayas have the highest mountain ranges in the world which show that the Himalayas are young folded mountains.

Q2.) Consider these pairs –

- Cardamom Hills Coromandel Coast
- Kaimur Hills Konkan Coast
- Mahadeo Hills Central India
- Mikir Hills North-East India

Which of the above pairs are correctly matched? (2014)

1. 1 and 2 only
2. 2 and 3 only
3. 3 and 4 only
4. 2 and 4 only

Answer: C Kaimur Range is the eastern portion of the Vindhya Range, about 483 kilometres (300 mi) long, extending from around Katangi in Jabalpur district of Madhya Pradesh to around Sasaram in Rohtas district of Bihar.

- It passes through the Rewa and Mirzapur divisions.
- The Cardamom Hills or Yela Mala are a mountain range of southern India and part of the southern Western Ghats located in southeast Kerala and southwest Tamil Nadu in India.
- Their name comes from the cardamom spice grown in much of the hills' cool elevation, which also supports pepper and coffee.
- The Western Ghats and Periyar Sub-Cluster including the Cardamom Hills, are UNESCO World Heritage Sites.

Q3.) Which one of the following pairs of islands are separated from each other by the 'Ten Degree Channel'? (2014)

(a) Andaman and Nicobar
(b) Nicobar and Sumatra
(c) Maldives and Lakshadweep
(d) Sumatra and Java

Answer: A This is a map based question, please refer to map.

Q4.) Which of the following pairs of States of India indicates the easternmost and Westernmost State? (2015)

(a) Assam and Rajasthan
(b) Arunachal Pradesh and Rajasthan
(c) Assam and Gujarat
(d) Arunachal Pradesh and Gujarat

Answer: D

- Go through the Map

Q5.) Consider the following pairs: Famous place Region

- Bodhgaya – Baghelkhand
- Khajuraho – Bundelkhand
- Shirdi: – Vidarbha
- Nasik (Nashik) – Malwa
- Tirupati – Rayalaseema

Which of the pairs given above are correctly matched? (2016)

1. 1, 2 and 4
2. 2, 3, 4 and 5
3. 2 and 5 only
4. 1, 3, 4 and 5

Answer: C

- Malwa region includes districts of Western Madhya Pradesh and parts of South-Eastern Rajasthan. Baghelkhand covers the North-Eastern regions of Madhya Pradesh and a small area of Western Uttar Pradesh. Vidarbha is the Eastern region of the Indian State of Maharashtra, comprising Nagpur division and Amravati division.

Q6.) If you travel by road from Kohima to Kottayam, what is the minimum number of States within India through which you can travel, including the origin and the destination? (2017)

- 6
- 7
- 8
- 9

Answer: B

- States – Kerala, Tamil Nadu, Andhra Pradesh, Odisha, West Bengal, Assam, Nagaland

Q7.) Consider the following statements:

1. In India, the Himalayas are spread over five States only.
2. The Western Ghats are spread over five States only.
3. Pulicat Lake is spread over two States only.

Which of the statements given above is/are correct? (2017)

(a) 1 and 2 only
(b) 3 only
(c) 2 and 3 only
(d) 1 and 3 only

Answer: B

- In India, the Himalayas are spread over more than 5 states.
- The Western Ghats are spread over 6 states, including Gujarat.
- Pulicat Lake is spread over Andhra Pradesh and Tamil Nadu.
- The Indian Himalayan Region (IHR) is the section of the Himalayas within India, spanning the states of Jammu & Kashmir, Himachal Pradesh, Uttarakhand, Sikkim, Arunachal Pradesh, as well as the hill regions of two states – Assam and West Bengal.
- Western Ghats are a chain of mountains running parallel to India's western coast, approximately 30–50 km inland.
- The Western Ghats traverse the States of Kerala, Tamil Nadu, Karnataka, Goa, Maharashtra and Gujarat.
- Pulicat Lake formerly PralayaKaveri is the second largest brackish water lake or lagoon in India, after Chilika Lake.
- It straddles the border of Andhra Pradesh and Tamil Nadu states with over 40% of it in Andhra Pradesh and 60% in Tamil Nadu situated on the Coromandel Coast in South India.
- The lake encompasses the Pulicat Lake Bird Sanctuary. The barrier island of Sriharikota separates the lake from the Bay of Bengal and is home to the SatishDhawan Space Centre.

Q8.) Consider the following statements:

- The Barren Island volcano is an active volcano located in Indian territory.
- Barren Island lies about 140 km east of Great Nicobar.
- The last time the Barren Island volcano erupted was in 1991 and it has remained inactive since then.

Which of the statements given above is/are correct? (2018)

(a) 1 and 3 only
(b) 2 and 3
(c) 3 only
(d) 1 only

Answer: D

- According to the researchers at Goa-based National Institute of Oceanography (NIO), the Barren Island volcano, India's only live volcano in the Andaman and Nicobar islands.
- The only live volcano in the Andaman and Nicobar islands is erupting once again. The Barren Island volcano, located 140-km north-east of Port Blair in Great Nicobar Island, dormant for more than 150 years, started erupting in 1991 and has since then shown intermittent activity.
- Last eruption was as recent as 2022.

Q9.) Among the following cities, which one lies on a longitude closest to that of Delhi? (2018)

(a) Bengaluru
(b) Hyderabad
(c) Nagpur
(d) Pune

Answer: A

- Both Delhi and Bangalore fall on the 77° E longitude.

Q10.) Siachen Glacier is situated to the: (2020)

(a) East of Aksai Chin
(b) East of Leh
(c) North of Gilgit
(d) North of Nubra Valley

Answer: D

- Siachen Glacier, one of the world's longest mountain glaciers, lying in the Karakoram Range system of Kashmir near the India–Pakistan border, extending for 44 mi (70 km) from north-northwest to south-southeast.
- The Siachen Glacier is a glacier located in the eastern Karakoram Range in the Himalayas.

Q11.) Consider the following pairs: Peak Mountains

- Namcha Barwa – Garhwal Himalaya
- Nanda Devi – Kumaon Himalaya
- Nokrek – Sikkim Himalaya

Which of the pairs given above is/are correctly matched? (2022)

(a) 1 and 2
(b) 2 only
(c) 1 and 3
(d) 3 only

Answer: B

- Namcha Barwa is in an isolated part of southeastern Tibet. It stands inside the Great Bend of the Yarlung Tsangpo River.
- On other hand the Garhwal Himalayas are mountain ranges located in the Indian state of Uttarakhand
- Kumaun Himalayas rise to 7,817 metres at Nanda Devi, the range's highest peak.
- The Nokrek Biosphere Reserve is located in the northeast of India on the Tura Range, which forms part of the Meghalaya Plateau

Q12.) Which one of the following is the best example of repeated falls in sea level, giving rise to present-day extensive marshland?

(a) Bhitarkanika Mangroves
(b) Marakkanam Salt Pans
(c) Navpada Swamp
(d) Rann of Kutch

Answer: D

- Rann of Kutch is a perfect example of shoreline of emergence.

Q13.) Consider the following statements:

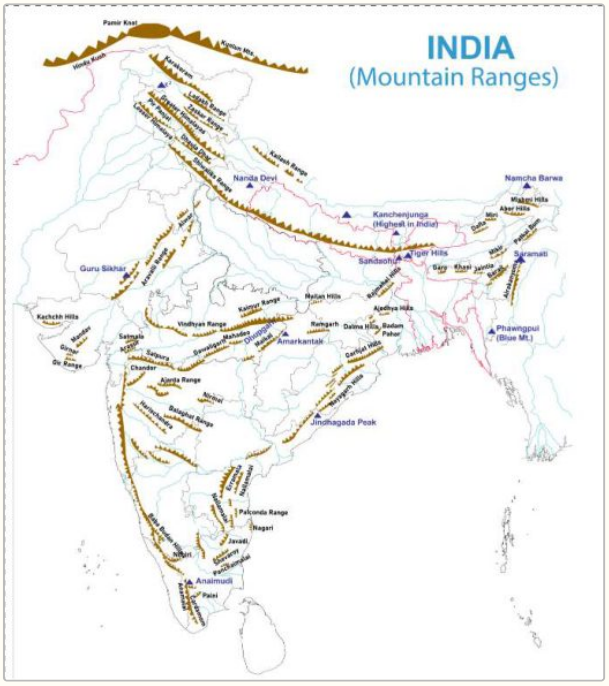
- Amarkantak Hills are at the confluence of Vindhya and the Sahyadri Ranges.
- Biligirirangan Hills constitute the easternmost part of Satpura Range.
- Seshachalam Hills constitute the southernmost part of Western Ghats.

How many of the statements given above are correct? (2023)

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: D

- Amarkantak is at the confluence of Vindhyas and Satpuras.
- Biligiriranjan hills are in South East Karnataka.
- Seshachalam Hills are a part of the Eastern Ghats.



Q14.) Which of the following geographical features or phenomena is/are associated with the Peninsular Block of India?

1. Submergence of parts of the western coast due to tectonic activity
2. Presence of residual mountain ranges such as the Veliconda Hills and Mahendragiri Hills
3. Deep, V-shaped river valleys formed by fast-flowing rivers

Select the answer using the code given below:

(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only

Answer: B The Peninsular Block is an ancient stable cratonic landmass, but parts of the western coast experienced submergence due to tectonic movements. Eastern Ghats are denuded mountains. Most rivers of Peninsular India are mature rivers flowing over old hard rocks.

Q15.) Which of the following with reference to Indian States is/are not correct?

1. Uttar Pradesh shares its boundary with the highest number of other Indian States.
2. Rajasthan shares the longest international border among all Indian States.
3. Sikkim is the only State that shares its boundary with just one other Indian State.

Select the answer using the code given below:

(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only

Answer: A West Bengal has longest international border. Meghalaya also shares border with only 1 state.

Q.) The Brahmaputra, Irrawaddy and Mekong rivers originate in Tibet and flow through narrow and parallel mountain ranges in their upper reaches. Of these rivers, Brahmaputra makes a "U" turn in its course to flow into India. This "U" turn is due to – (2011)

(a) Uplift of folded Himalayan series
(b) Syntaxial bending of geologically young Himalayas
(c) Geo-tectonic disturbance in the tertiary folded mountain chains
(d) Both (a) and (b) above

Ans: B About 70 million years ago, there was a sea called Tethys sea where the Himalayas stand today. The sea started to recede during the early Eocene which was marked by the coming together of Indian and Asian plates.

- This trend of marine regression continued until the final emergence of the Himalayan Mountains by the end of the Tertiary Period.
- The collision of the Indian Plate with the Asian Plate retarded the pace of northerly drift of the Indian Plate.

- During the upliftment of the Himalayas, the maximum push was offered by the Aravalli mountains in the West and Shillong Plateau in the East.
- Arakanyoma and Sulaiman ranges are regarded as an offshoot of Himalaya, Arakanyoma bends near Namcha Barwa (Arunachal Pradesh East), while Sulaiman bends near Nanga Parbat.
- Both these mountains/offshoots give the appearance of a hairpin.
- Due to this structure Brahmaputra river makes U turn into the Indian subcontinent.

Q.) The Narmada river flows to the west, while most other large peninsular rivers flow to the east. Why? (2013)

1. It occupies a linear rift valley.
2. It flows between the Vindhyas and the Satpuras.
3. The land slopes to the west from Central India.

Select the correct answer using the codes given below.

(a) 1 only
(b) 1 and 3 only
(c) 2 and 3 only
(d) None

Answer: A Narmada and Tapi flow through rift valleys caused due to the upheaval of the Himalayas.

- The Narmada rises at an elevation of about 3,500 feet (1,080 metres) in the Maikala Range in eastern Madhya Pradesh state on the border with Chhattisgarh state.
- It first follows a tortuous course through the hills near Mandla and then turns northwest to pass the city of Jabalpur.
- There it turns southwest and then enters the structural trough between the Vindhya and Satpura ranges at Marble Rocks Gorge.
- Turning more westward, the river continues across Madhya Pradesh until it passes into Gujarat state.
- The Narmada enters the Gulf of Khambhat through an estuary 13 miles (21 km) wide, just below Bharuch.
- The Narmada is flowing through the area where the land is not sloping towards the west from central India, but it is flowing west because of the rift valleys only.

Q.) Consider the following rivers:

1. Barak
2. Lohit
3. Subansiri

Which of the above flows / flow through Arunachal Pradesh? (2014)

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: B The Barak is a 900 km river flowing through the states of Nagaland, Manipur and Assam in India and into the Bay of Bengal via Bangladesh.

- The Barak River flows 900 kilometres through the states of Manipur, Nagaland, Mizoram and Assam in India. Further it enters Bangladesh where it is known by the name of the Surma and the Kushiyara and later called the Meghna before receiving the combined flow of the Ganga and the Brahmaputra.
- Barak rises in the Manipur hills and enters the plains near Lakhipur. The river enters Bangladesh as Surma and Kushiyara. Later, the river is called the Meghna and receives the combined flow of the Ganga and Brahmaputra.
- It is being developed as one of the Inland waterways of India.
- The environment of Barak's basin hosts a wide variety of flora and fauna.

Q.) Consider the following rivers :

1. Vamsadhara
2. Indravati
3. Pranahita
4. Pennar

Which of the above are tributaries of Godavari? (2015)

(a) 1,2 and 3
(b) 2,3 and 4
(c) 1,2 and 4
(d) 2 and 3 only

Answer: D The Godavari is India's second longest river after the Ganga.

- Its source is in Trimbakeshwar, Maharashtra
- The major tributaries of the river can be classified as the left bank tributaries which include the Purna, Pranhita, Indravati and Sabari River covering nearly 59.7% of the total catchment area of the basin and the right bank tributaries Pravara, Manjira, Manair together contributing 16.1% of the basin.
- River Vamsadhara or River Vamsadhara is an important east flowing river between Rushikulya and Godavari, in Odisha and Andhra Pradesh states in India.

- The Penna (also known as Pennar, Penner, Pennеru or North Pinākinī) is a river of southern India.
- The Penna rises in Nandi Hills in Chikballapur District of Karnataka state and runs north and east through the states of Karnataka and Andhra Pradesh to empty into the Bay of Bengal.

Q.) Which of the following is/are tributary/tributaries of Brahmaputra?

1. Dibang
2. Kameng
3. Lohit

Select the correct answer using the code given below: (2016)

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: D The Brahmaputra (meaning the son of Brahma):

- The Brahmaputra's source is the Chemayungdung glacier in southwestern Tibet. It's source is very close to the sources of Indus and Satluj.
- Mariam La separates the source of the Brahmaputra from the Manasarovar Lake.
- In Tibet, it passes through the depression formed by the Indus-Tsangpo Structure Zone between the Great Himalayas in the south and the Kailas Range in the north.

Brahmaputra Luit, Dilao(Basin features):

Tributaries

- **Left** – Lhasa River, Nyang River, Parlung Zangbo, Lohit River, Dhansiri River, Kolong River
- **Right** – Kameng River, Manas River, Beki River, Raidak River, Jaldhaka River, Teesta River, Subansiri River

Q.) Recently, linking of which of the following rivers was undertaken? (2016)

(a) Cauvery and Tungabhadra
(b) Godavari and Krishna
(c) Mahanadi and Son
(d) Narmada and Tapti

Answer: B

- India took a step forward in its ambitious but long-pending goal to interlink major rivers to form a national water grid. The Pattiseema project lifts flood water from the river Godavari and pumps it into the Polavaram right canal that empties into the river Krishna in Vijayawada.
- Andhra Pradesh Government is interlinking Krishna-Godavari rivers through the Pattiseema lift irrigation scheme.

Q.) With reference to river Teesta, consider the following statements:

1. The source of river Teesta is the same as that of Brahmaputra but it flows through Sikkim.
2. River Rangeet originates in Sikkim and it is a tributary of river Teesta.
3. River Teesta flows into Bay of Bengal on the border of India and Bangladesh.

Which of the statements given above is/are correct? (2017)

(a) 1 and 3 only
(b) 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: B Teesta River (or Tista River) is a 315 km (196 mi) long river that rises in the eastern Himalayas, flows through the Indian states of West Bengal and Sikkim through Bangladesh and enters the Bay of Bengal

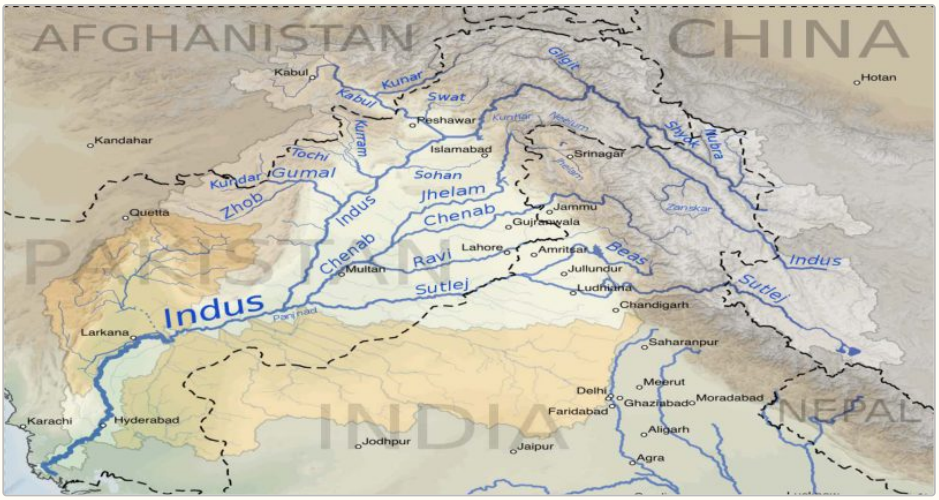
- The Teesta River originates from the Pahunri (or Teesta Kangse) glacier above 7,068 metres (23,189 ft) and flows southward through gorges and rapids in the Sikkim Himalaya.
- It is fed by rivulets arising in the Thangu, Yumthang and Donkha mountain ranges.
- The river then flows past the town of Rangpo where the Rangpo River joins, and where it forms the border between Sikkim and West Bengal up to Teesta Bazaar.
- Just before the Teesta Bridge, where the roads from Kalimpong and Darjeeling join, the river is met by its main tributary, the Rangeet River.

Q.) With reference to the Indus River system, of the following four rivers, three of them pour into one of the joins the Indus direct. Among the following, which one is such river that joins the Indus direct? (2021)

(a) Chenab
(b) Jhelum
(c) Ravi
(d) Sutlej

Answer: D

- The Indus is a transboundary river of Asia and a trans-Himalayan river of South and Central Asia. The 3,180 km river rises in Western Tibet, flows northwest through the Ladakh and Gilgit-Baltistan.
- Because Jhelum, Ravi and Chenab join theStulej, which joins the Indus directly.



Q.) Consider the following Rivers:

1. Brahmani
2. Nagavali
3. Subarnarekha
4. Vamsadhara

Which of the above rise from the Eastern Ghats? (2021)

(a) 1 and 2
(b) 2 and 4
(c) 3 and 4
(d) 1 and 3

Answer: B

- Statement 1 is incorrect. The Brahmani, known as South Koel in its upper reaches, rises near Nagri village in Ranchi district of Jharkhand at an elevation of about 600 m.
- Statement 2 is correct. The Nagavali river originate in the Eastern Ghats near Lakhbahal, located at an altitude of 1,300 metres in the Kalahandi district of the Indian state Odisha. Langulya is another name for Nagavali River.
- Statement:3 is incorrect. Subernarekha River originate in the Chhotnagpur Plateau near Nagri village in the Ranchi District of Jharkhand at an elevation of 600 m.
- Statement 4 is correct. Vamsadhara river rises in the Eastern Ghats of Orissa State and enters Srikakulam District in Bhamini Mandal and finally falls into the Bay of Bengal near Kalingapatnam.
- Vamsadhara: It is an east-flowing river that originates in the Kalahandi district of Odisha, Eastern Ghat, flows in Odisha, along its boundary with Andhra Pradesh and finally joins the Bay of Bengal at Kalingapatnam, Andhra Pradesh. Hence, 4 is correct.

Q.) Gandikota canyon of South India was created by which one of the following rivers? (2022)

(a) Cauvery
(b) Manjira
(c) Pennar
(d) Tungabhadra

Answer: C Gandikota is a small village in the Kadapa district of Andhra Pradesh. The village is majorly known for housing the spectacular gorge which is famously adjudged as the Grand Canyon of India. The stunning gorge has been created by the waters of the famous river Pennar that streams from the Erramala hills.

Q.) Consider the following statements:

1. Jhelum River passes through Wular Lake.
2. Krishna River directly feeds Kolleru Lake.
3. Meandering of the Gandak River formed Kanwar Lake.

How many of the statements given above are correct? (2023)

(a) Only one
(b) Only two
(c) All three
(d) None

Answer: A

- Wular Lake is a part of the Jhelum River Drainage.
- Kolleru lake lies between Krishna and Godavari.
- Kanwar lake is created by the meandering of Burhi Gandak (Sikrahna) River.

Q.) Identify the river of the Indian sub-continent on the basis of the following information:

1. It has an antecedent drainage system.
2. It flows through three countries.

3. It originates in the Tibetan Plateau and is an important river for irrigation.
4. It does not form distributaries.
Select the answer from the following:
(a) Brahmaputra
(b) Indus
(c) Sutlej
(d) Teesta
Answer: C Brahmaputra has distributaries. Teesta originates in Sikkim. Indus has distributaries.

Q.) La Nina is suspected to have caused recent floods in Australia. How is La Nina different from El Nino?

1. La Nina is characterised by unusually cold ocean temperature in the equatorial Indian Ocean whereas El Nino is characterised by unusually warm ocean temperatures in the equatorial Pacific Ocean.

2. El Nino has an adverse effect on the south-west monsoon of India, but La Nina has no effect on monsoon climate.

Which of the statements given above is/are correct? (2011)

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Answer: D La Niña is a climate pattern that describes the cooling of surface ocean waters along the tropical west coast of South America (It means when Peru/Chile cold current reestablished itself).

- La Nina is considered to be the counterpart to El Nino, which is characterised by unusually warm ocean temperatures in the equatorial region of the Pacific Ocean.
- Over the Indian subcontinent, El Nino during winter results in the development of warm conditions. During summer, it leads to dry conditions and deficient monsoon.
- It also leads to drought in Australia.
- On the other hand, La Nina results in better than normal monsoon in India.
- At the same time, in Australia, it has caused floods.
- La Nina is characterised by unusually cold ocean temperature in the equatorial Indian Ocean whereas, El Nino is characterised by unusually warm ocean temperature in the equatorial Pacific Ocean.El Nino has an adverse effect on the south-west monsoon of India, but La Nina has no effect on the monsoon climate.

Q.) With reference to 'Indian Ocean Dipole (IOD)' sometimes mentioned in the news while forecasting Indian monsoon, which of the following statements is/are correct?

1. IOD phenomenon is characterized by a difference in sea surface temperature between tropical Western Indian Ocean and tropical Eastern Pacific Ocean.

2. An IOD phenomenon can influence El Nino's impact on the monsoon.

Select the correct answer using the code given below: (2017)

(a) 1 only

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

Answer: B The Indian Ocean Dipole (IOD) is defined by the difference in the sea surface temperature between the two equatorial areas of the Indian Ocean – a western pole near the Arabian Sea (in western Indian Ocean) and an eastern pole closer to the Bay of Bengal (in the eastern Indian Ocean).

- The IOD affects the climate of Southeast Asia, Australia and other countries that surround the Indian Ocean Basin. The Indian Monsoon is invariably influenced by the IOD.
- A positive IOD occurs when the sea surface temperatures are greater than normal in the Arabian Sea and less than normal in the tropical eastern Indian Ocean.
- When the reverse is the case, a negative IOD is said to have developed.
- A positive IOD leads to greater monsoon rainfall and more active (above normal rainfall) monsoon days while negative IOD leads to less rainfall and more monsoon break days (no rainfall).
- El-Nino conditions normally result in diminished monsoonal rainfall.
- However, if there is a positive IOD, it can bring good rains to India despite an El Nino year.

Q.) With reference to the climate of Andaman and Nicobar Islands, which of the following statements is/are correct?

1. The climate can be defined as a humid, tropical coastal climate.

2. It receives rainfall from both South-west monsoon and North-east monsoon.

3. Maximum precipitation is between December and May.

Select the answer using the code given below:

(a) 1 only

(b) 1 and 2

(c) 2 and 3

(d) 3 only

Answer: B The Andaman and Nicobar Islands experience a humid tropical maritime/coastal climate. In Ocean, so both SW and NE winds are onshore. Maximum rainfall generally occurs during the South-West Monsoon months, especially May to September. (Heat=LP=rain, and NH has heat in March-Sept)

Q.) Consider the following statements: Statement-I : The soil in tropical rain forests is rich in nutrients. Statement-II : The high temperature and moisture of tropical rain forests cause dead organic matter in the soil to decompose quickly.

Which one of the following is correct in respect of the above statements? (2023)

- Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I
- Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-I
- Statement-I is correct but Statement-II is incorrect
- Statement-I is incorrect but Statement-II is correct

Answer: D

- The soil of Tropical Rainforests is highly leached and poor in nutrients. Though there is a high availability of leaf litter and high decomposing activity, the trees in the region

consume these nutrients as fast as they are produced. This leaves the soil devoid of nutrients.

Q.) The 2004 Tsunami made people realise that mangroves can serve as a reliable safety hedge against coastal calamities. How do mangroves function as a safety hedge? (2011)

- The mangrove swamps separate the human settlements from the sea by a wide zone in which people neither live nor venture out\
- The mangroves provide both food and medicines which people are in need of after any natural disaster
- The mangrove trees are tall with dense canopies and serve as art excellent shelter during a cyclone or tsunami
- The mangrove trees do not get uprooted by storms and tides because of their extensive roots.

Answer: D Benefits of mangroves:

- Coastal protection: The dense root systems of mangrove forests trap sediments flowing down rivers and off the land.
- This helps stabilize the coastline and prevents erosion from waves and storms.
- In areas where mangroves have been cleared, coastal damage from hurricanes and typhoons is much more severe.
- By filtering out sediments, the forests also protect coral reefs and seagrass meadows from being smothered in sediment.
- Fisheries: Mangrove forests are home to a large variety of fish, crab, shrimp, and mollusk species.
- These fisheries form an essential source of food for thousands of coastal communities around the world.
- The forests also serve as nurseries for many fish species, including coral reef fish.
- This makes mangrove forests vitally important to coral reef and commercial fisheries as well.
- Timber and plant products: Mangrove wood is resistant to rot and insects, making it extremely valuable.
- Many coastal and indigenous communities rely on this wood for construction material as well as for fuel.
- These communities also collect medicinal plants from mangrove ecosystems and use mangrove leaves as animal fodder. Recently, the forests have also been commercially harvested for pulp, wood chip, and charcoal production.
- Tourism: Given the diversity of life inhabiting mangrove systems, and their proximity in many cases to other tourist attractions such as coral reefs and sandy beaches, it is perhaps surprising that only a few countries have started to tap into the tourism potential of their mangrove forests.

Q.) Which of the following statements regarding laterite soils of India are correct?

1. They are generally red in colour.

2. They are rich in nitrogen and potash.

3. They are well-developed in Rajasthan and UP.

4. Tapioca and cashew nuts grow well on these soils.

Select the correct answer using the codes given below. (2013)

(a) 1 only

(b) 1 and 2 only

(c) 1 and 4 only

(d) 1, 2, 3 and 4

Ans: C Laterite is soil and rock type rich in iron and aluminium but poor in nitrogen, potash, potassium, lime and organic matter and is commonly considered to have formed in hot and wet tropical areas.

- Nearly all laterites are of rusty-red colouration , because of high iron oxide content.
- They develop by intensive and prolonged weathering of the underlying parent rock.
- Tropical weathering (alteration) is a prolonged process of chemical weathering which produces a wide variety in the thickness, grade, chemistry and ore mineralogy of the resulting soils.
- The majority of the land area containing laterites is between the tropics of Cancer and Capricorn.
- Laterite soil is commonly found in areas of high altitude and heavy rainfall in Karnataka, TN, MP, Bihar, Orissa, Assam and Meghalaya.
- Plants like Cashew grow well along with Tapioca in this type of soil.

Q.) Which one of the following regions of India has a combination of mangrove forest, evergreen forest and deciduous forest? (2015)

(a) North Coastal Andhra Pradesh

(b) South-West Bengal

(c) Southern Saurashtra

(d) Andaman and Nicobar Islands

Answer: D

- The Andaman and Nicobar Islands have a tropical rainforest canopy, made of a mixed flora with elements from Indian, Myanmar, Malaysian and endemic floral strains. So far, about 2,200 varieties of plants have been recorded, out of which 200 are endemic and 1,300 do not occur in mainland India.
- The South Andaman forests have a profuse growth of epiphytic vegetation, mostly ferns and orchids.
- The Middle Andamans harbours mostly moist deciduous forests.
- North Andamans is characterised by the wet evergreen type, with plenty of woody climbers.
- The North Nicobar Islands (including Car Nicobar and Batti Malv) are marked by the complete absence of evergreen forests, while such forests form the dominant vegetation in the central and southern islands of the Nicobar group.
- Grasslands occur only in the Nicobars, and while deciduous forests are common in the Andamans, they are almost absent in the Nicobars. The present forest coverage is claimed to be 86.2% of the total land area.
- The Andamans and Nicobars have nearly 1136 Sq. Km. and 35 Sq. Km. (450 and 13.8 square miles) respectively of mangrove forests.
- They develop in areas covered by brackish water at high tide, such as the mud flats along tidal streams.

Q.) Consider the following States:

1. Arunachal Pradesh

2. Himachal Pradesh

3. Mizoram In which of the following states do 'Tropical Wet Evergreen Forests' occur? (2015)

(a) 1 only

(b) 2 and 3 only

(c) 1 and 3 only

(d) 1, 2 and 3

Answer: C In India, evergreen forests are found on the eastern and western slopes of the Western Ghats in such states as Tamil Nadu, Karnataka, Kerala and Maharashtra.

- Tropical Wet Evergreen Forests are found in Assam Arunachal Pradesh Meghalaya Nagaland Tripura West Bengal and Andaman and Nicobar Island and on the eastern and western slopes of the Western Ghats in such states as Tamil Nadu Karnataka Kerala and Maharashtra.
- Tropical wet evergreen forests: these forests occur in the states of Maharashtra, Karnataka, Tamilnadu, Kerala, Andaman and Nicobar islands in the south and in the entire north-eastern region including the sub-mountain division of West Bengal.

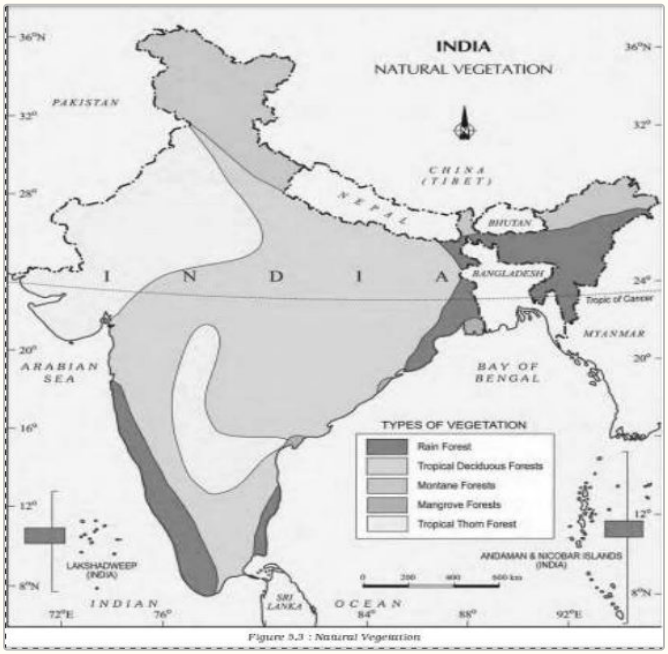


Figure 5.3 : Natural Vegetation

Q.) In India, in which one of the following types of forests is teak a dominant tree species? (2015)

(a) Tropical moist deciduous forest

(b) Tropical RainForest

(c) Tropical thorn scrub forest

(d) Temperate forest with grasslands

Answer: A Tropical moist deciduous forests are found in areas of moderate rainfall of 100 to 200 cm per year, mean annual temperature of about 27°C and the average annual relative humidity of 60% to 75%.

- The trees of these forests drop their leaves for about 6 to 8 weeks during the spring and early summer when sufficient moisture for the leaves is not available.
- Teak trees are the most dominant species of trees found in these forests.
- Bamboos, sal, shisham, sandalwood, khair, Kusum, Arjun, mulberry are some of the other commercially important species found here.

Q.) With reference to 'Red Sanders', sometimes seen in the news, consider the following statements:

1. It is a tree species found in a part of South India.

2. It is one of the most important trees in the tropical rainforest areas of South India.

Which of the statements given above is/are correct? (2016)

(a) 1 only

(b) 2 Only

(c) Both 1 and 2

(d) Neither 1 nor 2

Answer: A Pterocarpus santalinus, with the common names red sanders, red sandalwood, and sanders wood, is a species of Pterocarpus endemic to the southern Eastern Ghats mountain range of South India.

- This tree is valued for the rich red colour of its wood.
- The wood is not aromatic.
- The tree is not to be confused with the aromatic Santalum sandalwood trees that grow natively in South India.
- It's found in Tropical Dry Deciduous Forests.

Q.) The black cotton soil of India has been formed due to the weathering of (2021)

(a) Brown forest soil

(b) Fissure volcanic rock

(c) Granite and schist

(d) Shale and limestone

Answer: B

- Most pedologists believe that these black soils have been formed due to the solidification of lava spread over large areas during volcanic activity in the deccan plateau, thousands of years ago.
- Black soil is of volcanic origin. These soils have been formed by the solidification of lava that has been spread over a large area during volcanic activities in the Deccan plateau (Deccan and the Rajmahal trap) occurred over thousands of years ago.

Q.) Consider the following trees:

1. Jackfruit (Artocarpus heterophyllus)

2. Mahua (Madhuca indica)

3. Teak (Tectona grandis)

How many of the above are deciduous trees? (2023)

(a) Only one

(b) Only two

(c) All three

(d) None

Answer: B

- Jackfruit is a tree of Tropical Evergreen Forests. It is an evergreen tree. Mahua and Teak are deciduous trees.

Q.) Which of the following best explain(s) the rationale for protecting mangrove ecosystems in the context of climate resilience?

1. Mangroves reduce tidal energy and store freshwater, making them ideal sites for paddy cultivation in saline estuarine belts.

2. Their salt-sensitive roots filter seawater, making mangroves key to converting coastal land into freshwater aquaculture zones.

3. By withstanding tidal surges and offering biomass resources, mangroves function both as natural bio-shields and livelihood bases for rural communities.

Select the answer using the code given below:

(a) 1 only

(b) 1 and 2

(c) 2 and 3

(d) 3 only

Answer: D Do not store salt water. Roots are salt tolerant not salt sensitive.

🔤 ***ABBREVIATION COLUMN (full forms as the handout gives them where it does; the one-line meanings are plain-language aids, verify)***

AC = air conditioner
CG = Chhattisgarh
DBH = diameter at breast height — standard way to measure a tree trunk
ELLA = El Niño + La Niña — the handout's word for phases when the Pacific moves between El Niño and La Niña
ENSO = El Niño Southern Oscillation
ESA = Ecologically Sensitive Area
ESCAP = Economic and Social Commission for Asia and the Pacific — UN regional body
FAO = Food and Agriculture Organization — UN body
FW = freshwater
GFRA = Global Forest Resources Assessment — FAO report
HP = Himachal Pradesh
ICAR = Indian Council of Agricultural Research
IHR = Indian Himalayan Region
IMD = India Meteorological Department
IMD-IITM = India Meteorological Department and Indian Institute of Tropical Meteorology
IOD = Indian Ocean Dipole
ITCZ = Inter-Tropical Convergence Zone — low-pressure belt where the trade winds meet
LP = low pressure
MH = Maharashtra
MJO = Madden-Julian Oscillation
MP = Madhya Pradesh
NH = Northern Hemisphere (in highway lists: National Highway)
NIO = North Indian Ocean
NP = National Park
NPD = Nimmu-Padum-Darcha route
NTFPs = non-timber forest products
NW = north-west
NWDA = National Water Development Agency — plans river-linking
PDO = Pacific Decadal Oscillation
PoK = Pakistan-occupied Kashmir
PYQ = previous year question
RSMC = Regional Specialised Meteorological Centre
SKM-ST = handout mnemonic for Brahmaputra tributaries
SST = sea surface temperature
SSTs = sea surface temperatures — plural of SST
SW = south-west
TEJ = Tropical Easterly Jet stream
TN = Tamil Nadu
UAE = United Arab Emirates
UNCCD = United Nations Convention to Combat Desertification
UNESCO = United Nations Educational, Scientific and Cultural Organization
USA = United States of America
UTs = Union Territories
WB = West Bengal

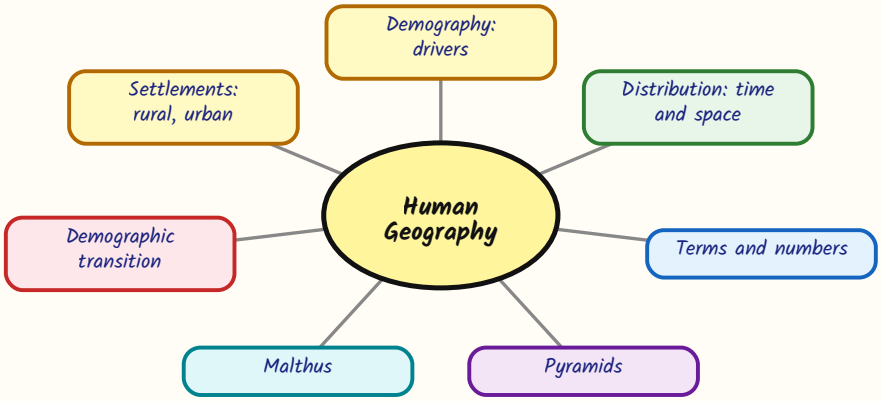
WD = Western Disturbance
WDs = Western Disturbances
WiSAR = handout mnemonic for the four seasons
WMO = World Meteorological Organization

HUMAN GEOGRAPHY

PYQ Prelims 2013 (tribe-state pairs), 2014 (Changpa) + 2 undated (TFR definition; low birth rate / ageing countries)

✎ *Population terms and numbers, pyramids, Malthus, demographic transition, tribes, rural and urban settlements: short, definition-heavy, 4 Prelims PYQs.*

At a glance: the Human Geography set



Pattern: definitions (TFR, sex ratio, dependency) and where-is-which-tribe; theory: Malthus, Demographic Transition. Prepare: five DTT stages as a birth/death table; state-wise tribes. Mains hooks (none quoted): demographic dividend, ageing, urbanisation, gentrification, climate-driven settlement shifts. No teacher ink: priority = PYQ count. ⭐

1 · DEMOGRAPHY

Understand it first: Demography studies how many people, where, and how fast the number changes. Distribution is uneven because of land, climate, water and jobs; the key terms are simple ratios, so know exactly what each measures.

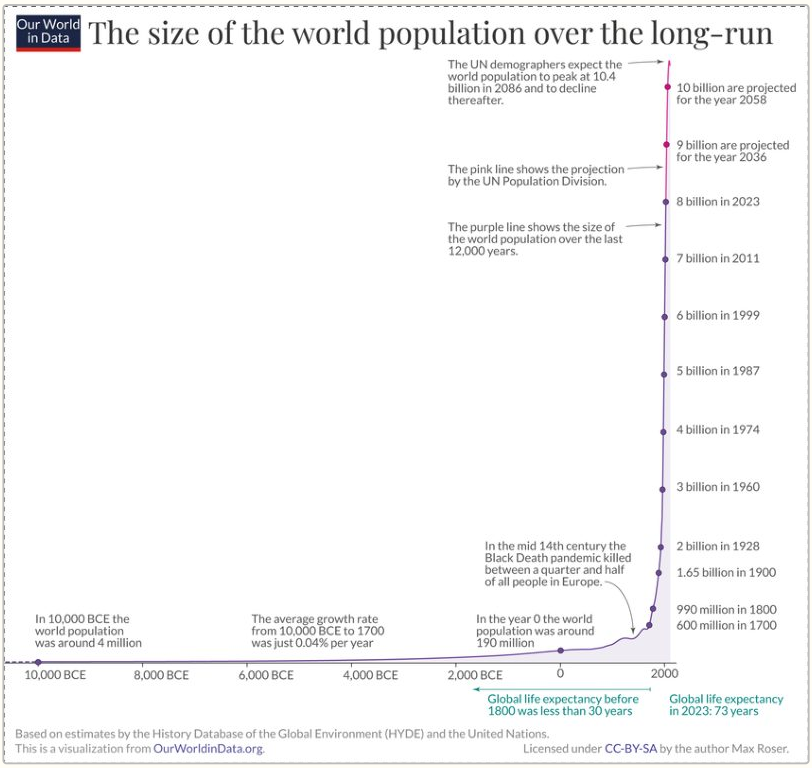
Demography = study of population size, structure, distribution, dynamics (birth, death, migration, age, fertility). 💡

Drivers of growth

→ Birth rate: live births per 1,000 people per period
→ Death rate: deaths per 1,000; low = growth
→ Migration: immigration (in) raises, emigration (out) lowers
→ Fertility rate: live births per 1,000 women of childbearing age; TFR = average children per woman
→ Age structure: youthful = rapid growth; ageing = slow/decline
→ Economic and social: health care, education, jobs, cultural norms

Temporal distribution

→ World: UN, peaks 10.3 billion in mid-2080s
→ 2025 shares: Asia 62.18% · Africa 19.36% · Europe 9.3% (-0.09% vs 2024) · N. America 7.71% · S. America 5.47% · Oceania 0.58%
→ India: most populous; 1.46 billion (UN); peaks early 2060s at 1.7 billion; latest SRS: TFR 1.9



Temporal Distribution of Population

Handout alert: "1.31 billion according to the 2011 census" is printed next to UN's 1.46 billion. Check the census figure before quoting.

Spatial distribution: drivers

- → Geography/climate: Indo-Gangetic plain, temperate zones
- → Resources: river valleys (Nile, Ganga)
- → Economic opportunity: Mumbai, New York, Tokyo
- → Political/social stability: Canada, Australia
- → History: coasts, trade routes
- → Patterns: urban > rural; Asia most populous; coast > inland

Term	Meaning / India number
TFR	average children a woman has in her reproductive years at present birth rates (PYQ: "average live births a woman would have by end of child-bearing age") · 2.1 = replacement rate
IMR	infant (<1 yr) deaths per 1,000 live births · India 25.5 (2021)
Sex ratio	males per 1,000 females (world); India counts females per 1,000 males · 2011: 943 · NFHS-5: 1020
👫 Child sex ratio	juvenile sex ratio, age 0–6 · 2011: 914 · NFHS-5: 920
Life expectancy	average years if current mortality continues · 70.19 yrs (2022)
Wasting / Stunting / Underweight	low weight-for-height / low height-for-age / low weight-for-age
Dependency ratio	dependants (0–14 and >65) ÷ working-age (15–64)
Demographic dividend	working-age ≫ dependants · 56% in 2022–23 → peak 68.9% in 2030
Second dividend	lower fertility + longer life → older people with few dependants and big savings fund investment, boosting growth

Big Mummy's Up-to-Juggle Mania — Replacement 2.1 ("two parents → two kids + a little extra"). India's TFR: NFHS-5 2.0 → SRS 1.9 → projected 1.29 by 2050. Only 5 states have TFR > 2.1: Bihar · Meghalaya · UP · Jharkhand · Manipur.

Stunt = Short — Stunting = short for age; Wasting = thin for height (Waste = Weight-for-height); Underweight = light for age (Under = Weight-for-age).

Factors Affecting Population Growth

→ Migration: The movement of people from one place to another . Immigration (inward migration) can increase population size, while emigration (outward migration) can decrease it.

→ Fertility Rate: The number of live births per 1000 women of childbearing age in a population . The total fertility rate is the average number of children born to women of childbearing age in a population. High fertility rates lead to population growth, while low fertility rates can lead to population decline.

→ Age Structure : The distribution of individuals in different age groups within a population . A youthful age structure (with a high proportion of young people) can lead to rapid population growth, while an aging population structure may lead to slower growth or decline.

Spatial Distribution of Population

Factors affecting population distribution:

→ Geography and Climate: Areas with favorable climates, such as temperate regions, tend to have higher population densities. Harsh climates, such as deserts or polar regions, generally have lower population densities. Example: The Indo-Gangetic Plain in India is densely populated due to its fertile land and favorable climate.

→ Economic Opportunities: Regions with more economic opportunities, such as jobs and industries, attract people. Example: Urban areas like Mumbai, New York, and Tokyo have high population densities due to abundant job opportunities and economic activities.

→ Political and Social Factors: Stability, government policies, and social services also influence population distribution. Example: Countries with political stability and good social services, like Canada and Australia, attract migrants and have specific regions with high population densities.

→ Historical Factors: Historical events, colonization, trade routes, and historical settlements impact population distribution. Example: Coastal regions and ancient trade routes often have higher population densities due to historical trade and settlement patterns.

Patterns of Population Distribution

→ Urban vs. Rural: Urban areas tend to have higher population densities due to concentrated economic activities, better infrastructure, and more services. Rural areas generally have lower population densities. Example: Metropolitan cities like Delhi, Mumbai, and Kolkata in India are densely populated compared to the rural hinterlands.

→ Continental Distribution: Population density varies significantly across different continents. Example: Asia is the most populous continent, home to major population centers like China and India, while continents like Australia and Antarctica have very low population densities.

→ Coastal vs. Inland: Coastal areas are often more densely populated than inland areas due to access to trade, transportation, and resources. Example: Coastal cities like Mumbai, Chennai, and New York are densely populated compared to many inland areas.

Indian Population

→ (d) However, the latest SRS report, released last month by India's Office of the Registrar General and Census Commissioner, said that India's Total Fertility Rate (TFR) had dropped to 1.9 children per woman.

Important Terminologies

→ Total Fertility Rate: average number of children a woman is expected to have during her reproductive years, based on the current Birth Rate. 2.1 is the replacement rate.

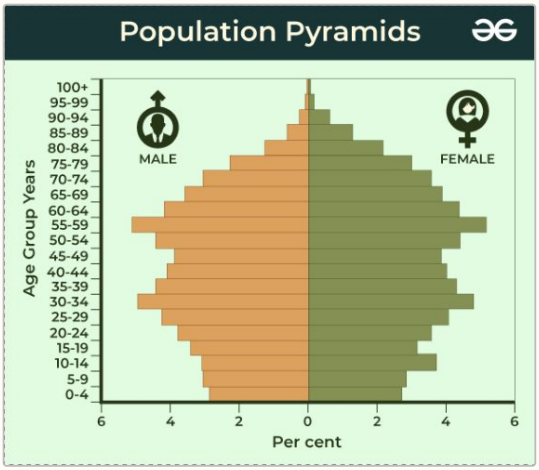
→ – Only 5 states TFR > 2.1; Bihar, Meghalaya, Uttar Pradesh, Jharkhand and Manipur

→ Second Demographic Dividend : Lower fertility and increased life expectancy generally lead to greater wealth. Older people with fewer dependents and large savings can improve the money available for investment in an economy, thereby boosting growth.

2 · POPULATION PYRAMIDS

Understand it first: A population pyramid is a bar chart of age groups, males on one side and females on the other. Its shape tells the story: a wide base means many children and a growing population, a narrow base means an ageing one.

Pyramid = graph of age and sex distribution; shows growth pattern, development, future challenges.



Population Pyramids

Shape	Features	Example
Expansive ▲	broad base, narrow top; rapid growth; high birth rates and dependency; low life expectancy	Nigeria; India = expansive with a stabilising base
Constricted	narrow base (low birth rate), bulging middle (working age), narrow top; growth slowing; top wider than base = ageing, long life	Japan
Stationary ▮	even across ages, narrow only at the very old; birth and death rates low; stable or very slow growth	USA

Population Pyramids

→ (b) It is a useful tool for understanding the demographic structure of a population, which can provide insights into the population's growth patterns, economic development, and potential future challenges.

→ (c) Population pyramids are typically divided into different shapes based on the characteristics of the population they represent.

→ (d) Expansive Pyramid (i) Broad base, indicating a high number of young people. (ii) Narrow top, indicating a lower number of older people. (iii) Rapid population growth. (iv) Example: Nigeria 's population pyramid has a wide base, showing a high proportion of children and young people. This reflects high birth rates and a high dependency ratio. The pyramid narrows sharply at older ages, indicating lower life expectancy. (v) Indian Pyramid is expansive but with a stabilizing base. (e) Constricted Pyramid (i) Narrow base, indicating lower birth rates. (ii) Bulging middle, indicating a higher proportion of working-age individuals. (iii) Narrow top, reflecting fewer elderly people compared to the working-age population.

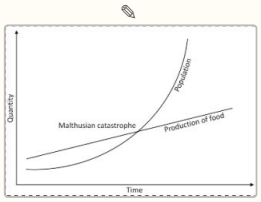
3 · MALTHUS

Understand it first: Malthus argued that population grows faster than food supply, so shortage must follow unless growth is checked. Learn his two kinds of checks and the criticisms.

Thomas Robert Malthus (English cleric)

→ "An Essay on the Principle of Population", 1798

→ Population grows geometrically (2–4–8–16); food arithmetically (1–2–3–4) → population outruns food



Population Theory of Malthus

Checks

→ Preventive: moral restraint, delayed marriage, celibacy, contraception (under our control)

→ Positive: famine, disease, war (after the catastrophe, not in our control)

⚠ Criticisms

→ Technology (Industrial and Green Revolution)

→ Ignores social/economic complexity; "equated sociological and biological needs"

→ Modern reading: environmental sustainability and resource depletion, not just food

Multiply vs Add — Babies multiply (×2), bread adds (+1). Preventive = "Pre" = before birth; Positive checks = "Pos-t-mortem", death does the checking.

▶ Population Theory of Malthus

💡 Key Principles of Malthus's Theory

→ Imbalance : According to Malthus, over time , the potential for population growth outstrips the potential for food production , leading to a situation where the population exceeds the food supply.

→ Checks on Population : Malthus proposed two types of checks to keep the population in balance with the food supply:

→ – Positive Checks : Occur after the Malthusian catastrophe takes place. They are not under our control. These increase the death rate through famine, disease, war, and other calamities that reduce the population.

⚠ Criticisms

→ Technological Advancements : Critics argue that Malthus underestimated human ingenuity and the capacity for technological advancements to increase food production. The Industrial Revolution and later the Green Revolution significantly increased food supply, countering Malthus's predictions.

4 · DEMOGRAPHIC TRANSITION THEORY

Understand it first: This theory says every society moves through stages as it develops: first both births and deaths are high, then deaths fall while births stay high (population explodes), and finally both are low. Place India on that path.

Stage	Name	Birth	Death	Growth	Example
1	High stationary (pre-industrial)	High	High	slow, fluctuating	tribal societies, pre-18th c.
2	Early expanding (transitional)	High	Falling	rapid	many African countries
3	Late expanding (industrial)	Falling	Low	slowing	India, Brazil

4	Low stationary (post-industrial)	Low	Low	stable/ declining	USA, much of Europe
5	Decline (proposed)	Very low	Low	declining	Japan, South Korea, Germany, Italy

Death Drops first, Birth Boards later – Stage 1: both high. Stage 2: death falls → boom. Stage 3: births fall → boom slows. Stage 4: both low. Stage 5: births below deaths.

Handout alert: the assumption line says "fertility and mortality reduce over time. Former before latter", but the stage table shows mortality falls first (Stage 2), fertility next (Stage 3). Exam rule: follow the stage table.

Till 1921
stagnant

1921-51
increasing

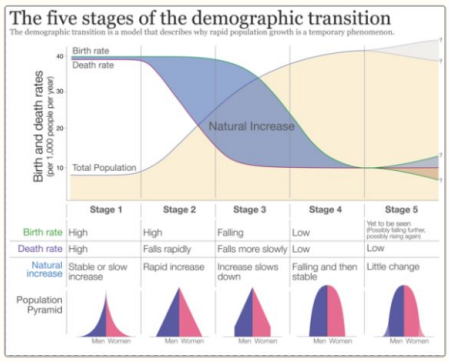
1951-81
explosion

1981 on
late expanding (growth with stability)

India's transition: 1921–81 = early expanding stage; 1981 onwards = late expanding.
Why DTT matters: development ↔ lower birth/death; ageing and pensions; urbanisation; women's education and rights lower fertility; environmental pressure vs labour shortages.

Demographic Transition Theory

→ The theory is divided into four to five stages, each characterized by distinct demographic patterns.



Demographic Transition Theory

Stage 1: High Stationary Stage (Pre-Industrial Stage)

→ Characteristics: This stage is marked by high fertility and high mortality due to limited access to medical care, poor nutrition, and lack of sanitation. Populations remain relatively stable but can fluctuate due to famines, diseases, and wars.

Stage 2: Early Expanding Stage (Transitional Stage)

→ Characteristics: Improved healthcare, sanitation, and food supply lead to a significant decline in mortality rates, particularly among infants and children. Birth rates remain high, resulting in rapid population growth.

Stage 3: Late Expanding Stage (Industrial Stage)

→ Death Rates: Low
→ Characteristics: Birth rates begin to decline due to changes in social norms, increased access to contraception, improved women's education and employment opportunities, and urbanization. Population growth rates begin to slow down.

Stage 4: Low Stationary Stage (Post-Industrial Stage)

→ Birth Rates: Low
→ Death Rates: Low
→ Characteristics: Both birth and death rates are low, leading to a stable or slowly growing population. Societies in this stage often face challenges related to aging populations and the economic impacts of a shrinking workforce.

→ Birth Rates: Very low

Stage 5: (Proposed) Decline Stage

→ Death Rates: Low

→ Characteristics: In this speculative stage, birth rates fall below death rates, leading to a decline in the overall population. This stage is associated with highly developed economies, where the cost of raising children is high and social attitudes towards family size have changed significantly.

Significance of the Demographic Transition Theory

→ **Economic Development** : The DTT suggests that economic development is associated with lower birth and death rates. As countries develop, they move through the stages, which has implications for planning and policy-making.

→ **Population Aging** : In the later stages, countries face challenges related to aging populations, such as increased healthcare costs and the need for pension systems.

→ **Urbanization** : The transition often accompanies increased urbanization, changing patterns of settlement and resource use.

→ **Education and Women's Rights** : The decline in birth rates is closely linked to improvements in women's rights, education, and employment opportunities.

→ **Environmental Impact** : Population dynamics affect environmental sustainability. Rapid population growth can lead to increased pressure on natural resources while declining populations might lead to labor shortages and economic challenges.

5. TRIBES AND PYQ

PYQ 2013 PYQ 2014

Understand it first: Tribe questions are match-the-pair questions: tribe with State or region. Learn them State by State from the handout list.

PYQ	Asked	Answer
2013	Limboo-Sikkim ✔ · Karbi-Himachal ✘ · Dongaria-Odisha ✔ · Bonda-Tamil Nadu ✘	A (1 and 3). Limbu = Kirati, Limbuwan (E Nepal, N Sikkim, W Bhutan). Dongria Kondh of Niyamgiri hills, Odisha; god Niyam Raja; Niyam Dongar ~4,000 m
2014	Changpa: live mainly in Uttarakhand ✘; rear pashmina goats ✔; Scheduled Tribe ✔	B (2 and 3). Semi-nomadic Tibetan people, Changtang, Ladakh; yaks, horses, sheep; pashmina/cashmere; smaller number in W Tibet, partly relocated for Changtang Nature Reserve; half a million nomads in Changtang (1989)
Undated	Total fertility rate is defined as	D average live births a woman would have by end of child-bearing age (not birth rate, not births per couple, not birth rate minus death rate)
Undated	Low birth rate / ageing / declining: Italy, Japan, Nigeria, South Korea, South Africa	A = Italy, Japan, South Korea (1, 2, 4)

Changpa = Chang-Pa-shmina — Ladakh's high plateau herders (Changtang). Dongria = "Dongar" (hill) in Niyamgiri hills. Karbi ≠ Himachal; Bonda ≠ Tamil Nadu.

Table No. 1A: State-wise major tribes of India

Sl_No.	States/UTs	Major Tribes
1	Andaman and Nicobar Islands	Jarawa , Oraons, Onges, Sentinelese, Shompens
2	Andhra Pradesh	Andh, Sadhu Andh, Bhagata, Bhil, Chenchus (Chenchawar), Gadabas, Gond, Goundu, Jatapus, Kammara, Kattunayakan, Kolawar, Kolam, Konda, Manna Dhora, Pardhan, Rona, Savaras, Dabba Yerukula, Nakkala, Dhulia, Thoti, Sugalis.
3	Arunachal Pradesh	Apatanis, Abor, Dafla, Galong, Momba, Sherdukpen, Singpho.
4	Assam	Chakma, Chutiya, Dimasa, Hajong, Garos, Khasis, Gangte.
5	Bihar	Asur, Baiga, Birhor, Birjia, Chero, Gond, Parhaiya, Santhals, Savar.
6	Chhattisgarh	Agariya, Bhaina, Bhattra, Biar, Khond, Mawasi, Nagasia.
7	Goa	Dhodia, Dubia, Naikda, Siddi,Varli.
8	Gujarat	Barda, Bamcha, Bhil, Charan, Dhodia, Ganta, Paradhi, Patelia.
9	Himachal Pradesh	Gaddis, Gujjars, Khas, Lamba, Lahaulas, Pangwala, Swangla.
10	Jammu and Kashmir	Bakarwal, Balti, Beda, Gaddi, Garra, Mon, Purigpa, Sippi.
11	Jharkhand	Birhors, Bhumij, Gonds, Kharia, Mundas, Santhals, Savar.
12	Karnataka	Adiyan, Barda, Gond, Bhil, Iruliga, Koraga, Patelia, Yerava.
13	Kerala	Adiyan, Arandan, Eravallan, Kurumbas, Malai arayan, Moplahs, Uralis.
14	Madhya Pradesh	Baigas, Bhils, Bharia, Birhors, Gonds,Katkari, kharia, Khond, Kol, Murias.
15	Maharashtra	Bhaina, Bhunjia, Dhodia, Katkari, Khond, Rathawa, Warlis.
16	Manipur	Aimol, Angami, Chiru, Kuki, Maram, Monsang, Paite, Purum, Thadou.
17	Meghalaya	Chakma, Garos, Hajong, Jaintias Khasis, Lakher, Pawai, Raba.
18	Mizoram	Chakma, Dimasa, Khasi, Kuki, Lakher, Pawai, Raba, Synteng.
19	Nagaland	Angami, Garo, Kachari, Kuki, Mikir, Nagas, Sema.
20	Odisha	Gadaba, Gharu, Kharia, Khond, Matya, Oraons, Rajuar, Santhals.
21	Rajasthan	Bhils, Damaria, Dhanka, Meenas(Minas), Patelia, Sahariya.
22	Sikkim	Bhutia, Khas, Lepchas.
23	Tamil Nadu	Adiyan, Arnadan, Eravallan, Irular, Kadar, Kanikar, Kotas, Todas
24	Telangana	Chenchus.
25	Tripura	Bhil, Bhutia, Chaimal, Chakma, Halam, Khasia, Lushai, Mizel, Namte.
26	Uttarakhand	Bhotias, Buksa, Jannsari, Khas, Raji, Tharu.
27	Uttar Pradesh	Bhotia, Buksa, Jaunsari, Kol, Raji, Tharu.
28	West Bengal	Asur, Khond, Hajong, Ho, Parhaiya, Rabha, Santhals, Savar.

Source: Indian Census

6 · SETTLEMENTS

Understand it first: A settlement is where people live together. Rural ones depend on land-based work and are classed by how close the houses are and by the pattern they make; urban ones are classed by size and function, with their own set of terms.

Rural: factors

- → **Physical**: relief, climate, **water**, soil, resources
- → **Economic**: farm productivity (Punjab), jobs, roads/rail
- → **Social-cultural**: kinship, ritual, land ownership (zamindari)
- → **Political**: policies, security; Green Revolution reshaped Punjab/Haryana
- → **Environmental**: floods, cyclones (coastal Odisha)
- → **Climate change**: coastal villages shift inland; drought/flood displacement (parts of Brazil, Burkina Faso)

Huddled vs scattered — Good relief + good climate → huddled together; bad → scattered. **Compact** (north Indian plains, wet points of Rajasthan, Bundelkhand, Nagaland) · **Semi-compact** (Indus basin, Gujarat plains, parts of Rajasthan) · **Hamleted** (socially separate units: **panna, para, palli, nagla, dhani**; Ganga plains, lower Himalayan valleys, **Chhattisgarh plains (tribal)**, parts of south) · **Dispersed** (Uttarakhand, HP, Ladakh, Meghalaya).

Rural patterns (shape)

- → **Linear**: most common; roads, forest paths, rivers; T-shaped, star or checkerboard at crossings
- → **Oval/circular**: around temple, oasis, pond
- → **Triangular**: deltas, doabs
- → **Rectangular/grid**: plains
- → **Step**: hills

Urban term	Meaning	Handout example
Town	bigger than village, smaller than city; local administrative, commercial functions	–
City	large, dense, diverse economy, advanced infrastructure	Mumbai, Lucknow, Guwahati
Metropolis	major economic, political, cultural centre	New York, Delhi
Megacity	>10 million	Tokyo, Greater Mumbai, Greater Chennai
Conurbation	several towns/cities + suburbs merged	Delhi NCR
Megalopolis	several big cities grown together; labour-market separation	BosWash (Boston-Washington)
Suburb	residential, outskirts, less dense, depends on city	Bandra; Gurugram earlier
Urban sprawl	uncontrolled, low-density, car-based spread into rural land	–
Slum	densely populated, substandard housing, squalor, no basic services	Dharavi

CBD	commercial and business centre ("downtown"); high land values	Chandni Chowk, Connaught Place
Edge city	business, shopping, entertainment outside the CBD in former suburb/rural land	Gurgaon (for Delhi)
Gentrification	redevelopment → higher property values → lower-income residents displaced	Dharavi redevelopment
Satellite city	planned city to relieve a major city	–

Climb the ladder – Town → City → Metropolis → Megacity (>10 million) → Conurbation → Megalopolis. CBD = the city's wallet (downtown); Edge city = the mall that moved to the edge; Gentrification = the café that raised the rent.

Factors Affecting Settlements

→ Rural settlements are influenced by a variety of factors that determine their location, size, structure, and pattern.

Physical Factors

→ Topography:

→ – Flat and fertile plains are more conducive to agriculture and human habitation.

→ – Hilly and mountainous regions often lead to dispersed settlements due to difficult terrain and limited arable land.

→ Climate:

→ – Areas with moderate climates are more likely to have higher population densities.

→ – Extreme climates (too hot, too cold, or too wet) can hinder large-scale settlements.

→ – Example: The Thar Desert in Rajasthan, India, has sparse rural settlements due to harsh climatic conditions.

→ Water Supply:

→ – Proximity to rivers, lakes, or groundwater sources is crucial for drinking water, irrigation, and other daily needs.

→ – Settlements are often concentrated near reliable water sources.

→ Soil Fertility:

→ – Fertile soils support agriculture, which is the primary livelihood in rural areas.

→ – Areas with rich, alluvial soils are densely populated.

→ Natural Resources:

→ – Availability of resources like minerals, forests, and fisheries can attract settlements.

→ – Regions rich in natural resources often see clustered settlements.

→ – Example: The mining regions in Jharkhand, India, have developed settlements around mining activities.

Economic Factors

→ Agricultural Productivity:

→ – Regions with better irrigation, technology, and crop yields tend to have more stable settlements.

→ Employment Opportunities:

→ – The availability of non-agricultural jobs (e.g., cottage industries, mining) can attract and retain population.

→ Accessibility and Transportation:

→ – Proximity to roads, railways, and other transport networks facilitates trade, and access to markets, and services.

→ – Better connectivity generally leads to more populated and prosperous settlements.

→ – Example: Villages near major highways or railway lines tend to be larger and more developed.

Social and Cultural Factors

→ Community and Kinship:

→ – Social bonds and kinship ties can influence the location and growth of rural settlements.

→ – Communities with strong social networks often cluster together.

→ – Example: Traditional villages in India often have extended families living in close proximity.

→ Cultural Practices:

→ – Cultural and religious practices can dictate settlement patterns, such as the layout and structure of homes.

→ – Specific cultural requirements (e.g., space for religious rituals) influence the design of rural settlements.

→ – Example: Some rural areas in Rajasthan, India, have unique architectural styles due to cultural influences.

→ Land Ownership Patterns:

→ – Areas with fragmented landholdings may have dispersed settlements.

Political Factors

→ Government Policies:

→ – Policies related to land use, agriculture, infrastructure development, and rural development impact settlement patterns.

→ – Programs aimed at rural development can lead to the establishment of new settlements or the growth of existing ones.

→ Security and Stability:

→ – Political stability and security influence where people choose to settle.

→ – Areas prone to conflict or instability may see depopulation or fragmented settlements.

→ – Example: Rural areas in conflict zones may have sparse and dispersed settlements.

Environmental Factors

→ Natural Disasters:

→ – Areas prone to natural disasters (floods, earthquakes, hurricanes) may have fewer stable settlements.

→ – Repeated exposure to natural disasters can lead to migration and changes in settlement patterns.

→ – Example: Coastal villages in Odisha, India, are often affected by cyclones, influencing their settlement structure and resilience measures. (e) Climate Change (i) In future, climate change might force the coastal settlements to be removed to more interior locations. (ii) Some villages might get displaced due to extreme drought of flood conditions. (iii) E.g. some regions of Brazil and Burkina Faso have started to feel these problems.

Types of Rural Settlements

Compact/Nucleated Settlements:

→ Favourable relief + favourable Climate .

→ Houses and buildings are closely packed .

→ Common in fertile plains, valleys, and areas with abundant natural resources.

→ Sometimes in locations where resources are concentrated or for security reasons.

→ Houses are somewhat clustered but with more space between them compared to compact settlements.

Semi-Compact Settlements:	→ Found in areas with moderate agricultural productivity .
Hamleted Settlements:	→ Fragmentation of settlements in socially/ethically motivated separate units.
Dispersed Settlements:	→ Himalayan Region: Uttarakhand, Himachal Pradesh, Ladakh; Meghalaya, etc.

Patterns of Rural Settlements

→ Step: hilly regions

Terminologies associated with Urban Settlements

→ Metropolis : A large and densely populated urban area that serves as a significant economic, political, and cultural center of a country or region.

→ Conurbation: An extended urban area formed by the merging of several towns or cities and their suburbs.

→ – Large, continuous urban area.

→ – Multiple urban centers within a single metropolitan area.

→ – Integrated economic and social networks.

→ Megalopolis : A region in which several large cities and surrounding areas grow together, forming a massive, interconnected urban complex.

→ – Extensive urbanized region.

→ – Integrated transportation and economic systems.

→ Suburb: A residential area on the outskirts of a city , typically less densely populated than the urban core.

→ – Primarily residential.

→ Urban Sprawl : The uncontrolled expansion of urban areas into the surrounding rural land.

→ – Low-density, automobile-dependent development.

→ – Expansion into rural areas.

→ – Often leads to environmental and infrastructural challenges.

→ Slum : A densely populated urban area characterized by substandard housing and squalor.

→ – Poor housing conditions.

→ – Inadequate access to basic services (water, sanitation, electricity).

→ Central Business District (CBD): The commercial and business center of a city , often referred to as the "downtown" area.

→ – High concentration of retail and office buildings.

→ – Major transportation hubs.

→ Edge City: A concentration of business, shopping, and entertainment outside a traditional downtown (CBD) in what had previously been a suburban residential or rural area .

→ – Significant commercial and business activity.

→ – Often characterized by new office buildings, shopping malls, and entertainment facilities.

RECALL DRILL: COVER AND ANSWER

TFR replacement; India NFHS-5 and SRS?	2.1; 2.0 and 1.9	States with TFR above 2.1?	Bihar, Meghalaya, UP, Jharkhand, Manipur (Big Mummy's Up-to-Juggle Mania)
Sex ratio 2011; child sex ratio 2011?	943; 914 (NFHS-5: 1020 / 920)	Working age; dependants; dividend peak?	15–64; 0–14 and >65; 68.9% in 2030

Pyramids: Expansive, Constricted, Stationary?	Nigeria (India stabilising) · Japan · USA	Malthus: growth, work, year?	population geometric, food arithmetic; Essay on the Principle of Population, 1798
DTT: which falls first; India stage?	Death rate (Stage 2); late expanding (Stage 3) since 1981	Changpa; Dongria Kondh?	Changtang, Ladakh, pashmina; Niyamgiri, Odisha
Rural types and urban ladder?	Compact, semi-compact, hamleted, dispersed; town, city, metropolis, megacity (>10 m), conurbation, megalopolis	Low birth rate PYQ answer?	Italy, Japan, South Korea

🎯 PYQs OF THIS CHAPTER

🎯 Exam lens (from the handout only)

- Priority: MEDIUM (short chapter, concept-heavy). 4 Prelims PYQs printed: 2013 (tribe–state pairs) · 2014 (Changpa) · two undated (definition of Total Fertility Rate; countries known for low birth rate / ageing). The handout also reprints five "missed" Indian-Geography PYQs at the end – they are covered in the Indian chapter. Mains quoted: none quoted in the material.
- Pattern: definitions (TFR, sex ratio, dependency) and where-is-which-tribe. Plus theory: Malthus, Demographic Transition.
- What to prepare: Prelims – learn the five DTT stages as a birth-rate / death-rate table; learn state-wise tribes (table figure below). Mains hooks – demographic dividend, ageing, urbanisation, gentrification, climate-driven settlement shifts.
- No teacher ink in this PDF; priority = PYQ count in the handout.

★ ***PYQ answers in one line (asked year and the key)***

MISSED PYQS (INDIAN GEOGRAPHY) 💡

Asked	Answer / explanation
Andaman and Nicobar climate: humid tropical coastal ✔; SW + NE monsoon ✔; max precipitation Dec-May ✘	B (1 and 2): humid tropical maritime; both SW and NE winds onshore; max rain May-Sept (SW monsoon). (Heat = LP = rain; NH has heat Mar-Sept)
👫 Peninsular Block: W-coast submergence by tectonics ✔; residual Veliconda, Mahendragiri hills ✔; deep V-shaped valleys ✘	B: ancient stable craton; Eastern Ghats denuded; peninsular rivers mature on old hard rock
Mangroves for climate resilience: store freshwater/paddy ✘; salt-sensitive roots ✘; bio-shield + livelihood ✔	D (3 only): roots are salt-tolerant, do not store salt water
🏛️ River: antecedent, 3 countries, Tibetan origin, irrigation, no distributaries	C Sutlej (Brahmaputra and Indus have distributaries; Teesta originates in Sikkim)
🗺️ States: UP most neighbours · Rajasthan longest int'l border · Sikkim only one-neighbour state: which not correct?	A as printed; explanation: West Bengal has the longest int'l border; Meghalaya also shares border with only 1 state

Source conflict

→ the last PYQ's printed answer (A, statement 1 wrong) disagrees with its own explanation (statements 2 and 3 wrong). Check atlas / class notes. Traps: TFR ≠ birth rate

→ Changpa = Ladakh, not Uttarakhand

→ sex ratio in India = females per 1,000 males. Connects to: Indian Geography (Changpa/Ladakh, Dongria/Niyamgiri, Thar/Himalaya settlements), Economic Geography.

✍️ ***The PYQs as printed in the handout, with answers***

Q.) Consider the following pairs :

1. Limboo (Limbu): Sikkim
2. Karbi: Himachal Pradesh
3. Dongaria: Odisha
4. Bonda: Tamil Nadu

Which of the above pairs are correctly matched? (2013) 1 and 3 only 1 and 2 only 2 and 3 only 1, 2, 3 and 4

Answer: A The Limbu are Kirati people indigenous and native to the Himalayan Limbuwan region of the Indian subcontinent, what is now modern-day Eastern Nepal, Northern Sikkim, India and Western Bhutan.

- The Niyamgiri hill range in Odisha state, eastern India, is home to the Dongria Kondh tribe .
- Niyamgiri is an area of densely forested hills, deep gorges and cascading streams. To be a Dongria Kondh is to farm the hills' fertile slopes, harvest their produce, and worship the mountain god Niyam Raja and the hills he presides over, including the 4,000 metre Mountain of the Law, Niyam Dongar.

Q.) With reference to 'Changpa' community of India, consider the following statement:

- They live mainly in the State of Uttarakhand.
- They rear the Pashmina goats that yield fine wool.
- They are kept in the category of Scheduled Tribes.

Which of the statements given above is/are correct? (2014)

(a) 1 only
(b) 2 and 3 only
(c) 3 only
(d) 1, 2 and 3

Answer: B The Changpa or Champa are a semi-nomadic Tibetan people found mainly in the Changtang in Ladakh and in Jammu and Kashmir.

- The Changpa or Champa are a semi-nomadic Tibetan people found mainly in the Changtang in Ladakh, India. A smaller number resides in the western regions of the Tibet Autonomous Region and were partially relocated for the establishment of the Changtang Nature Reserve.
- Most of the regions that the Changpas reside in is quite inhospitable and unsuitable for farming. Thus, the primary occupation for the tribe is rearing of yaks, horses and sheep. Cashmere, also known as pashmina, is majorly sourced from the pashmina sheep that the tribe rears.
- A smaller number resides in the western regions of the Tibet Autonomous Region and were partially relocated for the establishment of the Changtang Nature Reserve.
- As of 1989, there were half a million nomads living in the Changtang area.

Q.) The total fertility rate in an economy is defined as:

- The number of children born per 1000 people in the population in a year.
- The number of children born to a couple in their lifetime in a given population.
- The birth rate minus death rate.
- The average number of live births a woman would have by the end of her child-bearing age.

Answer:

(d) The average number of live births a woman would have by the end of her child-bearing age.

Q.) Consider the following countries:

- Italy
- Japan
- Nigeria
- South Korea
- South Africa Which of the above countries are frequently mentioned in the media for their low birth rates, or ageing population or declining population?
- (a) 1, 2 and 4
- (b) 1, 3 and 5
- (c) 2 and 4 only
- (d) 3 and 5 only

Answer: A MISSED QNS OF INDIAN GEOGRAPHY

Q.) With reference to the climate of Andaman and Nicobar Islands, which of the following statements is/are correct?

1. The climate can be defined as a humid, tropical coastal climate.
2. It receives rainfall from both South-west monsoon and North-east monsoon.
3. Maximum precipitation is between December and May.

Select the answer using the code given below:

(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only

Ans: (b)

Explanation: The Andaman and Nicobar Islands experience a humid tropical maritime/coastal climate. It is in the ocean, so both SW and NE winds are onshore. Maximum rainfall generally occurs during the South-West Monsoon months, especially May to September. (Heat=LP=rain, and NH has heat in March-Sept)

Q. Which of the following geographical features or phenomena is/are associated with the Peninsular Block of India?

1. Submergence of parts of the western coast due to tectonic activity
2. Presence of residual mountain ranges such as the Veliconda Hills and Mahendragiri Hills

3. Deep, V-shaped river valleys formed by fast-flowing rivers
Select the answer using the code given below:
(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only
Ans: (b)
Explanation: The Peninsular Block is an ancient stable cratonic landmass, but parts of the western coast experienced submergence due to tectonic movements. Eastern Ghats are denuded mountains. Most rivers of Peninsular India are mature rivers flowing over old hard rocks.
Q. Which of the following best explain(s) the rationale for protecting mangrove ecosystems in the context of climate resilience?
1. Mangroves reduce tidal energy and store freshwater, making them ideal sites for paddy cultivation in saline estuarine belts.
2. Their salt-sensitive roots filter seawater, making mangroves key to converting coastal land into freshwater aquaculture zones.
3. By withstanding tidal surges and offering biomass resources, mangroves function both as natural bio-shields and livelihood bases for rural communities.
Select the answer using the code given below:
(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only
Ans: (d)
Explanation: Do not store salt water. Roots are salt tolerant not salt sensitive. (Played with words)
Q. Identify the river of the Indian sub-continent on the basis of the following information:
1. It has an antecedent drainage system.
2. It flows through three countries.
3. It originates in the Tibetan Plateau and is an important river for irrigation.
4. It does not form distributaries.
Select the answer from the following:
(a) Brahmaputra
(b) Indus
(c) Sutlej
(d) Teesta
Ans: (c)
Explanation: Brahmaputra has distributaries. Teesta originates in Sikkim. Indus has distributaries.
Q. Which of the following with reference to Indian States is/are not correct?
1. Uttar Pradesh shares its boundary with the highest number of other Indian States.
2. Rajasthan shares the longest international border among all Indian States.
3. Sikkim is the only State that shares its boundary with just one other Indian State.
Select the answer using the code given below:
(a) 1 only
(b) 1 and 2
(c) 2 and 3
(d) 3 only
Ans: (a)
Explanation: West Bengal has longest international border. Meghalaya also shares border with only 1 state

🔤 **ABBREVIATION COLUMN (full forms as the handout gives them where it does; the one-line meanings are plain-language aids, verify)**

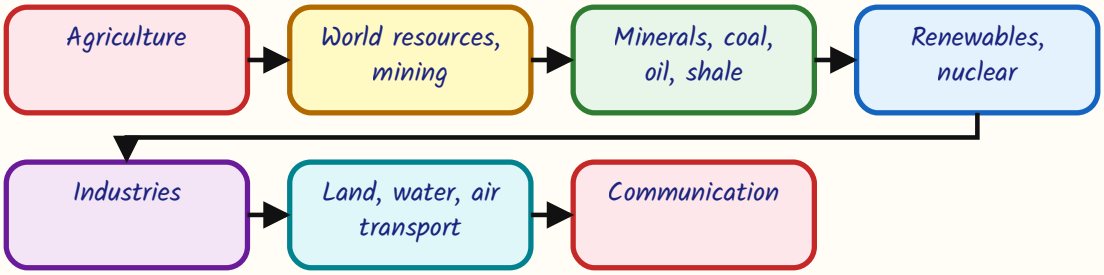
CBD = Central Business District — commercial core of a city
DTT = Demographic Transition Theory
HP = Himachal Pradesh
IMR = Infant Mortality Rate — infant deaths per 1,000 live births
LP = low pressure
NCR = National Capital Region
NFHS-5 = National Family Health Survey, round 5
NH = Northern Hemisphere (in highway lists: National Highway)
PYQ = previous year question
SRS = Sample Registration System
SW = south-west
TFR = Total Fertility Rate — average children per woman
USA = United States of America

ECONOMIC GEOGRAPHY

PYQ Prelims 31 PYQs: agriculture 19 (2011 x4, 2012 x2, 2013, 2014 x2, 2015, 2017, 2018, 2019, 2020, 2021 x2, 2022, 2024, 2025) · minerals and energy 9 (2012, 2013 x2, 2016, 2018, 2022, 2023 x2 + 1 undated on rare earths) · transport 3 (2014, 2015, 2023) · industries 0 · Mains none quoted

✎ *Which crop where, which mineral where, which energy source and how much, then industries, transport and communication: the heaviest handout (54 pages) and a VERY HIGH PYQ bank.*

At a glance: the Economic Geography route



Pattern: crop-to-region, mineral-to-state, "statement" questions on fertiliser, salinisation, permaculture, conservation agriculture, plus current affairs (cocoa, turmeric, REE, NCMM). No teacher ink: priority = PYQ count. Image-only pages (crop map, waterways) kept as tables. ⭐

1 · AGRICULTURE: SEASONS

PYQ 2013

Understand it first: Indian farming runs on three seasons set by the monsoon: kharif (sown with the rains), rabi (sown in winter) and zaid (the short summer gap). Know which crops belong to which.

Arable land: India has the largest arable area in the world, ≈ 1.597 million km² (394.6 million acres). 💡

	Kharif	Rabi	Zaid
Time	Jun–Oct; sow with SW monsoon, harvest Sep–Oct	Oct/Nov–Mar/Apr; sow Oct–Dec, harvest Mar–Apr	Mar–Jun; harvest Jun–Jul
Weather	hot, humid, heavy rain	cool dry growth, warm at harvest; Western Disturbance rain helps	warm, dry; irrigation needed
Crops	paddy, maize, jowar, bajra, pulses (moong, urad, arhar), cotton, groundnut, soybean, sugarcane	wheat, barley, oats, gram, mustard, peas, linseed	watermelon, muskmelon, cucumber, vegetables, fodder, some moong/urad

📌 **Kar-Rain · Rest-of-winter · Zara-garmi** — Kharif = Kar-Rain (monsoon crops) · Rabi = Rest-of-winter (WDs help) · Zaid = Zara-garmi (summer gap).

Pairing pegs

- → Cotton + black soil (Maharashtra, Gujarat, Telangana)
- → Jute + West Bengal (wet, 120–200 cm)
- → Tea = acidic slopes (Assam, WB, TN, Kerala)
- → Coffee = Karnataka (shade; Arabica and Robusta)
- → Rubber = Kerala
- → Coconut = Kalpavriksha (Kerala)
- → Bajra = toughest (tolerates 42 °C), Rajasthan.

Trap (2013 PYQ): wheat is Rabi, not Kharif. Kharif = Arabic for autumn; generally Jun–Sep (state-wise differs); sown ~June, harvested Sep/Oct. Kharif also: rice, maize, bajra, ragi, soybean, groundnut, cotton.

Agriculture

→ Understand it first

Agriculture: Seasons: Indian farming runs on three seasons set by the monsoon: kharif (sown with the rains), rabi (sown in winter) and zaid (the short summer gap). Know which crops belong to which.

Crop Table (Temp °C · Rain Cm): Each crop needs its own temperature, rainfall and soil. Read the table across: once the conditions are known, the growing States follow.

Cropping Seasons of India

Kharif Season

→ Time Period : June to October .
→ Sowing: Beginning with the onset of the southwest monsoon in June.
→ Harvesting: September to October .
→ Climatic Conditions: Requires hot and humid weather with substantial rainfall . Millets, pulses, and oilseeds can grow even in less humid areas.

Rabi Season

→ Time Period: October/November to March/April .
→ Sowing: After the monsoon rains, from October to December.
→ Harvesting: March to April.
→ W inter rainfall from Western disturbances supports growth.
→ Climatic Conditions: Requires cool and dry weather for growth and warmer weather during the harvesting period .

Zaid Season

→ Time Period: March to June .
→ Climatic Conditions: Requires warm and dry weather with the availability of irrigation .

CROP TABLE (temp °C · rain cm)

Understand it first: Each crop needs its own temperature, rainfall and soil. Read the table across: once the conditions are known, the growing States follow.

Crop	Temp · rain	Soil	Top Indian states	Top world	Remarks
Rice	20–32 · 150–300	deep clayey, alluvial	UP, Telangana, WB	India, China, Bangladesh, Indonesia, Vietnam	staple; Aus, Aman, Boro in WB; high humidity
Wheat	10–25 · 50–100	loamy, well-drained	UP, MP, Punjab	China, EU, India, Russia, USA	winter crop; bright sun at ripening
Jowar	20–27 · 50–75	light sandy–loamy	Maharashtra, Karnataka, MP	India, Niger, China, Mali, Nigeria	Kharif and Rabi; fodder + biofuel
Bajra	20–27 · 50–75	sandy loam	Rajasthan, UP, Gujarat	India, Niger, China, Nigeria	drought-resistant, up to 42 °C
Millets (jowar, bajra, ragi)	20–27 · 50–75	wide range	Rajasthan, Karnataka, MP	India, Niger, China, Mali, Nigeria	antioxidants, diabetes/ heart, protein; low cost; marginal farmers

Crop	Temp · rain	Soil	Top Indian states	Top world	Remarks
Maize	15–27 · 50–80	well-drained loam	MP, Karnataka, Bihar	USA, China, Brazil, EU, Argentina	food, alcohol, feed, biofuel
Cotton	21–30 · 50–100	black (regur)	Maharashtra, Gujarat, Telangana	China, India, Brazil, USA, Australia	cash crop; ≥ 210 frost-free days
Jute	24–35 · 120–200	alluvial	WB, Bihar, Assam	India, Bangladesh, China	retting; plastic alternative
Sugarcane	20–27 · 100–150	loamy	UP, Maharashtra, Karnataka	Brazil, India, China, Thailand, Pakistan	very water-intensive; ethanol + sugar
Pulses	15–27 · low rain	well-drained	MP, Maharashtra, UP	India, Canada, Myanmar, Brazil, Ethiopia	N-fixing; intercropping; India 766 kg/ha vs world 1,015
Oilseeds	15–30 · 30–75	loamy–sandy	Rajasthan, MP, Maharashtra	USA, China, Brazil, India	soybean, mustard, groundnut
🎯 Palm oil PYQ 2021	25–28 · >200	tropical well-drained	AP, Telangana, Kerala	Indonesia, Malaysia, Thailand, Colombia, Nigeria	deforestation concern; push in NE
🎯 Tea PYQ 2022	16–32 · 150–300	acidic, slopes	Assam, WB, TN, Kerala (+ Tripura, HP)	China, India, Kenya, Sri Lanka, Vietnam	plantation; labour-intensive
Coffee	15–28 · 150–200	acidic, fertile	Karnataka, Kerala, TN	Brazil, Vietnam, Colombia, Ethiopia, Indonesia	shade-grown; Arabica and Robusta
🎯 Cocoa PYQ 2024	18–32 · 100–250	laterite	AP, Kerala, Karnataka, TN	Ivory Coast, Ghana, Indonesia, Nigeria, Ecuador	shade (under coconut/areca); ~70% of world beans from W. Africa (Ivory Coast, Ghana, Nigeria, Cameroon)
Rubber	25–35 · 200–450	laterite, acidic	Kerala, TN	Thailand, Indonesia, Vietnam, India, China	latex
Cashew	20–30 · moderate	sandy loam	Maharashtra, AP, Odisha, Karnataka, Kerala	Vietnam, India, Ivory Coast, Brazil	export cash crop
🎯 Turmeric PYQ 2025	20–30 · >150	sandy loam	Maharashtra, Telangana, TN, Karnataka	India, China, Myanmar, Bangladesh	>30 varieties, >20 states
Mango	24–30 · 75–250	deep loamy	UP, AP, Telangana, Karnataka, Bihar	India, China, Thailand, Indonesia, Mexico	India largest; Alphonso, Dasheri, Langra
Coconut	20–32 · 100–200	sandy, coastal alluvial, laterite	Kerala, TN, Karnataka, AP	Indonesia, Philippines, India, Sri Lanka, Brazil	Kalpavriksha; humid coast

Red gram (tur)	20–30 · 60–100	loam–black	Maharashtra, Karnataka, UP, MP	India, Myanmar, Malawi, Tanzania, Kenya	Kharif; deep-rooted; N-fixing
Mustard	10–25 · 25–75	alluvial–loamy	Rajasthan, UP, Haryana, MP	Canada, China, India, Germany, France	Rabi; frost-sensitive; sarson, rai

In the news · 2025 (verify dates) National Turmeric Board: HQ Nizamabad (Telangana), inaugurated June 2025.

Sugarcane moved south: tropical climate, higher sucrose, longer crushing season (Oct–May south vs Nov–Feb north; no extreme summer heat), cooperatives more successful in the south. ⭐

TYPES OF FARMING PYQ 2011 PYQ 2012

Understand it first: Farming types are sorted by purpose and intensity: for the family or for the market, with much labour on small plots or with machines on large ones. Shifting, plantation and mixed farming are the special cases.

Type	Rainfall	Growing season	Focus
Dry farming	<75 cm	<75 days	moisture conservation, drought-resistant crops
Dryland	75–115 cm	75–120 days	soil moisture retention, drought-tolerant crops, adaptive strategies
Wetland	>115 cm	>120 days	rainwater harvesting, moisture conservation, crop selection
Irrigated	–	–	protective (saves crops from drying) vs productive (raises yield)

Types of Farming in India · Irrigated

75–115–120 ladder. Dry/dryland farming gives ~80% maize, 90% bajra, ~95% pulses, 75% oilseeds. Groundnut, sesamum, pearl millet = predominantly rainfed (2012 PYQ, D): sesamum on marginal dry sloping land; ~80% groundnut area rainfed; main rainfed cereals sorghum, bajra, ragi. ⭐

Micro-irrigation PYQ 2011

→ Drip, sprinklers, foggers; improves water-use efficiency

→ Pros: saves water, stops soil erosion, no levelling, any soil, uniform watering, fertigation and chemigation, labour saving, low pressure (drip), higher yield, bigger fruit

→ Cons: high investment, pipe clogging, spacing needed, technical installation

→ 2011 (C): reduces nutrient loss ✔; checks falling groundwater table ✔; "only means of irrigation in dryland farming" ✘

Micro Irrigation · No need to level the fields .

→ Drip: water drop by drop at root zone; emitters inline, online, microtubes, pressure-compensated; for wide-spaced crops · Micro-sprinkler: sandy/loamy soils, horticulture, small grasses, nurseries and lawns

SRI (System of Rice Intensification)

→ Madagascar, 1980s; raises yield while using fewer resources
→ AWD (Alternate Wetting and Drying) instead of continuous flooding in nurseries
→ Organic manures; yields +30–50%; less methane

			alluvial)			
Coconut	20–32 °C	100–200 cm (well distributed)	Sandy loam, coastal alluvial, laterite (well-drained)	Kerala, Tamil Nadu, Karnataka, Andhra Pradesh	Indonesia, Philippines, India, Sri Lanka, Brazil	Coastal crop; called **"Kalpavriksha"** (tree of life); used for oil, coir, food; requires high humidity
Red Gram (Tur / Pigeon Pea)	20–30 °C	60–100 cm (can tolerate low rainfall)	Well-drained loamy to black soils	Maharashtra, Karnataka, Uttar Pradesh, Madhya Pradesh	India, Myanmar, Malawi, Tanzania, Kenya	Kharif pulse; drought-tolerant; deep-rooted crop; important for intercropping; nitrogen-fixing legume
Mustard (Rapeseed & Mustard)	10–25° C	25–75 cm (low rainfall)	Alluvial to loamy soils (well-drained)	Rajasthan, Uttar Pradesh, Haryana, Madhya Pradesh	Canada, China, India, Germany, France	Rabi oilseed crop; frost-sensitive; important edible oil source; includes varieties like sarson, rai

System of Rice Intensification

System of Rice Intensification

Types of Farming in India

→ In India, almost 80 percent of maize, 90 percent of Bajra, approximately 95 percent of pulses, and 75 percent of oilseeds are obtained from dry or dryland agriculture.

Micro Irrigation

→ Advantages:
→ – No need to level the fields .
→ – Fertilizers/Pesticides etc can be added to the water and provided to the plants.
→ – Drip irrigation: Needs lesser pressure to operate, reducing energy costs.
→ Disadvantages:
→ – Clogging of pipes can render certain regions devoid of water.
→ – Drip irrigation needs defined spacing between crops , so not feasible in regions where
→ – Technical installation : needs expertise, otherwise waste of resources (time, money, pipes, etc.)

The shift of sugarcane industry to the South

→ Tropical Climate better for sugarcane yield

→ Higher sucrose content

→ Longer crushing season (does not get extremely hot in the summers): only Nov-Feb in North; October to May in South

→ Cooperatives more successful in South

Crop	Temperature	Precipitation	Soil	Major Producing States (India)	Major Producing Countries (World)	Remarks
Rice	20–32 °C	150–300 cm	Deep clayey & alluvial	Uttar Pradesh, Telangana, West Bengal	India, China, Bangladesh, Indonesia, Vietnam	Staple food; Aus, Aman, Boro crops in WB; needs high humidity
Wheat	10–25° C	50–100 cm	Loamy, well-drained	Uttar Pradesh, Madhya Pradesh, Punjab	China, EU, India, Russia, USA	Winter crop; needs bright sunlight during ripening. India grows winter wheat.
Jowar (Sorghum)	20–27° C	50–75 cm	Light sandy to loamy	Maharashtra, Karnataka, Madhya Pradesh	India, Niger, China, Mali, Nigeria	Kharif & Rabi crop; fodder + biofuel use
Bajra (Pearl Millet)	20–27° C	50–75 cm	Sandy loam	Rajasthan, Uttar Pradesh, Gujarat	India, Niger, China, Nigeria	Highly drought-resistant (can tolerate upti 42 degrees Celcius)
Millets (General)	20–27° C	50–75 cm	Wide range (clay to sandy)	Rajasthan, Karnataka, Madhya Pradesh	India, Niger, China, Mali, Nigeria	Benefits: Health (anti-oxidants, helps in diabetes, cardio vascular diseases, protein), Economic (Low cost as do not need many inputs), Social (grown mainly by marginal farmers); includes jowar, bajra, ragi, etc.

Important Crops across the World

Maize	15–27° C	50–80 cm	Well-drained loamy	Madhya Pradesh, Karnataka, Bihar	USA, China, Brazil, EU, Argentina	Used for food, Alcohol, feed, biofuel
Cotton	21–30° C	50–100 cm	Black soil (regur)	Maharashtra, Gujarat, Telangana	China, India, Brazil, USA, Australia	Cash crop; needs atleast 210 frost-free days
Jute	24–35 °C	120–200 cm	Alluvial	West Bengal, Bihar, Assam	India, Bangladesh, China	Retting required; packaging fibre, being promoted as a plastic alternative for packaging.
Sugarcane	20–27° C	100–150 cm	Loamy	Uttar Pradesh, Maharashtra, Karnataka	Brazil, India, China, Thailand, Pakistan	Highly water-intensive crop. Used for ethanol and sugar production.
Pulses	15–27° C	Low rainfall	Well-drained	Madhya Pradesh, Maharashtra, Uttar Pradesh	India, Canada, Myanmar, Brazil, Ethiopia	Nitrogen-fixing; key for intercropping; Source of protein, need less water. India's productivity at 766 kg/ha is far below the world average productivity of 1015 kg/ha.
Oilseeds	15–30° C	30–75 cm	Loamy to sandy	Rajasthan, Madhya Pradesh, Maharashtra	USA, China, Brazil, India	Includes soybean, mustard, and groundnut. Need less water.
Palm Oil	25–28 °C	>200 cm	Well-drained tropical	Andhra Pradesh, Telangana, Kerala	Indonesia, Malaysia, Thailand, Colombia, Nigeria	Deforestation concerns; govt push in India especially in North-East.
Tea	16–32° C	150–300 cm	Acidic, well-drained (slopes)	Assam, West Bengal, Tamil Nadu, Kerala. Minor states: Tripura, Himachal Pradesh	China, India, Kenya, Sri Lanka, Vietnam	Plantation crop; labour-intensive
Coffee	15–28° C	150–200 cm	Acidic, fertile	Karnataka, Kerala, Tamil Nadu	Brazil, Vietnam, Colombia, Ethiopia, Indonesia	Shade-grown; Arabica & Robusta. Native to Africa & Arabia.

Important Crops across the World

Cocoa	18–32	100–250	Well-drained laterite soil	Andhra Pradesh, Kerala, Karnataka, Tamil Nadu	Ivory Coast, Ghana, Indonesia, Nigeria, Ecuador	The plant requires **s**hade and is commonly grown under taller trees like coconut or areca. About 70 percent of the world's cocoa beans come from four West African countries: Ivory Coast, Ghana, Nigeria and Cameroon. Sustainability concerns and initiatives are growing in the cocoa industry due to issues like deforestation, child labor, and environmental degradation in major producing countries. Native to Central America.
Rubber	25–35 °C	200–450 cm	Laterite, acidic	Kerala, Tamil Nadu	Thailand, Indonesia, Vietnam, India, China	Latex-based crop; biodiversity concerns
Cashew	20–30 °C (tropical)	Moderate rainfall	Sandy loam	Maharashtra, Andhra Pradesh, Odisha, Karnataka, Kerala	Vietnam, India, Ivory Coast, Brazil	Export-oriented cash crop
Turmeric	20–30 °C	>150 cm	Sandy loam	Maharashtra, Telangana, Tamil Nadu, Karnataka	India, China, Myanmar, Bangladesh	More than 30 varieties of turmeric are grown in over 20 states across India. National Turmeric Board HQ in Nizamabad, Telangana (inaugurated in June 2025).
Mango	24–30 °C (frost-free)	75–250 cm	Deep, well-drained loamy (can tolerate laterite &	Uttar Pradesh, Andhra Pradesh, Telangana, Karnataka, Bihar	India, China, Thailand, Indonesia, Mexico	Tropical fruit; requires dry weather during flowering; India is the **largest producer**; varieties like Alphonso, Dasheri, Langra

Important Crops across the World

AGRICULTURE REVOLUTIONS

Understand it first: Each "revolution" is a colour tied to one product. Learn them as pairs.

Revolution	When · who	Win	Cost
Green	late 1960s–early 1980s · M.S. Swaminathan	HYV seeds + fertiliser + irrigation; wheat/rice jump; food self-sufficiency	regional (Punjab, Haryana, W. UP), social (big farmers), pollution, water-table fall

White (Operation Flood)	1970s · Verghese Kurien	world's largest milk producer; cooperatives (Amul)	quality, cold chain, processing
Blue	1980s	fish output; ~17% of agri exports (2021–22)	overfishing, pollution
Yellow	1980s	oilseeds; fewer edible-oil imports	climate and price swings
Evergreen	Swaminathan's proposal	sustainable mix of traditional + modern (ecology + technology)	scale, policy support

Colour = crop

→ Green grains
→ White milk
→ Blue fish
→ Yellow oilseeds
→ Golden fruit/honey/horticulture
→ Golden fibre jute
→ Silver eggs
→ Silver fibre cotton
→ Pink onion/pharma/prawn
→ Red tomato/meat
→ Round potato
→ Brown coffee/leather/cocoa
→ Grey fertiliser
→ Protein lentils.

Agriculture Revolutions in India

Green Revolution

→ Key Figure: Dr. M.S. Swaminathan.
→ Main Focus : Introduction of high-yielding varieties (HYVs) of seeds, increased use of fertilizers, pesticides, and irrigation.
→ Successes:
→ – Reduced the import dependency for grains.
→ Drawbacks:
→ – Regional disparities , with major benefits concentrated in Punjab, Haryana, and Western Uttar Pradesh.
→ – Social disparities , as only the farmers with significant land holdings benefitted.
→ – Environmental degradation due to overuse of chemical fertilizers and pesticides.

White Revolution (Operation Flood)

→ Period: 1970s onwards .
→ Main Focus: Development of the dairy industry .
→ Successes:
→ – Improved rural incomes and employment .
→ – Created a robust cooperative network (e.g., Amul).
→ Drawbacks:
→ – Challenges in maintaining quality standards .
→ – Inadequate cold chain infrastructure in many parts of the country.
→ – Inadequate processing capabilities.

Blue Revolution

→ Period: 1980s onwards .
→ Main Focus: Development of the fisheries sector .
→ Successes:

→ – Increase in fish production , making India one of the top fish-producing countries.
→ – Enhanced export opportunities. Around 17% of India's agricultural exports in 2021-22.
→ – Improved nutritional standards and employment in coastal regions.
→ Drawbacks:
→ – Overfishing and depletion of marine resources .
→ – Environmental issues such as water pollution and habitat destruction.

Yellow Revolution

→ Period: 1980s onwards .
→ Main Focus: Increase in oilseed production.
→ Successes:
→ – Significant increase in the production of oilseeds like mustard, sunflower, and soybean.
→ – Reduced dependency on edible oil imports .
→ Drawbacks:
→ – Inconsistent growth and production challenges due to climatic variability.
→ – Market fluctuations affecting farmer incomes.

Evergreen Revolution

→ Main Focus: Sustainable agriculture through a blend of traditional and modern practices.
→ Successes:
→ – Focus on genetic diversity and conservation of natural resources .
→ Drawbacks:
→ – Implementation challenges on a large scale.
→ – Need for significant policy support and farmer education .

Other Revolutions:

→ Protein Revolution: encourages farmers to adopt scientific agricultural techniques to boost food grain production and reduce the import of protein-rich lentils.

AGRICULTURAL REGIONS OF THE WORLD

Understand it first: World agricultural regions follow climate. For each region hold where it is, how farming is done there and what is produced.

Type	Key points
Nomadic herding	4th-world communities; camels–Bedouins · cattle–Masai · horses–Kazakhs · reindeer–Chukchis
Livestock ranching	subsistence: African savannah; commercial: Australia (sheep, cattle), Brazil (beef)
Shifting agriculture	mostly 4th-world communities; sedentary = rotation of crops, not fields
Intensive subsistence	small holdings, multiple seasons; with paddy = wetter margins, without paddy = drier; primitive (manual, no surplus) vs non-primitive (tech, surplus)
Mixed subsistence / mixed farming PYQ 2012	crops + livestock together (2012, C); not "mixed cultivation"; Europe (Ireland–Russia), N. America east of 98° meridian, Argentine pampas, SE Australia, S. Africa, NZ; dense, urbanised, industrial societies
Extensive commercial grain	breadbasket of Anglo-America, Wheat Triangle of Eurasia; temperate grasslands
Commercial dairy	wetter temperate margins; India = tropical yet leading milk producer
Plantation	tropics; years of yield from same stand: coffee (Brazil), cocoa (Ghana), rubber (Indonesia, Malaysia), spices (India), tea (China); non-traditional (2–3 yrs): cotton, cane

Mediterranean	gardens and vineyards
Horticulture / truck	Bulgaria (roses), Netherlands (tulips); now India (marigold, rose, lily, brinjal)
Cooperative	pool resources except land; Belgium, Netherlands, Denmark; IFFCO, Amul
Collective	USSR Kolkhoz; state owns land; free-riding problem

Agricultural Regions of the World

→ Developing World:

Nomadic Herding: → Mainly 4th world communities in unfavourable envt. Regions.

Mixed Subsistence: → Evolving

Temperate Developed

Extensive Commercial Grain Farming:
→ Dual benefit
→ Largest geographical expanse of temperate regions

Commercial Dairy Farming: → Intensive Commercial: needs manual + machine labour.

Specialized Agriculture

Plantation:
→ Same stand gives yield for years, so special care
→ Specialized harvesting

Horticulture: → Special fertilizers and harvesting

Cooperative Farming:
→ Farmers come together to form cooperatives through resource pooling (except land).
→ Cooperatives help farmers get access to farm inputs at affordable rates.
→ Increases the negotiating powers of the farmers.
→ Successful in many West European nations like Belgium, Netherlands, and especially Denmark.

Collective Farming:
→ Farmers work together on collective lands.
→ Small land parcels provided to farmers only for their personal needs.

AGRICULTURAL TERMS PYQ 2017 PYQ 2020

Understand it first: These are definition questions: one clean line for each term is enough.

Term	Meaning
Organic	no synthetic pesticides/fertilisers
Permaculture	mimics natural ecosystems; mulching; multicropping; agroforestry example
Agroforestry	trees and shrubs in farmland
Regenerative	renew soil health and ecosystems
Conservation agriculture	minimum tillage, residue cover, rotations
Hydroponics · Aquaponics · Aeroponics	nutrient water, no soil · fish + hydroponics · air/mist
Vertical farming	stacked layers, often indoors

Precision · Smart	GPS and IoT field management · AI, drones, sensors
Crop rotation · Intercropping	alternate crops for fertility · crops in proximity for pest control, resource use
Cover cropping	protects and enriches soil between crops: mustard, clover, radish
📚 Silviculture · Viticulture · Apiculture · Sericulture	forest plantations · grapes · bees · silkworms
📚 Horticulture · Floriculture	fruits, vegetables, flowers, ornamentals · flowers for the floral industry
Climate-smart	raises productivity and resilience, cuts emissions
Biopharming	GM plants/animals making pharmaceuticals
Agrivoltaic	solar power + farming: land use, rural energy access

Natural farming (Fukuoka method)

→ Minimal human intervention, works with nature; no synthetic chemicals, fertilisers, pesticides, excess tilling

→ Biodiversity, soil health, local natural resources

→ Masanobu Fukuoka (Japan), The One-Straw Revolution, 1975

→ ZBNF (Zero Budget Natural Farming): low-cost, local preparations, no external inputs; weeds, insects, animals not enemies; emulates natural conditions

💰 Land use: 9 categories (Land Revenue records)

→ Forests · Barren and wasteland · Non-agricultural use

→ Permanent pastures (mostly Panchayat/govt) · Misc. tree crops and groves (mostly private)

→ Culturable wasteland (fallow >5 yrs) · Current fallow (≤1 yr) · Fallow other than current (1–5 yrs) · Net sown area

→ Records: Land Revenue Dept (Survey of India); Digital India Land Records Modernization Programme; land degradation: see Soils of India

Agricultural Terminologies

→ Conservation Agriculture : Practices that preserve and improve soil health.

→ Aquaponics : Combining aquaculture (raising fish) with hydroponics.

→ Precision Farming : Using technology (like GPS and IoT) to optimize field-level management.

→ Viticulture : Cultivation of grapevines for winemaking.

→ Apiculture : Beekeeping for honey and other bee products.

Natural Farming

→ The concept promotes biodiversity, soil health, and sustainable farming practices using locally available natural resources.

→ It was established by Japanese farmer and philosopher Masanobu Fukuoka in his 1975 book The One-Straw.

💡 Key concepts

→ Minimal Human Intervention : The core idea is to let nature follow its course as closely as possible.

Land Use Categories of India

→ According to the Land Revenue records , land use in India is mainly of 9 types:

→ – Barren & Wasteland: Area that cannot be brought under cultivation on the basis of the available technologies.

→ – **Land put to Non-Agri Use:** Settlements, Commercial centres, Industries, Infrastructure etc.

→ – **Area under Permanent Pastures and Grazing Lands:** Most of such land is owned by the Panchayat (Common Property Resource) or government. (Rarely Private)

→ – **Current Fallow:** Land left without cultivation for less than or equal to 1 agricultural year. (Land resting for re-couping)

Land Use Change:

→ **Changes due to changes in the economic activities.**

→ Over time, land under cultivation reduces (improvement of agricultural production, diversion of land for non-agricultural activities)

→ Land under forests can also reduce due to diversion.

→ As land is a finite resource, wastelands are also either reclaimed for agriculture or other works. Similarly Barren lands start getting used for non-agricultural activities.

2 · WORLD RESOURCES, MINING

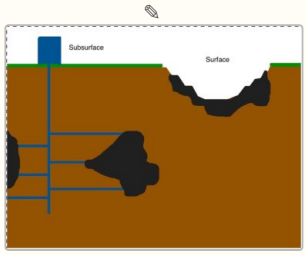
Understand it first: A resource is anything that can be used to meet a need. They are classed by origin, by whether they renew, by ownership and by stage of development. Mining is how the mineral ones are taken out.

Resource = availability + utility

→ Anything usable from the environment for needs and wants

→ **Availability:** accessible (technology, economic feasibility, cultural acceptance)

→ **Utility:** some benefit



Types of Resources · Based on Ownership

Classes

→ Origin: natural / human-made · biotic / abiotic

→ Exhaustibility: renewable / non-renewable

→ Ownership: individual / community / national / international (high seas, outer space)

Mining condition: cost of extraction + purification < value of final product. Types: **open-cast** (near surface), **shaft** (deep).

Mineral	Major Ores	Uses	Top producers and reserves in world	Top producers and reserves in India	Extra Info
Iron ore	Hematite, Magnetite, Limonite, Siderite	It is mainly used **for the production of Steel**.	**Producers:** Australia, Brazil, India, China, Russia; **Reserves:** Australia, Brazil, Russia, China, Mauritania.	**Reserves:** 95% in Odisha, Jharkhand, Chhattisgarh, Karnataka, Goa, Telangana, Andhra Pradesh, Tamil Nadu; **Producers / major belts:** Odisha (Mayurbhanj, Jhar, Sundargarh),	Acts as a feedstock for many capital industries

Mining · Shaft: for resources at a deeper level

				Jharkhand (East & West Singhbhum), Chhattisgarh (Durg, Dantewada, Bailadila), Karnataka (Ballari, Hospet, Kudremukh etc.), Maharashtra, Telangana, Andhra Pradesh, Tamil Nadu, Goa.	
Manganese	Pyrolusite, Psilomelane, Rhodochrosite	Manganese, a critical industrial metal, is an essential component in the **manufacturing of steel, batteries, and various alloys.**	**Producers:** South Africa, Gabon, Ghana, Australia, Brazil; **Reserves:** South Africa, Australia, Brazil, China, Gabon.	**Reserver**: Odisha, Karnataka, Madhya Pradesh, and Maharashtra. **Producers:** • Odisha: near the iron belt. Kendujhar, Sundergarh, Koraput, Kalahandi • Madhya Pradesh: Balaghat-Chhindwara-Nimar-Mandla and Jhabua districts • Karnataka: Dharwar, Ballari, Belagavi, North Canara, Chikkmagaluru, Shivamogga, Chitradurg and Tumakuru • Maharashtra: Nagpur, Bhandara and Ratnagiri Minor Producers: Telangana, Goa, Jharkhand	The **Manganese Node of India** encompasses MP, Maharashtra and Chhattisgarh.
Copper	Chalcopyrite, Bornite, Malachite, Cuprite	**Extensive applications** in industries ranging from **construction to electronics**, is a crucial resource in the global economy	**Producers:** Chile, DR Congo, Peru, China, Russia; **Reserves:** Chile, Australia, Peru, DR Congo, Russia.	**Reserves:** • Jhunjhunu, Khetri and Alwar districts in Rajasthan (52.25 per cent of deposits) • Balaghat district in Madhya Pradesh (23.28 per cent) • Singhbhum district in Jharkhand (15.14 per cent) • Minor in Andhra Pradesh (Guntur), Karnataka (Chitradurg, Hasan) & Tamil Nadu (South Arcot). Also found in very small quantities in Gujarat, Haryana, Maharashtra, Meghalaya, Nagaland, Odisha, Sikkim, Telangana, Uttarakhand, and West Bengal. **Production:** Madhya Pradesh, Rajasthan, Jharkhand	Hindustan Copper Limited has mines in Chile.
Bauxite (Aluminium)	Gibbsite, Boehmite, Diaspore	Crucial mineral supporting modern	**Producers:** Guinea, Australia, China, Brazil, India; **Reserves:** Guinea,	**Reserves:** Odisha, Andhra Pradesh, Gujarat,	China: ~ 75% production. Aluminium production requires

Mining · Shaft: for resources at a deeper level

		infrastructure, transportation, and various industrial applications	Australia, Indonesia, Jamaica, Brazil.	Jharkhand, Maharashtra, and Chhattisgarh. Recently, the Geological Survey of India (GSI) has identified massive deposits Bauxite deposits in Meghalaya. **Production:** • Odisha: Kalahandi, Sambalpur, Bolangir, Koraput • Jharkhand: Lohardaga Patlands • Gujarat: Bhavnagar, Jamnagar • Chhattisgarh: Amarkantak plateau • Madhya Pradesh: Katni-Jabalpur area, Balaghat • Maharashtra: Kolaba, Thane, Ratnagiri, Satara, Pune Minor Producers: Tamil Nadu, Karnataka, Goa	a huge amount of electricity. That is why, most of the Aluminium smelting industries of India are located in regions in the vicinity of Power Plants
Gold	Native gold, Auriferous quartz veins, Placer deposits	Jewellery, Financial Systems, Electronics.	**Producers:** China, Russia, Australia, Canada, USA; **Reserves:** Australia, Russia, Indonesia, China, Canada.	**Reserves:** Bihar (preliminary, possibly very large), Jharkhand, Chhattisgarh, Rajasthan, Andhra Pradesh; recent discovery in Odisha (Sundargarh, Nabarangpur, Keonjhar, Deogarh). **Producers / mines:** Karnataka (Kolar, Hutti), Lava mines in Jharkhand, Sonbhadra (U.P.), Jonnagiri (A.P.).	India to restart mining at Kolar Gold Mines, 24 yrs after its closure.
Silver	Argentite, Galena (by-product), Native silver	Silver's **industrial applications**, including solar panels, electronics, and medical devices, drive a significant portion of its demand.	**Producers:** Mexico, Peru, China, Bolivia, Chile; **Reserves:** Peru, Russia, Australia, China, Poland.	**Reserves:** limited; some deposits in Rajasthan, potential reserves in Karnataka, Jharkhand, Andhra Pradesh. **Producers:** Rajasthan contributes about 60% of total production, especially Zawar mines (Udaipur) and Rampura Agucha; also produced in Andhra Pradesh, Jharkhand, Karnataka as a by-product of zinc/lead/copper/gold smelting.	**Recycling of silver** from old electronics and other sources is **becoming increasingly important**. Recovered as a **by-product from lead & zinc concentrates, copper slime, and as a coproduct of gold refining.**
Uranium	Pitchblende (Uraninite), Carnotite	Uranium is a crucial element used primarily as a fuel in **nuclear reactors for electricity generation.**	**Producers:** Kazakhstan, Canada, Namibia, Australia, Uzbekistan; **Reserves:** Australia, Kazakhstan, Canada, Namibia, Russia.	**Potential reserves:** Rajasthan (Udaipur, Alwar, Jhunjhunu), Chhattisgarh (Durg), Maharashtra (Bhandara), Himachal Pradesh (Kullu, Hamirpur, Una); possible reserves in Andhra, Telangana, Meghalaya, Karnataka. **Producers / mines:** UCIL mines at Jaduguda, Jharkhand.	India in talks with Namibia for a Uranium partnership. U235 is fissile not U238. Natural uranium contains approximately 0.72% of the U-235 isotope.

Mining · Shaft: for resources at a deeper level

					ENRICHMENT is done to increase the concentration.
Thorium	Monazite (important in India), Thorite	Much cleaner than Uranium. More energy-efficient. Very crucial for Clean Energy Production. Also, U233 produced via Thorium is more fissile than U235.	**Production data:** not publicly available; **Reserves:** India, Brazil, USA, Australia, China.	**Reserves:** India has about 25% of global deposits; richest monazite deposits in Palakkad and Kollam (Kerala), near Visakhapatnam (A.P.), and Mahanadi delta (Odisha). **Producer:** Kerala is the key producer.	While uranium reactors typically utilize less than 1% of the fuel, thorium reactors, particularly in well-designed reprocessing systems, can utilize up to 99% of the fuel. China has recently created world's first thorium-powered plant.
Tin	Cassiterite	Tin is a silvery-white metal revered for its **high malleability and corrosion resistance**, embarks on a captivating journey across the globe. Its is **used to make alloys** like Bronze.	**Producers:** China, Indonesia, Peru, Brazil, DR Congo; **Reserves:** Indonesia, China, Brazil, Myanmar.	**Reserves:** Chhattisgarh (Dandkaranya); possible deposits in Odisha (Koraput). **Producer:** Chhattisgarh (Dandkaranya).	
Lead	Galena	Used in batteries, radiation shielding, ammunition, cable sheathing, and some construction materials.	**Producers:** China, Australia, Peru, USA, Russia; **Reserves:** Australia, China, Russia, Mexico, Peru.	**Reserves:** primarily Rajasthan; also Andhra Pradesh and Madhya Pradesh; resources also in Bihar, Maharashtra, Gujarat, Meghalaya, Odisha, Sikkim, Tamil Nadu, Uttarakhand, West Bengal. **Producer:** Rajasthan is the largest lead-producing state.	
Zinc	Sphalerite (Zinc blende), Smithsonite	Primarily used in **galvanization**, where it protects iron and steel from corrosion. Key **component in alloys like brass** and is essential for making	**Producers:** China, Peru, Australia, India, Mexico; **Reserves:** Australia, China, Russia, Peru, Mexico.	**Reserves:** Gujarat, Meghalaya, Odisha, Sikkim, Tamil Nadu, Uttarakhand, West Bengal. **Producers:** mainly Rajasthan via HZL mines—Rampura Agucha, Sindesar Khurd, Rajpura Dariba, Zawar, Kayad; other producers include Andhra Pradesh, Madhya Pradesh, Bihar, Maharashtra.	Stable Zinc-Sulphus batteries might be a good alternative to Lithium batteries.

Mining · Shaft: for resources at a deeper level

		batteries, fertilizers, paints, and pharmaceuticals			
Nickel	Pentlandite, Garnierite	Key ingredient in stainless steel and rechargeable batteries.	**Producers:** Indonesia, Philippines, Russia, Canada, New Caledonia; **Reserves:** Indonesia, Australia, Brazil, Russia, New Caledonia.	**Reserves:** about 0.2% of global reserves; mainly Odisha (Sukinda Valley, Jajpur; Mayurbhanj, Keonjhar), Jharkhand (Singhbhum East), Nagaland (Kiphire); also associated with uranium deposits at Jaduguda. **Producer:** no active mines.	Produced as a **byproduct of copper smelting**. Also, focus on **recycling**
Cobalt	Cobaltite, Smaltite	Powering our electronics and enabling the clean energy transition.	**Producers:** DR Congo, Indonesia, Russia, Madagascar, Philippines, Australia; **Reserves:** DR Congo, Australia, Russia, Indonesia, Philippines.	**Potential reserves:** Singhbhum (Jharkhand), Kendujhar and Jajpur (Odisha), Jhunjhunu (Rajasthan), Tuensang (Nagaland), Jhabua and Hoshangabad (M.P.). **Producer:** no domestic production currently.	**Focus on re-cycling** (re-extraction from electronics).
Lithium	Spodumene, Lepidolite, Petalite, Brines	Powering our electronics and enabling the clean energy transition.	**Producers:** Australia, China, Chile, Argentina, Brazil; **Reserves:** Chile, Australia, China, USA. The text snippet repeats Australia once; I have removed the duplicate in the final rendering.	**Reserves:** Jammu & Kashmir, Rajasthan, Karnataka (Mandya, Yadgiri); recent discoveries in Meghalaya and Jharkhand. **Producer:** no operational mines currently; first mine expected in Korba, Chhattisgarh.	Lithium mining is causing a lot of water pollution.
Titanium	Ilmenite, Rutile	Used in aircraft, spacecraft, medical implants, paints, and corrosion-resistant equipment due to its light weight and strength.	**Producers:** China, Japan, Russia, Kazakhstan, Saudi Arabia; **Reserves:** Australia, China, South Africa, Canada, Mozambique.	**Indian occurrence/deposits:** Kerala, Tamil Nadu, Andhra Pradesh, Odisha via ilmenite, rutile, brookite, leucoxene. **Production:** Kerala, Tamil Nadu, Odisha	
Rare Earth Elements	Monazite, Bastnäsite, Xenotime	After the table	**Producers:** China, USA, Australia, Myanmar, Thailand; **Reserves:** China, Brazil, Australia, Russia, Vietnam, Greenland (Denmark).	**Reserves:** about 13.07 million tonnes of in-situ monazite resource; 5th largest global deposits; coastal placer sands in Kerala, Tamil Nadu, Odisha, Andhra Pradesh, Maharashtra, Gujarat, and inland placers in Uttar Pradesh, Jharkhand, West Bengal, Tamil Nadu. **Producers:** Odisha, Kerala, Tamil Nadu,	15 Lanthanides + Scandium & Yttrium. The **near monopolistic position of China** has led to the rise of concerns amongst the Western nations. **Minerals Security Partnership** is the

Mining · Shaft: for resources at a deeper level

				Andhra Pradesh, Maharashtra, Gujarat.	result of this concern.
Mica	Muscovite, Biotite, Phlogopite	Mica is a group of silicate minerals prized for their unique properties like flexibility, and heat resistance, which play a vital role in various industries. From **electronics and construction to cosmetics and pharmaceuticals**.	**Producers:** India, China, Russia, Brazil, USA; **Reserves:** India, China, Republic of Korea, Turkey.	**Reserves / occurrence:** Jharkhand (Hazaribagh plateau, Koderma), Andhra Pradesh (Nellore), Rajasthan (Jaipur–Bhilwara–Udaipur belt), Karnataka (Mysuru, Hassan), Tamil Nadu (Coimbatore, Tiruchirapalli, Madurai, Kanyakumari), Kerala, Maharashtra, West Bengal. **Top Producer:** Jharkhand	In Feb 2025, classified as a Major Mineral along with Feldspar, barytes and Quartz.
Tantalum	Columbite–Tantalite (Coltan)	Used for **electronics & semiconductors**, from computers to ACs (for making capacitors mainly)	**Top Producers** Democratic Republic of Congo, Nigeria, Rwanda, Brazil & China **Top Reserves** Australia, China, Brazil, Russia. For the rest of the countries, data is not available	Potential found in Sutlej River sands in Punjab. No Production in India.	
Silicon	Quartz, Silica sand	Widely used in **electronics** (as a semiconductor in chips and solar cells) and in **construction** (as silica in glass, cement, and ceramics).It is also crucial in **metallurgy (ferrosilicon)**, **silicones (lubricants, sealants)**, and in **cosmetics and medical implants**	**Top producers:** Silicon metal–China, Brazil, Norway; Ferrosilicate–China, Russia, Norway, Brazil; **Reserves:** abundantly available in most countries.	**Reserves / sources:** derived from quartz; significant quartzite/quartz resources in Haryana, Rajasthan, Andhra Pradesh, Tamil Nadu, Maharashtra, Jharkhand, Uttar Pradesh. **Producers:** Haryana, Rajasthan, Andhra Pradesh.	2nd most abundant element in Earth's crust.

Mining · Shaft: for resources at a deeper level

Tungsten	Wolframite, Scheelite	**Dense, hard metal** known for its high **melting point**, which is the **highest of all metals.** **Production of hard materials**, including tungsten carbide, used for cutting tools and wear-resistant applications. **Used in the aerospace and electronics industries**, including for filaments in light bulbs and electrodes	**Producers:** China, Vietnam, Kazakhstan, Russia, North Korea; **Reserves:** China, Australia, Russia, Vietnam, Spain.	**Reserves:** Karnataka, Rajasthan, Andhra Pradesh, Maharashtra; smaller resources in Haryana, Tamil Nadu, Uttarakhand, West Bengal; notable deposit at Degana, Rajasthan. **Producers:** Rajasthan and Karnataka.	Recently Tungsten mining was scrapped in Madurai due to protest.
Antimony	Stibnite	Used to create many compounds and alloys, which are then used to make flame-retardant materials, paints, enamels, glass, and pottery	**Top Producers:** China, Russia, Tajikistan, Australia, Bolivia **Top Reserves:** China, Russia, Bolivia, Kyrgyzstan, Myanmar	Minor deposits in the Lahul & Spiti district of HP and Damoh district of MP. Complete import dependency.	China limited the exports in July 2024.
Gallium	By-product of Bauxite (Aluminum ores)	**Soft, silvery-white strategic metal** predominantly used in electronics	**Top Producers:** China (>90 per cent), Russia, Japan, South Korea, Ukraine **Top Reserves**: Data on Global reserves not available in public domain	NALCO planning a 10 tpy gallium extraction plant at its Damanjodi alumina refinery. Used to be produced by Hindalco Industries Ltd. and National Aluminium Co. Ltd. (NALCO), but not anymore.	China limited exports in July 2023
Germanium	By-product of Zinc ores (Sphalerite)	**Electronics and solar applications**, **fiber-optic systems, infrared**	**Top Producers:** The producers do not publicly give data regarding their production. But the biggest producer is	No known reserves. 100 per cent import dependency (from China, Germany & South Africa).	

Mining · Shaft: for resources at a deeper level

		optics, and polymerization catalysts. Other uses included chemotherapy, metallurgy, and phosphors.	China.Other producers include the USA, Belgium, Canada, Russia, South Africa, and Germany. **Top Reserves:** Data on Global reserves not available in public domain		
Tellurium	By-product of Copper refining	Used primarily in **thermoelectric devices**, solar panels (CdTe solar cells), and as an additive in **steel and copper alloys.** Also in **optical discs, semiconductors**, and **vulcanization of rubber**.	**Producers:** China, Russia, Japan, Sweden, Canada; **Reserves:** China, Russia, Japan, Sweden, Uzbekistan.	Both Tellurium and Selenium are rarely found in their native form and are typically extracted as byproducts of other mining operations. **Occurrence / producer:** found as by-product of copper and gold ores; Ghatshila copper smelter, Jharkhand; minor tellurium from Dahej, Gujarat; no recent production data cited.	Recovered as by-products, mostly from anode mud or slime obtained during electrolytic refining of copper
Selenium	By-product of Copper refining	Used in **photocells, solar panels, and glass manufacturing** (to decolorize or tint glass). It is also important in **electronic rectifiers**, and in **agriculture and animal nutrition** as a micronutrient.	**Top Producers:** China, Japan, Russia, Belgium, Canada **Top Reserves:** China, Japan, Russia, Belgium, Mexico		
Indium	By-product of Zinc ores	Mainly used in **Indium Tin Oxide (ITO)** for making **touchscreens, LCDs, and solar panels** due to its transparency and conductivity. It is also used in	**Top Producers:** China, South Korea, Japan, Canada, France **Top Reserves:** Detailed Reserves data not available publicly	Complete import dependency. NIL production.	Byproduct of Zinc and Lead Mining

Mining · Shaft: for resources at a deeper level

		semiconductors, solders, and alloys that require low melting points and good thermal properties.			
Helium	Natural gas fields (radioactive decay origin)	Helium is used in **balloons/airships, cryogenic cooling of MRI machines, deep-sea diving gas mixtures, welding, leak detection, and as a protective gas in rockets and scientific instruments**.	**Top Producers:** USA, Qatar, Russia, Algeria, Australia, Canada **Top Reserves:** USA, Qatar, Algeria, Russia	No major reserves. 100% Import dependency	
Diamonds	Kimberlite pipes, Alluvial deposits	Used in jewellery, cutting/drilling tools, abrasives, and precision instruments due to its hardness.	**Top Producers:** Russia, Botswana, Canada, Angola, Democratic Republic of Congo, South Africa **Top Reserves:** Russia, Botswana, Democratic Republic of Congo, Angola	Approximately 90% concentrated in Madhya Pradesh, specifically in the Panna belt. Other smaller, diamond-bearing regions include Andhra Pradesh (Krishna River basin) and Chhattisgarh. Production only in Panna, M.P.	
Crude Oil	Sedimentary basins (organic origin)	Used as fuel, lubricants, asphalt, and raw material for petrochemicals like plastics, fibres and fertilizers.	**Top Producers:** USA, Saudi Arabia, Russia, Canada, China **Top Reserves:** Venezuela, Saudi Arabia, Canada, Iran, Iraq	Major reserves and production centres: Mumbai High, Rajasthan (Barmer Basin), Gujarat (Cambay Basin), Assam (Assam-Arakan Belt), The Krishna-Godavari Belt has reserves, but no production. Recent discovery in the Andaman Sea region. Still being evaluated.	
Natural Gas	Associated & non-associated gas fields	Used for cooking, power generation, fertilizers, industries, and as CNG/PNG fuel.	**Top Producer:** USA, Russia, Iran, China, Canada **Top Reserves:** Russia, Iran, Qatar, USA, Saudi Arabia	Assam (Digboi and Rudrasagar), Mumbai High, Gujarat, Rajasthan, Tripura, KG Basin. Reserves discovered in Andaman Sea as well.	

Mining · Shaft: for resources at a deeper level

World Resources

→ Resources are essentially anything that we can use from our environment to fulfill our needs and wants . An object/entity should fulfill these 2 criteria to be called a resource :

→ Utility: It must be something that can be used for some kind of benefit.

Types of Resources

→ Resources can be classified based on various parameters:

Based on Exhaustibility

→ Renewable Resources: Resources that can be replenished naturally over time. E.g. Solar energy, wind energy, biomass, hydropower.

→ **Non-renewable Resources:** Resources that are finite and cannot be replaced within a human timescale once depleted. E.g. Fossil fuels (coal, oil, natural gas), minerals (gold, iron ore).

Based on Origin (Biotic and Abiotic)

→ **Biotic Resources:** Resources derived from living organisms. E.g.: Forests, animals, plants, and fossil fuels (originated from ancient biological matter).

→ **Abiotic Resources:** Resources derived from non-living elements. E.g. Water, air, minerals, metals.

Based on Ownership

→ **Individual Resources:** Resources owned by private individuals or companies. E.g. Private land, personal vehicles, privately owned mines.

→ **Community Resources:** Resources accessible to all members of a community. E.g. Public parks, community forests, common grazing lands.

→ **National Resources:** Resources owned by the nation and managed by the government. E.g. National highways, public sector industries, forests.

Mining

→ **Types** of Mines:

→ – **Shaft:** for resources at a deeper level

→ **Applications of REE:**

→ – **Electronics:** Rare earth metals are essential components in the production of electronics, including smartphones, computers, and TVs.

→ – **Magnets:** Neodymium magnets, which contain rare earth elements, are crucial for various applications, such as electric motors in electric vehicles and wind turbines.

→ – **Catalysis** : Some rare earth metals serve as catalysts in chemical reactions, impacting processes like refining petroleum and automobile emissions control.

→ – **Lighting** : Phosphors containing rare earth metals are used in the production of energy-efficient lighting, including fluorescent lamps and LED lights.

→ – **Medical Imaging** : Gadolinium is used in contrast agents for magnetic resonance imaging (MRI) in medical diagnostics.

→ – **Defense Technologies** : Rare earth elements are used in the manufacturing of components for defense technologies, including guided missiles and radar systems.

MINERALS OF INDIA

PYQ 2013 PYQ 2018 PYQ 2022 PYQ 2023

Understand it first: Minerals are tested as **mineral with State or mine**, and ore with metal. Learn them belt by belt.

State pegs

→ Iron = Odisha–Jharkhand–Chhattisgarh–Karnataka–Goa belt

→ Copper = Rajasthan (Khetri)

→ Bauxite = Odisha

→ Gold = Kolar and Hutti (Karnataka)

→ Diamond = Panna (MP) only

→ Mica = Jharkhand

→ Monazite/thorium = Kerala

→ Uranium = Jaduguda (Jharkhand)

→ Lead-zinc-silver = Rajasthan (Zawar, Rampura Agucha)

→ Chromite = Odisha >96%.

Mineral	Where in India	Hooks
Iron ore	Odisha (Mayurbhanj, Keonjhar, Sundargarh); Jharkhand (Singhbhum); Chhattisgarh (Durg, Dantewada, Bailadila); Karnataka (Ballari, Hospet, Kudremukh); Goa, Maharashtra, Telangana, AP, TN	hematite, magnetite, limonite, siderite; steel
Manganese	Odisha, Karnataka, MP, Maharashtra	steel, batteries
Copper	Rajasthan 52.25% (Jhunjhunu, Khetri, Alwar) · MP 23.28% (Balaghat) · Jharkhand 15.14% (Singhbhum)	chalcopyrite; world leaders Chile, DR Congo, Peru
Bauxite	Odisha (Kalahandi, Koraput), Gujarat, Jharkhand (Lohardaga), Maharashtra, Chhattisgarh (Amarkantak), MP; new finds in Meghalaya	aluminium smelters near power plants (huge electricity)
Gold	Karnataka (Kolar, Hutti), Jharkhand, Chhattisgarh, Rajasthan, AP (Jonnagiri); Bihar preliminary	Kolar to restart 24 yrs after closure
Silver	by-product; Rajasthan ≈ 60% (Zawar, Rampura Agucha)	lead–zinc by-product
Uranium	Jharkhand (UCIL Jaduguda); potential Rajasthan, Chhattisgarh, Maharashtra, HP, Andhra	natural U has 0.72% U-235; U-235 fissile, U-238 not; enrichment raises concentration
Thorium (monazite)	India ≈ 25% of global deposits; Palakkad and Kollam (Kerala), Visakhapatnam, Mahanadi delta	cleaner; U-233; reactors can use up to 99% of fuel
Tin	Chhattisgarh (Dandkaranya)	bronze
Lead / Zinc	Rajasthan largest (HZL mines: Rampura Agucha, Sindesar Khurd, Rajpura Dariba, Zawar, Kayad)	zinc: galvanising; Zn–S batteries as lithium alternative
Nickel	Odisha (Sukinda), Jharkhand, Nagaland; ~0.2% of global reserves; no active mine	stainless steel, batteries
Cobalt	potential: Jharkhand, Odisha, Rajasthan, Nagaland, MP; no domestic production	
Lithium	J&K, Rajasthan, Karnataka (Mandya, Yadgiri); first mine expected in Korba, Chhattisgarh; no operating mine	
🎯 Titanium PYQ 2023	Kerala, TN, AP, Odisha (ilmenite, rutile)	ilmenite $FeO.TiO_2$, rutile TiO_2
Rare earths	monazite, bastnäsite, xenotime; coastal placers in Kerala, TN, Odisha, AP; 13.07 million t monazite (5th largest)	17 elements = 15 lanthanides + scandium + yttrium; China near-monopoly → Minerals Security Partnership
Mica	Jharkhand (Hazaribagh, Koderma), AP (Nellore), Rajasthan	Feb 2025: Major Mineral with feldspar, barytes, quartz
Tantalum, silicon	no production; silicon from quartz in Haryana, Rajasthan, AP	
💰 Tungsten	Karnataka, Rajasthan (Degana)	Madurai mining scrapped after protest
Antimony, germanium, indium, helium	~100% import dependent (antimony complete; germanium 100%; indium complete; helium 100%)	China limited exports (antimony July 2024; gallium July 2023)
Gallium, tellurium, selenium	by-products of bauxite/copper	ALCO plans 10 tpy gallium plant at Damanjodi
Diamond	~90% Panna (MP); AP (Krishna basin), Chhattisgarh	production only in Panna
Limestone	Rajasthan, MP (Satna, Katni), AP, Gujarat	$CaCO_3$ sedimentary rock; world: China, India, USA, Russia, Brazil
Others	Chromite (>96% Odisha) · Pyrite (none since 2003) · Borax (small, Puga valley) · Cryolite (nil)	

S.No	Ore	Product	Presence in India
1	Apatite and Rock Phosphate	Phosphorous, Fluorine, Uranium (secondary product)	Apatite (very less): WB, JH, Meghalaya Rock Phosphate: JH, RJ, MP
2	Borax and Sassolite	Boron	No economically viable deposits. Small deposits in Puga Valley.
3	Chromite	Chromium	>96% in Odisha. Rest in Manipur, Meghalaya, KR, JH, MH, TN, Andhra P.
4	Cryolite	Fluorine, Sodium (very less)	Nil
5	Diatomite (Dead Diatom Deposits)	Silica	Gujarat, Rajasthan
6	Fluorite	Fluorine	GJ, RJ, CG, MH

Some Important Ores and their Products

Handout alert (2018 PYQ): explanation prints both "states can auction" and "states cannot", and says AP/Jharkhand "do not have gold mines" while the table lists them as gold states. Take answer C and the gold table.

In the news · 2025 (verify dates) NCMM: National Critical Mineral Mission launched 2025 for critical-mineral self-reliance; REE and China export curbs are the context.

Limestone

→ Limestone, a sedimentary rock composed primarily of calcium carbonate ($CaCO_3$), is a ubiquitous and versatile resource with immense economic and geologic significance.

→ From building materials and cement production to agriculture, soil amendment, and environmental applications, limestone underpins numerous industries and plays a crucial role in shaping landscapes and ecosystems worldwide.

Top Producers: China, India, USA, Russia, Brazil

→ China: Hubei, Guangxi, and Shandong.
→ India: Rajasthan, Madhya Pradesh, Andhra Pradesh, and Gujarat
→ USA: Texas, Florida, and Kentucky
→ Russia: Caucasus Mountains and the Ural Mountains
→ Brazil: Minas Gerais and Bahia

India:

→ Rajasthan : Jaisalmer, Jodhpur, Nagaur, and Bikaner
→ Andhra Pradesh : Kurnool and Anantapur
→ Gujarat : Bhavnagar, Amreli, and Kachchh

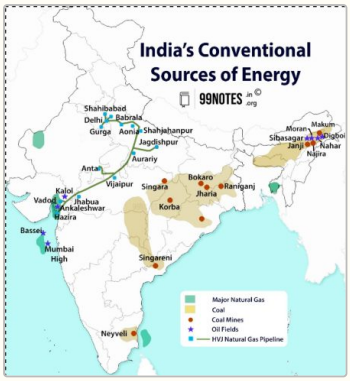
COAL, OIL, GAS, SHALE PYQ 2012 PYQ 2013 PYQ 2016

Understand it first: Fossil fuels form from buried organic matter. For coal learn the ranks by carbon content and the fields; for oil and gas the basins; shale gas is gas trapped in fine-grained rock that needs fracturing to release.

Coal

→ Plant remains → coalification; world: China (Shanxi, Inner Mongolia), India, USA, Australia, Indonesia

→ Indian coal: non-coking (non-metallurgical), high ash 40–45%, low sulphur ~0.5–1% (~0.7%), high ash-fusion ~1,500 °C; ~75% of electricity from coal thermal plants



Coal

Gondwana (bituminous)

→ Valleys: Damodar, Mahanadi, Godavari, Son

→ Jharkhand-Bengal: Raniganj, Jharia, Bokaro, Giridih, Karanpura

→ Singrauli (MP/UP) · Korba (CG) · Talcher, Rampur (Odisha)

→ Chanda-Wardha, Kamptee, Bander (Maharashtra) · Singareni (Telangana) · Pandur (AP)

Tertiary

→ Anthracite: Kalakot (J&K)

→ Assam (Makum, Jaipur, Nazira) · Arunachal (Namchik-Namphuk) · Meghalaya (Darangiri, Cherrapunji, Mewlong, Langrin)

→ Nagaland: rat-hole mining despite SC ban, Art. 371A

→ Lignite: Neyveli (TN), Gujarat, Puducherry

7	Ilmenite, Rutile, Brookite, Leucoxene	Titanium	Kerala, Tamil Nadu, Andhra P, Odisha
8	Kyanite, Sillimanite and Andalusite	Aluminium	Kyanite: TL, Andhra P; Sillimanite: OD, TN; Andalusite: possible reserves in UP
9	Magnesite	Magnesium	UK, TN,
10	Monazite	Thorium, Lanthanum, Cerium	130 deposits in the coastal beach placer sands in Kerala, Tamil Nadu, Odisha, Andhra Pradesh, Maharashtra & Gujarat and in the inland alluvium in parts of Jharkhand, West Bengal and Tamil Nadu
11	Pyrite	Iron, Sulphur	No production since 2003
12	Vermiculite	Similar to Mica	TN, Andhra P, KR
13	Zircon	Zirconium	State-wise breakup not available. Occurs in association with Ilmenite, Rutile, Monazite

Coal

Coal

Oil and gas

→ Crude: Mumbai High, Barmer (Rajasthan), Cambay (Gujarat), Assam-Arakan; KG belt has reserves but no production; new find in Andaman Sea region (being evaluated)

→ Natural gas: Assam (Digboi, Rudrasagar), Mumbai High, Gujarat, Rajasthan, Tripura, KG; also Andaman Sea

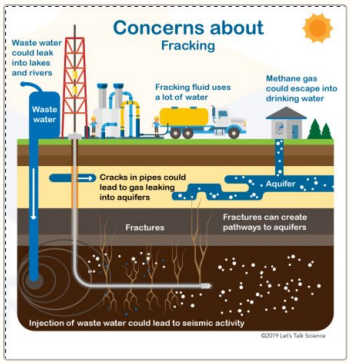
🎯 Shale oil and gas PYQ 2016

→ Unconventional, in low-permeability shale; hydraulic fracturing; world: USA, China, Argentina, Canada, Russia

→ India basins: Cambay, Ganga valley, Assam-Arakan, KG, Kaveri

→ 2016 PYQ (D): Cambay, Cauvery, KG all ✔; six basins: Cambay, Assam-Arakan, Gondwana, KG onshore, Cauvery onshore, Indo-Gangetic

→ Problems: water contamination, induced seismicity, land disturbance, methane leaks



Shale Oil And Gas

Coal

→ Coal, a cornerstone of the global energy sector for centuries, plays a crucial role in powering industries and generating electricity.

→ Coal is primarily formed from the remains of plants that lived and died millions of years ago by the process called Coalification.

→ Indian coal is of non-coking grade (non-metallurgical coal, has higher impurities, so cannot be converted into coke).

Top Producers: China, India, USA, Australia, Indonesia.

→ USA: Appalachians (mainly Anthracite)
→ Australia: Queensland, New South Wales
→ Indonesia: Kalimantan, Sumatra

India:

→ Gondwana (Bituminous) : River Valleys (Damodar, Mahanadi, Godavari, Son)
→ – Madhya Pradesh : Singrauli (part in UP, but CIL plant in MP)
→ – Korba in Chhattisgarh

→ Nagaland (minor rat-hole mining allowed despite a ban by the supreme court, under article 371A)

Shale Oil And Gas

→ Shale oil (tight oil) and shale gas, unconventional oil and gas resources trapped within shale rock formations (less permeable and difficult to drill through), have revolutionized the energy landscape in recent decades.

→ These resources, previously inaccessible, are now extracted through innovative technologies like hydraulic fracturing , unlocking vast new energy reserves and impacting global energy markets significantly.

→ Problems related to shale oil & gas mining : Shale oil and gas extraction, particularly through hydraulic fracturing (fracking), faces several environmental and public health concerns:

→ – Water Contamination : Fracking uses large volumes of water mixed with chemicals, raising concerns about potential contamination of groundwater resources through spills or improper disposal of wastewater. The underground aquifers might also get contaminated.

→ – Induced Seismicity : The process of injecting water underground during fracking can trigger earthquakes in some areas.

→ – Land Disturbance : Shale oil and gas development requires significant land use for well pads, pipelines, and infrastructure, disrupting ecosystems and impacting local communities.

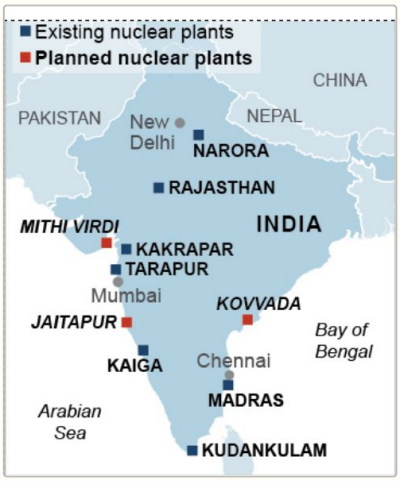
→ – Air and Climate Impacts : Methane leaks during extraction and processing contribute to greenhouse gas emissions. Additionally, air pollution from drilling and associated activities can harm local air quality.

🎯 ENERGY: RENEWABLES, NUCLEAR PYQ 2023

Understand it first: Non-conventional sources are renewable: sun, wind, biomass, the Earth's heat, tides and ocean temperature difference. For each know how it works in a line and where it is found; nuclear is taken alongside.

Source	India facts	World / notes
All renewables	283.46 GW up to March 2026	
Solar	157 GW by May 2026; target 280 GW by 2030; top: Rajasthan, Gujarat, Maharashtra, TN, Karnataka	China, USA, India, Japan, Germany; thermal + electric; desert panels: warm area (darker than sand), logistics and maintenance costs, sand cover
Wind	56 GW; top: Gujarat, TN, Karnataka, Rajasthan, Maharashtra; no offshore farm	China, USA, India, Germany, Brazil; offshore winds faster (no friction) but cost cancels gain
Bio-energy	10.75 GW; Maharashtra, Karnataka, UP, Punjab, Bihar	USA, Brazil, Indonesia, China, India; waste-to-energy (incineration → steam), biogas (anaerobic digestion: methane + CO_2)
Geothermal	potential 10,600 MW; Manikaran (HP); target 10 GW by 2030; not yet tapped; SCCL 20 kW pilot, Manuguru, Bhadradri Kothagudem (Telangana)	USA, Indonesia, Philippines, Turkiye, New Zealand
Tidal	Kachchh (Gujarat) 50 MW; Durgaduani Creek (WB) 3.75 MW; started, stopped due to cost	S. Korea, France, Canada, China, Russia
OTEC	65 kW at Kavaratti (Lakshadweep) for desalination	only 2 pilot plants: Japan (Okinawa), USA (Makai, Hawaii)
Nuclear	8,180 MW → 22,480 MW by 2031-32; +70% in last 10 years	fusion and fission; only fission used; 2021: USA, China, France, Russia, S. Korea

Geothermal Energy



Nuclear Energy

📝 3 stages of India's nuclear programme
- → 1: PHWR, natural uranium (U-235)
- → 2: Fast Breeder, plutonium (Pu-239), Kalpakkam
- → 3: thorium, U-233, KAMINI, Kalpakkam

U → Pu → Th – Use uranium, Produce plutonium, Then thorium. Stages 2 and 3 are Kalpakkam-linked.

Source conflict (handout vs mind map): mind map prints older figures, renewables 220.10 GW and solar 105.65 GW (31 Mar 2025, MNRE); handout text says 283.46 GW (Mar 2026) and 157 GW (May 2026). Use the handout; 280 GW by 2030 is the same in both.

Why not cover deserts with panels? Just 1.2% of the Sahara could power the world, but dark panels absorb more heat, raise temperature and change rain and wind nearby; plus high logistics and maintenance costs (sand cover).

Solar Energy

→ India being a tropical nation has a huge solar energy potential.

→ By May 2026, total installed capacity 157 GW

→ Top Producers: Rajasthan, Gujarat, Maharashtra, Tamil Nadu, Karnataka

Problems of covering sub-tropical deserts with Solar Panels:

→ Solar panels , which are darker in colour compared to the sands, may absorb more heat -> increase in temperature -> changes in rainfall and wind patterns in surrounding areas.

Wind Energy

→ Onshore and Offshore

→ In offshore locations, due to the lack of frictional forces, the speed of the winds is higher . However, the cost of these plants with the current technology nullifies the increased electricity production due to higher wind speeds.

Bio-Energy

→ Waste to Energy: Waste to Energy (WtE) is the process of generating energy in the form of electricity or heat from the primary treatment of waste. It typically involves incineration, where municipal solid waste is burned to produce steam that drives turbines to generate electricity.

→ Biogas: Biogas is produced through the anaerobic digestion of organic materials such as agricultural waste, manure, and food scraps. The process results in a mixture of methane and carbon dioxide, which can be used as a renewable energy source for heating, electricity, and as a vehicle fuel.

→ India total production: 10.75 GW

Geothermal Energy

→ Indian potential: 10600 MW

→ India plans to harness 10 GW by 2030 especially at Manikaran, HP.

→ Currently not tapped.

Otec (Ocean Thermal Energy Conversion)

Nuclear Energy

→ Currently, only Fission is used for energy generation by humans.

→ According to Government of India, the present installed nuclear power capacity is set to increase from 8180 MW to 22480 MW by 2031-32.

→ 3 stages of India's Nuclear Power Program :

→ – Natural uranium fuelled Pressurized Heavy Water Reactors (PHWRs) (U235)

→ – Fast Breeder Reactors (FBRs) utilizing plutonium -based fuel. (Pu239): Kalpakkam, TN (achieved criticality)

→ – Advanced nuclear power systems for utilization of thorium . (U233 from Thorium): KAMINI-Kalpakkam Mini Reactor, Kalpakkam, TN (experimental)

→ Map shows the locations of stage 1 reactors in India:

🎯 4 · INDUSTRIES (no PYQ)

Understand it first: Industry is classed by raw material, size and ownership. Where an industry settles depends on a few pulls: **raw material, power, labour, market, transport and policy.** Then learn the main industrial regions.

Sector	What	Examples
Primary	extraction/harvest	agriculture, fishing, forestry, mining
Secondary	processing/ manufacture	light (consumer goods), heavy (capital goods: steel, ships), construction
Tertiary	services	trade, tourism, transport, finance, health, education, IT, professional
Quaternary	knowledge	R&D, information services, advanced training
Quinary	top decisions	government, NGOs, culture and arts

P-S-T-Q-Q – Dig → Make → Serve → Think → Decide.

👥 Size

→ (investment / turnover): Micro <₹2.5 cr / <₹10 cr
→ Small ₹2.5–25 cr / ₹10–100 cr
→ Medium ₹25–125 cr / ₹100–500 cr
→ Large above. **Ownership:** public, private, joint, cooperative. **Use:** consumer vs capital goods.

Location factors

→ Raw material · labour · market · transport/infrastructure · energy · capital · government policy
→ **Climate/environment:** circular-economy push, co-locate near recycling centres, "urban mining"
→ Social/cultural · **technology:** reshoring (robots replace cheap labour, plants return to high-income countries)
→ **Global:** nearshoring, friend-shoring · ecological sustainability

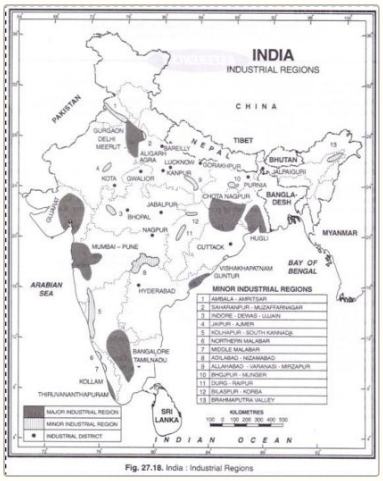
Weber's weight rule

→ **Weight-losing** (bulky raw material → light product, e.g. ore) → sit at the raw material
→ **Weight-gaining** (product heavier/bulkier than inputs) → sit near the market
→ **Footloose:** weight does not matter
→ **Isotim** = equal transport-cost line · **Isodapane** = equal total-cost line (transport of inputs + outputs + labour); best site on the lowest isodapane

World industrial regions	Handout lines
North America	NE USA (steel, autos) · West Coast (tech, aerospace) · South USA (oil, aerospace) · Ontario–Quebec
Europe	**Ruhr, Alsace–Lorraine, Benelux** · UK Midlands · Poland–Czechia–Slovakia–Hungary (Western, Northern, Eastern Europe)
Asia	East Asia (China, Japan, S. Korea, Taiwan) · **South Asia** (Maharashtra, Gujarat, TN, Karnataka) · SE Asia (Singapore, Malaysia, Thailand, Indonesia)
Africa	N. Africa · **Gauteng and Western Cape** (Southern) · W. Africa (Nigeria, Ghana)
Australia · S. America	east Australia · Brazil (São Paulo–Rio–Minas Gerais), Argentina, Chile

Indian region	Industries	Key cities
Mumbai–Pune	textiles, autos, chemicals, engineering, pharma, IT	Mumbai, Pune, Thane, Nashik
Kolkata–Hooghly	jute, textiles, chemicals, engineering, iron and steel, paper	Kolkata, Howrah, Durgapur, Haldia
Delhi–NCR	IT, electronics, autos, consumer goods	Delhi, Gurugram, Noida, Faridabad

Ahmedabad–Vadodara	textiles, chemicals, pharma, petrochemicals	Ahmedabad, Vadodara, Surat, Bharuch
Chennai–Bengaluru	autos, electronics, IT, heavy engineering	Chennai, Bengaluru, Hosur, Coimbatore
Kollam–Thiruvananthapuram	cashew, coir, software, biotech	Kollam, Thiruvananthapuram
Chotanagpur	iron and steel, coal, heavy engineering	Jamshedpur, Ranchi, Bokaro, Dhanbad
Visakhapatnam–Guntur	shipbuilding, petrochemicals, fertiliser, steel	Visakhapatnam, Guntur



Important Industrial Regions of India

Types of Industries

→ **Primary Industries:** These industries involve the extraction and harvesting of natural resources.

→ – **Agriculture:** Farming, horticulture, and animal husbandry.

→ – **Fishing:** Commercial fishing and aquaculture.

→ – **Forestry:** Logging, timber production, and forest management.

→ – **Mining and Quarrying:** Extraction of minerals, ores, and fossil fuels.

→ **Secondary Industries:** These industries involve the processing and manufacturing of raw materials into finished goods.

→ – **Construction:** Building infrastructure such as roads, bridges, buildings, and other structures.

→ **Tertiary Industries (Service Sector):** These industries provide services rather than goods.

→ – **Retail and Wholesale Trade:** Selling goods directly to consumers and businesses.

→ – **Hospitality and Tourism:** Hotels, restaurants, travel agencies, and tour operators.

→ – **Transport and Logistics:** Transportation services (road, rail, air, sea), warehousing, and logistics management.

→ – **Financial Services:** Banking, insurance, investment, and real estate.

→ – **Healthcare:** Hospitals, clinics, pharmaceuticals, and medical services.

→ – **Education:** Schools, colleges, universities, and vocational training.

→ – **Professional Services:** Legal, accounting, consulting, and advertising.

Quaternary Industries

→ These industries involve the **intellectual activities** often associated with technological innovation and knowledge-based services.

→ **Research and Development (R&D):** Scientific research and technological innovation.

→ **Information Services:** Data processing, information management, and media.

Quinary Industries

→ These industries include the highest levels of decision-making and societal advancement .

→ Government: Public administration and defense.

→ Nonprofit Organizations: Charities, NGOs, and other community services.

→ Culture and Arts: Museums, theaters, and cultural heritage management.

Other Classifications of Industries

→ Based on Size:

→ Medium: Mid-sized businesses with moderate resources. Investment between 25-125 crore; turnover between 100-500 crore.

→ Large: Large corporations with substantial resources and workforce .

→ Based on Ownership:

→ Private Sector: Owned and operated by private individuals or companies.

→ Cooperative Sector: Owned and operated by a group of individuals for mutual benefit.

→ Based on Market Orientation:

→ Consumer Goods: Products sold directly to consumers (e.g., food, clothing, electronics).

Factors Affecting the Location of Industries

→ The location of industries depends on various factors that influence their operational efficiency, cost-effectiveness, and overall success.

Raw Materials

→ Availability and Quality of Raw material: The type, quality, and availability of raw materials can significantly impact the location decision.

Labor

→ Availability of Labor: Access to a sufficient and skilled labor force is crucial for industries requiring intensive manual or specialized labor.

→ Labor Costs: Regions with lower labor costs may attract labor-intensive industries.

→ Labor Laws and Conditions: Favorable labor laws and conditions can also influence industrial location.

Market Access

→ Market Size and Demand: Larger markets with higher demand can attract industries looking to maximize their sales and distribution.

Transport and Infrastructure

→ Transportation Facilities: Good connectivity through roads, railways, ports, and airports facilitates the efficient movement of raw materials and finished goods.

→ Infrastructure Availability: Adequate infrastructure, including power supply, water supply, and communication networks, is essential for industrial operations.

Energy Supply

→ Access to Power: Reliable and affordable energy sources are crucial for energy-intensive industries like manufacturing and mining.

→ Types of Energy: The availability of specific types of energy, such as electricity, natural gas, or renewable sources, can influence industrial location.

Capital and Financial Factors

→ Access to Capital: Proximity to financial institutions and access to credit can impact the decision on where to locate an industry.

→ Investment Incentives: Government incentives, such as tax breaks and subsidies, can attract industries to specific regions.

Government Policies and Regulations	→ **Industrial Policies:** Government policies, including zoning laws, environmental regulations, and industrial promotion policies, play a significant role in industrial location decisions. → **Taxation:** Favorable tax policies and lower tax rates can attract industries to particular areas.
Climate and Environmental policies	→ **Environmental Regulations:** Stringent environmental regulations can deter industries from locating in certain regions.
Social and Cultural Factors	→ **Community Attitudes:** Local community support or opposition can influence industrial location. → **Quality of Life:** Areas offering a higher quality of life, including good housing, education, and healthcare, may attract industries that value employee well-being.
Availability of Technology and Innovation	→ **Access to Technology:** Proximity to technology hubs and research institutions can benefit high-tech industries. → **Innovation Ecosystem:** A robust ecosystem that supports innovation and research and development can attract industries looking to stay competitive. → **Reshoring:** Advanced robotics negates the need for cheap human labor. This causes industries to "reshore" back to high-income countries closer to the final consumer.
🌐 Global and International Factors	→ **Trade Agreements:** International trade agreements and export incentives can influence the location of industries targeting global markets. → **Political Stability:** Regions with stable political environments are more attractive to industries, reducing the risk of disruptions. → **Nearshoring and Friendshoring:** As the cost of production in a particular country increases, the industry tries to relocate to a nearby country, so that the markets are still nearby. This is called nearshoring. If two countries sign FTAs, etc., then companies of one country are free to establish their plants in the other country, if the costs of production increase in the former (friend-shoring)
Ecological and Sustainability Considerations	→ **Sustainability Practices:** Industries increasingly consider ecological footprints and sustainability practices, favoring locations that support green initiatives and offer renewable resources.
Isotim:	→ It represents the boundaries of equal transport costs for a specific commodity. → Isotims form concentric circles around the source, with each circle representing a higher transportation cost.
Isodapane:	→ It depicts the combined impact of transportation costs for both inputs and outputs on a firm's total cost.

Important Industrial Regions of the World

🗺️ North-Eastern United States:	→ Regions: New York, New Jersey, Pennsylvania, Ohio, Illinois, and Michigan. → Key Cities: New York City, Chicago, Detroit, Pittsburgh, and Cleveland.
🗺️ West Coast of the United States:	→ Regions: California, Oregon, and Washington. → Key Cities: San Francisco, Los Angeles, Seattle, and San Diego.
	→ Regions: Texas, Florida, Georgia, and North Carolina.

Southern United States:	→ Key Cities: Houston, Dallas, Atlanta, Miami, and Charlotte.
Ontario and Quebec in Canada:	→ Regions: Toronto, Montreal, and Ottawa.
Europe	→ Western Europe: → – Regions: Ruhr Valley (Germany), Alsace-Lorraine (France), and Benelux countries (Belgium, Netherlands, Luxembourg). → – Key Cities: Berlin, Hamburg, Paris, Brussels, and Amsterdam. → Northern Europe: → – Regions: United Kingdom, particularly the Midlands and Northern England. → – Key Cities: London, Manchester, Birmingham, and Glasgow. → Eastern Europe: → – Key Cities: Warsaw, Prague, Bratislava, and Budapest.
Asia	→ East Asia: → – Key Cities: Shanghai, Beijing, Tokyo, Seoul, and Taipei. → Southeast Asia: → – Key Cities: Singapore, Kuala Lumpur, Bangkok, and Jakarta.
Africa	→ North Africa: → – Regions: Egypt, Morocco, and Tunisia. → – Key Cities: Cairo, Casablanca, and Tunis. → Southern Africa: → – Key Cities: Johannesburg, Cape Town, and Durban. → West Africa: → – Key Cities: Lagos, Accra, and Abuja.
Australia	→ Eastern Australia: → – Regions: New South Wales, Victoria, and Queensland. → – Key Cities: Sydney, Melbourne, Brisbane, and Adelaide.
South America	→ Brazil: → – Key Cities: São Paulo, Rio de Janeiro, and Belo Horizonte. → Argentina: → – Regions: Buenos Aires and Córdoba. → – Key Cities: Buenos Aires and Córdoba. → Chile: → – Regions: Santiago and Valparaíso. → – Key Cities: Santiago and Valparaíso.

🎯 5 · ROADS AND RAILWAYS PYQ 2014

Understand it first: Land transport: roads for flexibility and short hauls, railways for bulk over long distance, pipelines for liquids and gas. Learn the classification and the named networks.

Roadways	→ Network >63 lakh km (Jan 2025), among the largest in the world → NH = 2% (NHAI, MoRTH) · State highways 4% (state PWDs) · District roads 14% · rural roads (Gram Panchayats, PMGSY) · urban roads (municipalities)

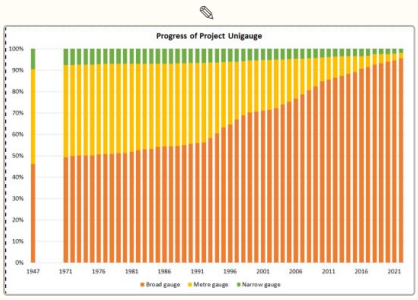
→ NH44 (earlier NH7) Srinagar–Kanyakumari = longest · NH48 (earlier NH8) Delhi–Mumbai

→ Since 1 April 2018 length counts each lane separately

→ NH numbers: even = North–South (rise east → west: NH2 east, NH68 west) · odd = East–West (rise north → south: NH1 north, NH87 south) · sub-highway = prefix (NH101 from NH01); new scheme after 2010

Railways

→ >68,000 km; state-owned, 19 zones (General Manager each), Ministry of Railways



Railways · Standard Guage

Gauges

→ Broad 1,676 mm: most used, long-distance, freight

→ Metre 1,000 mm: rural; Unigauge launched 1 April 1992

→ Narrow 762 or 610 mm: hills, being phased out

→ Standard 1,435 mm: metro, monorail, tram

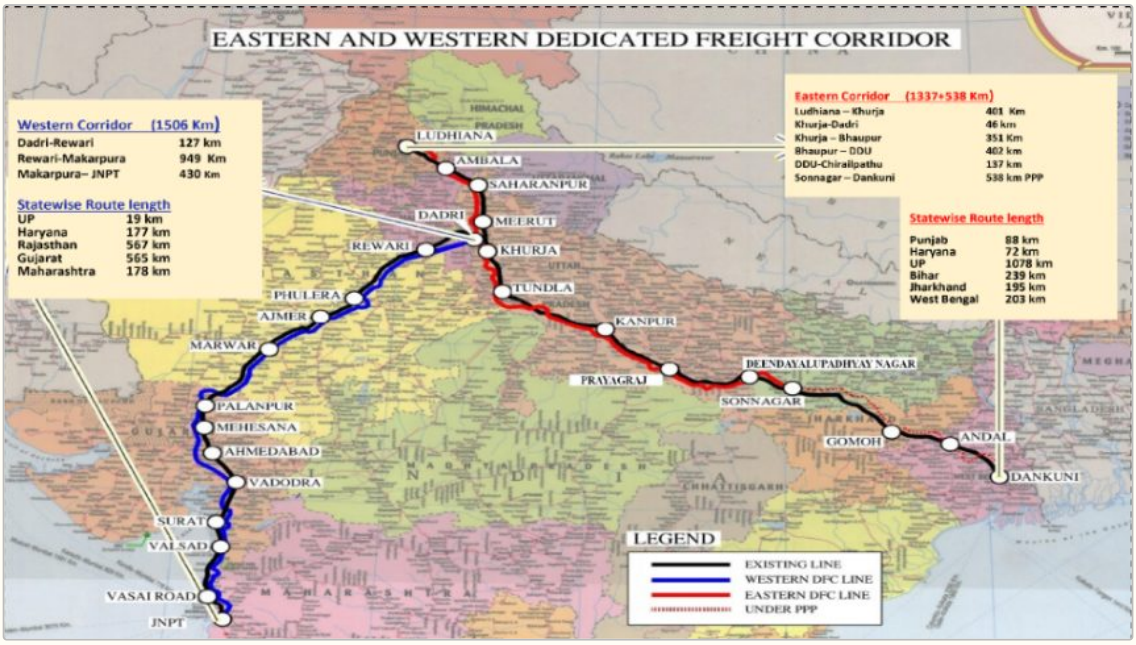
DFCCIL dedicated freight corridors

→ Eastern: Ludhiana (Punjab) → Dankuni (near Kolkata), ~1,839 km (coal, steel)

→ Western: Dadri (UP) → JNPT (Mumbai), ~1,506 km

→ Future: East-West (Kolkata-Mumbai), North-South (Delhi-Chennai), East Coast (Kharagpur-Vijayawada), Southern (Chennai-Goa)

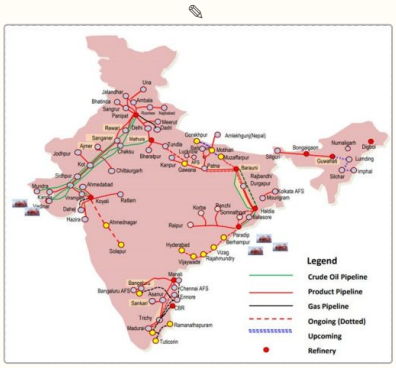
→ Also suburban (Mumbai, Delhi), metro (DMRC, BMRCL), proposed Mumbai–Ahmedabad high-speed rail



Railways · Metro Rail Corporations

Metre = 1,000 – Metre = 1,000 mm (name = number). Broad = the biggest, 1,676. Standard 1,435 = city metros.

Pipelines



Pipelines

Pipelines

→ Liquids and gases only; less flexible than rail and road; safe, cost-effective over long distances

→ Naharkatia–Noonmati–Baravni (crude) · Mumbai High–Mumbai–Ankleshwar–Koyali (crude + gas) · Salaya–Koyali–Mathura (crude; to Panipat, Jalandhar)

→ Hazira–Bijapur–Jagdishpur (GAIL): largest underground pipeline in the world; extended to Delhi via Dadri

→ Also carry ores (slurry), water, local milk

Ropeways

→ Suspended cables for difficult terrain; early 20th c. (mining, logging); first major passenger ropeway Auli, 1969; also Darjeeling, Gulmarg, Vaishno Devi

→ Parvatmala (Budget 2022–23, PPP, MoRTH): 8 projects ~60 km initially; Jan 2024: 400 projects, ₹1.25 lakh crore in 5 yrs

→ Varanasi urban ropeway · Gaurikund–Kedarnath (9.7 km, 3,584 m)

NH numbering (new system 2010; earlier under NH Act 1956)

→ North–South = even numbers, rising from east to west (NH2 east, NH68 west)

→ East–West = odd numbers, rising from north to south (NH1 north, NH87 south)

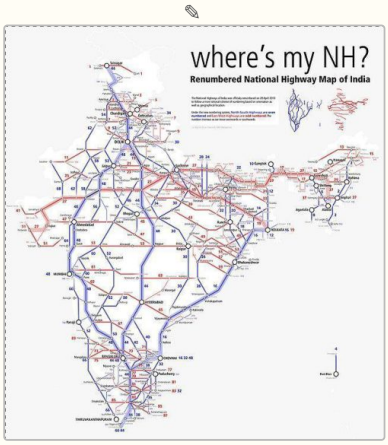
→ Branch (sub-highway): prefix on the parent number, e.g. NH101 from NH01

→ 1 April 2018: lane system, each lane's length counted separately

Roadways

→ It plays a crucial role in the country's transportation system, connecting remote areas with cities, facilitating trade, and contributing to economic growth.

→ The roadways in India can be categorized into various types based on their importance and the authority responsible for their construction and maintenance.



Roadways · Types of Roadways in India

Types of Roadways in India

→ National Highways (NH)

→ – These are the primary roadways connecting major cities, ports, state capitals, and large industrial areas. They facilitate long-distance travel and play a vital role in the country's economic integration.

→ – Authority: Constructed and maintained by the National Highways Authority of India (NHAI), under the Ministry of Road Transport and Highways (MoRTH).

→ State Highways (SH)

→ – These roads connect important cities within a state and provide links to the national highways and other state highways.

→ – Authority: Constructed and maintained by the respective State Public Works Departments (PWDs).

→ – Examples: Maharashtra State Highway 4 (connecting Pune and Ratnagiri), Karnataka State Highway 1 (connecting Bangalore and Mysore).

→ District Roads

→ – Authority: Constructed and maintained by the Zilla Parishads (District Councils) and the District PWDs.

→ – Examples : MDRs in Tamil Nadu and Uttar Pradesh that connect rural areas to district centers.

→ Rural Roads

→ – These are roads connecting rural areas and villages to nearby towns or district roads. They are crucial for local connectivity and agricultural activities.

→ – Authority: Constructed and maintained by the local bodies (Gram Panchayats) and under the Pradhan Mantri Gram Sadak Yojana (PMGSY).

→ – Examples: Roads constructed under the PMGSY program in various states, improving rural connectivity.

→ Urban Roads

→ – These roads are found within the limits of towns and cities. They include arterial roads, sub-arterial roads, and local streets.

→ – Authority: Constructed and maintained by Urban Local Bodies (Municipal Corporations or Municipalities).

→ – Examples: Major city roads like Marine Drive in Mumbai, and Rajpath in Delhi.

Naming of National Highways in India:

→ Up until 2010, all of our National Highways were numbered on the basis of the National Highways Act of 1956.

→ However, these numbers of National Highways did not give any indication of its location and direction. Hence, a new system was brought into place by the Ministry of Road Transport and Highways.

→ North-South Highways: These highways have even numbers. They run primarily in a north-south direction.

→ – The numbering increases from east to west. Highways close to the eastern part of India have smaller even numbers, and the numbers increase as one moves westward. For example, NH2 is in the eastern part, while NH68 is further west.

→ East-West Highways: These highways have odd numbers. They run primarily in an east-west direction.

→ – The numbering increases from north to south. Highways closer to the northern part of India have smaller odd numbers, and the numbers increase as one moves southward. For example, NH1 is in the northern part, while NH87 is further south.

Railways

→ India has one of the largest and most complex railway networks in the world, spanning over 68,000 km, playing a critical role in the country's transportation system.

→ The Indian Railways network is extensive and serves both passenger and freight needs across diverse geographical regions.

Types of Railways in India

→ Broad Gauge (BG)

→ – The most widely used gauge in India, with a track width of 1,676 mm (5 ft 6 in).

→ – Predominantly used for long-distance and heavy freight traffic, as well as passenger services.

→ Meter Gauge (MG)

→ – A narrower gauge with a track width of 1,000 mm (3 ft 3 3/8 in).

→ – Mostly found in some rural and less developed areas, but is gradually being converted to broad gauge under the Unigauge project.

→ Narrow Gauge (NG)

→ – Used in hilly and mountainous regions where constructing broad gauge is difficult. Also being phased out or converted to broad gauge.

→ Suburban Railways

→ – Serve the metropolitan and larger urban areas, facilitating daily commutes.

→ Metro Railways

→ – Urban transit systems operating in major cities to reduce congestion and provide efficient public transportation.

→ High-Speed Railways

→ – Proposed and under-construction projects aimed at providing high-speed travel between major cities.

Construction and Maintenance Authorities

→ Indian Railways (IR)

→ – A state-owned national railway company, responsible for the construction, maintenance, and operation of the majority of the rail network in India.

→ – Divided into 19 zones, each headed by a General Manager. These zones are further divided into divisions.

→ – Under the Ministry of Railways, Government of India.

→ Dedicated Freight Corridor Corporation of India Limited (DFCCIL)

→ – A public sector unit under the Ministry of Railways, responsible for planning, construction, and operation of dedicated freight corridors to improve the efficiency of freight movement.

→ Metro Rail Corporations

→ – Autonomous bodies or special purpose vehicles (SPVs) formed to plan, construct, and operate metro rail systems in cities.

→ Dedicated Freight Corridors: Dedicated freight corridors (DFCs) in India are specialized railway tracks designed exclusively for freight transportation. These corridors aim to improve the efficiency, speed, and capacity of freight movement across the country, reducing congestion on the existing railway network and facilitating economic growth.

→ – Eastern Dedicated Freight Corridor (EDFC)

→ · Length: Approximately 1,839 kilometres.

→ · Purpose: To facilitate the movement of coal, steel, and other bulk goods from eastern India to northern India, improving the efficiency of goods transportation in the region.

→ · Status: Under construction, with several sections already operational.

→ – Western Dedicated Freight Corridor (WDFC)

→ · Route: Dadri in Uttar Pradesh to Jawaharlal Nehru Port Trust (JNPT) near Mumbai in Maharashtra.

→ · Length: Approximately 1,506 kilometres.

→ · Purpose: To enhance the movement of goods from the industrial heartland of Delhi-NCR to the major port in Mumbai, thus facilitating export-import trade and reducing logistics costs.

→ · Status: Under construction, with significant sections already operational.

Pipelines

→ Pipelines in India form a crucial part of the country's infrastructure for transporting a variety of commodities, including crude oil, natural gas, petroleum products, and water. They provide a safe, efficient, and cost-effective means of transportation over long distances.

What They Transport

→ Crude Oil: Transported from oil fields to refineries.
→ Natural Gas: Transported from gas fields to processing plants and consumers.
→ Petroleum Products: Transported from refineries to distribution points, including petrol, diesel, kerosene, and aviation fuel.
→ Ores: Mixed with water to allow them to flow.

Important Examples

→ Mumbai High–Mumbai–Ankaleshwar–Koyali Pipeline
→ – Crude oil & Natural Gas

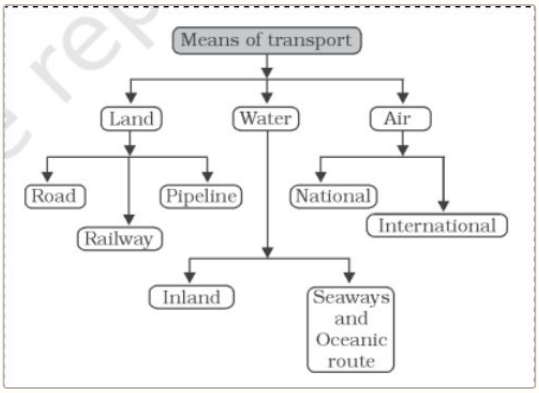
Ropeways

→ Ropeways use suspended cables to transport goods and passengers, typically over difficult terrain. They are especially popular in hilly, mountainous, and remote areas where conventional transportation modes like roads and railways are challenging to construct .

→ Ropeways in India date back to the early 20th century , with industrial applications in mining and logging. The first major passenger ropeway in India was established in 1969 in Auli , Uttarakhand, which opened the door for further developments across the country. Since then, several ropeways have been established in popular tourist destinations , such as in Darjeeling, Gulmarg, and Vaishno Devi.

PARVATMALA SCHEME

→ The Parvatmala Scheme , announced in the 2022-23 Union Budget by India's Union Finance Minister, is a National Ropeways Development Programme focused on improving connectivity in India's hilly regions.
→ The program, launched under a Public-Private Partnership (PPP) model, offers an eco-friendly alternative to conventional roadways in difficult terrains. It aims to enhance commuter convenience, promote tourism, and potentially relieve congestion in crowded urban areas.
→ The scheme initially targets states like Uttarakhand, Himachal Pradesh, Manipur, Jammu & Kashmir, and other Northeastern regions, with eight ropeway projects covering 60 kilometers planned for 2022-23.
→ The Ministry of Road Transport and Highways (MORTH) oversees the program, supported by a regulatory framework to ensure standards and efficiency.
→ The scheme is significant for its economic benefits (lower land acquisition costs due to direct routes), speed (aerial transport bypasses road obstacles), environmental friendliness (reduced dust emissions), and potential for last-mile connectivity .

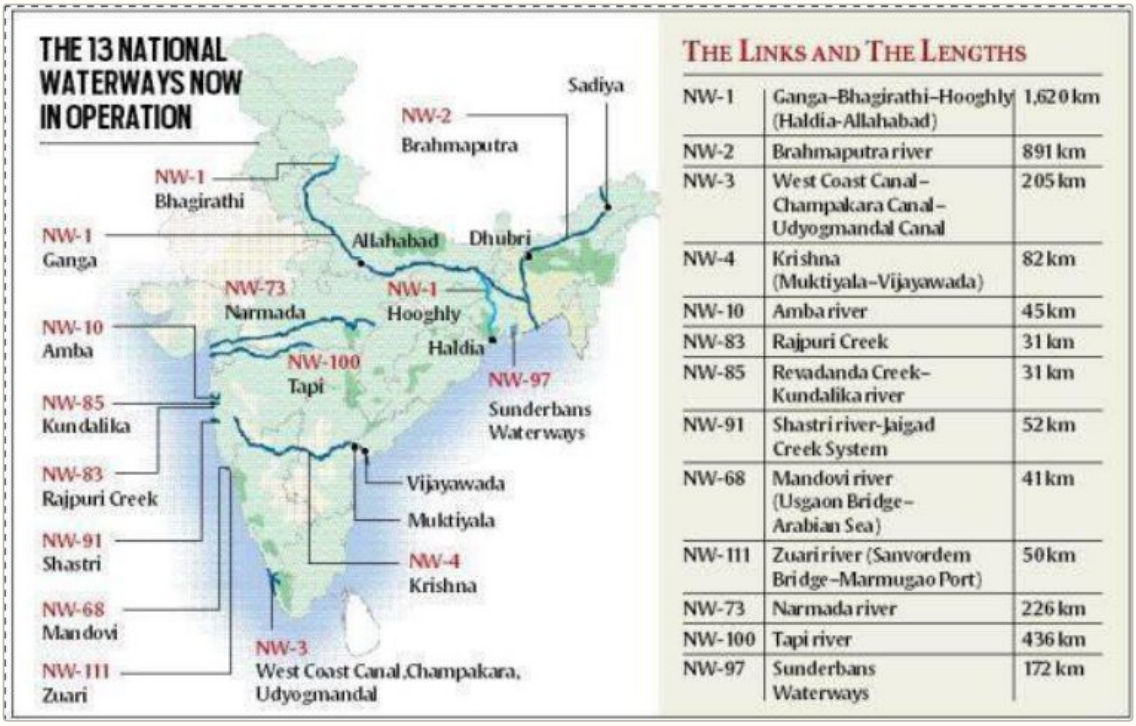


Means Of Transportation

🎯 6 · WATER TRANSPORT PYQ 2023

Understand it first: Water transport is the cheapest for heavy, bulky goods. Two parts: inland waterways on rivers and canals, and ocean routes with their ports.

Why cheap: low friction, fuel efficient; no separate infrastructure except terminals. 💡
IWAI established 1986; National Waterway Act 1986; 111 notified inland waterways (13 in operation).



THE LINKS AND THE LENGTHS

NW-1	Ganga-Bhagirathi-Hooghly (Haldia-Allahabad)	1,620 km
NW-2	Brahmaputra river	891 km
NW-3	West Coast Canal-Champakara Canal-Udyogmandal Canal	205 km
NW-4	Krishna (Muktiyala-Vijayawada)	82 km
NW-10	Amba river	45 km
NW-83	Rajpuri Creek	31 km
NW-85	Revadanda Creek-Kundalika river	31 km
NW-91	Shastri river-Jaigad Creek System	52 km
NW-68	Mandovi river (Usgaon Bridge-Arabian Sea)	41 km
NW-111	Zuari river (Sanvordem Bridge-Marmugao Port)	50 km
NW-73	Narmada river	226 km
NW-100	Tapi river	436 km
NW-97	Sunderbans Waterways	172 km

Inland Waterways · Construction and Development

Waterway	Route	Length
NW-1	Ganga–Bhagirathi–Hooghly (Haldia–Allahabad/Prayagraj)	1,620 km
NW-2	Brahmaputra (Dhubri–Sadiya)	891 km
NW-3	West Coast Canal: Champakara–Udyogmandal	205 km
NW-4	Krishna (Muktiyala–Vijayawada)	82 km
NW-73 / NW-100	Narmada / Tapi	226 / 436 km
NW-97	Sunderbans	172 km
NW-10, 68, 83, 85, 91, 111	Amba 45 · Mandovi 41 · Rajpuri creek 31 · Revdanda–Kundalika 31 · Shastri–Jaigad 52 · Zuari 50	km

Ocean transport

→ ~95% of foreign trade by volume, 70% by value

→ 13 major ports + 200+ minor/ intermediate; newest major Galathea Bay (Great Nicobar, transshipment)

→ Vadhavan (14th, near Mumbai, Palghar): approved June 2024, ₹76,220 crore, yet to be built

→ Major ports: Port Trusts (Major Port Trusts Act 1963) · minor: State Maritime Boards

Open Ocean Transport · History

Major Port	Location	Important facts / achievements / significance
Deendayal Port	Kandla, Gulf of Kachchh, Gujarat	Formerly **Kandla Port**; one of India's busiest west-coast ports; important for petroleum, chemicals, salt, fertilisers and dry cargo. Strategically serves North-West India.
Mumbai Port	Mumbai Harbour, Maharashtra	One of India's oldest modern ports; a natural deep-water harbour; historically important for western India's trade, petroleum and general cargo.
Jawaharlal Nehru Port Authority (JNPA)	Nhava Sheva, Navi Mumbai, Maharashtra	India's leading container port; developed to decongest Mumbai Port; key gateway for containerised trade of western and northern India.
Vadhavan Port	Palghar district, Maharashtra	**Notified/approved greenfield major port**; planned as a deep-draft, all-weather mega port. The Government approved it in June 2024 at an estimated cost of **₹76,220 crore**.
Mormugao Port	Mormugao, Goa	Important west-coast port; historically significant for iron ore export from Goa; also handles coal, petroleum products and general cargo.

Open Ocean Transport · History

New Mangalore Port	Panambur, Mangaluru, Karnataka	Only major port of Karnataka; important for petroleum products, crude oil, LPG, coal, fertilisers, iron ore, coffee and cashew exports.
Cochin Port	Kochi, Kerala	Major port on the Arabian Sea; has **Vallarpadam International Container Transshipment Terminal**; important for container cargo, petroleum, fertilisers, cruise tourism and maritime trade of Kerala.
V.O. Chidambaranar Port	Thoothukudi/Tuticorin, Tamil Nadu	Important port in southern Tamil Nadu; handles coal, salt, fertilisers, containers and general cargo. It became the **first Indian major port to implement a Digital Twin platform** for port management.
Chennai Port	Chennai, Tamil Nadu	One of India's oldest major ports on the east coast; important for automobiles, containers, petroleum products and general cargo; historically a major trade gateway for Tamil Nadu.
Kamarajar Port	Ennore, near Chennai, Tamil Nadu	India's first corporatised major port; important for coal, automobiles, LNG, petroleum products and bulk cargo. It helps reduce pressure on Chennai Port.
Visakhapatnam Port	Visakhapatnam, Andhra Pradesh	Natural harbour and major east-coast port; important for iron ore, coal, petroleum, fertilisers and steel-sector cargo; strategically linked with eastern and central India.
Paradip Port	Paradip, Odisha	Major bulk-cargo port on the east coast; important for coal, iron ore, crude oil and fertilisers. It has gained recognition for high bulk-cargo performance.
Syama Prasad Mookerjee Port	Kolkata-Haldia, West Bengal	Formerly **Kolkata Port**; India's only major riverine port system on the Hooghly; consists of **Kolkata Dock System** and **Haldia Dock Complex**; crucial for eastern and north-eastern India.
Galathea Bay Port	Great Nicobar Island, Andaman & Nicobar Islands	**Notified major port**, planned as an **International Container Transshipment Port** under the Great Nicobar project; strategically located near the east-west international shipping route.

Kind Men Join Very Many New Clubs — West coast, north → south: Kandla (Deendayal) · Mumbai · JNPA (Nhava Sheva, leading container port) · Vadhavan · Mormugao (Goa, iron ore) · New Mangalore (only major port of Karnataka) · Cochin (Vallarpadam transshipment).

Very Clever Kids Visit Pretty Kolkata — East coast, south → north: V.O. Chidambaranar (Tuticorin) · Chennai · Kamarajar (Ennore) · Visakhapatnam (natural harbour) · Paradip (bulk) · Kolkata–Haldia (Syama Prasad Mookerjee Port, only riverine major port, Hooghly).

Handout alert: the 2023 PYQ explanation says "Kamarajar Port at Tuticorin", but the port list puts Kamarajar at Ennore and V.O. Chidambaranar at Tuticorin. Use the list.

Water Transport

Why is water transport cheap ?

→ Water does not provide as much friction as land, so **fuel efficiency is higher** .

→ There is **no need to create separate infrastructure** for sea travel **except the terminals** . E.g. roads, rails, and pipes are needed for land-based transportation. This adds to the savings.

Inland Waterways

→ Inland waterways in India have a long history dating back to ancient times, when rivers were utilized as crucial arteries for trade and transportation.

→ – **Ancient Importance:** Rivers like the Ganges, Brahmaputra, and Godavari have historically served as major trade routes, facilitating commerce and cultural exchange.

→ – **Colonial Legacy:** The British colonial administration developed some inland waterways for transportation of goods during the colonial period, recognizing the strategic importance of river navigation.

Construction and Development

→ **Inland Waterways Authority of India (IWAI):** Established in 1986, IWAI is the primary authority responsible for the regulation and development of inland waterways in India.

→ **National Waterway Act:** Enacted in 1986, this legislation identified specific stretches of rivers, canals, and backwaters as national waterways for development and regulation.

→ **Modernization Efforts:** Recent years have seen increased investment in the modernization and expansion of inland waterways infrastructure, including dredging, construction of terminals, and installation of navigation aids.

→ There are 111 officially notified Inland Waterways in the country. Some Important Examples :

Open Ocean Transport

→ India's ocean-based water transport has a **rich history** and plays a vital role in the country's economy and connectivity.

History:

→ **Ancient Times:** India's maritime history **dates back to the Indus Valley Civilization** , with evidence of trade links with Mesopotamia and Egypt. Ancient ports like Lothal were significant centers of commerce.

→ **Medieval Period:** During the medieval period, Indian ports facilitated **trade with Africa, the Middle East, and Southeast Asia** . Notable ports included Calicut, Surat, and Cochin.

→ **Colonial Era:** The colonial period saw the establishment of **modern ports by the British** , such as Mumbai (Bombay), Kolkata (Calcutta), and Chennai (Madras), to facilitate trade and control.

→ **Post-Independence:** After independence, India **expanded and modernized its port infrastructure** to support economic growth and international trade.

Port Management

→ **Ministry of Ports, Shipping, and Waterways:** The central government body responsible for formulating policies and overseeing the development and regulation of ports in India.

→ **Private Sector Participation:** In recent years, there has been significant private sector participation in port development and operations through Public-Private Partnerships (PPPs). Private companies invest in port infrastructure, container terminals, and logistics services.

7 · AIR TRANSPORT

Understand it first: Air transport is the fastest and costliest, best for difficult terrain and long distances.

18 Feb 1911
first commercial flight, Allahabad–Naini

1948
govt takes 49% of Tata-owned Air India

1953
nationalised: Indian Airlines domestic, Air India international

1990s
liberalised

2021
Air India sold to Tata Group (handed over Jan 2022)

157 operational airports, 133 with AAI (Airports Authority of India). >90% traffic with two companies (duopoly concern). Private operators: GMR, Adani, GVK (Delhi, Mumbai, Bengaluru). ⭐

Airways in India

History:

→ Early Beginnings: Air transport in India began in the early 20th century. The first commercial flight took place on February 18, 1911, from Allahabad to Naini, covering a distance of about 6 miles.

→ Post-Independence Era: After independence, India established its own national carrier, which played a key role in developing the country's aviation sector. In 1948, the government acquired a 49% stake in Tata-owned Air India

→ Liberalization and Growth: The aviation sector was liberalized in the 1990s, leading to the entry of private airlines and significant growth in the industry. This period saw the emergence of several domestic carriers, including Jet Airways, IndiGo, SpiceJet, and others.

→ Modern Era: Today, India boasts a rapidly growing aviation market, with a large number of domestic and international flights connecting major cities and regions. The sector continues to expand with increasing passenger traffic and investment in airport infrastructure.

→ State exit: A proposal for the sale of Air India was floated in 2017. After multiple failures and delays, it was finally sold to the Tata Group in 2021 (officially handed over in Jan 2022).

Airport Management in India:

→ Airports Authority of India (AAI): AAI is the statutory body responsible for creating, upgrading, maintaining, and managing civil aviation infrastructure in India. It operates over 100 airports, including international, domestic, and civil enclaves.

→ Private Sector Participation: In recent years, several major airports have been developed and managed through Public-Private Partnerships (PPPs). Private entities such as GMR Group, Adani Group, and GVK Group operate major airports like Indira Gandhi International Airport in Delhi, Chhatrapati Shivaji Maharaj International Airport in Mumbai, and Kempegowda International Airport in Bengaluru, in conjunction with the AAI (except Mumbai).

→ State Governments: Some state governments also play a role in managing smaller airports within their territories, often in collaboration with AAI or private players.

8 · COMMUNICATION

Understand it first: Communication moved from post and telegraph to telephone, radio, television, then digital and satellite links. Learn each with its key facts.

1837 Indian Post Office Act

mid-19th c. Railway Mail Service

1851 telegraph Kolkata–Diamond Harbour

1854 Post Office Act, uniform rates; telegraph links Kolkata, Bombay, Madras, Peshawar

1881 first telephone exchange, Kolkata

1927 Indian Broadcasting Company (Mumbai, Kolkata)

1936 All India Radio

1959 Doordarshan

1982 colour TV

1986 Speed Post

1990s private telecom and TV

1995 Internet

2013 telegraph ended

2015 Digital India

Notes

→ Post: before it, communication went with transport or messengers; first organised system late 18th c. (British); Act 1854 set urban and rural post offices; Speed Post (1986) to compete with private couriers; network among largest after 1947

→ Radio: private clubs early 1920s; British took over 1936 as AIR (key in WWII); Quit India used radio; multilingual after independence

→ TV: private channels broke Doordarshan monopoly (1990s liberalisation)

→ Telephone: urban and government offices at first; early 2000s mobile revolution; one of the fastest 5G rollouts

→ Digital India (2015): infrastructure, connectivity, e-governance, online education, telemedicine

→ Satellite: ISRO communication satellites for TV, telephony, data, internet (see Science and Tech)

Traps

→ wheat is Rabi

→ palm oil native to Guinea hills, not SE Asia

→ excess N = acidity + nitrate leaching

→ chemical farming hard in semi-arid

→ uranium ~5% not 60%

→ REE = 17, China first not India second

→ Mundra/JNPT lead containers, not Visakhapatnam

→ even NH = N–S, odd = E–W. Connects to: Indian Geography (soils, Gondwana coal, rivers, ports)

→ Climatology

→ Human Geography.

Communication

→ Prior to the introduction of the Postal System in India, communication was mostly in conjunction with transportation. Either the person who wanted to communicate travelled to the receiver, or they sent some messengers.

→ The British colonial administration recognized the importance of an efficient communication system for governance and trade. The first organized postal system was established in the late 18th century.

Postal System

→ Indian Post Office Act of 1837: This act laid the foundation for the modern postal system in India. It standardized postal services and rates across the country.

→ Indian Post Office Act of 1854: This act reorganized the postal system, making it more efficient and widespread. It introduced uniform postage rates and established post offices in urban and rural areas.

→ Expansion and Modernization: After independence in 1947, the Indian postal system expanded rapidly to cover rural and remote areas. The postal network became one of the largest in the world.

Telegraph

→ Expansion: By 1854, telegraph lines connected major cities like Kolkata, Mumbai (then Bombay), Chennai (then Madras), and Peshawar. The telegraph network expanded rapidly, covering vast distances and enhancing communication across the British Empire in India.

Telephones

→ Expansion: Telephone services gradually expanded to other major cities, but their reach remained limited primarily to urban areas and government offices.

→ Liberalization and Privatization: The liberalization of the Indian economy in the 1990s led to the entry of private telecom companies, which transformed the sector. This period saw the launch of mobile phone services and significant growth in the number of subscribers.

→ Mobile Revolution: The early 2000s marked the mobile revolution, with affordable handsets and competitive pricing making mobile phones accessible to a large population.

→ Currently, India has become one of the fastest countries to roll out 5G.

Radio

→ All India Radio (AIR): The British government took over radio broadcasting in 1936, establishing All India Radio (AIR) as the national broadcaster. AIR played a key role in disseminating news and information during World War II.

→ The Quit India Movement was also energised by the use of this mass media.

→ All India Radio expanded its reach post-independence, with more radio stations set up across the country, broadcasting in multiple languages.

Television

→ Introduction of Television: Doordarshan, India's public service broadcaster, began television services in 1959. Television rapidly gained popularity, especially with the introduction of color broadcasts in 1982.

Digital Communication

→ Introduction of the Internet: The Internet was introduced in India in 1995. The subsequent years saw a rapid increase in internet users, driven by affordable data plans and smartphones.

→ Digital India Initiative: Launched in 2015, the Digital India initiative aimed to transform India into a digitally empowered society and knowledge economy. It focused on improving digital infrastructure, enhancing internet connectivity, and providing digital services to citizens.

Satellite Communication

→ ISRO, has been instrumental in developing and deploying a series of communication satellites that cater to various needs, including television broadcasting, telephony, data transmission, and internet services. More details in Science & Tech.

RECALL DRILL: COVER AND ANSWER

Kharif, Rabi, Zaid months?	Jun–Oct · Oct/Nov–Mar/Apr · Mar–Jun	Dry, dryland, wetland farming?	<75 cm/<75 d · 75–115/75–120 · >115/>120
Dry/dryland share: maize, bajra, pulses, oilseeds?	80% · 90% · 95% · 75%	Least water-efficient crop (2021)?	Sugarcane (1,800–2,200 mm)
Who: Green, White revolution?	M.S. Swaminathan; Verghese Kurien	Copper shares: Rajasthan, MP, Jharkhand?	52.25%, 23.28%, 15.14%
Indian coal traits; why import (2012)?	non-coking, high ash, low sulphur; short supply + coking coal	Renewables, solar, wind, nuclear?	283.46 GW; 157 GW (280 by 2030); 56 GW; 8,180 → 22,480 MW
Weber: weight-losing, isotim, isodapane?	at raw material; equal transport cost; equal total cost	Gauges; NH rule?	1,676 / 1,000 / 762–610 / 1,435 mm; even N–S, odd E–W
Ports: count, newest, only riverine?	13 major + Vadhavan; Galathea Bay; Kolkata–Haldia (SPM)		

🎯 PYQs OF THIS CHAPTER

🎯 Exam lens (from the handout only)

- Priority: HIGH. 31 Prelims PYQs printed (counted from the handout text): agriculture 19 (2011×4, 2012×2, 2013, 2014×2, 2015, 2017, 2018, 2019, 2020, 2021×2, 2022, 2024, 2025) · minerals & energy 9 (2012, 2013×2, 2016, 2018, 2022, 2023×2, one undated on rare earths) · transport 3 (2014, 2015, 2023) · industries 0 – none quoted in the material. Mains quoted: none quoted in the material.
- Pattern: which crop where, which mineral where, which state does what. Plus "statement" questions on fertiliser, salinisation, permaculture, conservation agriculture, and current affairs (cocoa, turmeric, REE, NCMM).
- Heavy handout, thin on teacher stress. No teacher ink in this PDF; priority = PYQ count in the handout. Two image pages (crop table, waterways) were checked visually.

★ *PYQ answers in one line (asked year and the key)*

AGRICULTURE PYQ BANK 💡

Year	Asked	Answer / rule
2011	Micro-irrigation statements	C (1 and 3)
2011	Lower Gangetic plain, humid-hot: best crop pair	Paddy and jute (C): jute >25 °C, humidity 70–90%; rice 21–37 °C, tolerates max 40–42 °C
2011	State: arid north, cotton centre, cash crops > food crops	Gujarat (B)
2011	Effect of salinisation	Makes some soils impermeable (B): salt breaks aggregates, pores collapse; advanced stage = barren, desertified land
2012	Mixed farming	Crops + livestock (C)
2012	Rainfed: groundnut, sesamum, pearl millet	All three (D)
2013	Kharif: cotton, groundnut, rice, wheat	1, 2, 3 (B); not wheat
2014	Kinnaur–areca? Mewat–mango? Coromandel–soya?	None (D); MP = soybean
2014	Maize: starch, biodiesel oil, alcohol	All three (D)
2015	Excess nitrogen fertiliser	Acidity ↑ and nitrate leaching (C); not more N-fixing microbes
2017	Conservation agriculture	Minimum tillage, residue cover, rotations (D: 2, 4, 5)
2018	Organic matter ↓ water holding? Soil no role in sulphur cycle? Irrigation salinises?	Only 3 (B)
2019	Agri soils emit nitrogen oxides; cattle ammonia; poultry reactive N	All three (D); ~90% N_2O from soils
2020	Permaculture vs chemical farming	1, 2, 4 (B); chemical farming not easy in semi-arid areas
2021	Palm oil: native to SE Asia ✘; lipstick/perfume raw material ✔; biodiesel ✔	B (2 and 3); native to hills of Guinea

2021	Least water-efficient crop	Sugarcane (A): 1,800–2,200 mm/season; sunflower 672.4; pearl millet 350
2022	Tea states: AP, Kerala, HP, Tripura	All four (D); Kerala major, rest minor
NEW 2024	Two largest cocoa producers	Côte d'Ivoire and Ghana (~70% of world cocoa)
NEW 2025	Turmeric 2022–23: India largest producer/exporter; >30 varieties; Maharashtra, Telangana, Karnataka, TN	All three (D)

Source conflict: 2024 cocoa PYQ prints "Answer: D" but the correct pair is option (c) Côte d'Ivoire and Ghana (explanation: ~70% of world cocoa). Take the pair.

2013 PYQ (B): natural gas in Gondwana beds ✘; mica abundant in Kodarma ✔; Dharwars famous for petroleum ✘ (Dharwars: iron, manganese, lead, zinc, gold; Champion reef, Kolar). Natural gas areas: KG basin, Assam, Gulf of Khambhat, Cuddalore (TN), Barmer. Kodarma = ruby mica. **2018 (C):** Bhilwara (Rajasthan) has iron ore; MMDR Amendment Act 2015 (from Jan 2015): states auction non-coal mines, first auction Dec 2015; Rajasthan ~57 minerals, sole producer of lead-zinc, selenite, wollastonite; copper 809.09 mt = 54% of resources, 41% of output (2nd leading producer). ✅

2022 PYQ (b: 1, 2, 4): monazite = rare earths + thorium (prescribed substance, DAE); only govt bodies process/export: IREL (under DAE); in-situ monazite in Odisha, AP, TN, Kerala, WB, Jharkhand (AMD); not the entire coast. **REE PYQ (a: 1, 3):** NCMM launched 2025 ✔; China first but India not second ✘; REE = 17 elements, not 13. ✅

2012 (B): India imports coal because power plants lack domestic supply AND steel needs coking coal; not "saving reserves". 2013 (B: 1, 2): high ash ✔, low sulphur ✔, low ash-fusion ✘. Also: India 5th largest reserves, 3rd largest producer and importer; sulphur removed by washing. ✅

2023 PYQ (C): India depends on coal (~75% power from thermal) despite uranium ✔; "uranium enriched ≥60% needed" ✘, ~5% enrichment is enough. ✅

2014 PYQ (D): NH4 Chennai–Hyderabad? NH6 Mumbai–Kolkata? NH15 Ahmedabad–Jodhpur? All ✘: old NH4 became NH48; NH6 (Jorabat–Selling) is in the NE; NH15 starts at Baihata, Assam. ✅

PYQ 2014 (answer D, none matched): NH 4 = Andaman Trunk Road, 230.7 km (old NH 4 Mumbai–Pune–Hubli–Bangalore–Chennai is now NH 48); NH 6 = Jorabat (Meghalaya) to Selling (Mizoram); NH 15 = Baihata (Assam) to Wakro (Arunachal).

2023 PYQ (B, two pairs): Kamarajar = first major port registered as a company (Companies Act, started 2001) ✔; Mundra (Adani) = largest private port, also overtook JNPT as largest container port ✔; Visakhapatnam largest container port ✘. ✅

2015 PYQ (A: 1, 2, 3): IRS satellites used for crop productivity, groundwater, mineral exploration ✔; telecom, traffic ✘. IRS-1A launched 17 March 1988 (polar sun-synchronous, from Baikonur); data used under NNRMS. ✅

✍ ***The PYQs as printed in the handout, with answers***

Q.) With reference to micro-irrigation, which of the following statements is/are correct?

1. Fertiliser/nutrient loss can be reduced.
2. It is the only means of irrigation in dryland farming.
3. In some areas of farming, the receding of the groundwater table can be checked.

Select the correct answer using the codes given below: (2011)

(a) 1 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: C Micro-irrigation is a modern method of irrigation; by this method, water is irrigated through drippers, sprinklers, foggers and by other emitters on the surface or subsurface of the land.

- Major components of a micro irrigation system are as follows.
  - Water source, pumping devices (motor and pump), ball valves, fertigation equipment, filters, control valves, PVC joining accessories (Main and sub-main) and emitters.
- In this system, water is applied drop by drop nearer the root zone area of the crop.
- The drippers are fixed based on the spacing of the crop.
- Many different types of emitters are available in the market. They are classified as Inline drippers, online drippers, Microtubes, and Pressed compensated drippers.
- Drip irrigation is most suitable for wider spacing crops.
- The micro sprinkler irrigation system is mostly followed in sandy or loamy soils.
- This system is most suitable for horticultural crops and small grasses.
- In this method, water is sprinkled at a lower height in various directions.
- It is used for preparing nursery and lawns in soils with low water holding capacity.

Advantages of drip irrigation system-

- Water saving and higher yield
- High quality and increased fruit size
- Suitable for all types of soil
- The easy method of fertigation and chemigation
- Saving in labour and field preparation cost

Q.) The lower Gangetic plain is characterised by a humid climate with high temperatures throughout the year. Which one among the following pairs of crops is most suitable for this region? (2011)

(a) Paddy and cotton
(b) Wheat and Jute
(c) Paddy and Jute
(d) Wheat and cotton

Answer: C The lower deltaic plains of the Ganges and Brahmaputra rivers that are Characterised by swamps and Sundarbans.

1. The suitable climate for growing jute is a warm and wet climate, which is offered by the monsoon climate during the fall season, immediately followed by summer. Temperatures ranging to more than 25 °C and relative humidity of 70%–90% are favourable for successful cultivation.

2. Rice crops need a hot and humid climate. It is best suited to regions which have high humidity, prolonged sunshine and an assured supply of water. The average temperature required throughout the life period of the crop ranges from 21 to 37° C. Maximum temp which the crop can tolerate is 400C to 42 0C.

Q.) A state in India has the following characteristics:

1. Its northern part is arid and semi-arid.
2. Its central part produces cotton.
3. Cultivation of cash crops is predominant over food crops.

Which one of the following states has all of the above characteristics? (2011)

(a) Andhra Pradesh
(b) Gujarat
(c) Karnataka
(d) Tamil Nadu

Answer: B

1. Gujarat possesses varied landforms with the dry deciduous forests, majestic grasslands, marine ecosystems, wetlands and rich moist deciduous forests.

2. These habitats are home to rare wildlife inhabitants.

3. The three main sources of growth in Gujarat's agriculture are from cotton production, the rapid growth of high-value foods such as livestock, fruits and vegetables, and from wheat production, which saw an annual average growth rate of 28% between 2000 and 2008 (According to the International Food Policy Research Institute).

Q.) Salinization occurs when the irrigation water accumulated in the soil evaporates, leaving behind salts and minerals. What are the effects of salinization on the irrigated land? (2011)

(a) It greatly increases the crop production
(b) It makes some soils impermeable
(c) It raises the water table
(d) It fills the air spaces in the soil with water

Answer: B Soil salinization occurs when water-soluble salts accumulate in the soil to a level that impacts on agricultural production, environmental health, and economics.

- Salt at higher levels causes soil aggregates to break down, collapsing soil pores and reducing the movement of water and air.

- In the early stages, salinity affects the metabolism of soil organisms and reduces soil productivity, but in advanced stages it destroys all vegetation and other organisms living in the soil, consequently transforming fertile and productive land into barren and desertified lands

Q.) Which of the following is the chief characteristic of 'mixed farming'? (2012)

(a) Cultivation of both cash crops and food crops

(b) Cultivation of two or more crops in the same field

(c) Rearing of animals and cultivation of crops together

(d) None of the above

Answer: C Mixed farming is a type of agriculture which involves both crops and livestock.

(a) This is not to be confused with mixed cultivation, implying merely a series of different crops.

(b) It is found throughout Europe— from Ireland in the west through Central Europe to Russia.

(c) It is also found in North America, east of the 98° meridian, in the 'pampas' of Argentina, southeast Australia, South Africa and New Zealand.

(d) Mixed farming is essentially associated with the densely populated, urbanised and industrialised societies, dependent upon high incomes for the sale of its products and upon manufacturing industry for the provision of its inputs.

Q.) Consider the following crops of India :

1. Groundnut
2. Sesamum
3. Pearl millet

Which of the above is/are predominantly rainfed crop/crops? (2012)

(a) 1 and 2 only

(b) 2 and 3 only

(c) 3 only

(d) 1, 2 and 3

Answer: D Sesamum is mainly a rainfed crop and is grown on marginal, dry and sloppy lands.

- Nearly 80% of the area sown to groundnuts in India is rain-fed and relies entirely on summer monsoon rainfall.
- Among cereals, the main rainfed cereals grown in India are sorghum, pearl millet (Bajra), Finger Millet (Ragi).

Q.) Consider the following crops

1. Cotton
2. Groundnut
3. Rice
4. Wheat

Which of these are Kharif crops? (2013)

(a) 1 only

(b) 1, 2 and 3 only

(c) 2 and 3 only

(d) 1, 2, 3 and 4

Answer: B The word "Kharif" is Arabic for autumn since the season coincides with the beginning of autumn or winter.

- Kharif crops also are known as monsoon crops. These are the crops that are cultivated in the monsoon season.
- The Kharif season differs in every state of the country but is generally from June to September.
- These crops are usually sown at the beginning of the monsoon season around June and harvested by September or October.
- Rice, maize, bajra, ragi, soybean, groundnut, cotton are all Kharif types of crops.

Q.) Region Well-known for the production of

1. Kinnaur (Himachal) Areca nut
2. Mewat (Haryana) Mango
3. Coromandel (TN-Andhra) Soya bean

Which of the above pairs is/are correctly matched? (2014)

(a) 1 and 2 only

(b) 3 only

(c) 1, 2 and 3

(d) None

Answer: D Portions of Kinnaur are situated high in the Himalaya, where vegetation is sparse and consists primarily of hardy grasses. Alpine species such as juniper, pine, fir, cypress, and rhododendron can be found at elevations between 3,500 and 5,000 metres, primarily in Middle Kinnaur. At lower altitudes, temperate-climate trees are found, including oak, chestnut, maple, birch, alder, magnolia, apple, and apricot.

- Yaks and dzos are reared by local farmers in the higher areas. Scattered populations of the Himalayan black bear and small ponies may also be found.

- The main occupation in the Mewat district is agriculture along with allied and agro-based activities.
- The Meos are the predominant population group and are all agriculturists.
- Agriculture is mostly rain-fed except in small pockets where canal irrigation is available.
- Agriculture production measured in terms of crop yield per hectare is low in comparison to the other districts of the State.
- Animal husbandry, particularly dairy, is a secondary source of income for the people and those who live closer to the hilly ranges of Aravali also keep sheep and goats.
- MP is famous for the production Soybean

Q.) Consider the following statements :

1. Maize can be used for the production of starch.
2. Oil extracted from maize can be a feedstock for biodiesel.
3. Alcoholic beverages can be produced by using maize.

Which of the statements given above is/are correct? (2014)

(a) 1 only
(b) 1 and 2 only
(c) 2 and 3 only
(d) 1, 2 and 3

Answer: D In Central America, the alcoholic beverage brewed from malted maize is known as chica fuerte.

- Corn starch has a wide variety of industrial applications. Biodiesel can be produced from corn oil.
- Maize can be used for the production of starch.Oil extracted from maize can be a feedstock for biodiesel. Alcoholic beverages can be produced by using maize.
- The type of alcohol in the alcoholic drinks we drink is a chemical called ethanol. To make alcohol, you need to put grains, fruits, or vegetables through a process called fermentation. The common source of ethanol is maize and other grains.

Q.) What can be the impact of excessive/inappropriate use of nitrogenous fertilisers in agriculture?

1. The proliferation of nitrogen-fixing microorganisms in the soil can occur.
2. Increase in the acidity of soil can take place
3. Leaching of nitrate to the ground-water can occur.

Select the correct answer using the code given below. (2015)

(a) 1 and 3 only
(b) 2 only
(c) 2 and 3 only
(d) 1,2 and 3

Answer: C Farmers occasionally need to add nitrogen fertiliser to their farms and gardens to make available just the precise nutrients for their plants' growth.

- The applications of inorganic nitrogen fertilisers to various crops have been continuously increasing since the last many decades globally.
- Excessive use of N (urea) encourages climate change (when lost through denitrification) and groundwater pollution (when lost through leaching).
- Increase in nitrate content of groundwater in some intensively-cropped areas has been reported, which is obviously due to leaching of nitrates beyond the crop root zone. Increase in nitrate content of groundwater is potentially harmful, as it is used for drinking purposes in most of the rural areas.
- Although nitrogen fertiliser contributes substantially to yield enhancement, excessive use of this manure has posed serious threats to the environment and human health.
- The rate of nitrogen fertiliser application has a close relationship with nitrate accumulation in the surrounding environment, groundwater, as well as leafy and root vegetables.
- Excessive use of urea leads to several adverse implications on soil, crop quality and overall ecosystem.
- Nitrogen applied in excess of crop demand is lost through volatilization, acidification, denitrification and leaching.

Q.) With reference to the circumstances in Indian agriculture, the concept of "Conservation Agriculture" assumes significance. Which of the following fall under the Conservation Agriculture?

1. Avoiding monoculture practises
2. Adopting minimum tillage
3. Avoiding the cultivation of plantation crops
4. Using crop residues to cover the soil surface
5. Adopting spatial and temporal crop sequencing/crop rotations

Select the correct answer using the code given below: (2017)

(a) 1, 3 and 4
(b) 2, 3, 4 and 5
(c) 1, 2, 3 and 5
(d) 2, 4 and 5

Answer: D

- Conservation agriculture (CA) technologies involve minimum soil disturbance, permanent soil cover through crop residues or cover crops, and crop rotations for achieving higher productivity.
- Main features of Conservation agriculture (CA) are; Least interference with natural processes, No-till or drastically reduced tillage (biological tillage, Low wind and soil erosion, Surface retention of residues (permanently covered), Infiltration rate of water is high, Use of in-situ organics/composts, Brown manuring/cover crops (surface retention),Diversified and more efficient rotations etc.
- The principal indicators of non-sustainability of agricultural systems include soil erosion, soil organic matter decline, salinization. These are caused mainly by
  - intensive tillage-induced soil organic matter decline, soil structural degradation, water and wind erosion, reduced water infiltration rates, surface sealing and crusting, soil compaction,
  - insufficient return of organic material, and
  - Monocropping.

Q.) With reference to agricultural soils, consider the following statements :

1. The high content of organic matter in soil drastically reduces its water holding capacity.
2. Soil does not play any role in the sulphur cycle.
3. Irrigation over a period of time can contribute to the salinization of some agricultural lands.

Which of the statements given above is/are correct? (2018)

(a) 1 and 2 only
(b) 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: B

- Plants obtain nutrients from two natural sources: organic matter and minerals. Organic matter includes any plant or animal material that returns to the soil and goes through the decomposition process. In addition to providing nutrients and habitat to organisms living in the soil, organic matter also binds soil particles into aggregates and improves the water holding capacity of the soil.
- The sulphur cycle is the collection of processes by which sulphur moves to and from rock, waterways and living systems. The isotopic composition of sedimentary sulphides provides primary information on the evolution of the sulphur cycle.
- Too much salt can reduce crop production and water infiltration on soils that have been irrigated for 20 years or more. Too little salt can also result in a chemically compacted soil. The level of sodium (Na) in the soil can result in soil particles or even layers of differing soil profiles forming a compacted layer that the roots do not penetrate.

Q.) Consider the following statements:

1. Agricultural soils release nitrogen oxides into the environment.
2. Cattle release ammonia into the environment.
3. Poultry industry releases reactive nitrogen compounds into the environment.

Which of the statements given above is/are correct? (2019)

(a) 1 and 3 only
(b) 2 and 3 only
(c) 2 only
(d) 1, 2 and 3

Answer: D Statement 1 is correct:

- Nitrous oxide is emitted into the atmosphere as a result of biomass burning, and biological processes in soils. Biomass burning is not only an instantaneous source of nitrous oxide, but it results in a longer term enhancement of the biogenic production of this gas.
- The current belief is that 90% of the emissions come from soils. Nitrous oxide is formed in soils during the microbiological processes of nitrification and denitrification. Because nitrous oxide is a gas it can escape from soil during these transformations.

Statement 2 is correct:

- Cattle release both ammonia and methane into the atmosphere. Ruminant animals do not efficiently utilise dietary nitrogen.

Statement 3 is correct:

1. Nitrogen emissions in chicken production occur in several forms but mainly ammonia can contribute directly or indirectly to several environmental and public health hazards.
2. Chicken production also contributes to some extent to climate change through the emissions of nitrous oxide, fine particulate matter, and methane.

Q.) How is permaculture farming different from conventional chemical farming ?

1. Permaculture farming discourages monocultural practices but in conventional chemical farming, monoculture practices are predominant.
2. Conventional chemical farming can cause increase in soil salinity but the occurrence of such phenomenon is not observed in permaculture farming.

3. Conventional chemical farming is easily possible in semi-arid regions but permaculture farming is not so easily possible in such regions.
4. Practice of mulching is very important in permaculture farming but not necessarily so in conventional chemical farming.

Select the correct answer using the code given below. (2020)

(a) 1 and 3
(b) 1, 2 and 4
(c) 4 only
(d) 2 and 3

Answer: B

- Permaculture farming discourages monocultural practices but in conventional chemical farming, monoculture practices are predominant.
- 2. Conventional chemical farming can cause an increase in soil salinity but the occurrence of such phenomenon is not observed in permaculture farming.
- Statement 1 is correct. Permaculture is a totally integrated design system that's modelled on nature. Permaculture farming promotes multicropping and integrated farming systems. Agroforestry is its example.
- Chemical farming is more suitable for monoculture cropping due to use of crop specific inputs like irrigation, chemical fertiliser and harvesting methods.
- Statement 2 is correct. Chemical fertilizers will make the soil less fertile over time, which can lead to serious problems like soil salinisation. But such problems are not seen in permaculture farming as it relies on organic fertilizers.
- Statement 3 is incorrect. Chemical farming is based on intense use of inputs like Chemical fertilizers and irrigation that make the soil less fertile over time, thus restricting its practise in semi-arid region. Thus, conventional farming is not suitable for semi-arid regions.
- The concept of permaculture involves well-designed systems don't produce waste and permaculture tries to mimic well-designed systems. Thus, permaculture tries to take into account the local conditions like arid climate. This helps in developing a suitable system to ensure sustainable production.
- Statement 4 is correct. Mulching is an important component of Permaculture systems for promoting maximum efficiency.

Q.) With reference to 'palm oil', consider the following statements:

1. The palm oil tree is native to Southeast Asia.
2. The palm oil is a raw material for some industries producing lipstick and perfumes.
3. palm oil can be used to produce biodiesel.

Which of the statements given above are correct? (2021)

(a) 1 and 2 only
(b) 2 and 3 only
(c) 1 and 3 only
(d) 1, 2 and 3

Answer: B

- The palm oil tree is native to hills of guinea.
- It is used in cosmetics products like lipstick and perfumes and cleaning products. Palm oil is one of the best options and most cost-effective cosmetics ingredients. It is versatile. Due to its high fatty acid content, it can be used to make surfactants, emulsifiers and emollient
- Palm oil can become a sustainable option for biodiesel production because it is a bye product of palm oil (Pannilawithana and Pathirana, 2017). Both unrefined palm oil (UPO) and palm kernel oil are primary oil sources from oil palm trees.

Q.) Among the following, which one is the least water-efficient crop? (2021)

(a) Sugarcane
(b) Sunflower
(c) Pearl millet
(d) Red gram

Answer: A

- Option a is correct. Sugarcane is the least water efficient crop. Among the given options, its water requirement is 1800-2200 mm/season which is highest.
- Option b is incorrect. Sunflower's water requirement was estimated at 672.4 mm/season.
- Option c is incorrect. Pearl millet is a drought tolerant crop.
- Water requirement: 350 mm/season.

Q.) Consider the following States:

1. Andhra Pradesh
2. Kerala
3. Himachal Pradesh
4. Tripura

How many of the above are generally known as tea-producing States? (2022)

(a) Only one State
(b) Only two States
(c) Only three States
(d) All four States
Answer: D
- All of these states produce tea. Kerala is a major, while the rest are minor tea producing states.

Q.) Which of the following countries are well known as the two largest cocoa producers in the world? (2024)
(a) Algeria and Morocco
(b) Botswana and Namibia
(c) Côte d'Ivoire and Ghana
(d) Madagascar and Mozambique
Answer: D
Explanation: These two countries produce about 70% of world's cocoa.
Q.) Consider the following statements about turmeric during the year 2022–23: I. India is the largest producer and exporter of turmeric in the world. II. More than 30 varieties of turmeric are grown in India. III. Maharashtra, Telangana, Karnataka and Tamil Nadu are major turmeric producing States in India.
Which of the statements given above are correct? (2025)
(a) I and II only
(b) II and III only
(c) I and III only
(d) I, II and III
Answer: D
Explanation: India is the largest turmeric producer in the world, we also produce the maximum varieties.

Q.) Despite having large reserves of coal, why does India import millions of tonnes of coal ?

1. It is the policy of India to save its own coal reserves for the future, and import it from other countries for the present use.

2. Most of the power plants in India are coal-based and they are not able to get sufficient supplies of coal from within the country.

3. Steel companies need a large quantity of coking coal which has to be imported.

Which of the statements given above is/are correct? (2012)

(a) 1 Only

(b) 2 & 3 Only

(c) 1 & 3 Only

(d) 1,2, & 3

Ans. B Coal is and will continue to be the backbone of India's economic and development growth. India has the world's fifth largest coal reserves and is also the world's third largest producer and importer of coal.

- India's total thermal coal imports in 2021 may have edged higher from 156.09mn t in 2020. India imported 116.57mn t of thermal coal in January-September 2021, according to customs data, and going by the average fourth-quarter imports in the last five years, the receipts could have been slightly over 160mn t in 2021.
- The lack of coking coal reserves that are used as a raw material in steelmaking and allied industries is the main reason India imports millions of tonnes of coal. Coal in India is used either from domestic sources, mostly mined by coal India, or is imported.
- The issue for India is that the gap between domestic supply and demand is growing at a phenomenal rate.This gap is being filled by imported coal.

Q.) Consider the following statements:

1. Natural gas occurs in the Gondwana beds.

2. Mica occurs in abundance in Kodarma.

3. Dharwars are famous for petroleum.

Which of the statements given above is/are correct? (2013)

(a) 1 only

(b) 2 only

(c) 2 and 3 only

(d) None

Ans. B

- KG basin, Assam, Gulf of Khambhat, Cuddalore district of Tamil Nadu, Barmer in Rajasthan etc. are the areas where natural gas can be found.Nearly 80% of the world's total proven natural gas reserves are located in ten countries. Russia tops the list, holding about a quarter of the world's total gas reserves, followed by Iran and Qatar in the Middle East.
- 4. Kodarma mines in Jharkhand are one of the important mica producing areas. Kodarma district of Jharkhand state has been famous worldwide for mica mining,especially for ruby mica. The district had the monopoly in producing mica for years and themica was exported to many countries worldwide. The main reserve of mica is found under the forest of mild life sanctuary of Kodarma.
- 5. The Dharwar rocks are rich in iron ore, manganese, lead, zinc, gold, silver etc. The Champions series containing gold mines lie within this system. This Champion system is named after the Champion reef in the Kolar Gold Fields. The Kolar Gold Fields contain one of the deepest gold mines of world
- 6. The Dharwar rocks are rich in iron ore, manganese, lead, zinc, gold, silver etc. The Champions series containing gold mines lie within this system.

Q.) Which of the following is/are the characteristics/ characteristics of Indian coal?

1. High ash content

2. Low sulphur content

3. The low ash fusion temperature

Select the correct answer using the codes given below. (2013)

(a) 1 only

(b) 1 and 2 only

(c) 2 and 3 only

(d) 1, 2 and 3

Ans. B Indian coals like any other Gondwana coals are of high ash content.

- Indian coals by their very nature are high in ash content but low in sulphur content. Indian coal has a high ash fusion temperature of about 1,500°C.
- India's domestic coal reserves have high ash content—up to 40 to 45 percent. As a result more coal is required to generate one kilowatt hour of electricity. This in turn means higher emissions.

- Sulphur content of Indian coal is, in general, around 0.5 – 1%. Studies show that the sulphur content in Indian coal is hardly 0.7%, which is not uniformly distributed in the coal but concentrated in the pyritic bands which are visible to the naked eye. The best solution to get rid of sulphur from Indian coal is to wash it
- Ash fusion temperature of Indian coal is generally high.

Q.) In which of the following regions of India are shale gas resources found ?

1. Cambay Basin

2. Cauvery basin

3. Krishna-Godavari Basin

Select the correct answer using the code given below: (2016)

(a) 1 and 2 only

(b) 3 only

(c) 2 and 3 only

(d) 1, 2 and 3

Ans. D To strengthen its hydrocarbon resource base, India has identified six basins as areas for shale gas exploration: Cambay (Gujarat), Assam-Arakan (North East), Gondwana (Central India), Krishna Godavari onshore (East Coast), Cauvery onshore, and Indo-Gangetic basins.

- Shale Gas can emerge as an important new source of energy in the country. India has several Shale formations which seem to hold Shale Gas and oil. The Shale Gas formations are spread over several sedimentary basins such as Cambay, Gondwana, Krishna-Godavari and Kaveri on land.
- Shale gas is an unconventional source of energy found in non-porous rocks. It has become an important source of natural gas in the US.

Q.) Consider the following statements:

1. In India, State Governments do not have the power to auction non-coal mines.

2. Andhra Pradesh and Jharkhand do not have gold mines.

3. Rajasthan has iron ore mines.

Which of the statements given above is/are correct? (2018)

(a) 1 and 2

(b) 2 only

(c) 3 only

(d) 1 and 3

Ans. C

- Statement 1 is incorrect: State governments can auction Non-coal mines.
- Statement 2 is incorrect: Both Jharkhand and Andhra Pradesh have gold mines.
- Statement 3 is correct: Bhilwara in Rajasthan has an iron ore mine.
- In India, State Governments do not have the power to auction non-coal mines. As per the new mining law – Mines and Minerals (Development and Regulation) Amendment Act, 2015 – which came into effect from January 2015, the non-coal mines have to be

auctioned by the respective state governments. The first ever non-coal auction in the country began in December, 2015.
- Andhra Pradesh and Jharkhand do not have gold mines. Rajasthan has iron ore mines.
- Rajasthan is the richest state in terms of availability and variety of minerals in the country and produces about 57 different minerals. Rajasthan is the sole producer of lead & zinc ores, selenite and wollastonite.
- In the State copper deposits are located in the districts of Jhunjhunu, Sikar, Sirohi, Udaipur, Bhilwara, Ajmer, Alwar, Bharatpur, Chittorgarh, Rajsamand and Dungarpur. With resources of 809.09 million tonnes, the State possesses a substantial share of 54% of the total resources of the country. The State is the second leading producer of copper concentrate contributing 41% of the nation's output.

Q.) With reference to India, consider the following statements?

1. Monazite is a source of rare earths.

2. Monazite contains thorium.

3. Monazite occurs naturally in the entire Indian coastal sands in India.

4. In India, Government bodies only can process or export monazite.

Which of the statements given above are correct? (2022)

(a) 1, 2 and 3 only

(b) 1, 2 and 4 only

(c) 3 and 4 only

(d) 1, 2, 3 and 4

Ans. (b)

- Monazite is a mineral mainly containing rare earths and thorium – a prescribed substance to be handled by the Department of Atomic Energy (DAE).
- Accordingly, Indian Rare Earths Ltd. (IREL) which is wholly owned by the Govt. of India, under the administrative control of the Dept. of Atomic Energy (DAE), utilises monazite mainly for production of rare earth compounds, and thorium, as needed in the Department of Atomic Energy.
- In-situ monazite established by AMD so far are present in Odisha, Andhra Pradesh, Tamil Nadu, Kerala, West Bengal, and Jharkhand.

Q.) Ilmenite and rutile, abundantly available in certain coastal tracts of India, are rich sources of which one of the following? (2023)

(a) Aluminium

(b) Copper

(c) Iron

(d) Titanium

Ans: (d) Ilmenite is represented chemically as FeO.TiO2 and rutile as TiO2. They are both ores of Titanium.

Q.) Consider the following statements: Statement-I: India, despite having uranium deposits, depends on coal for most its electricity production. Statement-II: Uranium, enriched to the extent at least 60%, is required for the production of electricity.

Which one of the following is correct in respect of the above statements? (2023)

(a) Both Statement-I and Statement-II are correct and Statement-II is the correct explanation for Statement-I

(b) Both Statement-I and Statement-II are correct and Statement-II is not the correct explanation for Statement-1

(c) Statement-I is correct but Statement-II is incorrect

(d) Statement-I is incorrect but Statement-II is correct

Ans: C

- India gets around 75% of its power from thermal power plants.
- Uranium enriched to 5% is enough for nuclear power production.

Q.) Which of the following statements about Rare Earth Elements (REEs) and Critical Minerals is/are correct?

1. Modern technological innovations including Artificial Intelligence, robotics and space exploration extensively utilise Rare Earth Elements (REEs).

2. China has the highest share in mining of REEs followed by India.

3. The Government of India launched the National Critical Mineral Mission (NCMM) in 2025 to establish a robust framework for self-reliance in the critical mineral sector.

4. Rare Earth Elements are a set of 13 metallic elements.

Select the answer using the code given below:

(a) 1 and 3 only

(b) 3 only

(c) 1, 3 and 4

(d) 1, 2 and 4

Ans: (a)

Explanation: India isnt 2nd highest. REE=17 elements.

Q.1) Consider the following pairs :

National Highway	Cities connected
NH4	Chennai and Hyderabad
NH6	Mumbai and Kolkata
NH15	Ahmedabad and Jodhpur

Which of the above pairs is/are correctly matched ? (2014)

(a) 1 and 2 only

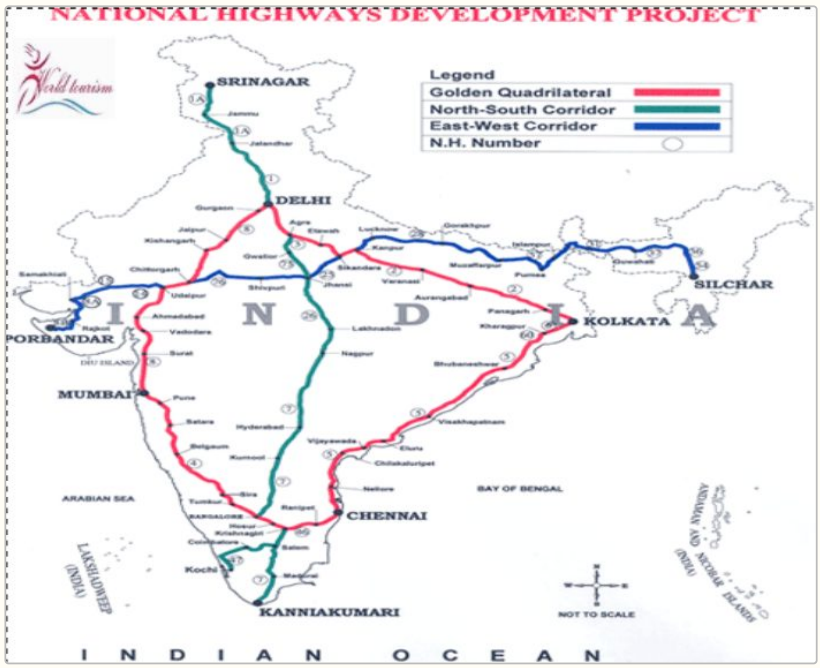
(b) 3 only

(c) 1, 2 and 3

(d) None

Answer: D National Highway 4 or NH 4, It is of 230.7 km of length.

- This highway is known as Andaman Trunk Road (The Great Andaman Trunk Road).
- A stretch of national highway from Mumbai to Pune to Hubli to Bangalore to Chennai was earlier called NH 4 before renumbering of national highways in the year 2010.
- The former NH 4 is now renumbered as NH 48.
- National Highway 6, is a national highway in India running from Jorabat in Meghalaya to Selling in Mizoram.
- The highway passes through the Indian states of Meghalaya, Assam, and Mizoram.
- National Highway 15 (NH 15) starts from Baihata in Assam and terminates at Wakro in Arunachal Pradesh.



Q.2) In which of the following activities are Indian Remote Sensing (IRS) satellites used?

1. Assessment of crop productivity

2. Locating groundwater resources

3. Mineral exploration

4. Telecommunications

5. Traffic studies

Select the correct answer using the code given below (2015)

(a) 1, 2 and 3 only

(b) 4 and 5 only

(c) 1 and 2 only

(d) 1, 2, 3, 4 and 5

Answer: A India's remote sensing programme under the Indian Space Research Organisation (ISRO) started off in 1988 with the IRS-1A, the first of the series of indigenous state-of-art operating remote sensing satellites, which was successfully launched into a polar sun-synchronous orbit on March 17, 1988, from the Soviet Cosmodrome at Baikonur.

- Data from Indian Remote Sensing satellites are used for various applications of resources survey and management under the National Natural Resources Management System (NNRMS). Following is the list of those applications:
  - Space Based Inputs for Decentralised Planning (SIS-DP)
  - National Urban Information System (NUIS)
  - ISRO Disaster Management Support Programme (ISRO-DMSP)
  - Biodiversity Characterization at landscape level- http://bis.iirs.gov.in
  - Preharvest crop area and production estimation of major crops.
  - Drought monitoring and assessment based on vegetation condition.
  - Flood risk zone mapping and flood damage assessment.
  - Hydro-geomorphological maps for locating underground water resources for drilling wells.
  - Irrigation command area status monitoring
  - Snow-melt run-off estimates for planning water use in downstream projects
  - Land use and land cover mapping
  - Urban planning
  - Forest survey
  - Wetland mapping
  - Environmental impact analysis
  - Mineral Prospecting
  - Coastal studies

Q.3) Consider the following pairs:

Port	Well known as
Kamarajar Port	First major port in India registered as a company
Mundra Port	Largest privately owned port in India
Visakhapatnam	Largest container port in Port India

How many of the above pairs are correctly matched? (2023)

(a) Only one pair

(b) Only two pairs

(c) All three pairs

(d) None of the pairs

Answer: B

- Kamarajar port at Tuticorin is the first corporate port of India, registered under the Companies Act. It started in 2001.
- The Mundra Port owned by Adani Ports is the largest private port of India. It has also overtaken JNPT port to become the largest container port of India.

🔤 ***ABBREVIATION COLUMN (full forms as the handout gives them where it does; the one-line meanings are plain-language aids, verify)***

AAI = Airports Authority of India
AI = artificial intelligence
ALCO = NALCO = National Aluminium Company Limited — expansion is general knowledge, verify
AMD = Atomic Minerals Directorate
AWD = Alternate Wetting and Drying — water-saving way of growing rice
BG = Broad Gauge — rail track width 1,676 mm; the most widely used in India
BMRCL = Bangalore Metro Rail Corporation Limited
CA = conservation agriculture
CaCO = $CaCO_3$ = calcium carbonate — what coral skeletons and shells are made of
CG = Chhattisgarh
CIL = Coal India Limited — the coal company; the handout mentions its plant at Singrauli
DAE = Department of Atomic Energy
DFCCIL = Dedicated Freight Corridor Corporation of India Limited
DFCs = dedicated freight corridors — rail lines kept only for goods trains
DMRC = Delhi Metro Rail Corporation
DR = Democratic Republic (of Congo)
EDFC = Eastern Dedicated Freight Corridor — freight line Ludhiana (Punjab) to Dankuni (near Kolkata)
EU = European Union
FBRs = Fast Breeder Reactors — plutonium-fuelled reactors (Kalpakkam)
FeO = iron oxide
FTAs = Free Trade Agreements — pacts that let firms of one country set up plants in the other
GAIL = Gas Authority of India Limited
GPS = Global Positioning System — satellite navigation
GW = gigawatt — 1,000 megawatts
HP = Himachal Pradesh
HQ = headquarters
HYV = high-yielding variety — seeds of the Green Revolution
HYVs = High-Yielding Varieties — seeds introduced with fertilizers and irrigation in the Green Revolution
HZL = Hindustan Zinc Limited
IFFCO = Indian Farmers Fertiliser Cooperative
IoT = Internet of Things
IREL = Indian Rare Earths Limited
IRS = Indian Remote Sensing (satellites)
ISRO = Indian Space Research Organisation
IWAI = Inland Waterways Authority of India
JNPT = Jawaharlal Nehru Port Trust — Nhava Sheva
KAMINI = Kalpakkam Mini reactor — experimental thorium (U233) reactor at Kalpakkam
KAMINI-Kalpakkam = Kalpakkam Mini reactor — experimental thorium (U233) reactor at Kalpakkam
MDRs = Major District Roads — roads connecting rural areas to the district
MG = Meter Gauge — narrower rail track, 1,000 mm
MMDR = Mines and Minerals (Development and Regulation) Act
MNRE = Ministry of New and Renewable Energy
MORTH = Ministry of Road Transport and Highways — oversees the roads programme
MoRTH = Ministry of Road Transport and Highways
MP = Madhya Pradesh
MRI = Magnetic Resonance Imaging — medical imaging that uses rare-earth contrast agents
MW = megawatt
NCMM = National Critical Mineral Mission
NCR = National Capital Region
NGOs = non-governmental organisations
NH = Northern Hemisphere (in highway lists: National Highway)
NHAI = National Highways Authority of India
NNRMS = National Natural Resources Management System
NPA = JNPA = Jawaharlal Nehru Port Authority
NUIS = National Urban Information System
NZ = New Zealand
OTEC = ocean thermal energy conversion — power from the temperature gap between warm surface and cold deep water
PHWR = Pressurised Heavy Water Reactor
PHWRs = Pressurized Heavy Water Reactors — natural-uranium-fuelled reactors
PMGSY = Pradhan Mantri Gram Sadak Yojana — rural roads scheme
PPP = public-private partnership
PPPs = Public-Private Partnerships — private firms invest in port infrastructure
PVC = polyvinyl chloride — a plastic
PWDs = Public Works Departments
PYQ = previous year question
REEs = rare earth elements
SC = Supreme Court
SCCL = Singareni Collieries Company Limited
SIS-DP = Space-based Information Support for Decentralised Planning
SPM = Syama Prasad Mookerjee Port — the Kolkata-Haldia port; expansion is general knowledge, verify
SPVs = Special Purpose Vehicles — autonomous bodies formed to build and run metro rail
SRI = System of Rice Intensification
SW = south-west
TiO = TiO_2 = titanium dioxide
TN = Tamil Nadu
TV = television
TVs = Televisions — electronics that use rare earth elements
UCIL = Uranium Corporation of India Limited
UK = United Kingdom
UPO = unrefined palm oil
US = United States
USA = United States of America
USSR = Union of Soviet Socialist Republics

WB = West Bengal
WDFC = Western Dedicated Freight Corridor – freight line Dadri (UP) to JNPT
WDs = Western Disturbances
WWII = Second World War
ZBNF = Zero Budget Natural Farming